# *A Player's Guide*

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The Markdown source of this guide lives next to the PDF in the mod's repository, at docs/player_guide.
Corrections and screenshots are welcome as issues or pull requests.

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CHAPTER 1

# Introduction

Vic3TimelineExtended carries Victoria 3 past its 1936 end date, through the twentieth century and into a speculative future; a campaign now runs to 2136. It adds seven eras of technology, new goods and buildings, over a hundred laws, and a set of systems that run alongside the base game: a banking cycle with monetary policy, a construction market, climate change, decolonization, nuclear weapons, a United Nations, a space race, cultural and covert competition between powers, and social movements. Most of these systems have their own journal entry, and most can be switched off with a game rule.

The mod adds one button to the sidebar, under Map List, in every game. It opens the Timeline Extended window: first the Nation tab, then a tab each for the space race, your Grand Monuments and your colonial empire, the same panels as their journal entries, and one for the seabed. Two more rows of tabs hold the mod's tabs in the Budget, Market, Population, Society, Diplomacy and Military panels, the same as there, so every system is in one place. Full Screen, at the window's top right, spreads it across the screen, each tab in columns; Side Panel, at its top left there, brings it back to the width of the other panels. The tab you chose stays chosen when you switch.

Each system has a Help page that explains how it works, with a heading for each topic. The "?" opens it: at the right of the tab's header, and at the top right of the system's journal entry and of a state's Demographics subtab. Press Escape or click outside the page to close it.

National Info, the base game's Census Data window under a new name, shows another country's panels. Open it from the foot of that country's Politics tab, or for your own country from the Population panel. A row of tabs over the window holds Pops, the base game's list, then a tab for each of the mod's systems a country shows the world, Demographics and the Seabed among them. For another country a tab shows its figures and none of its buttons; for your own it is the same tab as in the Timeline Extended window, buttons included. A tab is grayed where its system isn't running in that country, and the tab you chose stays chosen when you open another country's window, where it runs. Another country's charts open full size the same way, and an empty chart there says when the country keeps no history: only played countries and major powers do.

This guide explains what each system does and how to play it. It assumes you know the base game: pops, interest groups, laws, markets, construction and diplomatic plays are not explained here except where the mod changes them.

## 1.1 How to use this guide

Each chapter covers one area of the game and can be read on its own. Start with the chapters for the systems you meet first. In an 1836 start the construction market changes how building works from the first day (Economy and construction), and the banking cycle begins once you have researched Stock Exchange and have an Urban Center of level 5 or more (Banking and monetary policy). Most later systems unlock with technology, so you can read about them as they approach.

Numbers in this guide are there to help you plan: thresholds, tiers, durations and caps. They are rounded, and they can change between versions of the mod. The game's tooltips show the exact values for your country, and many of the mod's journal entries and panels break their numbers down line by line when you hover over them.

Many panels chart a system's history month by month or year by year. The round button beside a chart's title opens that chart alone, across most of the screen, with the same range buttons and hovers. Escape or a click outside closes it.

The Markdown source of this guide is kept in the mod's repository, and the chapters can be read there as web pages.

## 1.2 Setting up a game

Two things need attention before your first campaign: where the mod sits in your load order, and which of its game rules you want.

### Installing the mod

Subscribe to the mod on the Steam Workshop, or copy it into your Victoria 3 `mod` folder, then enable it in a playset in the launcher. The mod overrides the base game's ideologies, a number of its production methods, the state-region map files and several interface panels, so other mods that change those will conflict with it. Put Vic3TimelineExtended last in the load order so its versions win. Mods that only add content usually work alongside it.

Start a new campaign with the mod enabled. Don't add it to, or remove it from, a game already in progress. In multiplayer, every player needs the same version of the mod.

### Game rules

The mod adds nineteen game rules to the game setup screen. Each one turns a system on or off, and a few offer a reduced version of the system. You choose them when you start a campaign, and outside Ironman you can change them later.

Turning a system off hides its journal entry and stops its events, but the technologies, laws and buildings connected to it stay in the game and keep their ordinary effects. With the Banking System off, for example, the financial regulation laws still change Urban Center output, investment, innovation and government dividends; only their effects on the banking cycle go.

The mod is balanced with its default systems on, and switching one off removes its costs along with its benefits. A choice whose drawback lives in a disabled system loses that drawback: a financial regulation law that trades faster growth for a more volatile banking cycle is simply better with the Banking System off. Expect some laws, technologies and policies to become obvious picks when you turn a default system off.

| Rule | Default | Settings and what they control | Chapter |
|---|---|---|---|
| Banking System | Enabled | *Enabled*: the banking cycle with the full monetary-policy layer (policy rate, inflation, exchange rate, currency pegs and shared currencies). *Simplified*: the banking cycle only; you still borrow at your own interest rate but can't set it. *Disabled*: no banking cycle. | Banking and monetary policy |
| Free Market Construction | Enabled | *Enabled*: construction is a good bought on the market, and buildings use some as maintenance. *Without Retooling Costs*: as Enabled, but switching production methods doesn't raise that maintenance. *Without Maintenance*: construction is still a market good, but buildings don't consume it. *Disabled*: base-game construction sectors. | Economy and construction |
| Global Warming | Enabled | Greenhouse emissions, rising temperatures, climate events and climate policies. | Climate and pollution |

*Figure 1: The mod's rules on the Game Rules screen, with the Cultural Hegemony tooltip open. The text under each setting says what it turns on or off.*

| Rule | Default | Settings and what they control | Chapter |
|---|---|---|---|
| United Nations | Enabled | Founding and joining the UN, its votes, resolutions and agencies. | The United Nations |
| Nuclear Weapons | Enabled | Nuclear programs, arsenals, doctrine, crises, strikes and the nuclear taboo. | Nuclear weapons |
| Space Race | Full Colonies | *Races Only*: the space race milestones and their events, with colonies founded the month their fleet leaves and staying with their holder. *Simple Colonies*: as Races Only, and colony fleets travel for months or years and claim their site on arrival, colonies hold settlers sent from Earth and are built up with modules, Extraplanetary Bases work the room the near-Earth asteroids and colonies give, a colony can change hands by the Cede Colony treaty article, sold, given or demanded in a war, and colonies grow restless, can be abandoned or break away, and live on as free colonies. *Full Colonies*: as Simple Colonies, and Space Command builds warships that move between the planets, fight, | The space race |

| Rule | Default | Settings and what they control | Chapter |
|---|---|---|---|
| | | blockade colonies and seize or destroy them; an enemy fleet in Earth orbit stops your launches until you put up your reserve, and ground-to-orbit batteries fire on it. *Disabled*: no space race. | |
| Decolonization | Enabled | Colonial stability, the decolonization journal entry and its events. | Colonial empires and decolonization |
| Cultural Hegemony | Enabled | The competition for global cultural influence. | Cultural hegemony and covert warfare |
| Covert Warfare | Enabled | Intelligence agencies and covert operations against other countries. | Cultural hegemony and covert warfare |
| Social Movements | Enabled | The six social-movement journal entries (civil rights, human augmentation, morphological freedom, digital rights, mental health and post-scarcity) and their events. Feminism, LGBTQ+, religious, anti-war, transhumanist and environmental events fire either way. | Social movements |
| Internal Resettlement | Enabled | *Enabled*: the Settlement Authority and government resettlement programs. *AI Voluntary Only*: AI countries run only voluntary programs. *Disabled*: no resettlement. | States and population |
| Grand Monuments | Enabled | Grand Monuments, their dedications, the Monuments journal entry and contested monuments. | The extended timeline |
| Demographics | Full | *Full*: every state keeps a census of its people by age and sex, shown in a Demographics tab in the Population panel and National Info and a subtab on the state panel, and the census sets each state's births and deaths. *Display only*: the same census and panels, with births and deaths left to the base game. *Disabled*: no census and no tabs. Wealth Concentration keeps working. The census is experimental; a game saved before it was added keeps it off. | States and population |
| World War | Disabled | A journal entry for great powers that tracks ideological tension into a world war and its aftermath. | Military and war |
| Heir Education | Disabled | Educating your heir, and administrative, diplomatic and military aptitude traits for rulers and heirs. | Government, laws |

| Rule | Default | Settings and what they control | Chapter |
|---|---|---|---|
| | | | and characters |
| Universal Aptitude Traits | Disabled | Gives aptitude traits to every adult character, with or without Heir Education. | Government, laws and characters |
| Legislated Tax Code | Tax Code Disabled | Experimental. *Tax Code Enabled*: the five taxation laws can't be enacted; every country holds the Legislated Tax Code instead, and each tax rate changes only through a bill that enough interest groups commit to, taking effect on the 1st of its month. AI countries legislate their taxes the same way. *Tax Code and Customs Enabled*: as Enabled, and a country that owns its market sets its tariffs and subsidies by bill too. *Light Tax Code*: the taxation laws stay and set each tax's range; within it you set each tax at once from the Tax Code tab, your heaviest tax sets the tax level, and interest groups react to each change. AI countries keep choosing their tax level. *Tax Code Disabled*: base-game taxes. | Taxation (experimental) |
| Custom Religion Allowed | Not allowed | A journal entry that lets you design a religion of your own. | Government, laws and characters |

## 1.3 Reading the Budget breakdown

The Budget panel's Breakdown tab, also the Timeline Extended window's Budget Breakdown tab, shows weekly income and expenses as pie charts and expandable lists. Groups start collapsed; click an arrow to see their components. Entries are ordered from largest to smallest amount at each level. The tab is available under every game-rule setting. Colors and percentages match across the pies and lists. Hover a tax, tariff, welfare or interest amount for the Overview's graph of it, an institution for its share of the pool, and Administration for the pool itself. Investment Pool Transfer and the construction it funds are private spending, so they are excluded from the totals and Construction Goods.

Taxes groups the individual tax receipts. Military expands into Army and Navy, which each expand into Wages plus Materials and Support. Materials and Support includes logistics centers and naval fortifications, whose operating costs combine their own wages, goods and any slave upkeep. Barracks, conscription and naval administration stay in the native branch entries, while support buildings contribute their full upkeep. Navy also includes warship construction and maintenance. Shipping and diplomatic flows have their own expandable groups. A collapsed group occupies one pie slice; expanding it replaces that slice with its visible components.

Administration groups your institutions and General Administration. Government Administration's operating costs are divided between your institutions and General Administration. Institutions receive the share corresponding to the bureaucracy they use, then divide it in proportion to their current levels. General Admin-

istration covers the remainder, including unused capacity. Other civil buildings, the military, construction and welfare have their own rows.

Recurring costs from Banking, Covert Actions, Cultural Hegemony, the United Nations and other journal systems appear under Program Costs, and so does the Space Program building's running cost, in the Space Race row. Expand it and hover over a system to see its current sources and amounts. One-time treasury payments and costs in resources such as innovation are outside this weekly money breakdown. The Banking row's hover also names what the Foreign Exchange Surrender Requirement buys for the central bank's gold each month, which stays outside the weekly budget.

The charts show positive component amounts; negative adjustments stay in the list. Expanding a group preserves the chart total, including when costs and refunds offset each other inside that group. The "?" in the tab's header opens the Help page with the accounting details.

## 1.4 The Nation tab

The Timeline Extended window opens on its Nation tab, unless the tab you last chose is open. The Nation tab gathers what your country carries for good, and it is open in every game from the first day.

At the top, five boxes: Legislative Override Capacity, Wealth Concentration and the score it is heading for, Construction Cost, Global Technology, and Judicial Independence and its ceiling. The first names the strongest Forceful Legislation option open to you, which also depends on the authority you produce.

Below them are five sections: Governance, Society, Economy, Science and Space, and International. Each starts with its scores, if it has any. Legislative Override Capacity's figure names every source on hover, and the lines on its bar mark where Forceful Legislation's options open, at 0, 2 and 4. Once a United Nations exists, International starts with your place in it: a member, a permanent member, suspended, or outside. Once you have joined it adds your International Standing, kept if you leave, and while you are a member your UN Dues, with any months in arrears.

Under the scores, each section has a row for each kind of effect the mod's permanent effects give your country, grouped under small headings: "Arable Land: +45%" under Production, Prestige under Standing Abroad, each interest group's approval under Interest Groups. The figure is your country's whole total, and hovering it lists every source: the technologies, laws, institutions and principles that add to it as well as the mod's effects, such as agricultural diffusion, space race milestones, demographic policies and the UN conventions you are bound by. Hover the name for what the effect does. A row appears once one of those permanent effects adds to it. A state, building or interest group row is what your country gives each of them; their own panels add their local sources.

Where These Come From, collapsed at the bottom, lists the permanent effects themselves under the section whose systems granted them, and its heading gives how many there are. Hover one to see what it does at its current size.

The Modifiers list in the country panel's Information tab leaves these effects out, and a line under it gives how many it left out. On your own country an arrow beside that line opens the Nation tab; on another country's it opens that country's Nation tab in National Info, which shows all of it: the scores, the rows, the permanent effects and the UN lines. The list keeps the timed effects, such as what an event has just done to you, and a few others: Nuclear Power, and what a foreign operation is doing to you. A system's weekly costs are in the Budget panel's Breakdown tab. Tooltips that break down a figure, such as legitimacy, still name every source.

## 1.5 How the guide is organized

The chapters follow the game's own areas rather than the order systems appear.

| Chapter | Covers |
|---|---|
| The extended timeline | The seven new eras, their technologies, new goods and buildings, company buildings, wonders and megaprojects. |

| Chapter | Covers |
|---|---|
| Economy and construction | The construction market, construction costs, living-standard expectations, transport goods, the Strategic Reserve and wartime demand for munitions. |
| Banking and monetary policy | The banking cycle, crashes and contagion, the policy rate, inflation, exchange rates and international monetary arrangements. |
| Taxation (experimental) | The Legislated Tax Code rule: the Light Tax Code's rates within your taxation law; drafting tax bills, winning interest groups over, promises, passed bills and legislated customs. |
| Government, laws and characters | New laws and law groups, ministries, political movements and parties, elections, heir education, custom religions and state collapse. |
| Social movements | The movement journal entries, from civil rights to post-scarcity, and the movements carried by events. |
| States and population | Migration crowding, dynamic homelands, cultural acceptance, tourism, world city rankings and internal resettlement. |
| Diplomacy | New treaty articles, diplomatic play escalation, irredentism and reunification, power blocs and formable countries. |
| The United Nations | Founding the UN, its authority, the Security Council, votes, resolutions, conventions and missions. |
| Cultural hegemony and covert warfare | Soft power and intelligence operations. |
| Colonial empires and decolonization | Colonial stability, independence and what follows it. |
| Military and war | Combined arms, new units and ships, military bases and the World War journal entry. |
| Nuclear weapons | Building a bomb, doctrine and posture, crises, strikes, the nuclear taboo and disarmament. |
| Climate and pollution | Emissions, warming, climate policy and state pollution. |
| The space race | The milestones from suborbital flight to colonizing the solar system. |
| Quick reference | When each system appears, the journal entries at a glance, and a glossary. |
| Appendix: social movement details | The numbers and event lists behind the social movements chapter. |
| Appendix: events | The event lists behind the chapters, from resettlement to the space race, with when each event fires and what its options do. |
| Appendix: reference lists | The longer lists the chapters summarize: wonders, banking tools, government types, combat units, ships, ship modifications, mobilization options and military base production methods. |

## 1.6 Reporting problems

A mod this size has bugs. Report them on the Steam Workshop page or, better, on the issue tracker. The most useful report says what you did, what you expected and what happened, and attaches a save and the game's `debug.log` and `error.log` (in the `logs` folder under `Documents/Paradox Interactive/Victoria 3`). If a mechanic is confusing even when it works as intended, that is worth reporting too.

Three systems have had the least play-testing: the monetary-policy layer of the banking system, the United Nations, and nuclear deterrence and crises. This guide describes what their scripts are written to do. Where the game behaves differently, either the game or the guide is wrong, and a report helps fix whichever it is.

CHAPTER 2

# The extended timeline

The base game ends in 1936 after five eras of technology. This mod moves the end date to 2136 and adds eight more eras, numbered 6 to 13, with 191 new technologies. The base game has 179, so the technology tree roughly doubles. The new eras bring twelve new goods, dozens of new buildings and production methods, a flagship building for every company, 37 wonders and seven megaprojects. None of this is behind a game rule: the eras and everything they unlock are always in the game, even when you switch off the systems that some of their technologies start. The one exception is the International Space Station wonder, which needs a Moon landing from the space race. The repeatable Grand Monument has a game rule of its own.

## 2.1 Eras six to thirteen

The technology tree continues past the base game's fifth era. Eras 6 to 9 follow the twentieth century, and eras 10 to 13 run from the present into the future. The game shows eras only by number. The years for eras 6 to 9 are the periods the mod's technologies were chosen to fit, so treat them as a rough guide. The later eras are told apart by how far off their technologies are. Era 11's are on today's roadmaps. Era 12's are known to be possible, with no near-term path to them. Era 13's need breakthroughs nobody yet knows how to plan for.

*Figure 2: Part of the production tree in eras 6 to 8, from Modern Materials and Television down to Cellular Networks and Personal Computers. The numeral under each icon is its era.*

| Era | Rough period | Research cost per technology | Technologies | Broad theme |
|---|---|---|---|---|
| 6 | 1919–1945 | 20,000 | 38 | Synthetics, appliances, aviation, radar, rockets and the atomic bomb |
| 7 | 1946–1967 | 65,000 | 29 | Transistors, nuclear power, jets, the Green Revolution, missiles and satellites |
| 8 | 1968–1987 | 115,000 | 21 | Computer networks, robotics, containerization, pharmaceuticals, stealth |
| 9 | 1988–2012 | 210,000 | 28 | The internet, renewable energy, biotechnology, globalization, drones and cyber defense |
| 10 | the present day | 450,000 | 20 | Machine learning, electric vehicles, reusable rockets, hypersonic and cyber weapons |
| 11 | the near future | 800,000 | 26 | AI agents and robots, advanced materials, fusion power, deep-sea and asteroid mining, genetic engineering, augmentation |
| 12 | the medium-term future | 1,500,000 | 16 | General AI, nanofabrication, compact fusion, orbital industry, space colonization, life extension |
| 13 | the speculative future | 3,000,000 | 13 | Molecular assemblers, antimatter, the space elevator, immortality, mind uploading, post-scarcity |

## Research costs in the new eras

Each technology costs its era's base research cost, before the usual modifiers. The base game's last era costs 17,500 per technology; era 6 is only a small step up at 20,000, but era 7 more than triples it, and an era 13 technology costs 150 times as much as an era 6 one. The base game's surcharge for researching ahead of unresearched technologies in the same tree still applies, and it grows with the number of eras between them.

Your innovation has to grow to match. Many of the new technologies raise the weekly innovation cap by 10 to 300 each, adding between 250 and 1,000 per era through era 11, 500 in era 12 and 200 in era 13. Self-Driving Laboratories (era 11) and Artificial General Intelligence (era 12) give the largest steps. Five wonders raise it further, and new University production methods (National Labs, Academic Computing and AI Assisted Research) raise the innovation each university produces.

## What each new era brings

Era 6 opens synthetic industry and the consumer economy. Bergius Process and Isoprene unlock Synthetic Fuel Works and Synthetic Rubber Works, Personal Appliances unlocks Consumer Appliance Industries, Commercial Aviation the Airport, and Public Works Programs the Hydro Plant. Nine of the era's technologies each open a new kind of mine, such as Modern Tools (Manganese Mine). Radar, Rocketry, Nuclear Weapons, Combined Arms, Keynesian Economics (the Fiat Money law), Intergovernmental Organizations and Decolonization are here too.

Era 7 is the postwar boom. Transistor Technology unlocks the Electronic Components and Semiconductor Industry, Nuclear Energy the Nuclear Plant, and Green Revolution modernizes farms and plantations (Modern

Farming, Mechanized Farm) and unlocks the Artificial Sweeteners Plant. Guided Missiles unlocks the Space Program building, and Space Exploration, Satellite Communications and Intercontinental Ballistic Missiles follow. Plastic Mass Production, Modern Urban Planning, Pollution Control and the Civil Rights Movement round out the era.

Era 8 brings computers and automation. Computer Networks unlocks the Software Industry and Network Infrastructure, Industrial Robotics the Robotics Industry, and Modern Pharmaceuticals the Rational Drug Design method for Pharmaceutical Industries (Pharmaceutical Industries and Drugs). Gene Splicing opens the Ban on Genetic Modification, Corporate Genetic Licensing and Open-Source Genetics laws. Containerization, Fiber Optics, Cellular Networks, Microprocessor, Stealth Technology and Video Games are also here.

Era 9 is the internet age. World Wide Web opens the internet laws, Clean Energy Technologies unlocks the Renewable Energy Plant, Hydraulic Fracturing adds fracking to oil rigs, and Biotechnology brings GMO plantations and two synthetic plants. Globalization, Social Media, E-Commerce, Cybersecurity, Missile Defense Systems and Network Centric Warfare are also in this era.

Era 10 is the present day: Machine Learning, Generative AI, Additive Manufacturing, Electric Vehicles, Internet of Things, Reusable Rocketry, Hypersonic Weapons, Cyber Warfare, Universal Basic Income and Mental Health Awareness. Cislunar Infrastructure opens the Space Program's Solar Colonization method, and with it colonies on the Moon after a Moon Landing, and In-Situ Resource Utilization the Extraplanetary Base.

Era 11 is the near future. Modern Material Science unlocks the Advanced Material Fabricator, Abyssal Plain Mining the Deep-Sea Mine, Asteroid Mining the Extraplanetary Base's Asteroid Mine method, Fusion Power the Fusion Plant, and Advanced Fission a second, more productive base method for the Nuclear Plant. Genetic Engineering unlocks the Carbon Conversion Works and Integrated Biorefinery, Lab-Grown Food the Cultured Meat Plant and Cultured Produce Facility, and Synthetic Biology adds bioengineered production methods such as Designed Crops. Brain-Computer Interfaces and Human Augmentation bring augmentation laws. Era 11 also brings the AI that works on its own: AI Agents (the Automated Bureaucracy law and Fully Automated Content Creation), Self-Driving Laboratories (the University's AI Assisted Research), Humanoid Robotics (Robotic Service Providers and Neuromorphic Computing) and Autonomous Weapons (Robotic Soldiers). Artificial General Intelligence in era 12 needs Self-Driving Laboratories and Humanoid Robotics.

Era 12 is the medium-term future. Artificial General Intelligence adds AI-managed production methods to many industries, and Orbital Manufacturing and Advanced Nanofabrication add a late tier to many more. Compact Fusion Reactors brings the fusion torch, Space-Based Solar Power and Orbital Weapon Platforms each unlock a megaproject, and Bioenhanced Soldiers brings the Mandatory Augmentation law. Rejuvenation Therapies halves the pace of aging for those in longevity care, and Post-Work Society opens the Post-Scarcity Transition.

Era 13 is the speculative future. Molecular Assemblers and Programmable Matter add a final tier to many industries. Space Elevator, Antimatter Production, Mind Backups, Molecular Assemblers and Telepathic Communities each unlock a megaproject. Self-Directed Evolution brings the Clade Citizenship law, and Biological Immortality and Post-Scarcity Economy are among the last technologies in the tree.

Some of the mod's buildings arrive earlier, with base-game technologies: the Aerospace Industry with Military Aviation, the Highway with Paved Roads, and the Tourism Industry with Romanticism.

## Global technological development

The further the world's leading powers advance, the faster everyone else catches up. Once a year, every country checks a list of 55 mid-century and later technologies, from Television and Cryptography to Post-Scarcity Economy. Each one that any great power has researched adds a bonus to technology spread, the trickle of innovation that pulls you toward technologies others already have. The bonuses add up, and the later the technology, the more it is worth: the earliest tier adds a small step, era 12's (Artificial Intelligence, Space Colonization and their peers) ten times as much, and era 13's two (Biological Immortality and Post-Scarcity Economy) fifteen times. Only great powers count, so a technology first reached by a minor country does not move the world's pace until a great power follows. One law adds to the bonus too: each major power or greater that holds Open Weights (Who builds frontier AI) adds 100, up to 300.

The bonus appears on your country as the Global Technological Development modifier, and every country gets the same one, recognized or not. It is a catch-up mechanism, not a research boost for the leaders. A backward country gains the most from it, because spread only works on technologies you lack. The leaders gain little, since spread only ever feeds technologies someone has already researched. The further ahead the great powers pull, the larger the bonus, but it closes the gap less than that suggests. The bonus is a flat amount of spread, and technology costs climb much faster than its tiers do: an era 13 technology costs 150 times as much as an era 6 one. So the same bonus pays for less of each technology as the eras go on. The tooltip on the modifier shows your current figure. It sits alongside the usual sources of spread (literacy, laws and power bloc principles) and is separate from agricultural diffusion, which grants arable land rather than research.

## Agricultural diffusion

Five farming technologies spread around the world once someone invents them: Nitrogen Fixation (a base-game technology), Modern Chemical Processes, Green Revolution, Gene Splicing and Biotechnology. The first country to research one gets an event naming it the pioneer, a permanent bonus to arable land, and a decaying +5% prestige and authority bonus that lasts 20 years. Two weeks later every other recognized country gets an event and the same arable-land bonus, whether or not it has the technology. Each diffusion bonus is half the technology's own arable-land bonus: +15% for Nitrogen Fixation, Modern Chemical Processes and Gene Splicing, +25% for Biotechnology and +60% for Green Revolution. It stacks with the technology's bonus once you research it. Countries that appear later, through release or formation, receive the bonuses already in the world.

## Why some technologies lower education access

Four technologies take 10 points off education access in every state: Public Works Programs and Computing Machines in era 6, Digital Education in era 9 and Brain-Computer Interfaces in era 11. Literacy drifts toward education access, so with nothing else changed each one lowers the level your literacy settles at by 10 points, and literacy follows down over the next years. All four together take 40 points off.

This is deliberate. Each of these technologies raises the standard for what counts as literate: working with computers, networks and digital systems asks more of a person than reading a newspaper, so someone who counted as literate under the old standard may not under the new one. The technology tooltip shows the penalty. It also keeps education relevant in the late game, because the gain has to be kept up as the bar rises.

Each of the four also raises the Schools institution's maximum investment by one level. Funding schools and the laws that raise education access are how a country holds its literacy against the drop.

AI Tutors (era 11, after Digital Education) works the other way. A personal tutor for every child adds 10 points of education access, speeds literacy growth and raises qualifications by 10%, so pops retrain for new jobs sooner. With all five researched, education access ends 30 points down rather than 40.

What the tutors teach is the catch. AI Tutors raises the Ministry of Thought Control's maximum investment by one level, and two events can follow, each at most once. Under Unregulated Internet, a private tutoring platform is caught steering its pupils' politics. Under State-Controlled Internet, or with a Ministry of Propaganda or of Thought Control, the state has written the syllabus every tutor will teach. Each event trades loyalty against trust and research. Controlling the lessons turns a small share of your people loyalist every month for ten years, fading, at a cost in legitimacy and research speed. Opening them up gives legitimacy and research speed instead, pleases the Intelligentsia, and radicalizes whoever lost control of them: the platform's capitalist owners or the ministry's bureaucrats.

## Technologies that open the mod's systems

Several of the new technologies first bring a system from another chapter into view. Each system also needs its game rule, and some have further conditions.

| Technology | Era | What it opens | Chapter |
|---|---|---|---|
| Intergovernmental Organizations | 6 | The United Nations journal entry; the Power Bloc Headquarters and the Peace Palace | The United Nations |

| Technology | Era | What it opens | Chapter |
|---|---|---|---|
| Decolonization | 6 | The colonial empire journal entry | Colonial empires and decolonization |
| Nuclear Weapons | 6 | The nuclear program | Nuclear weapons |
| Rocketry | 6 | The first space race milestone, for great and major powers | The space race |
| Combined Arms | 6 | The Gathering Storm, the world war journal entry, for great powers | Military and war |
| Civil Rights Movement | 7 | The civil rights journal entry | Social movements |
| Automated Surveillance or Cybersecurity | 9 | The digital rights journal entry | Social movements |
| Mental Health Awareness | 10 | The mental health journal entry | Social movements |
| Brain-Computer Interfaces or Human Augmentation | 11 | The human augmentation journal entry | Social movements |
| Post-Work Society | 12 | The post-scarcity journal entry | Social movements |
| Morphological Freedom | 12 | The morphological freedom journal entry and the Bodily Autonomy laws | Social movements |

Some society technologies also start occasional narrative events (Second Wave Feminism, Contraceptive Pill, Social Media and others), covered in Social movements.

## 2.2 New goods

The new industries trade in thirteen goods the base game doesn't have, and the mod renames twelve base-game goods to fit a longer timeline.

### Goods the mod adds

| Good | Category | Made by | Used for |
|---|---|---|---|
| Consumer Appliances | Luxury | Consumer Appliance Industries | The pops' Convenience need; inputs for some high-tech buildings |
| Electronic Components | Industrial | Electronic Components and Semiconductor Industry | The most widely used modern input: over a hundred buildings take it in their later production methods |
| Industrial Robotics | Industrial | Robotics Industry | Automation production methods in factories and mines |
| Software | Luxury | Software Industry | The Convenience need; digital and automation production methods |
| Digital Access | Luxury, local | Network Infrastructure | The Convenience need and many modern production methods; like Transportation it is only available in the state that produces it |
| Launch Capacity | Industrial | Aerospace Industry (with rocket production methods), Space Elevator, Antimatter Engine | Space programs, satellites, orbital production methods and megaprojects |
| Advanced Materials | Industrial | Advanced Material Fabricator, Nanofabrication Center | Late-game production methods and megaproject construction; base price 4,000, the most expensive good in the game |
| Construction Services | Industrial | Construction Sector | The construction market (Economy and construction) |

| Good | Category | Made by | Used for |
|---|---|---|---|
| Tourism | Luxury | Tourism Industry, National Park and others | The pops' Tourism need |
| Tech-Critical Metals | Industrial | Nickel and Cobalt, Lithium, Rare Earth Metals and Platinum Group Metals Mines; also the Deep-Sea Mine and Extraplanetary Base | Electronics, robotics, modern power plants and other high-tech production methods |
| Motor Ships | Industrial | Shipyards, from Advanced Submarine Technology | Later production methods for ports, fishing wharves and whaling stations, and the Deep-Sea Mine's tenders; the pops' Leisure need, as Steamers |
| Magnetic Drive Ships | Industrial | Shipyards, from Modern Material Science | The last tier of the same production methods; the Leisure need too |
| Grid Power | Industrial | Grid Transfer Stations, from a state's spare Electricity | Grid Transfer Stations, which turn it back into Electricity in short states of the same power grid (Economy and construction) |

The Convenience and Tourism needs, which wealthier pops fill with these goods, are covered in Pop consumption at high wealth.

### Renamed base-game goods

Several base-game goods now stand for a broader family of materials. They are the same goods underneath, so base-game buildings still make and use them.

| Base-game name | Name in this mod |
|---|---|
| Coal | Energy and Carbon Minerals |
| Iron | Structural Metals |
| Lead | Conductive and Base Metals |
| Sulfur | Chemical and Industrial Minerals |
| Fertilizer | Chemicals |
| Glass | Glass and Plastics |
| Opium | Drugs |
| Telephones | Wired Telecommunication Gear |
| Radios | Wireless Telecommunication Gear |
| Fine Art | Art and Entertainment |

Chemicals is the base game's fertilizer. Farms and plantations still take it, and dozens of modern industrial production methods now take it too. Four buildings are renamed to match: Fertilizer Plants are Chemical Plants, Electrics Industries are Wired Telecommunications Industries, Synthetics Plants are Synthetic Dyes Industries, and the Arts Academy is Creative Industry.

## 2.3 New buildings and production methods

The mod adds new buildings in families that follow the goods above, and extends most base-game buildings with production methods for the new eras.

### New mines and extraction

Fourteen new mines work deposits of specific minerals. Ten of them produce one of the renamed resource goods, adding supply to existing chains; the other four produce the new Tech-Critical Metals. Resource deposits are covered in New mineral deposits.

| Mine | Produces |
|---|---|
| Manganese, Chromium, Bauxite, Specialty Alloy Metal | Structural Metals |
| Copper, Precious and Minor Base Metal | Conductive and Base Metals |
| Nickel and Cobalt, Lithium, Rare Earth Metals, Platinum Group Metals | Tech-Critical Metals |
| Graphite | Energy and Carbon Minerals |
| Phosphate, Potash, Industrial Mineral and Salt | Chemical and Industrial Minerals |

Two late buildings mine several goods at once. The Deep-Sea Mine (Abyssal Plain Mining) can be built in any coastal state, and each level needs a seabed license (Seabed licenses). Like a fishing wharf, it loses up to three quarters of its throughput while its state is blockaded. Its ships come from the Support Vessels group: Motor Ship Tenders run it on Motor Ships, and Magnetic Drive Tenders, from Modern Material Science, run it on fewer Magnetic Drive Ships for £1,000 a level less each week at base prices. Switching an existing mine costs about £260,000 a level in retooling, around 5 years of that saving, so the switch pays back sooner while Motor Ships are expensive. The Extraplanetary Base (In-Situ Resource Utilization) costs 20,000 construction per level. Its staff is the same at any size: a large base needs no more workers than a small one. It starts on Regolith Processing, which makes Structural Metals, Tech-Critical Metals and Glass and Plastics at a thin margin: about £1,600 a level a week at base prices. Asteroid Mining brings the Asteroid Mine method, and Space Colonization the larger Mining Colony, or the Resort Colony production method, which produces art and tourism instead. Von Neumann Machines (era 13) brings Self-Replicating Miners, which build their copies from the ore they dig: against a Mining Colony, a level uses a third less Launch Capacity and mines a quarter more of each good. With the Space Race rule on Simple Colonies or Full Colonies its levels need room, from the near-Earth asteroids (with Asteroid Mining) and your colonies, and work at −75% past it (see Extraplanetary Base room).

The Deep-Sea Mine costs 5,000 construction per level, a quarter of the Extraplanetary Base's, and at base prices earns more per level: about three times as much in era 11, and still more than a Mining Colony after Advanced Nanofabrication. Most of an Extraplanetary Base's running cost is Launch Capacity, and two late production method groups each cut that bill by 30%: Transport in Space, whose Fusion Tugs need Compact Fusion Reactors (era 12), and Launch to Orbit, whose Tethered Launch needs the Space Elevator technology (era 13; not an elevator of your own). A Mining Colony with Fusion Tugs already earns about as much per level as an Advanced Deep-Sea Mine, and with both upgrades half as much again. Self-Replicating Miners with neither upgrade earn more than a Mining Colony with both, and the upgrades add less to them, since they launch less. With the Global Commons rule off, the Deep-Sea Mine has no licenses and keeps its older, smaller output.

## Seabed licenses

With the Global Commons rule on (the default), each Deep-Sea Mine level needs a seabed license. The seabed is split into five ocean basins, each holding a fixed pool of licenses. A coastal state belongs to the basin its sea lane opens onto.

| Basin | Licenses | Includes |
|---|---|---|
| Pacific | 300 | the Sea of Japan and the East and South China Seas |
| Atlantic | 250 | the Mediterranean, Black Sea, Baltic, North Sea, Caribbean and Gulf of Mexico |
| Indian | 200 | the Red Sea, Persian Gulf and Arabian Sea |
| Southern | 150 | Patagonia, the Cape of Good Hope, southern Australia and the Tasman Sea |
| Arctic | 100 | the Barents Sea, Greenland and Hudson Bay |

Every January the pools are shared out by claim weight among every coastal country. A country without Abyssal Plain Mining counts at three quarters of its weight, and its share waits for it: no one can build on it. Researching the technology first gives you a head start, then, and every other coastal country keeps a share for when it catches up. A country that has just researched it gets its share as claims the month after, and that sharing divides every pool again, so your own share can move in the middle of a year. In a basin where you

have coastal states, your weight counts those states twice, their port levels, and your naval power projection, scaled to about one unit a warship (the weight hover shows the divisor), in proportion to the coast you have there; an unincorporated state counts half, its ports in full. In any other basin it is a twentieth of that naval term, so you can claim far from home, but only a small share. Power projection rather than fleet size counts, so a modern fleet outweighs a larger obsolete one.

An event announces the seabed rush when the first coastal country researches Abyssal Plain Mining. After that, a notification tells you when a country wins its first share in a basin on your coast where you have one, and when a basin you have coast or a share in has all its licenses held. Both can be muted in the message settings, under Seabed Licenses.

A license is held by a level, built or queued, and stays with it for good. It belongs to the state the mine stands in and passes with it: conquer a mined state and its licenses come with it. Built levels are never taken away. If your share in a basin falls below what you hold there, you keep every level and can't take more licenses from that basin until you are back under it; levels you downsize, or that are destroyed, give their licenses back. A new level takes its own basin's license first. Once your claims there are used up it takes one from another basin and works at −25% throughput (Long-Haul Seabed Operations, shown on the mine), until a local claim frees up and replaces it on its own. The building's requirements list how many licenses you have to spare and which kind the next level would take.

### *Claims, protection and the holding fee*

What your share entitles you to and no level holds yet are your seabed claims. Each January divides them again, so a rival's naval buildup can take them. On Sponsor Contractors the January sharing leaves you at least three quarters of each basin's claims, so they fade by a quarter a year at most; on the other policies they are divided again in full. A sharing in the middle of the year, when a new claimant appears, honors January's protection without adding to it. A Scramble ends all protection. When the Law of the Sea comes into force after open years, protected claims can come out of the reserved half, as the real convention's pioneer investors kept the sites they had registered; they fade by a quarter a year like any other, so the reserved half refills.

Queuing a level turns a claim into a license at once, so you can hold a basin's claims by queuing far ahead. That costs a holding fee: a license on a level still unbuilt a year after it was queued pays 2% of a level's base output value a week (about 900), half the royalty, until the level is built. Building at a normal pace never pays it, whoever queued the level. Under the Regulated regime a party's fee goes to the same place as its royalties; anyone else's is simply an expense. The Budget Breakdown lists it under United Nations.

You can Relinquish a basin's claims from your own row in that basin's list of claimants: they are gone at once, so Sponsor Contractors' influence falls with them, and the next sharing divides them among the other claimants. You claim nothing there until you press Claim Again, which counts from the sharing after. Licenses on your levels are untouched.

### *Leasing seabed claims*

A Lease Seabed Claims treaty article, one for each basin, leases some of one country's own claims there to another, up to the number agreed. They pass at once, so the lessee can build before January. Each sharing divides them again, and the lease then takes them back from the lessor's new share, as far as it goes, until the licenses built on the lease and its unused claims make up the number. Licenses built on a lease are the lessee's for good, but they count against the lessor's share, not the lessee's, even after the lease ends, for as long as both countries take part in the sharing; meanwhile the lessee can lease that basin's claims again only from the same lessor. When the lease ends, its unused claims go back at the next sharing. The usual price is a Money Transfer the other way. The article can be enforced, so a war's winner can take a rival's claims as a lease. No country barred by the Antarctic Treaty can lease Southern claims or receive them, and Sponsor Contractors protects no leased claims. AI countries offer claims they are unlikely to use soon, and ask for them when they are short of claims and Tech-Critical Metals sell above base price.

An overlord can take its subject's unused claims with Administer Seabed Claims, an action like Raise Subject Payments: no approval, and not on a protectorate or a personal union. While it stands, the subject's claims pass to the overlord at once and after every sharing, with no limit, and the licenses the overlord builds on them count against the subject's share, even after the action ends, while both take part. The subject keeps its

own licenses. Its liberty desire rises as under Raise Subject Payments, and relations between the two fall. Only the overlord starts or ends it; when it ends, the subject's claims are its own again from the next sharing. AI overlords use it on subjects with claims to spare and liberty desire below 25, and end it at 50.

### *The Seabed tab*

The Seabed tab of the Timeline Extended window answers four questions. Its overview says how many more Deep-Sea Mines you can queue (and how many at home, without the long-haul penalty), your licenses and any holding fee, your long-haul levels, your unused claims and what next January can take of them, your policy and when it can next change, and the regime. The Ocean Basins table has a row for each basin: Free is your claims there, green where you have coast to build on at full throughput; Pool is a bar of the whole pool, from your mines and claims to rivals' mines and the rivals' claims a buildup could win; Weight is your share of the basin's claim weight; +1 license is about how much more naval power projection would win you one more license at the next sharing, a figure that is enough and often more than needed. Click a basin to see its claimants, largest share first, each with its mines, claims and share of the weight in columns, and your own row with its Relinquish button. A Seabed Licenses tile on every coastal state's panel shows that state's basin, the licenses its mines hold and how many are long-haul, your unused claims in its basin (what a new level there would take at full throughput) and what the state adds to your claim weight there. The tile's tooltip has the national figures. Once you have researched Abyssal Plain Mining, every other coastal state, whoever owns it, shows its basin in an Ocean Basin tile, so you can see which pool a port there would weigh in. Another country's National Info has a Seabed tab with that country's overview and Ocean Basins table; the policy buttons and the claimant lists, with Relinquish and Claim Again, are only on your own.

| Policy | Influence held | Claim weight | When to use it |
|---|---|---|---|
| Scientific Survey (default) | none | ×0.75 | You aren't building Deep-Sea Mines yet, or you build your claims as fast as you get them and need no protection. |
| Sponsor Contractors | 50, plus 0.5 for each unused claim | ×1, claims protected | You want licenses, and want to keep the claims you can't build on yet: the lawful route under the Law of the Sea. |
| Unilateral Licensing | 1.5 times Sponsor Contractors' figure, at least 150 | ×1, naval power projection counted twice | A large navy and little coast. Relations fall by 10 a year with each other unilateral claimant in a basin where you both have coast, and while the Law of the Sea regulates the seabed, a party that licenses unilaterally defects from it (The United Nations). |

A new policy holds its influence from the day you choose it; its weight counts from the next sharing. Any change holds your policy for five years, whichever way you change. Two changes are open sooner: Unilateral Licensing, once a rival in one of your home basins has switched to it since your last change, and Scientific Survey, while your influence is in deficit. Sponsor Contractors' cost has no cap, so claiming a whole basin is expensive; to cut it, build on the claims, relinquish a basin, or drop to Scientific Survey when you can. AI countries take up Sponsor Contractors when it would bring them a license and Tech-Critical Metals sell above base price, and more readily the more unused claims they hold. A few great powers, mostly those with poor UN standing or a rival claimant, choose Unilateral Licensing. The AI keeps to the five-year lock too.

## New industries

| Family | Buildings |
|---|---|
| Synthetics | The base game's Synthetics Plants split into twelve buildings, one per product: Synthetic Dyes Industries, Synthetic Clothes Industries, Synthetic Fuel Works, Synthetic Rubber Works, Pharmaceutical Industries, Carbon Conversion Works, Phenolic Resin Panels Plant, Artificial Sweeteners Plant, Cultured Meat Plant, Cultured Produce Facility, Beverage Concentrates Plant and Integrated Biorefinery. |

| Family | Buildings |
|---|---|
| Electrics | The base game's Electrics Industries split into Wired Telecommunications Industries, Wireless Telecommunications Industries and Consumer Appliance Industries. |
| High technology | Electronic Components and Semiconductor Industry, Robotics Industry, Software Industry, Network Infrastructure, Advanced Material Fabricator, Aerospace Industry. |
| Power | Hydro Plant, Nuclear Plant, Renewable Energy Plant and Fusion Plant, alongside the base game's Power Plants. |
| Transport | Airport and Highway, both producing Transportation, and airports some Merchant Marine too (Transportation and shipping). |
| Leisure | Tourism Industry and National Park (States and population). |
| Government | Space Program (The space race), which counts as a monument like the wonders below; State Youth Centers, which need Pro-Natalist Policy, State-Sponsored Family Planning, Communal Child-Rearing, State Eugenics Program or Mandatory Augmentation; Military Base (Military and war); and Public Housing, which raises its state's crowding threshold (States and population). |

### Pharmaceutical Industries and Drugs

Drugs, the base game's opium, also come from Pharmaceutical Industries, which arrive with the base game's Pharmaceuticals technology and gain better production methods with later medical technologies. Opium plantations are the cheaper source of Drugs until Antibiotics brings the factory level with them; its later methods pull ahead. Pops buy Drugs for Intoxicants and, from wealth 16, for the mod's Healthcare need (Pop consumption at high wealth), so in a rich country most Drugs are bought as medicine. The Ministry of Health and the Ministry of Consumer Protection cut the throughput of your own opium plantations, and a rich country whose market runs short of Drugs can get a Medicine Shortage event. The base game's journal entry The Opium Trade is not offered after January 1912, and one still open then fails.

### Later production methods for existing buildings

Most base-game buildings gain production methods through the new eras, so a building you built in 1850 keeps upgrading. Mines move on to Continuous Miners, Smart Miners and Laser Excavation Technology; grain farms to Modern Farming, Automated Harvesters and Planters and Designed Crops; plantations to Mechanized Farm and GMO Plantation; and automation groups end in AI-managed methods such as AI Managed Fab. A few polluting buildings get pollution-control groups (Pollution-control methods), and many get a hidden maintenance group used by the construction market (Construction maintenance and retooling).

## 2.4 Company flagship buildings

The mod gives every company a unique flagship building: the Krupp Essen Works, the Standard Oil Refinery, the Ford Rouge Plant, and so on.

A flagship building works like this:

- You can build it only while you have the company, and only while the company's prosperity bonus is active, because that bonus supplies its level cap of one.
- Only the government can build it, at 5,000 construction, and only in your own states. Investment rights in another country don't let you build or expand its flagships.
- You can have each flagship in only one state.
- Most are highly profitable at base prices, and each gives its state modifiers that fit the company, such as extra Highway levels and migration pull from the Volkswagen Autostadt.
- If you lose the company, the building is demolished.
- Once it stands, you can privatize it to its own company. A building the company owns gets the company's throughput bonus, which is often worth having on a flagship.

The mod also adds extension buildings and prosperity bonuses to many base-game companies, and eighteen new prestige goods, from Luxury Automobiles to Heavy-Lift Launch Systems.

### Companies added by the mod

The mod adds 95 company types. 84 are flavored companies, mostly real firms of the twentieth and twenty-first centuries (Volkswagen, Intel, SpaceX, Pfizer, Saudi Aramco and many more) plus a few fictional far-future ones such as Tessier-Ashpool S.A. The other eleven are generic companies for the new sectors: Entertainment, Power, Electronics, Aerospace, Software, Advanced Materials, Biotechnology, Infrastructure, Megastructure, Synthetic Fuels & Rubber and Synthetic Dyes & Fibers.

The mod's flavored companies are not tied to a country. The base game's usually require you to hold the company's home state, but the mod's need only a technology and industrial conditions, such as a high production rank or being the leading producer of their good. Japan can found Saudi Aramco and the United States can found Volkswagen if they meet those conditions. Generic companies need a large enough building of their industry.

Biotechnology companies, the generic one and ten flavored ones such as Genentech and BioNTech, also need the Corporate Genetic Licensing law (Economic law groups), which Gene Splicing unlocks, a technology earlier than Biotechnology. Three of them come from fiction and unlock late: the Rosen Association's androids with Genetic Engineering, InGen's resurrected species with Synthetic Biology, and RejoovenEsense's longevity treatments with Rejuvenation Therapies.

## 2.5 Wonders

The mod adds 37 wonders: landmarks of the twentieth and twenty-first centuries. Each costs 5,000 construction, has one level, and exists once in the world. Wonders, like the base game's monuments and the Space Program, count as monuments: each one raises tourism output in its state by 10% and adds 3 cultural pull (Where cultural pull comes from).

Most wonders can only be built in one state, the landmark's real location, so only that state's owner can build them: the Golden Gate Bridge in California, the Burj Khalifa on the Trucial Coast, the Three Gorges Dam in Western Hubei and 26 more. Wonder list gives each one's state and the technology that unlocks it.

Each wonder employs 10,000 workers, and its effects scale with how fully it is staffed. The effects are modest and themed: the dams add 10 to 20 Hydro Plant levels to their state and 25% Hydro Plant throughput, towers and trade landmarks raise trade advantage or Urban Center, Trade Center or service throughput, religious landmarks please the Devout, and several give flat prestige.

Eight science and diplomacy wonders can be built in any state, so the first country to meet their conditions and finish one claims it:

| Wonder | Unlocked by | Also requires | Main effect |
|---|---|---|---|
| Peace Palace | Intergovernmental Organizations | Ministry of Foreign Affairs at level 3 | Foreign affairs ministry impact, prestige, +0.5 UN Authority Target |
| Palais des Nations | (no technology) | Major power or better, Ministry of Foreign Affairs at level 5 | Foreign affairs ministry impact, prestige, Government Administration throughput, +0.5 UN Authority Target |
| International Space Station | Satellite Communications | Major power or better, a completed Moon landing (The space race) | +200 innovation cap, university throughput, education |
| LIGO Observatory | Fiber Optics | Great power | +100 innovation cap, university throughput |
| Large Hadron Collider | World Wide Web | Ministry of Science at level 3 | +200 innovation cap, +50% university throughput |

| Wonder | Unlocked by | Also requires | Main effect |
|---|---|---|---|
| Svalbard Global Seed Vault | Biotechnology | Ministry of the Environment at level 3 | Agriculture throughput, prestige |
| Continental Union Headquarters | Globalization | Major power or better, Ministry of Foreign Affairs at level 3 | Foreign affairs ministry impact, +100 prestige |
| ITER Fusion Reactor | Fusion Power | Major power or better, Ministry of Science at level 3 | +150 innovation cap, Nuclear Plant throughput |

The Continental Union Headquarters is the exception to "one in the world": each continent can have one, built on the builder's home continent. The United Nations Headquarters and the Power Bloc Headquarters are government buildings, not wonders, and don't count as monuments; see Founding the United Nations and Other power bloc changes.

## 2.6 Megaprojects

Megaprojects are the wonders of eras 12 and 13, too large to build with construction points alone. Instead you feed a construction site vast quantities of goods and specialists for one to four years.

### How megaproject construction works

Every megaproject follows the same two-phase pattern.

1. Build the construction site. It costs only 400 construction and the government builds it.
2. Choose its pace with its production method: Paused, 4 years, 2 years or 1 year. Faster paces consume more goods per week. The site employs 50,000 to 100,000 engineers and academics (the Orbital Battlestation's also takes officers), and it uses Advanced Materials, Electronic Components and other high-tech goods, plus Launch Capacity for the Space Elevator, Orbital Solar Collector and Orbital Battlestation.
3. Progress builds monthly in proportion to how fully the site is staffed. It stops completely in any month the site is short of an input good, so a supply gap stalls the whole project. Hover over the site to see its progress.
4. When progress reaches 100%, the site disappears and the megaproject gains a level (or appears at level 1). To add another level, build a new site in the same state.

The finished buildings run on inputs of their own, most employ several thousand specialists per level, and their effects scale with how fully they are staffed.

### The seven megaprojects

| Megaproject | Technology | Levels | What it does |
|---|---|---|---|
| Space Elevator | Space Elevator | 20 | Produces a million Launch Capacity per level and adds space race progress. Can only be built in states near the equator: most of tropical Africa, northern South America and Panama, southern India and Ceylon, Southeast Asia, the Pacific islands and a few others. |
| Orbital Solar Collector | Space-Based Solar Power | 10 | Each level opens three slots for Solar Power Receivers, ordinary buildings you build in any state; each receiver level produces a million electricity. |
| Orbital Battlestation | Orbital Weapon Platforms | 5 | Raises unit offense, defense, morale recovery and morale damage by 5% per level and improves defense against nuclear strikes. |
| Antimatter Con- | Antimatter Production | 10 | Each level opens five slots, shared between Antimatter Engines (Launch Capacity, faster army and fleet movement, space race |

| Megaproject | Technology | Levels | What it does |
|---|---|---|---|
| tainment Facility | | | progress) and Antimatter Warhead Plants (nuclear strike success, unit offense). Each level of either uses one slot. |
| Mind Upload Nexus | Mind Backups | 5 | Produces software, services, tourism and art, and raises research speed by 5% per level. Under the Demographics rule's Full setting it hosts uploaded minds and produces only as far as they fill it (Life extension). |
| Nanofabrication Center | Molecular Assemblers | 10 | Produces Advanced Materials and lowers space race risk. |
| Consciousness Network | Telepathic Communities | 10 | Adds bureaucracy, innovation, infrastructure, tax capacity and education access. Its Network Mode is Open Network (research, innovation, influence, prestige, standard of living) or Social Control Network (authority, government approval, lower turmoil and radicalism). Social Control Network needs Secret Police, Single-Party State, Autocracy, Mandatory Augmentation or Intrusive Surveillance System, and those laws rule out Open Network. |

Each type of megaproject you complete adds 3 cultural pull (Where cultural pull comes from). The Space Elevator, Orbital Solar Collector, Orbital Battlestation and Antimatter Containment Facility have their own events during construction and after completion. The military and nuclear effects are covered in Military and war and Nuclear weapons.

## 2.7 Grand monuments

A Grand Monument can be raised in any state from the start of the game. A government raises one to what it stands for: its crown, its republic, its revolution, its leader, its faith or the nation. You build it as a **Grand Monument (Under Construction)**: each level costs 1,000 construction, and while it is being built it has no upkeep and no effects. Dedicating it finishes it: a Grand Monument stands in its place at level 1, keeping the levels it was built with. It costs nothing to maintain and needs only a small caretaker staff. A monument's **grandeur** starts at what its levels were worth when they were built and grows as the monument ages (see A finished monument and its age). The Grand Monuments game rule can switch the system off.

A monument is impressive for what it cost its time. In 1836 a level adds 5 grandeur, and in a world ten times as rich it adds a tenth as much: 5 times the world's GDP in 1836, divided by the world's GDP when the level is finished, updated each January, to the nearest 0.001 and never less than 0.001. So the same share of the world's wealth buys the same grandeur in any era. The rate shows on the last line of the Grand Monuments overview, on a construction site's card and in the build menu before you build. The monuments standing in 1836, the trophies abroad in 1836 and the monuments dug up count at the 1836 rate, 5 grandeur a level, split as if each had been dedicated in the year it was built and had aged ever since (see A finished monument and its age). The Brandenburg Gate (1791) starts with about half of its 50 grandeur in years, the Parthenon, like anything a century old, with two thirds of it.

Everything a monument gives grows with grandeur in steps, and each step takes more grandeur than the one before. Inside a step every point counts. There are two kinds of step:

- **Effects in the monument's own state, prestige and cultural pull** taper slowly. The first step takes 5 grandeur, the second 10, the third 15 and so on, so steps complete at 5, 15, 30, 50, 75 and so on. Effects in a state stop at 30 steps, at 2,325 grandeur, and two stop sooner (see the table below). Prestige and cultural pull never stop.
- **The other national effects and interest group approval** taper fast. Each step takes twice the grandeur of the one before, so steps complete at 5, 15, 35, 75, 155 and so on, up to the eighth at 1,275.

Legitimacy and cultural pull take steps of 10. National effects count all your monuments' grandeur together, so twenty small monuments give the same national effects as one tall one; local effects count each monument on its own.

Ninety-three monuments already stand when a game begins, among them the Arc de Triomphe in Paris, the Alexander Column in Saint Petersburg, the Temple of Heaven in Beijing, Topkapı Palace, La Scala in Milan and the Jantar Mantar in Jaipur. Nineteen stand where the great powers' armies and collectors took the most: the Uffizi and the Capitoline Museums, the Parthenon, Bodrum Castle, Xanthos, the Temples of Baalbek, Palmyra, the Mesha Stele, the Arch of Ctesiphon, the Obelisks of Karnak, the royal palaces of Benin, Kumasi and Abomey, Great Zimbabwe, Persepolis, the Red Fort in Delhi, the Bukgwan Victory Stele in Korea, the Imperial City of Huế and Klungkung Palace on Bali. Fifty-one are the great sites of the past that draw visitors to their states, among them the Pyramids of Giza, Petra, Abu Simbel, the Alhambra, Windsor Castle, Kinkaku-ji, the Potala Palace, Borobudur and the Registan in Samarkand. They stand To History and carry the tourism their state's Base Appeal leaves to them (see Tourism output). Most monuments to a crown or a republic stand in countries that still honor them; two start contested: the Lion of Saint Mark, the old Venetian Republic's, in Austrian Venice, and Sans-Souci, a king's palace in republican Haiti. Each is a historical monument: it keeps its own look and picture while it stands, through a contest, a change of owner or a rededication. It keeps its name too unless it is rededicated; then it is named again like any new monument, and the naming event can give the historical name back. Rename and Name It are grayed out while it carries its historical name. Only pulling it down, or demolishing it, ends it.

## The Monuments journal entry

The Monuments journal entry appears when you own a Grand Monument, an opportunity is open or on offer, you hold a trophy or one taken from your lands stands abroad, or you are digging a buried site or have one open to dig. After your last monument is gone, it stays until the fading approval and legitimacy it shows, and any boon it carries, have run out.

The same panels appear on the Grand Monuments tab of the Timeline Extended window, which the button under the sidebar's Map List opens, and a change made in one shows in the other. The tab is grayed until the journal entry is active, and it ends with an Open Journal Entry button. Another country's National Info has a Grand Monuments tab with that country's monuments, national effects, trophies, works abroad and digs; the buttons, an open opportunity and Interest Group Approval are only on your own.

The state panel of a state with a Grand Monument, yours or another country's, shows a Grand Monument card under the tourism card: the monument's status, name and dedication, its grandeur with a bar toward its next step, its levels and age once finished, and the tourism output and local effect it gives that state. Hover a line for its breakdown. The card's arrow, on your own states, opens the Grand Monuments tab of the Timeline Extended window, where the buttons are. While you can carry the monument off as a trophy, the card has a Carry It Off button. A Grand Monument under construction has a card of its own under it: the levels built, the grandeur they come to, whether levels are still being built, and on your own state its Dedicate button. A state with more than one monument lists the others in the same card, under these details, and a state where trophies stand shows a Trophies on Display card under it.

The overview at the top is always shown: four boxes, two to a row, each with an icon, a caption, a word that turns red when something waits on you, and a figure. An icon is dimmed while its count is zero. Hover a box for its breakdown; Heritage and Contested are also terms you can hover.

| Box | Icon | Shows |
|---|---|---|
| Our Monuments | A gold wreath on the stone | How many are Upheld, and how many are Heritage. |
| Contested | The stone cracked | How many wait for your decision, in red, and the grandeur they hold back meanwhile. |
| Sites | The bare stone | How many construction sites you can dedicate now, in red, or how many are still building, and the grandeur they come to. |

*Figure 3: The Grand Monuments tab of the Timeline Extended window: the overview's four counts (ten monuments Upheld), National Effects with Interest Group Approval, and the start of Our Monuments.*

| Box | Icon | Shows |
|---|---|---|
| Hard Times | A mallet and chisel by an empty bowl | Red while a finished level would cause vanity backlash, with the remedy: pause construction. |

Under the boxes, a line names your monument policy; hover it for what the policy does and when it can next change. The last line says what a level built this year adds; hover it for how the rate is worked out.

These sections follow:

| Section | Starts | Shows |
|---|---|---|
| Opportunity | Open | Only while an opportunity is open: the moment that brought it, who asks for which dedication, what the monument will be named for, a bar of the grandeur added against the target, the levels still to build at today's rate, the months left, what completing it gives, Raise It and Let It Pass. |

| Section | Starts | Shows |
|---|---|---|
| Awaiting a Decision | Open | Only while a monument of yours is contested: its row, which names its Old Supporters and the group it is Resented By, with Pull Down, Rededicate and Keep as Heritage under it. A contested monument that stands beside a state's own has its row here too. |
| Awaiting Dedication | Open | Only while one of your Grand Monuments is a construction site: a row for each site, with its levels built, the grandeur they come to and whether levels are still being built, and a Dedicate button under it. |
| Archaeological Expeditions | Open | Only while you dig a buried site or one is open to you: the digs under way and the innovation they divert each week, each with its state, its weekly innovation and, abroad, its terms; then the sites you can dig, with the levels each will raise, the terms if you would dig it abroad, and a button for each method you can use. |
| National Effects | Open | The four monument policy buttons first. Then a row for each national effect in force: Prestige and Prestige (%), Legitimacy, Cultural Pull, Education Access under Public Education, each dedication's own effect, such as Authority or Max Innovation, and each unique dedication's. Each row's bar shows how far the step now being filled has come. Hover the row for the full effect, the grandeur behind it and where its next step completes. Below them, Interest Group Approval lists each group with a total; hover a group for the monuments it approves of, those it objects to, the contested ones it resents and its fading record. Fading Legitimacy shows what remains from pulling monuments down (The Old Order Torn Down) and from levels finished in hard times (Palaces amid Hardship). Opportunities Seized lists each boon in force with its months left, and each unique dedication you hold for a monument still to come. |
| Our Monuments | Open | A row for each finished monument that is not contested: its status as an icon with the word beneath, its name and dedication, and its state and grandeur. An upheld monument's row has Rename and Name It under it, and Clear Name once you have typed a name. A monument shows Unsettled for a moment after a change of government, until the month's check decides whether it still fits. A monument you can carry off as a trophy has Carry It Off, and the least grand of six in one state has Clear It Away. A monument that stands beside a state's own has a row after the states' rows, with where it stands. |
| Trophies | Open | Only while you hold a trophy or The Homecoming runs: what your trophies count for, The Homecoming's months left, and a row for each trophy with its monument's picture, what it was, where it came from and when, its worth here, who claims it, and Return It or Bring It Home. |
| Held Abroad | Open | Only while a trophy taken from your lands stands abroad: a row for each, with its holder and where it stands. |

The "?" at the right of the tab's header, and at the top of the journal entry, opens the Help page, How Grand Monuments Work. It explains grandeur and steps, what counts where, contested monuments, hard times, opportunities, unique dedications, names, the monument policy, trophies and archaeology.

Every monument, whatever it honors, gives:

- +10 prestige and +1% prestige for each step of standing grandeur, counting every monument that is not contested.
- +10% tourism output in its state for each step of its own grandeur, so five grandeur matches one wonder.
- +1 cultural pull for each step of standing grandeur (steps of 10, 20, 30 and so on), with no ceiling.

## A finished monument and its age

A monument is **finished** once it is dedicated. Until then it stands under construction and every level built on it joins it, for as long as you like: you can pause its construction and come back to it decades later. Its **Dedicate** button, on its card in the state view and under Awaiting Dedication, holds the dedication ceremony. The button is grayed while any of its levels are under construction: let them finish or cancel them first. The dedication finishes the monument at the levels built: a Grand Monument stands in its place at level 1, and the state card and its row under Our Monuments show "Finished: 15 levels · Age: +3.2". Damage to its building never shrinks it.

A finished monument grows grander with the years: by 2% of the grandeur it was dedicated with a year, so it doubles its grandeur in fifty years and triples it in a hundred, and then it stops. A historical monument's earlier years are already counted: the Parthenon has finished aging, the Brandenburg Gate grows for another 55 years to 79, and the Arc de Triomphe, finished in 1836, grows from 75 to 225 by 1936.

| Grandeur at dedication | +25 years | +50 years | +100 years |
|---|---|---|---|
| 5 | 7.5 | 10 | 15 |
| 10 | 15 | 20 | 20 |
| 25 | 37.5 | 50 | 75 |

It ages while its message stands or while it is kept as heritage, and not at all while it is contested. A trophy abroad does not age; it starts again once home. Each anniversary's first option adds a tenth of the grandeur it was dedicated with, at least 1, and completing an opportunity adds 5. Pulling a monument down throws its years away with its stone, and rededicating it halves its levels and its years, so an old monument is worth more than a new one of the same size.

More building in the state of a finished monument is a new construction site, and once dedicated a **new monument beside it**, with its own name and picture, as below. A site can't be started there while the monument is contested, while a dedication, unveiling or naming is under way, or once six monuments stand there, the site among them; the build tooltip names the reason. Levels already under construction always go on. The AI raises new monuments in other states before it builds beside one.

Once six stand in one of your states, the least grand of them, contested ones aside, has a Clear It Away button on its row and on the state's monument card. It is gone for good, with its grandeur and everything it gave, and a site can be started again, so you can replace a small monument with a grander one. Nothing else follows: no interest group objects and no legitimacy comes of it, as Pull Down's does for a contested monument. If it was the state's first monument, the grandest of the others takes its place. A historical monument can be cleared away too, and its tooltip warns you. The AI never clears one away.

## Several monuments in one state

A state has one Grand Monument building, so a second monument there stands beside the first instead of as a second building. It is a monument raised beside a finished one, a trophy that came home to a state with a monument already standing, such as the Luxor Obelisk coming back to Upper Egypt, where the Obelisks of Karnak stand, or a buried site dug up where a monument stands. Each keeps its own name, picture and dedication.

The state's one Grand Monument building stands at level 1 whatever stands there; each monument keeps its own levels. Each monument counts on its own: its dedication gives its own effect in the state and in the nation while its message stands, its interest groups approve or object, and it ages and can be contested on its own. A civic obelisk beside a crown column adds to the Petty Bourgeoisie's approval, not to the crown's legitimacy, and tourism output counts every monument's grandeur together.

The first monument is the grandest: once a new monument raised beside it has its dedication and name, the grander of the two stands first. Each of the others has its own row in the state's monument card, under the first one's details, and under Our Monuments, with its status and grandeur. A contested one's row, under Awaiting a Decision, has Pull Down, Rededicate and Keep as Heritage; rededicating it makes it the state's first

monument, at half its levels. An army that holds the state carries monuments off one at a time, with the Carry It Off on the first one's details or on another's row, and the rest stay. If the first monument is pulled down, cleared away or carried off whole, the largest of the others takes its place.

### Monument dedications

A monument's Dedicate button holds its dedication ceremony, which asks what it honors. The choice is permanent, and the ceremony's tooltips list each dedication's effects. The ceremony lists every dedication: one you can't raise yet is grayed out, with what it needs. Choosing to leave it as it is for now closes the ceremony and keeps the button for later; a monument under construction has no effects until it is dedicated. Dedicating a monument finishes it (see A finished monument and its age). The AI dedicates its monuments once nothing more is queued for them.

| Dedication | Requires | Approves / objects | National, per step | In its state, per step |
|---|---|---|---|---|
| To the Crown | A crowned head of state | Landowners / Intelligentsia | +2 legitimacy | +10% loyalists from movements |
| To the Republic | A republic | Intelligentsia / Landowners | +2 legitimacy | +10% loyalists from movements |
| To the Revolution | Single-Party State or Council Republic | Trade Unions / Industrialists | +2 legitimacy | +10% loyalists from movements |
| To the Leader | Autocracy or Single-Party State, not crowned | The ruler's own group / the strongest group outside the government | +2 legitimacy, +25 authority | +10% loyalists from movements |
| Grand Shrine | No State Atheism | Devout | +5% Devout attraction | +10% conversion |
| To the Nation | | Petty Bourgeoisie | | +10% loyalists from movements |
| War Memorial | | Armed Forces | 3% less war support lost to casualties | +1% conscription rate, up to +10% |
| Grand Opera House | Romanticism | Intelligentsia | +5% Intelligentsia attraction | +10% Creative Industry throughput |
| Botanical Gardens | Romanticism | Rural Folk | | 5,000 less pollution |
| Grand Observatory | Empiricism | Intelligentsia | +3 innovation cap | Faster literacy growth |
| Great Exhibition Hall | Marketing Research, no Industry Banned | Industrialists | +5% Industrialists attraction | +10% migration pull |
| Grand Stadium | Television Broadcasting | Trade Unions | | 10% lower turmoil effects, up to 50% |
| To History | Only the historical monuments that carry it | | | |

To History is the dedication of fifty-one historical monuments standing in 1836 and of fourteen buried sites once they are dug up. No ceremony offers it. It gives the tourism, prestige and cultural pull every monument gives and nothing more: it honors nothing that can fall, so no change of government or conquest contests it, and it can't be pulled down or rededicated. An army can still carry it, or a piece of it, off as a trophy. Every monument policy counts it at half.

Legitimacy counts the grandeur of all your monuments to the Crown, the Republic, the Revolution and the Leader together, in steps of 10, and only while each still fits. Each interest group's approval counts the grandeur of the monuments it approves of, minus those it objects to, and moves by 1 per step either way. No number of monuments can buy a group outright: each group has one total.

## Unique dedications

Most historical monuments carry a **unique dedication** of their own on top of the dedication they stand on. The Temple of Heaven stands To the Crown and carries The Mandate of Heaven. A unique dedication does everything its dedication does and adds an effect of its own. Its name shows in place of the dedication's in the monument's row, on its state card and on its trophy; hover the name for the dedication it was given on, what it adds and whether it counts now.

It counts only while the monument stands on the dedication it was given on:

| The monument | What its unique dedication gives |
|---|---|
| Upheld | Its national effect, its effect in its state, and its group's approval |
| Contested or kept as heritage | Its effect in its state only |
| Rededicated to another dedication | Nothing, and the row shows the new dedication's name; it counts again if the monument is rededicated back |
| A trophy abroad | Its national effect, for whoever holds the trophy, at the trophy's grandeur |

National effects count every monument that carries the same unique dedication together, in steps of 5 like the dedications' own, and each in force has its own row in National Effects. Ancestral Antiquity's cultural pull tapers slowly, like the monuments' own pull; the others taper fast. Monuments of one kind share one:

| Unique dedication | Carried by | Per step |
|---|---|---|
| Ancestral Antiquity | The ruins, tombs and treasures of the ancient world: the Obelisks of Karnak, Persepolis, the Venus de Milo, Troy, the Tomb of Tutankhamun and 32 more | +1 cultural pull, on top of the monuments' own pull and with no ceiling, and +5% tourism output in its state; the Intelligentsia approve |
| Epigraphic Archive | Inscriptions and lost libraries: the Rosetta Stone, the Temple of Dendera, the Mesha Stele, the Library of Ashurbanipal, the Dead Sea Scrolls and 10 more | +2 innovation cap and +2% technology spread, and +10% University throughput in its state; the Intelligentsia approve |
| Seat of the Throne | Royal palaces: Topkapı Palace, the Red Fort, Sans-Souci Palace, the Royal Palace of Benin and 6 more | +3% authority |
| Royal Pantheon | Royal tombs and statues: the Alexander Column, the Basilica of Superga, Nikkō Tōshō-gū and 4 more | +5% loyalists from high legitimacy |
| Martial Glory | Monuments to victories: the Brandenburg Gate, the Lion Monument, Tipu's Tiger, Baba Merzoug and 2 more, and three historical opportunities | +5% morale recovery |
| Liberty | Baltimore's Washington Monument, the Liberty Column, the Lion of Saint Mark, and the Washington Monument | +3% migration pull |
| Heroes of the Nation | Walhalla | +5% assimilation |

Four monuments have one of their own, which counts as their kind's as well and adds a little more: The Mandate of Heaven (the Temple of Heaven, Seat of the Throne and +1 legitimacy), To the Grande Armée (the Arc de Triomphe, Martial Glory and +3% prestige), The Glory of Athens (the Parthenon, Ancestral Antiquity and +2% influence) and The First Emperor's Army (the Terracotta Army, Ancestral Antiquity and +10% tourism output in its state). La Scala, the Teatro di San Carlo, the Jantar Mantar, the Shalimar Gardens, the Jardim Botânico and the Muhammad Ali Mosque have none.

Some opportunities give the monument that completes them a unique dedication, worth a fifth of the opportunity's boon for each step, for as long as it stands:

| Unique dedication | On | National, per step | From |
|---|---|---|---|
| To Victory | War Memorial | +2% prestige | Victory |
| To the Unknown Soldier | War Memorial | −5% radicals from low legitimacy | Defeat |
| To the Concert of Nations | To the Nation | +2% influence | Among the Great Powers |
| To the Conquerors of Space | To the Nation | +2% innovation; the Intelligentsia approve | A space race first |
| Olympic | Grand Stadium | −1% mortality | Television Broadcasting researched |

Each completion gives one monument, so each Victory completed raises another memorial To Victory. A monument that already carries a unique dedication keeps its own, and then you hold the new one instead, shown as Held in National Effects: the dedication ceremony of your next monument of that dedication offers it, and nothing else does. The AI always takes a held one. Pull the monument down or rededicate it and its unique dedication is gone for good; carried off as a trophy, it goes with the trophy and comes home with it.

## Monument opportunities

Moments in your country's history bring **opportunities** to raise a monument, whatever your rank. Each comes as an event, An Opportunity, titled for the moment that brought it, and names the interest group that wants it, the dedication, where it stands and what completing it gives. Seize the Opportunity opens it; Seize It and Start Building opens it and queues the levels its grandeur takes at today's rate at once, as government construction, in the state its tooltip names; Let It Pass closes it. Passing costs nothing, and an offer left unanswered is seized. Only one opportunity is open or on offer at a time; a moment that comes while one is open passes without one.

While it is open it has its own section on the Grand Monuments tab: where it counts, the grandeur added, the months left and what completing it gives. Raise It queues the levels still missing on a Grand Monument under construction in the state its tooltip names, at 1,000 construction each, which count once it is dedicated; it is grayed while any of your Grand Monuments is under construction, so a second press can't queue them twice. Levels count only once they are finished, and the section says so while any are being built. Let It Pass closes the opportunity. A monument that counts toward it is marked Counts toward the opportunity in its row under Our Monuments and on its state card. A year before time runs out, a notice reminds you.

| Moment | Who asks | For | Where | Boon for ten years, and unique dedication |
|---|---|---|---|---|
| A New Reign: a new ruler takes power, but not a regent | Landowners, or the new leader's own group | To the Crown when the ruler is crowned, otherwise To the Leader under Autocracy or a Single-Party State, named for the new ruler; under any other government, such as a democratic republic electing its president, none | Anywhere | +10% authority |
| A New Order: a new crown, republic or revolution that has no fitting monument | Landowners, Intelligentsia or Trade Unions | To the Crown, To the Republic or To the Revolution | Anywhere | +25% loyalists from high legitimacy |

| Moment | Who asks | For | Where | Boon for ten years, and unique dedication |
|---|---|---|---|---|
| Among the Great Powers: you become a great power for the first time | Petty Bourgeoisie, or the strongest group in government if they are marginal | A column To the Nation | The capital | +10% influence; To the Concert of Nations |
| Victory: you end a war of a year or more as a war leader, having enforced a war goal | Armed Forces | A War Memorial, named for the victory | Anywhere | +10% prestige; To Victory |
| Defeat: you end such a war with a war goal enforced against you | Armed Forces, or the strongest group outside the government if they are in it | A War Memorial to the fallen | Anywhere | −25% radicals from low legitimacy; To the Unknown Soldier |
| A Centenary: the 100th or 200th year since your founding | Petty Bourgeoisie, or the strongest group in government | A column To the Nation | The capital | +25% assimilation |
| A Reign Remembered: a ruler who reigned 15 years or more dies | The late ruler's group | A mausoleum To the Nation, named for them | The capital | +5 legitimacy |
| A space race first: First into Orbit, First on the Moon, First Lunar Colony, First on Mars or First Interstellar Probe | Intelligentsia | An obelisk To the Nation | The capital | +10% innovation; To the Conquerors of Space |
| A New Faith: your state religion changes | Devout | A Grand Shrine | Anywhere | +25% conversion |
| Romanticism researched | Intelligentsia | A Grand Opera House | Anywhere | +10% Arts Academy throughput |
| Empiricism researched | Intelligentsia | A Grand Observatory | Anywhere | +10% technology spread |
| Marketing Research researched | Industrialists | A Great Exhibition Hall | Anywhere | +5% Manufacturing Industries throughput |
| Television Broadcasting researched | Trade Unions | A Grand Stadium | Anywhere | −5% mortality; Olympic |
| The Environmental Movement researched | Rural Folk | Botanical Gardens | Anywhere | +25% pollution health effects reduction |

An opportunity asks for 5 grandeur of its dedication within 60 months in a great power, 120 in a major power and 180 in any other. Monuments of the dedication you finish count for the grandeur they were dedicated with, a new one or one raised beside a finished one; years of age do not. Late in the game that takes many more levels than 5, and Raise It queues what it takes at today's rate, at most 100 levels a press. An opportunity for the capital counts only the capital's monuments. One for the capital can be met anywhere when the capital already holds a monument of another dedication. A monument that becomes contested stops counting. Its levels count once it is dedicated, so press its Dedicate button before the deadline. When a new monument's dedication ceremony could answer the open opportunity, its first choice is "As the opportunity asks", with its dedication, and that is what the ceremony picks if you let it pass.

Each moment that comes once comes once a game: the first time you are a great power, the space firsts, each technology. What you already have when a game starts brings nothing, so a great power of 1836 and a country that already knows Empiricism get no opportunity for them. When a long-reigning ruler dies, A Reign Remembered comes at once and the successor's A New Reign passes. Countries that won independence before 1836 count their founding from that year, from the United States (1776) to Belgium and Ecuador (1830); a country formed or released in play is founded the year it appears. Other countries of 1836 have no founding year and no centenary.

| Outcome | What follows |
|---|---|
| Completed | An event, An Opportunity Seized, names the monument and the boon. The monument that completed it gains 5 grandeur at no cost. The group that asked approves, about +2 at first and fading over several years. The moment's boon runs for ten years, under Opportunities Seized in National Effects. The monument takes the opportunity's name, and its unique dedication where the table above names one. |
| Let it pass, or time runs out | Nothing. The next moment can bring another. |
| Lapsed | Nothing. An opportunity lapses when its state is lost or its dedication can no longer be raised, as when a crown falls. |

A second boon of the same kind restarts its ten years. An offer that brings a unique dedication names it, and for a moment that comes once, says that letting it pass loses it for good.

The AI gets the same offers. It seizes one unless it is below a major power, at war, in hard times, has more than a year of government construction queued or is in debt, and then queues the levels itself, at most 100 at a time; once those are dedicated it queues what is still missing.

### *Historical opportunities*

Six monuments that history raised in the game's early years come as opportunities of their own. Each is offered once a game, to whoever holds its state, at any rank, from the year shown for forty years, while its holder matches the Holder column. When another opportunity is open or on offer, it waits and comes once that one closes, if its years have not run out.

| Monument | State | From | Who asks | For | Levels | Holder |
|---|---|---|---|---|---|---|
| Walhalla | Bavaria | 1836 | Petty Bourgeoisie | To the Nation | 5 | German-speaking culture |
| Bunker Hill Monument | Massachusetts | 1836 | Armed Forces | War Memorial | 5 | Anyone |
| Obelisk of the Second of May | New Castile | 1836 | Armed Forces | War Memorial | 5 | Anyone |
| Muhammad Ali Mosque | Lower Egypt | 1836 | Devout | Grand Shrine | 10 | Sunni state religion |
| Nelson's Column | Home Counties | 1840 | Armed Forces | War Memorial | 5 | Anyone |
| Washington Monument | District of Columbia | 1848 | Intelligentsia | To the Republic | 10 | Republic |

It asks for its levels in the state within the usual months, twice as long for 10 levels. It brings no boon: the monument that completes it becomes the historical monument, with its name, look and picture, and the unique dedication it carries (Heroes of the Nation for Walhalla, Martial Glory for the three war memorials, Liberty for the Washington Monument), on top of the 5 grandeur and the approval of any opportunity. It lapses when you lose the state or stop matching its Holder column, as when the republic falls. Letting it pass or running out of time costs nothing, but it does not come again. The AI decides as it does for any opportunity.

## Monument names and forms

A monument to the Crown, the Republic, the Revolution, the Leader or the nation, or a War Memorial, can take the style of your primary cultures' classical heritage: a forum and triumphal column for a Romance-

speaking country, a hall of dynasties for a Chinese, Japanese or Korean one, an Aksumite stele for an Amharic or Tigrinya one, and so on through the classical languages that language reform can revive. A Grand Shrine takes the style of your state religion's great building: a basilica, a great mosque, a great temple, a pagoda. An opera house, gardens, an observatory or a stadium can take a style where your heritage has a tradition of its own, such as a Greek open-air theater, a Persian garden, a Jantar Mantar or a Roman amphitheater; for these the style changes only the look. When more than one style fits, an event asks which, and each style's option names what makes it fit; the AI keeps the most specific style without being asked. The style changes the monument's name and look, not what it does. Each monument shows a picture of its style and form in its row of the journal entry and on its state's monument card. The picture's frame marks how many grandeur steps the monument has reached: plain below the first, then bronze, bronze and silver, silver and gold, gold, and from the fifth step gold set with rubies. If you have revived the matching language under a state-led language reform, the monument's row says its dedication is carved in it.

Every dedicated monument has a name, from what it is named for and its form: the Lisbon Gate, Victoria's Column, the Arch of 1856. After the style, two events ask: The Form It Takes, when the style offers more than one form (a column, an arch, a gate, a pagoda, a sun stone), and A Name for the Monument, which names it for its city, its state, the ruler, what it honors (the Crown, the Republic, the Revolution, the Nation, the Faith or the Fallen), the year, or the occasion an opportunity marks. Each event keeps what was proposed unless you pick another. An opera house, gardens, an observatory, an exhibition hall or a stadium takes its own building's form, and a Grand Shrine of your faith takes its building's name. The AI names its monuments for their city, or as their opportunity asks.

From an upheld monument's row, Rename asks both again at no cost, and Name It opens the game's rename box so you can type a name of your own. Confirming the name there names the monument; Clear Name removes it. The box belongs to a company that exists only while you name, so if your companies are at their limit an alert may flash as it opens and closes. A contested or heritage monument keeps its name.

## Monument policy

One policy covers all your monuments. You set it with the four buttons at the top of National Effects, and a line under the overview's boxes names the one in force. Each policy is the best at one thing, so the cost of choosing one is what the others would give. After a change the policy cannot change again for five years, but your first choice starts no wait.

| Policy | Gives | When it is worth it |
|---|---|---|
| Open to the Public | Tourism output from each monument ×1.5, from a monument To History ×1.25, the tourism of Ancestral Antiquity and the First Emperor's Army included. | Monuments that stand beside a Tourism Industry. |
| State Ceremonial | Legitimacy and the Leader's authority from monuments also count a quarter of the grandeur of every monument you hold that is not contested, heritage and trophies included, whatever it honors; a monument To History counts at half. | Low legitimacy. |
| Cultural Promotion | Cultural pull from monuments ×1.5, Ancestral Antiquity's included, and the share of it from monuments To History ×1.25. With the Cultural Hegemony rule off, prestige from monuments instead, on the same terms. | The policy until you choose another. |
| Public Education | +1% education access in every state for each step of standing grandeur, a monument To History counting half, in steps of 10, 20, 30 and so on. | Literacy still well short of what your people could reach. |

No policy changes maintenance, which Grand Monuments don't need, or aging, anniversaries or what a dedication does in its own state.

The national effects change at once, and each monument's tourism at the start of the next month. The AI picks each January: State Ceremonial with legitimacy under 40 when its regime grandeur and the quarter together

reach legitimacy's first step, Open to the Public with a monument in a state with a Tourism Industry, Public Education with literacy under 50%, and otherwise Cultural Promotion.

## Contested monuments

A monument to the Crown, the Republic, the Revolution, the Leader or a faith can outlive what it honors. It becomes **contested** when its crown, republic or revolution falls, when its leader stops ruling for any reason, when the country adopts State Atheism or changes its state religion, or when another country takes its state. A contested monument keeps its tourism and its effect in the state, but gives no prestige, legitimacy or approval, and the group that objected to its message resents it every month it stands undecided.

When it happens, a notice lets you decide for all of them at once or one at a time. You can also decide later from the monument's row under Awaiting a Decision on the Grand Monuments tab. Each choice's hover gives this month's figures: the legitimacy Pull Down brings and the approval each choice moves.

| Choice | What happens |
|---|---|
| Pull Down | The monument and all its grandeur are gone. The new government gains legitimacy that fades over about five years. The group that objected to the old message approves and the group that raised it resents it, both fading. |
| Rededicate | The ceremony asks first, under your current laws. Dedicating it again costs 500 per level, and it keeps half its levels, rounded up, and half its years; Keep It As It Is changes nothing. Its old supporters resent it a little. |
| Keep as Heritage | The monument stays and its prestige returns, but under this government it lends no legitimacy and gives no approval. The group that objected resents it, less each year. |

Pulling down many monuments at once gives what pulling down one of their combined grandeur would: each reward and resentment comes from a single total that fades month by month. If what a contested or heritage monument honors comes back, for example when the monarchy is restored, it counts in full again. Pulling it down is the only permanent choice. A shrine in a state that another country takes becomes heritage at once and cannot be pulled down. After a civil war the winner inherits the monuments the loser raised, and they are contested only if its laws no longer fit them. A revolt that takes part of a monument's state leaves the monument whole with the part that keeps it; the rebels' part gets none of it.

A Grand Monument cannot be downsized or demolished from its building panel, by you or by the AI. Pull Down is the only way to remove one yourself.

## Monument trophies

A country can carry off a Grand Monument, or part of one, as a **trophy**, in three ways:

- In a war, from a state where your armies hold half the provinces or more. The state view's Grand Monument card has a Carry It Off button while you hold it, and an event asks the first time your armies hold such a state.
- From your own unincorporated state, if you did not raise the monument: a recent conquest or a colony. The button is on the card and on the monument's row under Our Monuments. Once you incorporate the state, the monument stays.
- By treaty, with the Surrender a Monument article, which a war can enforce (see treaty articles by purpose).

What goes depends on the monument's form. A Grand Shrine is never taken, and a monument that loses a piece is spared for ten years.

| Form | What is carried off |
|---|---|
| Column, obelisk, statue, stele, pillar, high cross, rune stone, menorah, sun stone or stela | The whole monument |
| Arch, gate, tower, bell tower, round tower or rock-cut monument | A quarter of its levels, at least one, and the same share of its years. What stands keeps its name and dedication. |

| Form | What is carried off |
|---|---|
| Any other form: halls, a pantheon, forum, mausoleum, pyramid, pagoda or church, and the five civic-progress buildings | A tenth of its levels, at least one |

A trophy stands in its taker's capital. It counts for half the grandeur carried off, rounded up, in the standing grandeur that gives prestige and cultural pull, and the Armed Forces approve of it. Whoever holds that capital's state holds the trophy, so conquering the state takes its trophies too. A trophy keeps its monument's name and picture, and a piece is named for what was taken, such as the Brandenburg Gate's Quadriga, shown with the Gate. A trophy keeps its monument's unique dedication too, and its holder counts it at the trophy's grandeur, whatever its government; the monument a piece was taken from keeps its own at its smaller size.

| Taken | You pay | The other side |
|---|---|---|
| In a war | 2 infamy, plus 1 for every 5 grandeur carried off, at most 8; doubled once the UN's Cultural Heritage Program (UNESCO) is in force. Relations −30. | +10 war support in the war |
| From your own unincorporated state | A tenth of the state's people turn radical. Relations −30 with the country you took the state from, if it still exists. | |
| By treaty | Relations −30 | |

Your trophies have their own Trophies section on the Grand Monuments tab. A trophy taken from your lands shows under Held Abroad, with its holder. To get one back:

| Way home | How |
|---|---|
| Return It | Its holder's button on the trophy's row. Relations between the two rise by 40. |
| Bring It Home | The same button, when the holder owns the state it came from, as after conquering the state where it stands. |
| Occupation | Occupy the state where it stands in a war. Its Trophies on Display card in the state view has a button for each trophy, and one taken from your lands goes straight home. |
| Return Cultural Property | A treaty article: the holder returns every trophy taken from your lands. A war can enforce it. |

The Trophies on Display card has three views, chosen by the buttons in its header. Full shows each trophy with its picture, where it came from and its Grandeur, with aggregated effects on hover. Short is one line for each trophy, with the rest on hover. Collapsed shows only how many trophies stand in the state and what they count for together. The card opens in Full, and your choice lasts until you leave the game.

A trophy that comes home goes back where it was taken from, with all the levels and years carried off. A piece of the monument standing there rejoins it, as the Quadriga goes back on the Brandenburg Gate. Anything else stands beside the monument that is there, as several monuments in one state; with no monument standing, the monument stands again with its name, style and dedication. Its owner gains The Homecoming, +5 legitimacy for ten years, shown in the Trophies section, unless it carried the trophy off itself: bringing home what you took from your own colony only moves it.

Ten trophies are already abroad when a game begins: the Parthenon's marbles, the Rosetta Stone and Tipu's Tiger in London, and the throne of the kings of Kandy at Windsor; the Dendera Zodiac, the Luxor Obelisk and the Venus de Milo in Paris; Baba Merzoug, the great cannon of Algiers, at Brest; and the Piraeus Lion and the Horses of Saint Mark in Venice. Greece, Egypt, the Ottoman Empire, Mysore, Mascara and British Ceylon see theirs under Held Abroad from the start. A trophy from a historical monument has a Historical monument line in its row; hover it for how it came to stand abroad.

The AI carries off monuments in the states it occupies, more often when its governing interest groups favor Total War, Ethnostate or National Supremacy, when it is unrecognized, or for a monument of 10 levels or

more, and much less once UNESCO is in force. It brings home every trophy taken from it that it can, but keeps what it carried off itself, and now and then returns one to an ally or a power bloc partner.

## Buried sites

Fifty monuments lie buried when a game begins, and a country that holds the ground, or the right to dig it by treaty, can dig them up: Nineveh's library, Troy, Mycenae, Olympia, Delphi, Knossos, the Pergamon Altar, the Ishtar Gate of Babylon, Machu Picchu, the tombs of Tutankhamun and Philip II, the Terracotta Army, Göbekli Tepe, Ephesus, Teotihuacan, Hattusa and others. A dug-up site is a historical monument like those standing at the start. It keeps its name and look, and an army can carry it off as a trophy.

Pompeii is already being dug when the game begins: whoever holds Campania runs an Antiquarian dig there, a third done, which diverts 5 innovation a week from the first month. At 15 levels it is one of the largest sites. Suspend it to keep the innovation, or restart it by Systematic Stratigraphy once you have Empiricism to bring it out whole; the progress is kept either way.

A site opens when you own a part of its state and have its technology: Romanticism for Nineveh, Empiricism for Troy, Mycenae, Pergamon, Olympia and Sidon, and later ones for the finds of the twentieth century, up to E-commerce for Göbekli Tepe. Some are known and open at once, and The Scholars Know Where to Look tells you when one is. The rest are found by chance: once you have the technology, each month there is about a 1% chance that a plow, a railway cutting or a hiker turns one up, and Something Under the Spade tells you. Both events name the place and the state, say what was found, add a short note on how it was found in history, and offer the dig.

You choose a method for each dig, from the Archaeological Expeditions section of the Grand Monuments tab. A button appears for each method you have the technology for.

| Method | Needs | Months (5, 10, 15 levels) | Weekly innovation | Result |
|---|---|---|---|---|
| Antiquarian | Romanticism | 17, 25, 34 | −5 | A 35% chance the dig destroys half the levels, at least 3 |
| Systematic | Empiricism | 27, 40, 54 | −20 | The whole monument |
| Forensic | Computer Networks | 20, 29, 39 | −100 | 2 levels more than the site's own |

Innovation is the only cost: a dig takes no money from the treasury. Each dig lowers your weekly innovation by its method's figure for as long as it runs; the innovation cap stays. A site's size sets its months, so a 10-level site dug by Systematic Stratigraphy diverts 20 innovation a week for 40 months, a 5-level one for 27. The button's tooltip gives this site's months, and a dig's row gives the innovation it diverts. Nothing caps how many digs you run, but a method's button is grayed while you make less weekly innovation than it would divert, so a new dig never takes your innovation below zero. Suspend Dig stops a dig and keeps its progress. A dig also stops if you lose the state, or the treaty that let you dig there ends, and whoever may dig it next can take it up from where it stopped.

A finished dig stands in the state it was dug in, To the Nation, at the site's levels (5 to 15). Fourteen sites stand To History instead, at 3 or 6 levels: the ruins whose visitors their state's Base Appeal leaves out until they are dug, such as Ephesus, Tikal, Chichén Itzá, Teotihuacan, Leptis Magna and the library cave at Mogao. With no monument there it is a new one; otherwise it stands beside the one there as several monuments in one state. An event tells you how it went, again with the place, the find and its history. The AI, at peace and out of default, starts a dig now and then, never runs more than two, and digs only by a method whose innovation it makes four times over: 20 weekly innovation for Antiquarian, 80 for Systematic, 400 for Forensic.

### *Digging abroad*

Two treaty articles let you dig the buried sites in another country's state. Both name a state that has a site open to you: a known site whose technology you have, or a chance find already turned up. The country that owns it concedes, and you pay the influence. While the article stands, the site is listed in your Archaeological Expeditions with the terms, and you dig it as you would at home, diverting the same innovation.

| Article | You need | Influence | When the dig finishes |
|---|---|---|---|
| Excavation Concession | Romanticism | 75 | Half the levels, rounded down, go to your capital as a trophy; the rest stand in the host's state as its monument. A rock-cut site has nothing to carry: it stays whole, and you gain Foreign Scholarship. |
| Excavation Permit | Empiricism | 40 | The monument stands in the host's state at its full levels, and you gain Foreign Scholarship: +10 prestige and +1 cultural pull for ten years. |

A concession's trophy is like any other: it counts for half its grandeur in yours, and the host can ask for it back with Return Cultural Property or take it in a war. The host may still dig its own sites, and may end the treaty, which stops your dig with its progress kept; the host can then finish the dig itself. A war can enforce a concession but not a permit. An AI host signs a permit far more readily than a concession, and hardly ever a concession once it has researched Pan-nationalism. A major AI power with Empiricism asks for one now and then.

## Vanity backlash

Finishing monument levels while the country is in default, in famine or in recession angers people: 0.5% of the state's pops turn radical, and legitimacy falls by 1 for each point of grandeur the levels add, fading over about two years. While this would happen, the journal entry shows a red Hard Times under its overview, so you can pause construction first.

## Monument anniversaries

Every dedication but To History has an anniversary event that can fire once a monument of that dedication reaches level 3 and still fits. Each offers a choice between two benefits, usually for two interest groups, and the only cost is money. A contested or heritage monument holds no anniversaries.

The AI raises most of its monuments through opportunities. Without one, it raises them mainly as a great or major power, in a state with a Tourism Industry, or when its legitimacy is low, and never while at war or in hard times. It decides its contested monuments within a few months: small ones and those under a revolutionary government tend to come down, tall ones are kept as heritage, and it rededicates only with money in the treasury.

CHAPTER 3

# Economy and construction

This chapter covers how the mod changes the way your economy builds, consumes and stores. The largest change is the construction market: construction is a good that Construction Sectors sell and that your government and investors buy, and most industry uses a little of it as upkeep. It runs from the first day of the game under the Free Market Construction game rule, which also offers four alternative settings. The rest of the chapter covers what construction costs in rich countries, how living-standard expectations adapt, what wealthy pops buy, the transport goods, power grids, setting tariffs for all goods at once, the Strategic Reserve journal entry, wartime demand for ammunition and the mod's new mineral deposits.

## 3.1 The construction market

In the base game, each Construction Sector adds construction points straight to a national pool. Under Free Market Construction, the default setting of the game rule of the same name, construction is a tradeable good instead. Construction Sectors sell it on the market like any factory, your government and your investors buy it every week, and what they buy becomes the points your two construction queues spend. The system is based on TOGFan's Free Market Construction mod, which is worth a look on its own if you want only this mechanic.

### From Construction Sector to construction queue

Each week, construction moves through four steps.

1. Construction Sectors produce the good, which your market lists as Construction Services (base price £1,000). In a market game the sector is ordinary heavy industry: investors build it when it pays, and your government can build it too. Barracks under the Engineering and Logistics power bloc principle add a little.
2. The government buys the number of points you set in the construction panel, paid from the treasury.
3. Investors buy for private projects on their own, paid from the investment pool. Each week they spend a 24th of the pool. That settles at about the pool's weekly income, more when the pool has built up and less when it runs low, and they keep spending when the income stops.
4. Construction Sites turn the purchased good into construction points. They appear on their own in every state where construction is under way, and in your capital when nothing is under way anywhere else. Nobody builds them.

Each purchase is capped at what its queue can use that week: at most one week's maximum progress for each level in the queue. The share of points that goes to private projects is recalculated every week from the two purchases, and your economic-system law no longer sets it.

When all buyers together want more construction than is on sale, the price rises, as it does for any good in Victoria 3. You still get all the construction you buy, for as long as the shortage lasts; each point just costs more, so the treasury pays more for its weekly purchase and investors' money buys fewer points. Think of the higher price as supply stretching in the short run. Every country in your market buys from the same supply, and construction is imported and exported like other goods.

Each level of a Construction Sector produces more with better production methods:

| Production method | Weekly construction per level |
|---|---|
| Wooden Buildings | 1 |
| Iron-Frame Buildings | 2 |
| Steel-Frame Buildings | 3.5 |
| Arc-Welded Buildings | 5 |
| Reinforced Concrete Buildings | 6 |
| Advanced Construction | 10 |
| Nanomaterial Buildings | 16 |
| Self-Replicating Construction | 24 |

## Construction maintenance and retooling

Most industry, transport, power and service buildings use some construction each week as maintenance, per level, at a rate that depends on the kind of building:

| Rate per level | Buildings |
|---|---|
| 0.05 | Trade Centers, Arts Academies, State Youth Centers, Software Industries, the tech metals mines (lithium, nickel and cobalt, rare earths, platinum group metals), every other mine once it runs a mechanized pump; Railways, Ports and Power Plants on their early methods |
| 0.1 | Factories of every kind, military industry, synthetic and biotechnology plants, Shipyards, Skyscrapers, Tourism Industries, Public Housing; Railways, Ports and Power Plants on their middle methods |
| 0.15 | Highways, Airports, Network Infrastructure, Hydro Plants, Renewable Energy Plants, Grid Transfer Stations, HVDC Undersea Interconnectors, Oil Rigs; Railways and Ports on their latest methods |
| 0.2 | Nuclear Plants, Fusion Plants, Deep-Sea Mines, Extraplanetary Bases |

Railways, Ports and Power Plants pay by the method they run, because a level on an early method carries a fraction of the traffic or power of a level on a late one:

| Rate per level | Railway | Port | Power Plant |
|---|---|---|---|
| 0.05 | Early and steam trains | Anchorage, basic and industrial ports | Early and coal-fired plants |
| 0.1 | Electric and diesel trains, Centralized Traffic Control | Modern port | Oil-fired and both modern plants |
| 0.15 | Containerized cargo and later | Container ports and later | No method |

The rate follows the method by itself, up or down. Farms, plantations and urban centers pay none. The other mines (coal, iron, lead, sulfur and gold, and the bauxite, copper, graphite and similar mines that use the same equipment) pay none while they run picks and shovels or the atmospheric engine pump, and start paying when their equipment changes to a condensing engine pump or anything later; going back to the old equipment stops the upkeep. A few company buildings also use construction in their production. This upkeep competes with your queues for the same supply, so a growing economy needs a growing construction sector, and a country built around railways and power pays more than one built around trade.

When a building switches production method it gets the Production Method Retooling modifier, which raises its construction use by +1,000% right after the switch, falling steadily to nothing over five years (260 weeks); the building shows when the modifier expires. Just after switching methods, a 20-level factory goes from 2 construction a week to 22, before cost scaling.

That adds up. Over the five years the 20-level factory buys 10 extra construction a week on average, about £10,000 a week at base prices, so the switch costs around £2.6 million: 2,600 construction, or about what it

takes to build all 20 levels of the cheapest kind of building (100 construction a level) from scratch. You pay it, or the building's owners do. Switch production methods when the new one earns that back, not on a whim.

A building that has no finished level yet can switch for free: the mod removes the retooling modifier from it, once a week and whenever it changes method. Set a new building's production methods while it is still under construction. A building that is only adding levels already has finished ones, so it pays.

## Reading the construction panel

The Construction Market section sits at the top of the construction panel's National Queue tab.

| Row | What it shows |
|---|---|
| Fixed Quantity / Fixed Budget | Whether you set the government purchase as a number of points or as weekly spending. The lit button is the mode in force; click the other to switch. |

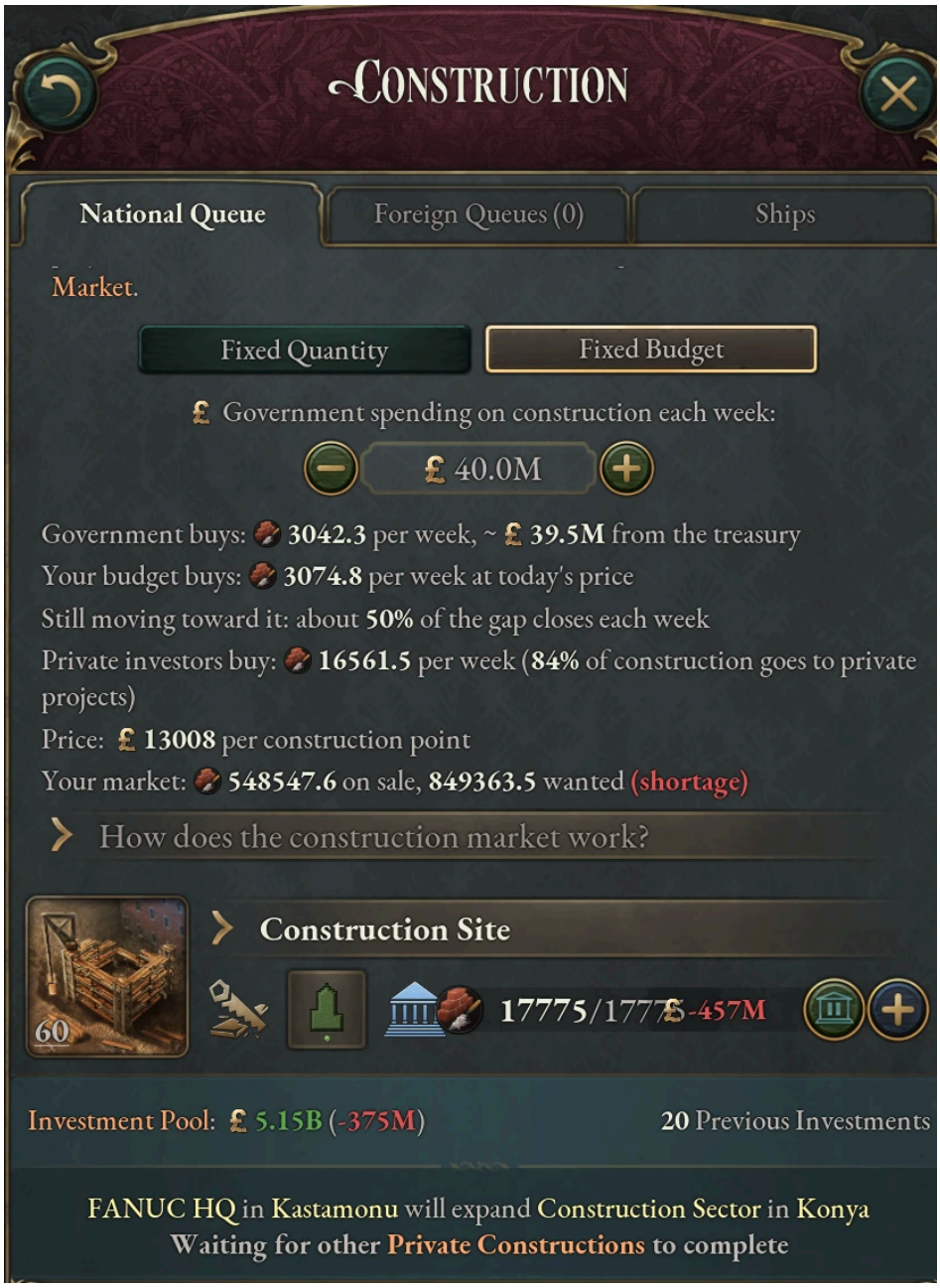


*Figure 4: The Construction Market section under Fixed Budget. The government spends £40.0M a week and buys 3,042 points, still closing on the 3,075 the budget buys at today's price; private investors take 84% of construction and the market is short.*

| Row | What it shows |
|---|---|
| Purchase control | The government purchase you have set. Fixed Quantity: click for ±1, Shift+Click ±10, Ctrl+Click ±100, Alt+Click ±1,000; right-click on the plus adds 10,000. Fixed Budget: click for ±£1,000, Shift+Click ±£10,000, Ctrl+Click ±£100,000, Alt+Click ±£1,000,000; right-click on the plus adds £10,000,000. In both, right-click on the minus sets 0. |
| Government buys | Points bought this week and the approximate treasury cost. A red "Capped" appears when your setting is higher than the queue can take (the Queue Cap). |
| Budget buys | Fixed Budget only: how many points a week your budget buys at today's price, which is where the government purchase is heading. |
| Still closing | Fixed Budget only, and only while the purchase has not yet reached what your budget buys: the share of the remaining gap that closes this week. Its tooltip explains why the purchase moves in steps. |
| Private buys | Points investors bought this week and the share of construction going to private projects. |
| Price | What one construction point costs now: the market price, raised by Construction Cost Scaling in rich countries with few idle workers. |
| On sale / wanted | Construction on sale (including imports) against construction wanted (every buyer, including maintenance and exports). A red "Shortage" marks demand above supply: the price rises, but you still get all the construction you buy. |

## Fixed quantity or fixed budget

The two buttons above the purchase control set what you fix and what follows the price.

| Mode | You set | What follows the price | Use it for |
|---|---|---|---|
| Fixed Quantity (default) | Points the government buys each week | The treasury cost, at once | A set build rate, such as finishing a queue by a date or rebuilding after a war |
| Fixed Budget | What the government spends each week | The points it buys, over a few weeks | A set cost: spending stays near your figure while the price moves |

Under Fixed Budget, the purchase moves each week toward what the budget buys at that week's price. A raise closes at most half of the remaining gap a week, so it gets most of the way in three or four weeks. A cut takes hold faster, often within a week or two. The delay on a raise is deliberate. The government's purchase moves the price, so a purchase that jumped straight to what the budget buys would push the price up, buy less the next week, let the price fall, and swing back and forth. How far the price moves depends on your purchase next to the market. A small purchase in a large market barely moves it, and a large one in a thin market, or one that lifts a very low price, takes longer. While the purchase is still moving, the panel shows how much of the gap closes that week. Setting the budget to 0 stops government buying at once.

Switching keeps the purchase in force. Fixed Budget starts at what your current purchase costs, to the nearest £1,000, and Fixed Quantity starts at the number of points your budget is buying. In both modes only what the queue can use is bought, so with a short queue a Fixed Budget spends less than you set.

## Free Market Construction settings

The game rule has five settings, fixed when you start the campaign.

| Setting | Construction good | Maintenance | Retooling | Private share of construction |
|---|---|---|---|---|
| Enabled (default) | Traded on the market | By building type | +1,000%, fading over five years | Follows the two purchases |
| Without AI Retooling Costs | Traded on the market | By building type | None for AI countries; normal costs for players | Follows the two purchases |
| Without Retooling Costs | Traded on the market | By building type | None | Follows the two purchases |
| Without Maintenance | Traded on the market; a few company buildings still use it in production | None | None | Follows the two purchases |
| Disabled | None | None | None | Set by your economic-system law |

Without AI Retooling Costs keeps normal maintenance for every country, but removes the retooling surcharge from AI countries after a production-method change, with a weekly cleanup for any lingering penalty. Every human player still pays it, including in multiplayer. AI subjects are exempt even when their overlord is a player. If you take control of an AI country, future changes pay normal retooling costs; penalties already removed stay removed.

With the rule disabled, construction works as in the base game, but the Construction Site takes the Construction Sector's place. The government builds, expands and downsizes Construction Sites like any government building, and each level provides 1 to 24 points a week by production method, the same tiers as the table above. The government pays their wages, and they buy materials for each point your queues spend. Construction Sites are built from the state view. Every country also gets Base Construction, 5 points a week, so a country without any sites can still build its first one. The construction panel shows a Direct Construction section with your construction per week and the private share, which the economic-system law sets: 25% under Traditionalism, 50% under Interventionism, Agrarianism, Extraction Economy and Industry Banned, 75% under Laissez-Faire, 35% under Cooperative Ownership and 10% under Command Economy.

## Running the construction market

- Your government purchase starts at 0 in a new game. Set it in your first week, or the government queue builds nothing.
- Only what the queue can use is bought, so a purchase set above your queue wastes nothing. With a long queue, though, a high setting buys all of it: that is expensive for your budget, and a government buying heavily pushes up the price for every buyer in your market, investors included. Set the purchase to what you can afford rather than to the maximum, or switch to Fixed Budget and set the weekly spending directly. To stop government construction while private building carries on, set it to 0; the sector keeps selling to investors.
- Watch the market line. A standing shortage keeps the price high, which means you need more Construction Sector levels or a better production method. Cheap wood, iron and steel make construction cheaper, because the sector buys them.
- A high price makes Construction Sectors profitable, and investors then build them on their own.
- Spread production-method switches out. Every switch adds five years of retooling upkeep, heaviest in the first months, and switching a whole industry at once can take the construction your queues were counting on.
- Don't downsize Construction Sites at all. They can be downsized only because the Disabled setting uses them as an ordinary government building. Under the market settings the game places and removes them as

needed, and if you downsize the last one, construction stops for about two weeks until a new one appears in your capital.

AI countries choose their own purchase each week, always as a fixed quantity. They spend roughly their net income, more when their gold reserves are full and less when they carry debt. They cut back when a war or banking stress meets thin reserves, and buy at least 10 points a week while their government queue has anything in it.

## 3.2 Construction costs in rich countries

Two mechanics make construction dearer as a country grows rich, and both work in every setting of the Free Market Construction rule.

### Construction Cost Scaling

Once a year, the Development Cost Scaling modifier raises the construction each point requires in a rich country, by two parts multiplied together:

- **The GDP part.** Above £1 of GDP per capita it rises in a straight line, about 10 percentage points for every £1, to +1,000% at £100.
- **The share of it that applies.** This depends on how much of your workforce is idle: peasants and unemployed together. With 3% or less idle, all of the GDP part applies. With 15% or more, none does. Between the two, the share falls in a straight line. It moves a third of the remaining way toward its new value each year, so the price follows your hiring over a few years rather than at once.

A rich country with workers to spare builds cheaply until it has put them to work. One at full employment pays the whole GDP part, which keeps it from building faster than its population can staff. The scaling applies to every use of construction: the points your queues spend and the maintenance your buildings pay. It combines by multiplication with other construction-cost modifiers, so another modifier that cuts construction costs by 20% still cuts 20% in a rich country. With Free Market Construction disabled, the same scaling raises the cost of the materials your construction uses instead.

It is also a catch-up mechanic: a poor country, with many peasants and a low GDP per capita, builds cheaply. The Nation tab shows the current scaling, and below it any excess private construction in force. Its tooltip gives the cost of a point as factors: the GDP part, the share that applies, your peasants and unemployed, and your other construction-cost modifiers.

### Excess private construction

This only matters for extremely rich countries, and you will rarely meet it before the late game. The private construction queue holds at most 1,000 buildings, so a huge investment pool can run out of projects to spend on; this modifier lets each project absorb more construction instead. You generally won't see it until your weekly investment pool income is in the billions.

Once a year, the game checks whether investors have more than 500 levels waiting in the private queue while the investment pool takes in more than it spends. If so, you get Excess Private Construction. Each project can then absorb more construction a week, so the pool can spend its money, at a cost in construction efficiency that grows with the modifier, up to −90% at its cap; the tooltip lists the two effects as Max Weekly Construction Progress and State Construction Efficiency. It starts small, at no more than +10 Max Weekly Construction Progress, and each year moves by at most about a fifth toward what the pool needs.

The modifier stays as long as it is needed: while 500 projects at normal speed still couldn't spend the pool's income and the private queue isn't empty. A pool that shrinks because the modifier is working doesn't end it. Once the need is gone, the modifier is removed outright, and the efficiency penalty goes with it. If it is needed again within five years, it comes back at its old strength or at what the pool now needs, whichever is less, so a short banking panic doesn't reset it. After longer it starts small again.

If the modifier passes +1,000 while the investment pool holds more than your yearly GDP, Overinvestment follows and most pops stop paying into the pool. It lasts until the pool is back down to a year's GDP, and

meanwhile Excess Private Construction holds its strength, neither growing nor shrinking. Investors keep spending the pool on the private queue throughout, a 24th of it a week or as much as the queue can absorb, whichever is less.

## 3.3 Adaptive standard-of-living expectations

In the base game, pops expect a standard of living set by their stratum, literacy and laws, and pops below that expectation turn radical over time. The mod drops the literacy part and adds Adaptive SoL Expectations: an extra expectation that follows your country's actual average SoL, with a delay. It aims at a level 10 SoL below your average SoL, moved up or down by some technologies and laws, and each month it closes part of the gap: about half of it in ten years and three-quarters in twenty. It only ever adds to the base-game expectation. While your SoL is too low for that level to reach the base-game expectation, expectations stay at the base-game level.

So the direction of your economy matters as much as its level. After a sudden rise in SoL, expectations lag behind for years, pops live better than they expect, and fewer turn radical. After a sudden fall, expectations stay high for years and pops sit below them, which feeds radicals long after the shock itself. Slow, steady growth is the calmest path. You see the adjustment as three country modifiers: Lower, Middle and Upper Class Expectations Shift.

Technologies, laws and literacy move expectations further:

- The society technologies Egalitarianism, Labor Movement, Socialism, Political Agitation and Mass Propaganda each raise the level expectations settle at by 0.5 SoL. They replace the literacy-based increase those technologies give in the base game. Several of the mod's later technologies and laws raise or lower it too, and Industry Banned lowers it by 1.
- Voting laws raise the expectations of the classes that hold power under them: Autocracy, Landed Voting and Wealth Voting raise the upper class, Universal Suffrage the lower class. Regulatory Bodies and Workers' Protections raise the lower class.
- Poor Laws, Wage Subsidies and Old Age Pension keep expectations at least 1, 2 or 3 SoL above the base-game level, however long hard times last. Universal Basic Income raises that floor to 10 SoL and the Post-Scarcity Economy to 15.
- Literate populations compare themselves with the rest of the world. Each month the gap between the world's average standard of living (weighted by population) and your country's average moves every stratum's expectations: for each point your average trails the world's, a fully literate population expects about 0.2 SoL more, and for each point it leads, 0.2 less. Literacy scales the effect, so an illiterate population doesn't compare at all. A country 10 points below the world average with 80% literacy expects 1.6 SoL more than it otherwise would, because its people know others live better; one 10 points above it expects 1.6 less, which makes a rich, educated country easier to keep content. This comparison is added after the floors above, so it can take expectations below the base-game level.

## 3.4 Pop consumption at high wealth

Pops in the base game stop at wealth 99; in the mod they can reach wealth 200. Four new needs appear as pops grow rich:

| Need | Starts at wealth | Goods that meet it |
|---|---|---|
| Convenience | 20 | Services, Consumer Appliances, Digital Access, Software |
| Healthcare | 16 | Drugs |
| Art | 25 | Art and Entertainment, some Services |
| Tourism | 25 | Tourism, some Transportation |

Healthcare takes about 3% of what a pop spends from wealth 29 to 50. Past wealth 50 it grows by the same amount at each level, so a richer pop buys more Drugs but they take a falling share of its spending. These Drugs come on top of the Drugs pops buy for Intoxicants (Pharmaceutical Industries and Drugs). The other needs,

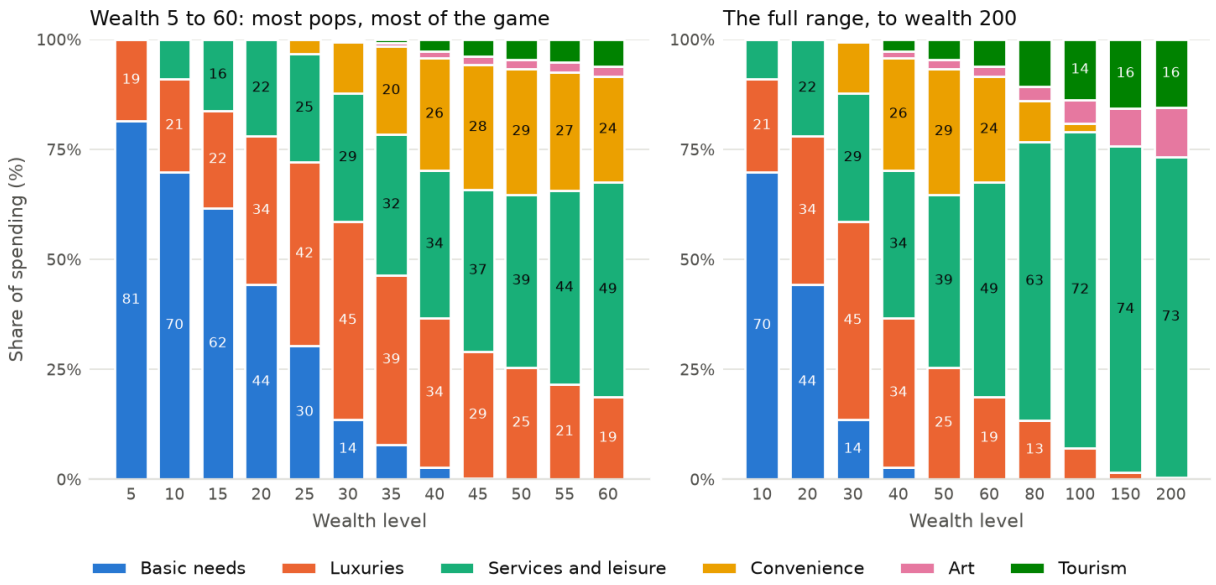


*Figure 5: How a pop's spending divides between needs as its wealth rises, at base prices. Left: wealth 5 to 60, where most pops spend most of a campaign. Right: the full range to 200.*

and Services, grow steeply with wealth. By wealth 60, services and leisure, Convenience, Art and Tourism take about four fifths of what a pop spends, and by wealth 100 nearly all of it. Luxuries peak at a little under half of spending around wealth 30 and then fade.

Tourism as an industry is covered in State tourism, and the new goods in The extended timeline, which also lists the base-game goods the mod renames. In the table above, Art and Entertainment is the base game's Fine Art. Chemicals, in the Strategic Reserve below, is its Fertilizer.

## 3.5 Transportation and shipping

The mod splits moving things the way the base game splits its goods, by distance:

- **Transportation** is all movement inside a state, of people and of goods. Like Electricity it is only available in the state that produces it.
- **Merchant Marine** is shipping between markets. It is traded across the market as in the base game.

Transportation is produced by:

- Railways, on every train method. In a power bloc with tier III or higher of the Transportation Infrastructure principle, each railway switches its steam, electric or diesel trains to a stronger version on its own, which carries 20–28% more and adds more infrastructure. That switch costs no retooling. The last train method, Vactrain Freight (from the era 12 technology Vactrains), runs on Electricity and burns no oil.
- Railways' Rolling Stock (passenger carriages up to Maglev Trains and Vactrains) and ports' Passenger Ships, which add more Transportation for a few more inputs.
- Highways and Airports, and Urban Centers' public transit.
- Railway, logistics and fulfillment company buildings.

Pops buy it to travel, and industry buys it to move its own goods: Construction Sectors from Iron-Frame Buildings onward, the rail-transport and tanker-car methods of logging camps, mines, plantations and oil rigs, the cold chain of fishing wharves, whaling stations and ranches, later mining and oil-well methods, and many late-game factory and retail methods.

Merchant Marine is produced by Ports, by Airports in a small amount, and by shipping and trading company buildings. Trade Centers buy it, and the base game spends it on port connections for overseas states and on convoys.

Railways and ports add infrastructure for every level, whether or not the level is staffed. The AI doesn't subsidize either: a state that needs more of them has dear Transportation, and they pay their way there.

Because Transportation can't be imported from another state, a state that buys a lot of it needs railways or highways of its own. The largest buyer is construction: a state with a large Construction Sector needs railways beside it, and its Transportation price shows when it is short. An agrarian economy barely touches the good.

## 3.6 Power grids

Electricity is only available in the state that produces it, as in the base game. A Grid Transfer Station, unlocked by Electrical Capacitors, joins its state to a power grid so that states can share it.

A grid starts as your stations in one market area. Once a month it moves Electricity from states where it is cheap to states where it is dear, until it sells at one price in the states that send it and at that price over their transmission efficiency (below) in the states that receive it. A sending state's station buys its spare power and sends it, and a receiving state's station delivers it. A grid with more power than it uses brings the price down in every state on it, the senders' too, so plants that make more than the grid needs earn less. A grid short of power raises the price in every state on it together, the same way. A grid takes out what it puts in, so power never crosses to a grid it isn't joined to. After a change, such as a new power plant or a new station, the grid takes a few months to settle.

Some of the power a short state receives is lost on the way. Grid Transmission Efficiency is the share that arrives. The last three technologies also set how far an undersea cable reaches (below):

| Technology | Era | Efficiency | Cable range |
|---|---|---|---|
| Electrical Capacitors | 4 | 75% | — |
| Rural Electrification | 6 | 85% | — |
| Transistors | 7 | 90% | — |
| Microprocessor | 8 | 92% | — |
| Clean Energy Technologies | 9 | 95% | 1 Sea Node |
| Smart Grids | 11 | 98% | 2 Sea Nodes |
| Quantum Materials | 12 | 100% | 3 Sea Nodes |

A receiving state pays the senders' price over its efficiency, so at 75% its Electricity costs a third more than theirs. A state whose price lies between the two neither sends nor receives, since losses would eat what the grid could move. An early grid evens prices out less than a late one.

You can build one station per state. Your first station in a market area can go in any of your states there, and every later one must border a state of yours that has a station, built or queued, so a grid grows outward from where it starts. A station costs 100 construction, the same as a level of Barracks, and joins its grid as soon as it is built. Engineers and clerks seek station jobs out and stay in them, so a station fills in two or three months and keeps its staff. A station moves its staff's share of what it is asked, so the monthly balancing asks a thinly staffed one for more, enough to move its full share down to a fifth staffed; while it is still hiring it moves a little more than its share. With no workers at all it moves nothing, and starts again as soon as it hires one.

Stations settle in Grid Power, a good whose price is fixed at 30, Electricity's base, and which can't be traded between markets. A sending station is paid in Grid Power what the Electricity it buys costs at the grid's price, and a receiving station pays in it what the Electricity it delivers sells for, so a station covers what its goods cost. The exception is a grid so short of power that a receiving state's price would pass its ceiling, about twice base: the receiving station then pays more for its power than it sells for. Stations are private infrastructure that is always subsidized: your government pays whatever a station's wages and upkeep cost beyond what it earns. That is about their whole cost, give or take a little while a grid is settling and prices are off the grid's.

Two things join grids that don't touch:

- With Clean Energy Technologies, you can build an HVDC Undersea Interconnector in a coastal state that has a station, built or queued: one per state, for 400 construction. Your government pays its engineers and Electronic Components. It lays an undersea cable with an Undersea Cable Range of 1 Sea Node. Smart Grids and Quantum Materials add 1 each, but only to interconnectors: a station without one has no cable. Two cables join their grids when either one's range reaches the other: at 1, states on the same Sea Node. Each point of range more crosses one more Sea Node, landing on a coast that touches both on the way. The cable joins your own grids across the sea, such as a mainland and a colony in your market, or a partner's under a Grid Interconnection Agreement. It needs a cable at both ends, and it joins nothing once its state's station is gone. The AI builds one at a coastal station when a grid of its own, or a partner's, is within its cable's reach, one at a time.
- The Grid Interconnection Agreement treaty article (from Intergovernmental Organizations) joins both countries' stations where they border, and their cables across the sea. It can be signed between two countries that share a border or a Sea Node, or whose cables already reach each other, which shows up to a month after both are laid. The two can be in one market or in two.

A grid that joins two markets sends power across the border from the market with more to spare to the one with less. Because Grid Power's price is fixed, the power is paid for at the grid's price on both sides: the sending market's stations are paid for all the Grid Power they sell, and the receiving market's pay for all they buy. One station in the receiving market, at the border, takes the import in: it buys the import and sells it into its market at the same price, which costs it nothing and keeps the market from running short of Grid Power.

The state panel shows a Power Grid tile on a state with a station: whether its station is sending or receiving, the grid's hub (its largest producer, which names the grid), the Electricity the station moves a week (its hover gives the price Electricity sells at in the state at that flow), how many states its grid has and your transmission efficiency. A state with a cable also shows Sea cables, how many stations from other grids its cable joins to this one; the hover names them and gives the cable's range. On a grid that joins two markets it also shows the Exports or Imports of the state's market across the border, and Pass-through on the station that takes the import in. Its hover has the grid's supply against its demand.

A station is worth building where power plants make more than the state uses, and where Electricity is dear. A state with no power uses none, because its buildings can't run the methods that need it; a station lets it receive power once a building there switches to one. The AI builds stations in every market area where it makes Electricity: it starts in a state with power plants and queues a station in one more bordering state each month, larger states first.

## 3.7 Tariffs for all goods at once

On your own market, the Trade tab of the Market panel opens with two All Goods rows above the goods list, one for imports and one for exports. Each row has the seven levels a good's own row has, from Maximum Tariffs to Maximum Subventions, and a click sets that level on every tradeable good, including goods your market doesn't trade yet. A good under a No Tariffs or No Subventions treaty article keeps its level when the level you pick is one the article forbids.

Only the market owner can use the rows, and only at levels your Trade Policy allows: under Free Trade the tariff levels are grayed out. After you use a row, you have a week to change that direction again to correct it; then the row waits 3 months, as each good's own buttons do after a change. Hover a level to see what it will do and what is stopping it; while the row waits, it says about how many days are left. Under *Tax Code and Customs Enabled* the rows are grayed out with the other tariff buttons.

## 3.8 The Strategic Reserve

The Strategic Reserve journal entry lets you stockpile strategic goods in your capital, buying them when they are cheap and releasing them when prices rise or supply fails. It is shown once you research Logistics and becomes active once you have a Strategic Reserve Hub. Build one, or take the decision Establish a Strategic Reserve, which places the hub in your capital at once with no construction cost. No game rule controls it.

## Reserve hub and silos

Two buildings make up the reserve, both unlocked by Logistics.

| Building | Where | Construction cost | Adds | Staff |
|---|---|---|---|---|
| Strategic Reserve Hub | Capital only; one per country, at one level | Very low | 5,000 storage for each good; each good may move 1,000 units a week | 5,000 |
| Strategic Reserve Silo | Any state, any number of levels | High | Per level: 1,000 storage for each good; +100 to each good's weekly limit | 500 per level |

The hub is the reserve: it holds the controls and does the buying and selling. It can't be expanded past its single level, downsized or demolished. Silos are how a reserve grows: they only add room and flow, and do nothing without a hub. The hub trades only when it is fully staffed. Below full occupancy the panel marks the hub Understaffed and every good Blocked, and the hub neither trades nor replaces what decays.

## Reserve goods and decay

The reserve holds eight goods. Stock decays every week at a yearly rate that technology lowers.

| Good | Available from | Decay per year at first | After all technologies |
|---|---|---|---|
| Grain | Always | 25% | 0% |
| Small Arms | Always | 1.5% | 0.7% |
| Artillery | Always | 1% | 0.4% |
| Ammunition | Percussion Cap | 2% | 0.5% |
| Oil | Fractional Distillation | 0.5% | 0.05% |
| Chemicals | Intensive Agriculture | 4% | 0.1% |
| Aeroplanes | Military Aviation | 4% | 2% |
| Tanks | Mobile Armor | 2.5% | 1.4% |

Aeroplane and tank decay rises for a time with jets, stealth aircraft and composite armor before later technologies bring it down. The hub buys each week's decay on top of whatever rate you set, so a rate of 0 holds a stockpile steady at a small ongoing cost. Hover a good's row for its decay rate broken down by source.

## The Strategic Reserve panel

The journal entry's panel has an overview and the Reserve Inventory, with a "?" at its top right that opens the Help page. Before you have a hub the entry shows only a line telling you how to get one. Once the hub stands there is no status line, because the overview shows how the hub is doing.

The same panels appear as a Reserve tab in the Market panel, and a change made in one shows in the other. The tab is there once you research Logistics and is grayed until you build a hub; hover it for what is still missing. It shows only while the Market panel shows your own market, the one your capital is in, whether you lead it or joined it, because that is where the reserve buys and sells. If you open another market with the tab still selected, it offers a button back to your own. The tab ends with an Open Journal Entry button.

The overview at the top is always shown, as four boxes. Hub has the hub's icon at full color while the hub is fully staffed, marked Hub Operating, and dimmed and marked Understaffed below that; hover it for the staffing. Under the word is the Cap, how many units of each good the hub may move in a week (hover it for where that comes from). Blocked counts the goods that are full, or empty while a release is set; it turns red while one is empty, and its hover names them. Purchases is what this week's buying costs at today's prices, and Sales Income what this week's releases earn.

The Reserve Inventory starts open. Its first line shows the Adjustment Step, with the Cycle Step and Reset Rates buttons beside it. The second line holds All: Stabilize, Military: Stockpile and All: Manual, which set the

*Figure 6: The Strategic Reserve tab of the Market panel: the overview (Hub Operating, Flow Cap, Sales Income) above the Reserve Inventory, with the Ammunition row expanded to its figures, the Fill by Month chart and the start of its Policy Settings.*

policy of every good at once (see Reserve policies and presets). Below them is a row for each good you have unlocked:

| Cell | What it shows |
|---|---|
| Good | The good's name. Click it to expand the row. |
| Stock | How full the good's reserve is. A marker shows the target stockpile when a buying policy stops short of full, and the protected stockpile when a selling policy keeps a floor. |
| Status | A crate icon: a yellow bar for Idle, a green arrow going in for Storing, an orange arrow coming out for Releasing, a striped barrier for Blocked. Hover it for the word and the reason. |
| Policy icon | A brass governor, lit while a reserve policy sets the good's rate and dimmed while you set it by hand. Hover it for the policy and what it is doing; click it to open the good's Policy Settings. |
| Rate | The decrease button, the good's rate, then the increase and stop buttons. Hover a button for the rate it would leave. |

Blocked means the stockpile is full, or empty while a release is set, or the hub is understaffed. Hover the Stock bar for what is stored and what its marker means.

An expanded row lists Stored, Rate Setting, Last Week (what actually moved), Decay, Market Price (against the good's base price, with the average the policy acts on) and Reserve Policy, then a line on what the policy is doing and why. Hover Rate Setting for what the hub actually trades this week, the rate plus the decay it replaces, and Decay for where the decay rate comes from. Under them, Fill by Month charts how full the reserve was each month, with 1 Year, 5 Years and 20 Years ranges. Each bar is a share of capacity, so building Silos lowers the bars of an unchanged stock. Policy Settings comes last, collapsed until you open it or click the policy icon.

The "?" at the right of the tab's header, and at the top of the journal entry, opens the Help page, How the Strategic Reserve Works. The page explains the hub and silos, rates, status, decay, reserve policies, presets and settings, sales income and the fill history.

## Storing and releasing reserve goods

You store and release goods from each good's row in the Reserve Inventory, as set out in The Strategic Reserve panel.

A positive rate buys that many units a week from the market into the reserve; a negative rate releases that many onto the market. Every press of a good's decrease or increase button moves its rate by the Adjustment Step, which the Cycle Step button sets to 1, 10, 100, 1,000 or 10,000 (10 at first). The stop button sets the good's rate to 0. Reset Rates sets every good's rate to 0 and switches every good to Manual. The game keeps each rate within the weekly flow limit and the room or stock left, and sets it to 0 when the reserve fills or empties.

The hub is a government building in your capital, and the goods pass through it as its inputs and outputs. Storing buys from your national market, paid from the treasury, and adds demand that raises the price. Releasing sells onto your market, adds supply that lowers the price, and pays the treasury; the overview shows this as Sales Income.

The overview's Purchases and Sales Income boxes show the week's two sides together. The budget panel shows them separately. Sales appear under Additional Income, as the Strategic Reserve journal entry's income; purchases appear with the hub's inputs under Goods for Government Buildings. They are not netted, so a reserve that buys small arms while selling grain can show +1,000 a week of income and −2,000 a week of expenses at the same time.

## Reserve policies and presets

Instead of setting a rate by hand, you can give each good a policy that decides its rate every week from the national market price, measured against the good's base price and averaged over several weeks. Click a good's policy icon, or expand its row and open Policy Settings, to choose one. The five policy buttons there use short names: Manual, Buy Cheap, Release High, Stabilize and Stockpile.

| Policy | What it does |
|---|---|
| Manual | You set the rate. The default for every good. |
| Buy When Cheap | Buys while the averaged price is below your purchase threshold. Never sells. |
| Release When Expensive | Sells while the averaged price is above your release threshold. Never buys. |
| Stabilize Prices | Does both. |
| Stockpile | Buys at full flow while the averaged price is at or below your purchase threshold. Sells above your release threshold while you are at war, and in peacetime only above +50%. |

Stockpile is for goods you want in hand before a war, such as ammunition. A munitions industry is rarely profitable enough to push the price far below base, so the other buying policies seldom fill a reserve of it. Choosing Stockpile loads the Stockpile preset, which buys at or below +5% of base price, and the response ramp does not slow its purchases. At war it sells into the price spike; in peacetime it keeps the stock unless

the price passes +50%, as it can when an army mobilizes for a diplomatic play. Between your release threshold and +50% in peacetime, the good's row says Holding.

Eight settings shape a policy, and four presets fill them all in one click:

| Setting | Conservative | Standard | Aggressive | Stockpile |
|---|---|---|---|---|
| Buy below (% of base price) | −20 | −10 | −5 | +5 (at or below) |
| Release above (% of base price) | +30 | +20 | +10 | +20 (at war) |
| Maximum weekly flow (units) | 2% of capacity | 5% of capacity | 12% of capacity | 5% of capacity |
| Weekly purchase budget (estimated) | 0.1% of weekly GDP | 0.3% of weekly GDP | 0.8% of weekly GDP | 0.3% of weekly GDP |
| Price memory (weeks averaged) | 8 | 4 | 2 | 4 |
| Response ramp (points) | 20 | 10 | 5 | 10 (sales only) |

The purchase threshold can be set from −50% to +25% of base price under any policy.

Every preset uses the whole capacity: a protected stockpile of 0% and a target stockpile of 100%. Set those two yourself if you want a floor the policy never sells. The response ramp spreads the reaction out: the flow rises from nothing at the threshold to the maximum that many points beyond it. At 0 the policy is a switch that runs at full flow once the price crosses the threshold.

A preset keeps its flow and budget in step with your country, recalculating them every week from your capacity and GDP, until you change either by hand. The policy and the preset in force are shown selected, and changing any setting by hand ends the preset. The budget is set for each good and caps each week's spending; what goes unspent doesn't carry over. A new reserve starts every good on Manual with the Standard settings loaded, so switching a good to a policy works at once. The row's own rate buttons, Reset Rates and All: Manual switch goods back to Manual. Policies keep running while the journal entry is closed.

The buttons on the Reserve Inventory's second line act on every good you have unlocked:

| Button | What it does | When to use it |
|---|---|---|
| All: Stabilize | Every good to Stabilize Prices with the Standard preset. | To lean against price swings in every market you stock. |
| Military: Stockpile | Ammunition, oil, small arms, artillery, aeroplanes and tanks to Stockpile with its preset. Grain and chemicals keep their policy. | To build a war reserve at about base price. |
| All: Manual | Every good to Manual. Each keeps its current rate and settings. | To take back control without stopping trade; Reset Rates also sets every rate to 0. |

## Moving or losing the reserve hub

The hub always follows your capital. If your capital moves, the next weekly update builds a new hub in the new capital for free and removes the old one. Your stock and your capacity carry over unchanged, since the hub has only one level and your silos stay where they are. The cost is time: the new hub hires its 5,000 workers from scratch and trades nothing until it is fully staffed, so the reserve neither buys, releases nor replaces decay in the meantime. Avoid moving your capital while you are counting on the reserve.

If a foreign power takes the hub's state, the hub is destroyed and the whole stockpile is lost. Since the hub always stands in your capital, protecting the reserve means keeping your capital state.

Losing the hub never closes the journal entry. Until you have a hub again the reserve waits: it buys, sells and loses nothing to decay, every good reads Blocked, its controls are grayed, and the entry offers a new hub. Build

one in your capital, or take the decision Establish a Strategic Reserve to place it at once. After a capture the new hub starts empty.

A revolution or secession that takes the hub's state does not cost you the stock. The rebels hold the building while the war lasts, and your stock waits for a hub. If you win, the old hub comes back to you, and if you placed a new one in the meantime, the next weekly update removes the spare. If a revolution wins, its new government continues the nation and keeps the reserve. If a secession wins, the new country keeps the building, and you keep your stock for a new hub in your capital. Silos in states you lose go with them: when a hub stands again, the reserve keeps only what the hub and your remaining silos can hold.

### Pledging grain to the World Food Reserve

Once the United Nations has founded the World Food Reserve, a member with a Strategic Reserve Hub can press Pledge Grain, under the World Food Reserve line in Our Obligations on the UN tab of the Diplomacy panel. When a UN aid mission opens in another country, the Reserve takes up to a quarter of your grain, at most 2,500 units, if you hold at least 500. The grain leaves your reserve through its own bookkeeping, and you gain standing for each draw. Withdraw Our Pledge stops it at any time (see The World Food Reserve and hunger).

### How the AI uses the reserve

The AI founds its reserve through Establish a Strategic Reserve: great and major powers take it, more readily in peacetime, and so does any country with a GDP above 10 million. AI reserves add silos once any good passes 75% of capacity. An AI reserve runs grain and chemicals on Stabilize Prices with the Conservative preset: it buys below −20% and releases above +30%. Its ammunition, oil, small arms, artillery, aeroplanes and tanks go on Stockpile with the Stockpile preset while its treasury allows: it must not be borrowing or in default, and it needs a quarter of its gold reserve limit in hand to start. It then buys them at about base price in peacetime and sells into a war's price spike. Once it borrows, those goods go back to Stabilize Prices on Conservative. The AI reviews this every week. It follows the same rules, limits and costs as yours.

## 3.9 Wartime demand for munitions

Peacetime armies train with half the base game's ammunition. When an army mobilizes, Basic Supplies, which no army can switch off, raises its ammunition use by +300% (+50% in the base game). A mobilized battalion therefore burns four times its peacetime ammunition, twice the base game's peacetime figure. The mod removes the extra ammunition that Extra Supplies and Luxurious Supplies add in the base game, so the supplies you choose no longer change how much ammunition an army uses. The Basic Supplies tooltip shows the ammunition figure. A few of the mod's own mobilization options add more on top; see Mobilization options.

The demand follows mobilization rather than war: an army mobilized for a diplomatic play spikes it even if no war follows, and the demand falls away as the army demobilizes. Where armies buy most of the ammunition, four times the demand can push its price to the maximum. Build munitions capacity, put ammunition on the Strategic Reserve's Stockpile policy (it buys at about base price and sells into the spike), or both, before you go to war.

## 3.10 New mineral deposits

The mod adds fourteen mine types, from Copper and Bauxite Mines to Lithium, Rare Earth Metals and Platinum Group Metals Mines, and places their deposits in the regions that hold them in reality, such as copper in the Atacama and Katanga. They start undiscovered and turn up over time, as undiscovered resources do in the base game. Twenty-one state traits for famous mining regions, among them the Bushveld Igneous Complex, the Central African Copperbelt, the Sudbury Basin and Bayan Obo, raise the throughput of the mines there. The goods these mines produce are covered in The extended timeline.

CHAPTER 4

# Banking and monetary policy

The banking cycle gives every economy with a stock exchange a financial cycle of its own: booms that build speculative bubbles, crashes that burst them, and contagion that carries a crash along trade routes into other countries. On top of the cycle sits a monetary layer: a central-bank policy rate, inflation, an exchange rate, gold reserves, and treaties that tie one currency to another. The Banking System game rule controls both. *Enabled* gives you everything in this chapter, *Simplified* keeps the cycle and drops the monetary layer, and *Disabled* removes the journal entry (see the Introduction).

Under every setting the mod replaces the base game's flat interest rate on government debt. You borrow at a rate built from a world reference rate, your country's credit standing and a risk premium, so a banking panic makes your loans dearer and a sound currency makes them cheaper.

## 4.1 The Banking Cycle journal entry

The cycle lives in a journal entry in the Domestic Affairs group. Its title follows your economic system: Boom & Bust Cycle in a market economy, Logistics & Allocation Gauge under Command Economy, and Cooperative Finance Dashboard under Cooperative Ownership. Its description changes with the title and explains what the readings mean in that economy. The model is the same in all three; the phase effects and the tools differ.

Every country has the entry from the start of the game, but its financial cycle starts only once you have researched Stock Exchange and own an Urban Center of level 5 or higher. Until then the entry shows the cycle as Not Started, with the two conditions on hover. You see your Monetary Policy readings, Credit Standing and Rate Paid among them, but the banking tools and the central bank's controls stay grayed, and nothing in the cycle happens: no crashes, banking events or contagion. When you meet both conditions during the game, a notice tells you the cycle has begun. It starts at Stable with no momentum or bubble, and keeps running for the rest of the game, even if the city later shrinks. If a revolution succeeds, the new government carries on the cycle and the central bank where the old one left them (see After a revolution), but policies you had switched on are lost and must be enabled again.

The same panels appear as a Banking tab in the Budget panel, and a change made in one shows in the other. The tab ends with an Open Journal Entry button, which opens the entry with its description. The banking panels describes what they show.

Another country's Banking tab, in National Info (Introduction), shows its phase, momentum and bubble pressure, the four bars, its Monetary Policy readings, the policies it has in force and its history. It has none of the buttons and no Available Interventions.

### Cycle value, momentum and bubble pressure

Three readings drive the cycle, each drawn as a bar at the top of the panels: Banking Cycle Value, Banking Momentum and Bubble Pressure. The cycle value runs from 0 to 100 and decides the phase. Momentum is added to the cycle value every month, so it sets how fast conditions change; it loses a tenth of itself each month. Bubble pressure, also 0 to 100, is accumulated speculation. It decides how likely a crash is and how hard it hits. A fourth bar, Policy Stance, runs from tight to loose and shows where your interest rate sits; it moves only under the full Banking System.

*Figure 7: The Banking tab of the Budget panel: the four bars, the overview's icons, Active Policies and the start of Monetary Policy.*

The overview's icons report momentum and bubble pressure as bands, not figures. Momentum reads Freefall (−5 or less), Collapsing (−5 to −3), Falling (−3 to −1), Steady (−1 to +1), Rising (+1 to +3), Surging (+3 to +5) or Overheating (+5 and up); the two outer bands are past the ends of the momentum bar. Bubble pressure reads Low (under 15), Building (15 to 29), Elevated (30 to 49), High (50 to 74) or Severe (75 and up).

Beside the phase, momentum and bubble pressure icons, a third line gives how far the cycle value, momentum and bubble pressure can move at the next monthly update, as a range such as −1.1 to +2.2. It is worked out from what the panels show: the bands, your modifiers, your policy stance and the random nudge. The ranges are wide because the bands are: Steady momentum lies anywhere from −1 to +1, so the cycle value's range is nearly two points wide before anything else moves it. Crashes and events can move a reading further. Hover a range for its parts, and hover a figure in it for the modifiers behind it.

## The seven banking cycle phases

Each phase applies a modifier every month. The low phases hurt output and drain bubble pressure; the high phases raise output and build it.

| Phase | Cycle value | Effects in a market economy |
|---|---|---|
| Panic | under 10 | Services output −50%, manufacturing throughput −20%, capitalists' investment-pool contribution −25%, +2 points of risk premium, bubble −7 a month |
| Downturn | 10–24 | Services −25%, manufacturing −10%, capitalists' contribution −10%, +1 point of risk premium, bubble −3 a month |
| Stagnation | 25–39 | Services −10%, manufacturing −5%, +0.4 points of risk premium, bubble −1 a month |
| Stable | 40–59 | None |
| Expansion | 60–74 | Services +25%, manufacturing +5%, capitalists' contribution +5%, bubble +3 a month |
| Boom | 75–87 | Services +35%, manufacturing +10%, capitalists' contribution +10%, bubble +4 a month |
| Frenzy | 88 and up | Services +50%, manufacturing +15%, capitalists' contribution +15%, bubble +8 a month |

The capitalists' contribution figures are percentage points, not percentages. Capitalists put a share of their income into the investment pool, and a Panic takes 25 points off that share rather than a quarter of it.

A command economy's phases change construction costs and bureaucracy instead of services and investment; a cooperative economy's add to the investment pool's weekly income or take from it, from a quarter more in a Frenzy to about a third less in a Panic. Both are far less exposed to crashes.

### What moves the banking cycle

Two forces pull the cycle back toward the middle: the phases, and a random monthly nudge that your laws scale (Banking Cycle Volatility).

Two others can push against that pull. In a market economy, bubble pressure feeds momentum, so a boom keeps climbing once a bubble forms; the tooltip shows this as the Speculative Inertia, Feedback and Euphoria modifiers. A cooperative economy feels a fifth of that pull (Reinvestment Drive, Overcommitment and Overcommitment Spiral), and a command economy none. A budget deficit lifts the cycle by a tenth of a point a month per 1% of GDP, up to one point, and a surplus lowers it (the Government Fiscal Policy Effect).

Policies, laws, events and, under the full Banking System, your policy stance push the readings directly. Fiat Money and Digital Currency add 0.2 bubble pressure a month, and Digital Currency and Decentralized Cryptocurrency make the cycle more volatile. Wealth Concentration moves it too: Great Family Fortunes adds up to 0.5 bubble pressure a month and Dispersed Wealth takes up to 0.53 off (see Who inherits).

## 4.2 Banking crashes and contagion

A crash is the cycle's failure state. It starts at home from your own bubble, or arrives from a trading partner's crash.

### How a banking crash starts

Every month the game rolls for a crash. The higher the phase, the lower the bubble at which risk begins and the faster it grows. Base monthly chances in a market economy, before the Banking Crash Likelihood your laws and policies add:

| Phase | Risk begins above bubble | At bubble 50 | At bubble 75 |
|---|---|---|---|
| Frenzy | 15 | about 11% | about 17% |
| Boom | 25 | about 4% | about 7% |
| Expansion | 50 | none | about 1% |
| Stable | 75 | none | none |
| Stagnation or lower | 90 | none | none |

A command economy rolls at about a third of these chances, a cooperative economy at about two thirds.

## Banking crash severity

Severity is about half the bubble pressure at the moment of the crash, multiplied by a random factor between a half and two. It decides the event and how far the cycle falls, and every crash wipes bubble pressure to 0.

| Severity | Event | Cycle value falls to | Momentum falls to |
|---|---|---|---|
| 80 and up | Systemic Collapse | 5 | −5 |
| 60–79 | Banking Panic | 10 | −4 |
| 40–59 | Financial Crash | 20 | −3 |
| 20–39 | Market Downturn | 30 | −2 |
| under 20 | Market Correction | 40 | −1 |

A crash never raises either figure. A country already below the tier's figure, such as one in a Downturn that an imported Market Correction reaches, keeps the cycle value and momentum it has; only the bubble pressure is wiped.

Climbing back from a Panic-level crash to Stable takes more than a year without a crash response. Command economies suffer 40% less severity and cooperative economies 20% less, and less still when the crash is imported.

## Responding to a banking crash

The crash event always offers Weather the storm, which is free but angers the upper and middle strata. Beside it you get one response set by your financial regulation law and up to four more, depending on your technology and laws: an emergency discount window (Central Banking), coordinated swap lines (Investment Banks; not the treaty article), a fiscal stimulus (Keynesian Economics) and emergency capital controls (Corporate Governance).

Each response adds back 3 to 12 cycle value and up to 3 momentum, but never lifts either above where it stood before the crash. It costs a GDP-scaled treasury expense that fades over six months, holds 1 to 4 intervention points for a year (see the intervention budget), and radicalizes some of the upper strata. If your intervention budget goes negative, for example after a law downgrade, the game switches off one crash response a month. Once those are gone, it lifts one of the five monetary interventions in External & Currency each month until the budget balances or none of those interventions remains.

## Banking contagion

A crash can spread to any country whose financial cycle has started and that is tied to the crashing one by a shared market, customs union, currency or border, economic dependence, or more than 1% of its market's trade, production or consumption. A country more than 20 times the crashing economy's size is immune, and the limit applies at every step of a cascade, so a crisis wave rarely climbs from small economies to large ones. Each exposed country has a 25% base chance to be reached, more when the source is at least your size, when the ties are close and when the crash was severe.

A week later, the reached country's own bubble pressure (plus a bump for the original crash's severity) decides whether it crashes outright or only suffers a scare of −3 cycle value and −1 momentum. A country with a small bubble is mostly safe, and one whose banks crashed in the last year, at home or by contagion, can only take the scare. A scare event reaches you at most once every two years: one that lands sooner passes with no event and no effect, and does not restart the two years. A crash always reaches you. One that does crash passes contagion on, so one crash can become a wave.

Command and cooperative economies are harder to infect. A Bank Holiday or a banking union lowers the chance of catching a crash, and a Lender of Last Resort guarantee makes an imported one start less deep. The contagion event lets you shut your markets to protect domestic banks or keep them open at the cost of more radicals.

## 4.3 The banking panels

The journal entry and the Budget panel's Banking tab show the same overview and sections, in the same order. The overview at the top is always shown: the four bars, then six boxes, two to a row, each with its icon on the left and three lines beside it: a caption, a word and a figure. Hover a caption for what the term means, and hover the box for the reading in detail; the cycle's tooltips also list what pushes it each month. Hover the word for what that reading does: the phase's monthly modifiers (they differ for a market, a command and a cooperative economy) and its crash risk, what a band of momentum or bubble pressure means for the cycle and the crash roll, what the stance and the price band put on your country, and what holds the budget's points. For the cycle value, momentum and bubble pressure, the third line is next month's range (see Cycle value, momentum and bubble pressure), with a tooltip of its own. With momentum Overheating the range has no top for the cycle value, which reads "+4.2 or more", and no bottom for momentum; with momentum in Freefall it is the reverse.

| Caption | Icon | Word | Third line |
|---|---|---|---|
| Cycle Phase | A bank front with a mark: arrows down in a Panic or Downturn, a level bar in Stagnation or Stable, arrows up from Expansion, and the bank gilded in a Boom or Frenzy | The phase | Next month's range for the cycle value |
| Momentum | An arrow, doubled from Collapsing and Surging outward | The momentum band | Next month's range for momentum |
| Bubble Pressure | Coins inside a bubble that grows band by band, its rim going from green to red, cracked at Severe | The bubble band | Next month's range for bubble pressure |
| Budget | A stack of gold tokens | How many intervention points your active policies hold | How many are free |
| Policy Stance | A tap: the more coins fall from it, the looser the stance, and a padlock at Very Tight | Very Loose, Loose, Neutral, Tight or Very Tight | The policy rate |
| Inflation | A price tag marked for the band, from a blue arrow down at Deflation to a flame at Very high; a wheelbarrow of banknotes at Hyperinflation | The price band, or Foreign money once you dollarize, or Set by plan in a command economy | Inflation over the past year |

Policy Stance appears only once you hold a dial, and Inflation only under the full Banking System. A warning mark on the bubble means a crash is at its most likely: the cycle value is 90 or more, or a Boom or Frenzy has momentum Surging or Overheating. The words are colored by how far the reading is from calm: green and white are safe, yellow and gold warn, and red marks an extreme (Panic, Frenzy, Freefall or Overheating momentum, Severe bubble pressure). A Boom reads blue.

Before your financial cycle starts, the overview has one box in place of the bars, the cycle's three boxes and Budget: Cycle Phase reads Not Started, its third line says how many of the two conditions you meet, and its hover lists them with ticks. Active Policies and History wait for the cycle. Monetary Policy and Available Interventions show their rows with every control grayed, under a line that says why.

The cycle's phase also shows in the top bar, as its icon right after your weekly balance, once your cycle has started. Hover it for the phase's word and the same tooltip, with a crash-risk warning while that risk is high, and click it to open the Banking tab.

Below the overview come four sections. Each heading opens or closes its section:

| Section | Starts | Shows |
|---|---|---|
| Active Policies | Open | The tools in force, each with its point cost and a Disable button. |
| Monetary Policy | Open | Under the full Banking System, the rate, gold, the exchange rate, what borrowing costs and prices (see Monetary policy under the full Banking System). Under Simplified, only the Open-Market Operations row, in a market economy. |
| Available Interventions | Open | Your economic system's other tools, in categories you can close one by one, each tool with its point cost and an Enable button. |
| History | Open | The charts in Banking history charts. |

The "?" at the right of the tab's header, and at the top of the journal entry, opens Banking's Help page, How Banking Works. It explains the cycle, crashes and the intervention budget, and under the full Banking System the policy rate and the stance, the mandates, gold and the peg, the exchange rate, international arrangements, what borrowing costs and prices.

Hover a tool's name for what the tool does, its full name and what it requires. A grayed button means you cannot use the tool now, and its tooltip says why. Enabling a tool moves it to Active Policies. Some rows carry a shortened name: under Directed Credit each row names only its sector, and Counter-cyclical Buffer stands for Enable Counter-cyclical Buffer. The tables in this chapter use the full names.

## The intervention budget

The intervention budget is the pool of points your active policies hold. Your financial regulation law gives 1 to 7 points, National Bank Established 1 more, and a power bloc's banking union 1 more. The overview's Budget box shows how many are in use and how many are free. Hover the free figure to see what each grant adds and what each active tool holds.

Switching a tool on has its own price: the lending tools charge the treasury 0.2% to 2.5% of GDP, most regulatory tools create radicals, and Capital Controls (Outflows) costs infamy and great-power relations. Sterilize Capital Inflows and Emergency Import Financing add a weekly government expense while they run, a quarter of their monthly rate, so a year costs 13 months' worth. It shows in the Budget panel under Banking, and their tooltips show the current weekly cost.

## Market economy banking tools

A market economy has twenty-two tools under the full Banking System, seventeen under Simplified. The leaning tools drain bubble pressure and cool a boom, the credit tools feed growth, the crisis tools shorten a slump, and the foreign capital tools manage gold and the exchange rate. The sections below say what each group buys, what it needs and when it is worth its points. Banking tool list gives every tool with its category, points and requirements in one table.

Because momentum loses a tenth of itself each month, a steady push settles at about ten times its monthly figure. A tool worth −0.05 momentum a month ends up taking about half a point a month off the cycle value.

## Leaning against a boom

The four leaning tools in Prudential Regulation drain bubble pressure and slow momentum every month they run. Switch them on once the cycle reaches Expansion and bubble pressure starts to build. Lift them when it falls back to Stable: in a slump they slow the recovery you want. The counter-cyclical buffer needs International Exchange Standards; the other three are available from the start.

| Tool | What you gain each month | What you pay or give up |
|---|---|---|
| Moral Suasion | −0.15 bubble pressure, −0.02 momentum | 1 point and nothing else. |
| Raise Reserve Requirements | −0.9 bubble pressure, −0.04 momentum; under the full | 2 points and 5% lower investment efficiency for capitalists. 2% of pops radicalize when you raise them, and again when you lift them. |

| Tool | What you gain each month | What you pay or give up |
|---|---|---|
| | Banking System, half a point less inflation pressure | |
| Enable Counter-cyclical Buffer | −1.5 bubble pressure, −0.05 momentum | 2 points. Capitalists pay 1 point and shopkeepers 5 points less of their income into the investment pool. Radicals when you enable it and when you lift it, as above. |
| Raise Margin Requirements | −1.2 bubble pressure, −0.05 momentum | 2 points and −0.5 standard of living for the middle strata. Radicals when you raise them and when you lift them. |

Measure them against the phases: Expansion adds 3 bubble pressure a month, Boom 4 and Frenzy 8. All four together drain about 3.75 and take 7 points, enough to hold an Expansion's bubble steady but not a Boom's. Under the full Banking System a Tight or Very Tight policy stance drains up to 3 more (see Monetary policy stance), so the rate and the tools together can hold a Boom. In a Frenzy the bubble outgrows everything you can drain, so lean early.

The AI uses Moral Suasion in an Expansion, reserve requirements and the buffer from Expansion upward, and margin requirements in a Boom or Frenzy. It uses any of them while momentum reads Surging or Overheating. It keeps the buffer, reserve requirements and margin requirements until the cycle is back below Expansion and no longer climbing, and drops Moral Suasion more freely, since switching it costs nothing.

## Directing credit into growth

Directed credit steers cheap loans to one sector. Each sector builds its buildings 10% faster, pleases some interest groups and angers others, and adds momentum and bubble pressure, so it belongs in a cycle at or below Stable, not in a boom. Switching one on costs a one-off 0.8% of GDP from the treasury, the state's share of the loans, and lifting it radicalizes 2% of pops. Only one sector runs at a time, two under Directed Credit & Development Banks, and Prudential / Narrow Banking forbids them all. Electrification needs Rural Electrification; the other four are available from the start.

The sectors are priced and built differently, so they are not interchangeable.

| Sector | Points | What you gain | Approve | Disapprove | Each month |
|---|---|---|---|---|---|
| Directed Credit to Infrastructure | 3 | Infrastructure buildings build 10% faster; infrastructure in every state +5%; decrees cost 10% less | Armed Forces +1, Trade Unions +1 | Industrialists −3 | +0.03 momentum, +0.1 bubble pressure |
| Directed Credit to Heavy Industry | 4 | Heavy industry builds 10% faster and runs at +5% throughput | Industrialists +2 | Landowners −3 | +0.06 momentum, +0.5 bubble pressure; greenhouse gas emissions +5% |
| Directed Credit to Agriculture | 2 | Farms, plantations and ranches build 10% faster; food security +5% | Landowners +3, Rural Folk +2 | Industrialists −3 | +0.03 momentum, +0.4 bubble pressure |
| Directed Credit to Armaments | 3 | Military industry and shipyards build 10% faster; military industry runs at +5% throughput; military goods cost 10% less | Armed Forces +3 | Intelligentsia −3 | +0.06 momentum, +0.05 bubble pressure |
| Directed Credit to Electrification | 1 | Power plants build 10% faster; electricity output +5% | Intelligentsia +2, Trade Unions +1 | Industrialists −3 | +0.04 momentum, +0.2 bubble pressure |

Heavy Industry is the strongest push and the one that overheats a cycle, so it costs the most, adds emissions, and is the sector to lift first when the cycle warms. Armaments pushes as hard with almost no bubble pressure, but it pays only where you build or buy military goods. Infrastructure adds the least bubble pressure of the five and is the one to hold into a recovery, though it is also the weakest rescue in a slump. Agriculture gives the least momentum for its bubble pressure, which is why it is cheap. Electrification covers only power plants, which makes it the cheapest sector and a modest one. Heavy Industry also builds high-tech industry faster, because high tech is part of it.

Under Directed Credit & Development Banks the second sector has to fit in its 5 points. Electrification fits beside any sector, and Agriculture beside Infrastructure or Armaments. A National Bank's sixth point adds Infrastructure with Armaments, and Heavy Industry with Agriculture. Heavy Industry never runs beside Infrastructure or Armaments.

Export Credit Facility, listed under External & Currency, is the credit tool for exporters. It gives +25% export advantage, +0.05 momentum and +0.15 bubble pressure a month, for 2 points, 0.2% of GDP at the start and 2% less tax collection while it runs. It needs Corporate Governance, and Prudential / Narrow Banking forbids it.

AI countries direct credit in a Downturn, in Stagnation, or while a Stable cycle has momentum Collapsing or in Freefall. They choose the sector by who governs: heavy industry for Industrialists, agriculture for Landowners or Rural Folk, armaments for the Armed Forces or in war, electrification for the Intelligentsia, infrastructure otherwise. They pay the same points as you, so a sector they cannot afford is not chosen. They lift it in a Panic or once a Stable cycle stops falling, and run Export Credit Facility from Stagnation down.

## Pulling out of a slump

Six tools help in a Downturn or Panic: the four in Crisis Response, Expand Deposit Guarantee from Prudential Regulation, and Open-Market Operations from Monetary Policy. They run alongside the crash event's responses (see Responding to a banking crash), which hold points of their own for a year. Emergency Liquidity needs Investment Banks, Asset Relief and Open-Market Operations Keynesian Economics, the bail-in Globalization and the deposit guarantee Consumer Credit.

| Tool | What you gain | What you pay or give up |
|---|---|---|
| Declare a Bank Holiday | Falling momentum stops at once and cannot drop below 0 for 90 days. Crash likelihood −90% for those days, which keeps contagion out. If the banks stay shut the full 90 days, +1 momentum when they reopen. | 2 points; −20% services output and −5% tax collection while it lasts. Radicals: 5% of middle-strata pops, 3% of upper and 2% of lower, about 2.5 to 3% of your population. Only in a Downturn or Panic, and once in five years. Ending it early gives up the +1. |
| Emergency Liquidity Program | +12 cycle value at once, once per crisis, then −0.8 points of risk premium and +0.08 momentum a month. | 4 points; 1.2% of GDP when you open it, 1% refunded when you close it. +10 bubble pressure at once, with the +12, and +0.8 a month. Only in a Downturn or Panic. Closes itself, with the refund, once the cycle has held at Stable or above for 12 months. |
| Asset Relief Program | +0.3 cycle value a month. | 5 points; 2.5% of GDP at once; −5% tax collection and −5% bureaucracy; 3% of pops radicalize. |
| Bail-in Regime | +0.15 cycle value and −0.3 bubble pressure a month. | 3 points and nothing from the treasury: the banks' creditors take the losses. +0.1 point of risk premium, and 3% of upper-strata pops radicalize. |
| Expand Deposit Guarantee | −0.2 points of risk premium, +0.03 momentum and −0.2 bubble pressure a month. | 4 points. 2% of pops turn loyalist when you expand it; 3% radicalize when you withdraw it. |

| Tool | What you gain | What you pay or give up |
|---|---|---|
| Open-Market Operations | +0.1 momentum a month and +5% services output. | 4 points and 0.5% of GDP at the start; +1.5 bubble pressure a month and, under the full Banking System, about a point of inflation pressure. |

A Bank Holiday is the first answer to a Panic: it stops the fall at once, holds it while the banks are shut and keeps a trading partner's crash out, but you get one every five years. It ends by itself after 90 days and the banks reopen with +1 momentum. You can end it sooner to stop paying for it, but then the banks reopen without the bounce. Emergency Liquidity lifts the cycle most at once, but only the first time you open it in a crisis: reopen it before the cycle has held at Stable or above for 12 months and you get neither the +12 nor the +10 bubble pressure. Closing it hands most of its cost back, so close it once you are out of the slump, before its bubble pressure feeds the next boom. If you leave it open, it closes itself, refund included, after those 12 months. A deposit guarantee is cheap to open and costly to withdraw, so keep it for a lasting slump.

Asset Relief Program and Bail-in Regime exclude each other. Asset Relief lifts the cycle twice as fast, at 2.5% of GDP and a standing cost in tax and bureaucracy; a bail-in costs the treasury nothing and angers the upper strata instead. Open-Market Operations is what is left when you can't cut any further: under the full Banking System it needs Fiat Money or Digital Currency with your policy rate at its floor. A bank running its mandate also buys assets itself at the floor (see When the bank cannot cut); that adds to this tool rather than replacing it.

The AI declares a Bank Holiday and opens Emergency Liquidity in a Panic, or in a Downturn that is still falling fast. It expands the deposit guarantee in a Downturn or Panic, uses Asset Relief or a bail-in from Stagnation down, and uses Open-Market Operations at the rate floor in a slump or deflation. It keeps each tool until the slump that prompted it is over, and lifts them as the cycle climbs back to Stable and beyond.

## Managing foreign capital and import credit

External & Currency groups export lending and outflow controls with five monetary interventions. These five appear only under the full Banking System. Each holds 2 points while it runs (Emergency Import Financing 3), leaving fewer for deposit guarantees, domestic credit or crisis relief. Restrict Speculative Inflows needs your own policy rate; Sterilize Capital Inflows and the surrender requirement need a convertible Gold Standard with your own policy rate; Limit Foreign-Currency Borrowing and Emergency Import Financing need National Bank Established. The tooltips show their current costs and restrictions.

| Intervention | What you gain | What you pay or give up |
|---|---|---|
| Restrict Speculative Inflows | Halves incoming speculative gold and the upward exchange-rate pressure from a positive real-rate gap. Gold can still leave at its usual speed. | Slower reserve accumulation, −3% bureaucracy and slightly slower credit growth. Bubble pressure also falls. |
| Sterilize Capital Inflows | Removes 75% of the inflation pressure from incoming gold. | 0.01% of annual GDP a month, charged weekly as a government expense, with slightly slower credit growth and less bubble pressure. Foreign lenders retain their claims on the vault, and outgoing gold still causes deflation. |
| Limit Foreign-Currency Borrowing | Gradually reduces the risk premium caused by a weak currency, reaching half that component after about 24 months. | Adds 0.25 percentage points of risk premium while active and slows credit growth. Protection takes time to build; it also takes about 24 months to disappear after you lift the limits. |
| Foreign Exchange Surrender Requirement | Buys gold for the central bank with treasury cash, without adding Borrowed Gold that foreign lenders can withdraw. | −10% export advantage and −3% bureaucracy, plus the cash spent on reserves. Purchases stop at the vault's capacity and cannot exceed available cash. |
| Emergency Import Financing | +15% import advantage while a financial crisis lasts. | A weekly government expense of a quarter of the monthly rate: initially 0.05% of annual GDP a month, rising to 0.10% after 12 funded months. Holds 3 intervention points. |

Surrender purchases depend on your market's exports. The monthly cap is 0.05% of annual GDP when export reliance is above 30%, half that amount above 10%, and a quarter with any smaller positive share. A market with no exports buys nothing. Cash needed for the other funded interventions reduces what you can spend on these purchases.

Emergency Import Financing is available during a Downturn or Panic, a gold crisis with Peg Confidence at 40 or below, or hyperinflation. War alone does not qualify. It supports imports generally; it does not guarantee supplies or get ships through a blockade. The tool ends when the crisis passes or you cannot fund it. Each inactive month removes one month of its accumulated surcharge; switching it off and back on does not reset the cost.

Sterilize Capital Inflows ends if the treasury cannot cover a month of it. The surrender requirement's purchases stay monthly: the treasury pays the same day, and the same sum, as the vault receives. Both it and Foreign Exchange Surrender Requirement end when you lose the conditions for convertible gold flows, including a suspension of convertibility. Other tools are lifted when their law or monetary requirements no longer hold. Restrict Speculative Inflows and the surrender requirement share the outflow control's wartime exception to restrictive financial regulation laws.

The AI restricts inflows when positive rate pressure accompanies an Expansion or hotter cycle, and sterilizes when gold is arriving during those phases. It uses borrowing limits when the Exchange Rate Index is below 95, surrender requirements when its vault is less than half full and it has cash, and import financing during a financial crisis. It favors lifting each tool when those reasons end.

## Command economy and cooperative tools

Under Command Economy the cycle measures planning strain, and the panels offer eight planning tools plus two transfers, which have a Transfer button and no point cost. Under Cooperative Ownership it measures the balance between what worker-owners pay themselves and what they reinvest, and there are eight council tools. Changing economic system switches the old system's tools off. An open Emergency Liquidity Program still refunds 1% of GDP when that closes it.

| Economy | Category | Tools (points) |
|---|---|---|
| Command Economy | Plan Management | Emergency Plan Revision (3), Production Target Reduction (2), Plant Consolidation Order (2) |
| Command Economy | Allocation & Supply | Emergency Resource Allocation (3), Strategic Material Stockpile (2), Central Distribution Upgrade (4) |
| Command Economy | Administration | Administrative Performance Campaign (3), Inter-Sectoral Coordination Protocol (1) |
| Command Economy | Investment Pool Transfers | Capital Injection moves 0.2% of GDP from the treasury into the investment pool. While private projects are queued, the pool then spends about a 24th of itself a week, and the treasury keeps topping it up as State Investment Funding. Capital Withdrawal moves 0.2% of GDP back, or the whole pool if it holds less |
| Cooperative Ownership | Surplus & Reserves | Dividend Restraint Resolution (2), Mutual Aid Fund Deployment (4) |
| Cooperative Ownership | Investment & Credit | Collective Capital Investment Plan (4), Cooperative Credit Union Expansion (3) |
| Cooperative Ownership | Council Direction | Federation Council Advisory Directive (1), Consumption Ceiling Resolution (2) |
| Cooperative Ownership | Membership Mobilization | Worker Solidarity Campaign (1), Worker Buyout Facilitation (3) |

### Banking history charts

The History section starts open in both the journal entry and the Banking tab. It charts the last 1 Year, 5 Years or 20 Years month by month: Banking Cycle Value, Banking Momentum, Bubble Pressure, Policy Rate and Rate Paid on Debt, plus Inflation and the Exchange Rate Index under the full Banking System. Markers flag policy changes, including tools a change of economic system switches off, and crashes, including imported ones. Hover a bar for that month's phase or band and the policies switched on or off. History is kept for the player and for major powers and above.

## 4.4 Financial regulation laws

The Financial Regulation law group sets how volatile your banking system is, how likely it is to crash, how large your intervention budget is, which tools you may use and which response the crash event offers.

| Law | Unlocked by | Points | Volatility | Crash likelihood | Locks |
|---|---|---|---|---|---|
| Unregulated Banking | default | 1 | +50% | +25% | Capital Controls (Outflows), Asset Relief Program |
| Free & Mutual Banking | Postal Savings | 3 | – | −10% | Capital Controls (Outflows), Asset Relief Program |
| Universal Banking (Light Prudence) | Central Banking | 4 | +10% | – | Capital Controls (Outflows), Asset Relief Program |
| Prudential / Narrow Banking | Modern Financial Instruments | 7 | −60% | −50% | Open-Market Operations, all directed credit, Export Credit Facility |
| Directed Credit & Development Banks | Central Planning | 5 | −20% | −20% | none (two credit sectors) |
| State-Owned Banking | needs Command Economy | 7 | −35% | −35% | Capital Controls (Outflows), Asset Relief Program |
| Central Bank Independence | Keynesian Economics, with National Bank Established | 7 | −10% | – | Capital Controls (Outflows) |

The capital-controls locks in the table also apply to Restrict Speculative Inflows and Foreign Exchange Surrender Requirement. All three are allowed in any war and lifted when it ends if your law forbids them. The laws also have effects outside the cycle, listed in their tooltips, and these apply under every setting of the Banking System rule, Disabled included: Urban Center throughput, the investment pool, innovation and government dividends. Central Bank Independence, for one, strengthens your economic defense against covert operations (see Covert defense).

The National Bank law group decides whether you have a central bank. National Bank Established (Central Banking) adds an intervention point and the National Bank institution, and gives you a policy rate under the full Banking System. Most countries start in 1836 with Unregulated Banking, Commodity Money and no national bank. Britain starts with Universal Banking (Light Prudence), a national bank and the Gold Standard; France, Sweden and the Netherlands with Universal Banking and a national bank; Austria and Denmark with a national bank; and the United States with Free & Mutual Banking.

## 4.5 Monetary policy under the full Banking System

With the rule on Enabled, every country runs the monetary model from 1836, whether or not its financial cycle has started. A dial needs a National Bank, not the cycle, so you can have one, or a pegged currency, before the cycle starts. Until then the Monetary Policy section is read-only: it shows every reading, but its controls are grayed, and your central bank runs the Dual Mandate (under Central Bank Independence, the mandate its act names). The controls open when the cycle starts.

*Figure 8: The Monetary Policy block of the banking dashboard: the rates and the Rate Target stepper, the exchange rate, what borrowing costs, prices, and the central bank's stance, delegation and mandate.*

The Monetary Policy section's rows are grouped under The Rate, Gold and the Peg (on a gold standard with a dial), Exchange Rate, International Arrangements (while you are anchored, party to a monetary treaty or in a bloc with a Monetary Union principle), What Borrowing Costs and Prices. Hover a row's label for what the term means and the row for its reading and what moves it; the Help page has the rules behind each group. Each mandate button says what its rule does. Your Policy Stance and price band are in the overview, with the policy rate and the past year's inflation under them.

## Who sets the policy rate

You have a dial, a policy rate of your own, with National Bank Established and a currency law of Commodity Money, Gold Standard, Fiat Money or Digital Currency, unless you run a Command Economy, have dollarized, or tie your currency to another country's. Everyone else still has a rate, and the Monetary Policy section says why it is out of your hands: without a national bank or under Decentralized Cryptocurrency it is the world rate plus expected inflation plus a point, a command economy's is fixed at 3%, and an anchored country takes its anchor's.

## Rate target, delegation and mandates

You choose a Rate Target, the stepper labeled Target beside your Policy Rate, and each month the policy rate moves a sixth of the distance toward it, never less than a third of a point; Digital Currency doubles both. The distance it counts is at most how far inflation is from its target, so with inflation on target the rate moves a third of a point a month however far you set the target, and a small change of course takes months. A bank chasing a runaway inflation from 25 points off covers most of the way in about a year. Click the stepper for a point, Ctrl-click for a tenth, Shift-click for the limit of your currency law. Changes are free and have no cooldown.

You start with the dial delegated: the central bank sets the target every month from its mandate. Take Control and Delegate switch between you and the bank, free of charge.

| Mandate | What the bank does |
|---|---|
| Inflation Targeting | Raises the rate more than a point for each point of inflation above target, and leaves the financial cycle to prices, so booms run further and slumps go deeper before it acts. Lenders believe it most, which holds expected inflation nearer the target |
| Dual Mandate | Reacts to inflation as hard, and also leans fully against where the cycle is heading. Every bank starts on it |
| Growth | A quarter point looser at target, half as responsive to the cycle, and blind to inflation below 4% |
| Peg Defense | Holds the world rate, up to a point above while gold is short, and ignores the domestic cycle; convertible gold standard only |

A mandate-run bank cuts at least a point a month (two under Digital Currency) once the cycle falls into Stagnation or worse.

A new mandate takes effect at the next monthly update, and is then fixed for a year. Lenders price a looser rule at once: for each step from Inflation Targeting to the Dual Mandate to Growth, expected inflation rises by half a point and the Looser Mandate modifier adds half a point of risk premium, fading over two years. A tighter rule earns nothing at once, and Inflation Targeting's extra credibility takes two years to build and is gone the month you leave it. An AI bank moves to Growth at war or deep in debt and back to the Dual Mandate in peace, and waits out the same year between changes.

Central Bank Independence makes delegation permanent, and you no longer set the rate or print money. Lenders reward it. Your credit standing improves by half a point and its floor drops from 0.5 to 0.25 points, and the bank's estimate of what your economy can bear (see Monetary policy stance) errs by less. Each level of the National Bank institution adds a tenth of a point of Inflation Anchoring, which absorbs that much standing wage and price pressure.

Under Central Bank Independence the mandate is an act on the law rather than a dashboard choice: the Inflation Targeting Act or the Dual Mandate Act. When the law has no act, because it has just passed or an act was repealed, the event What the Bank Is For asks you to pass one. The Industrialists gain approval for the Inflation Targeting Act and the Trade Unions for the Dual Mandate Act, and each loses it when the other's act passes. An interest group can also sponsor an act while the law is debated, and the Central Bank Independence Debate's compromise passes the Dual Mandate Act. The bank moves to the act's mandate 12 months after it passes. The dashboard shows the act in a Mandate by Law row in place of the mandate buttons, and a Takes Effect In row counts the months down. To change the act, repeal it in the law panel, which needs an interest group in government that opposes it. By default the Trade Unions and Rural Folk oppose the Inflation Targeting Act, and the Industrialists, Petite Bourgeoisie and Intelligentsia the Dual Mandate Act. On a convertible gold standard an independent bank defends the peg whatever its act says. AI countries pass and follow acts the same way, so an independent AI bank does not loosen for a war.

## When the bank cannot cut

A rate on its floor can't fall further, and a deflation can hold it there. Every mandate asks for far less than the floor allows, and falling prices keep the real rate high even at −3%, so the Policy Stance can read Neutral while prices fall. A bank that runs its mandate, whether delegated or independent, carries on cutting on paper.

It works out where it would put the rate if there were no floor, and buys assets on its own account for the difference. Each point it would cut below the floor adds 1 point of inflation pressure, up to 5 points at 5 points below, on top of Open-Market Operations.

The purchases grow and shrink at the speed the rate moves: up to a sixth of the distance a month and at least a third of a point, doubled under Digital Currency, and at least a point a month while the bank cuts with the cycle in Stagnation or worse. They never start or stop all at once. When the bank's rule asks for a higher rate again, the purchases wind down first, and the rate leaves the floor only once they reach zero.

The mandate decides when the bank buys:

| Mandate | Buys assets |
|---|---|
| Inflation Targeting, Dual Mandate | Only while inflation is under its 2% target. Once inflation reaches 2%, it winds the purchases down. |
| Growth | Whenever its rule asks for a rate below the floor, including a slump with inflation on target. |

The purchases cost no intervention points and no money, and you don't switch them on. While there are any, an Asset Purchases row below Price Pressure shows how much they add, together with Open-Market Operations, and its tooltip gives the rate the bank would set without the floor. They need Fiat Money or Digital Currency. A dial you set by hand gets none of them, because you have Monetize Deficit instead; while your rate is on its floor and prices are falling, the Delegation tooltip reminds you that a delegated bank would buy. AI countries are always delegated, so they get the purchases too.

Under Central Bank Independence you can't print for the treasury, so a long deflation leaves you the bank's purchases, Open-Market Operations and a budget deficit above 1% of GDP, which adds a little pressure of its own. Recovery from a deep deflation can still take years. Leaving independence for Monetize Deficit is possible but costly: you lose its credit-standing bonus and some of the intervention budget.

## Monetary policy stance

Policy Stance reports your rate as Very Loose, Loose, Neutral, Tight or Very Tight against what your economy can bear. The loose bands add momentum and bubble pressure every month, and the tight bands drain them. What the economy can bear is hidden even from your bank, and the band is the bank's estimate, so stepping the dial will not find the exact turning point.

The push is capped unevenly. On the loose side it tops out at about +0.25 momentum and +1.5 bubble pressure a month, so loosening much further adds nothing to the cycle. On the tight side it goes to about −0.5 momentum and −3 bubble pressure a month.

Six months on the tight side bring Dear Money Politics (Landowners and Petite Bourgeoisie approve; Industrialists, Rural Folk and Trade Unions disapprove), and six months on the loose side bring Cheap Money Politics, the reverse. Both fade after a return to Neutral.

## What your government pays to borrow

Rate Paid shows the yearly interest on your debt. It adds up the market yield (the higher of your policy rate and the Reference Rate plus expected inflation), your credit standing and your risk premium, less the inflation that actually happened, and never falls below 0.5%.

Credit standing is what your country is: rank, finance technologies and a stock exchange, institutions, your courts (the Judiciary law and Judicial Independence) and currency credibility. It cannot go below 0.5 points (0.25 under Central Bank Independence). Risk premium is what is happening: the cycle phase, crisis tools, banking events, bankruptcy, a debt load past a quarter of your credit limit (up to +4 points at the limit), expectations that have strayed from target, money printing and a weak currency.

The Reference Rate is 3% for every country, 2.5% once any great power has Macroeconomics and 2% once one has Globalization. The World Rate is the real interest rate of the great powers that set their own; gold flows and exchange rates are measured against it.

## Inflation and price bands

Inflation shows what prices did over the last year, and Expected Inflation what lenders and workers assume they will do. A loose stance, a boom, deficits, money printing, dearer goods and imports, and inflowing gold push it up, as do the Wage Pressure of your labor and welfare laws and any other standing Inflationary Pressure, such as a devaluation. A deficit adds at most 1.5 points of pressure (3 at war), and none in default or once dollarized, when nobody lends to cover it. Above 8% inflation, the wage pressure of your labor and welfare laws counts twice.

A slump pulls inflation down. Under Fiat Money or Digital Currency, Stagnation pulls it half a point below where it would otherwise settle, a Downturn 2 points and a Panic 4, and inflation closes most of that gap within a year. A Downturn alone outweighs the wage pressure of all your labor and welfare laws together, so prices slow through a recession and your bank can cut instead of holding its rate up against inflation. On metallic money and under Decentralized Cryptocurrency, prices already rest near zero and the pulls are 0.3, 0.8 and 1.5 points; more would hold them in Deflation. A dollarized economy takes the same smaller pulls, though its prices rest near the 2% target of the money it adopted.

Every country starts on metallic money, Commodity Money or the Gold Standard, and there expected inflation is fixed at 0%: money redeemable in metal promises stable prices, lenders take the promise at face value, and the money itself pulls inflation back toward zero. Expected inflation sets the floor under your borrowing, so on metal any inflation erodes your debt and any deflation adds to it.

Fiat Money and Digital Currency replace that promise with the bank's inflation target of 2%. Expected inflation then follows actual inflation, pulled toward the target the more lenders trust the bank: most under Central Bank Independence, less when you delegate the dial, less again when you set it yourself, and least under State-Owned Banking. That trust wears away as inflation strays from the target and is gone 10 points from it. Expectations adjust twice as fast under an independent bank, and only inflation that lenders did not foresee erodes your debt.

Inflation puts you in one of six price bands, each a modifier on your country; the overview's Inflation icon shows which. You leave a band only once inflation is a quarter point past its edge. Above Stable prices, each band is worse than the one below on every count; the modifier's tooltip gives the figures.

| Band | Inflation | Main effects |
|---|---|---|
| Deflation | below −1% | Momentum drains; Rural Folk and Trade Unions disapprove |
| Stable prices | −1% to 3% | None; labor laws pay the Real-Wage Dividend |
| Elevated | 3% to 8% | Lower strata expect more; Petite Bourgeoisie disapprove |
| High | 8% to 20% | Tax waste, weaker investment, more interest groups disapprove |
| Very high | 20% to 50% | Less minting, and bubble pressure builds |
| Hyperinflation | 50% and up | The hyperinflation crisis fires |

The Real-Wage Dividend turns your labor laws' wage pressure into a lower standard-of-living expectation for the lower strata (see Economy and construction) and a little momentum, while prices are stable. Command and dollarized economies never get it.

## Monetizing the deficit

Monetize Deficit, from 0 to 3, has your national bank print money for the treasury. Each level mints about 1% of a year's GDP, adds about 2.5 points of inflation pressure and half a point of risk premium. It needs Fiat Money or Digital Currency and a bank that takes instructions, so Central Bank Independence, Command Economy, dollarization and an anchored currency rule it out. An independent bank still buys assets by itself in a deflation (see When the bank cannot cut). The Sovereign Debt Scare's central bank option adds one level for a year (see Banking events).

## Currency regimes and the policy rate

Your Monetary Policy law sets the dial's range.

| Currency law | Unlocked by | Rate range | Notes |
|---|---|---|---|
| Commodity Money | default | World rate −1 to +3, with a national bank | Prices that outrun the world's are pulled back down; fixed at par |
| Gold Standard | Central Banking | 0% to 15% | A point off credit standing; gold reserve; Peg Defense; fixed at par |
| Fiat Money | Keynesian Economics, with National Bank Established | 0% to 60% | Open-Market Operations and money printing; floats |
| Digital Currency | Universal Digital Identity, with National Bank Established | −3% to 60% | As Fiat, and the rate moves twice as fast; floats |
| Decentralized Cryptocurrency | Cybersecurity | no dial | Borrows at the world rate plus a spread; floats |

## Gold reserves and the run on the vault

A gold-standard country with a dial gets a Gold and the Peg group in the Monetary Policy section. The Bank's Gold is the central bank's vault, not your treasury. Each point your rate sits above the World Rate draws in gold worth 0.2% of GDP a month, and each point below sends as much out. Gold drawn in is Borrowed Gold. It costs your budget nothing, but it leaves first, at double speed, once your rate is no longer above the world's. What the high rate costs you is a tighter Policy Stance and dearer borrowing for your government. Recapitalize the Bank moves a tenth of the reserve's limit from treasury to vault, in cash, for good.

Peg Confidence (0 to 100) reacts to your rate while the vault is under a tenth of its limit:

| While the vault is low | Peg Confidence a month |
|---|---|
| Each point your rate is below the World Rate | −3 |
| Downturn or Panic | −2 |
| Debt at half your credit limit or more | −2 |
| Rate at or above the World Rate | +2 (+1 at 70 or more) |

With more gold in the vault, it recovers while your rate is at or above the world's and no Borrowed Gold is leaving. Borrowed Gold leaves only once your rate is no longer above the world's, so a bank holding its rate above the World Rate keeps recovering while it draws gold in. The lower confidence is, the faster it comes back:

| Peg Confidence | Recovery a month |
|---|---|
| Under 40 (Doubted) | +3 |
| 40 to 69 (Watched) | +2 |
| 70 or more (Trusted) | +1 |

An overvalued currency drains it at any time and stops the recovery. So does a deep slump: a Downturn or Panic while your Policy Stance reads Tight or Very Tight costs 3 a month however full the vault is, because the peg is holding your rate up while the economy sinks. From full confidence that reaches 20 in a little over two years. Peg Defense holds the World Rate whatever the cycle does, so a bank left on it rides a long slump down to the run unless you cut the rate and pay for it in gold. At 20, The Run on the Vault fires:

| Option | Effect |
|---|---|
| Defend the peg: "Raise the rate until the gold comes back." | For a year the rate target stays at least 4 points above the World Rate; Peg Confidence +40 |

| Option | Effect |
|---|---|
| Suspend convertibility: "Suspend payment in gold. For the duration." | Five years as a paper currency with a free rate; +2 points of risk premium for five years; gold's credit-standing bonus lost for ten, or until you leave gold; then the choice below |
| Devalue: "The peg holds. The price of gold does not." | The vault gains 15% of its limit, Peg Confidence resets to 50, the exchange-rate index drops to 88 and recovers over five years; prices rise; the great powers resent it |

When the five years are up, The Suspension Lapses asks what the currency is now. The currency stays suspended until you answer; left unanswered for three months, the event restores convertibility by itself:

| Option | Effect |
|---|---|
| Restore convertibility | Back on gold at the old parity; Peg Confidence starts at 50 |
| Keep the paper | Fiat Money becomes law at once, ending any currency law you were enacting; the lost credit-standing bonus stops counting against you. Needs Keynesian Economics and National Bank Established, and a currency of your own; grayed out otherwise |

## The exchange-rate index and capital controls

The Exchange Rate Index is your currency's real value to foreigners, where 100 is par. On Commodity Money and the Gold Standard your currency is fixed at par and the index moves only after a devaluation. The pressure that would move it shows up as overvaluation instead, which pulls prices down on commodity money and drains Peg Confidence on gold.

Fiat Money, Digital Currency and Decentralized Cryptocurrency float, and nobody sets the index: it moves between 50 and 150, drifting toward a target that rises when your real interest rate beats the World Rate and falls when your inflation runs above the world's (running below it does not raise it), when lenders charge you a risk premium, or when events knock it; the tooltip lists each term. Every point below par gives +1.25% export advantage and −1.25% import advantage, and above par the reverse. A weak currency adds risk premium, and a fall below its three-year average raises prices for a while.

Capital Controls (Outflows), an External & Currency tool for 2 points, cut the effect of the gap between your rate and the world's on your exchange rate, gold and peg to a quarter. Each peacetime year they stay on adds a step of Trapped Capital, up to five, costing Industrialist and Petite Bourgeoisie approval and investment efficiency. The count freezes in a war or crisis and fades after the controls come off.

## Hyperinflation and dollarization

When inflation reaches 50%, Not Worth the Paper offers three answers. Bond fortunes vanish whichever you choose: every state's Wealth Concentration (Who inherits) falls 5 when the crisis fires.

| Option | Effect |
|---|---|
| Currency reform: "Call in every note. We begin again." | Inflation and expectations reset to 5%; printing stops; the investment pool is wiped out; Wealth Concentration falls 5 more; middle and upper strata radicalize; Currency Reform adds 5 points of risk premium for ten years |
| Dollarize: "Let them keep the foreign money. They already have." | A Dollarized Economy: inflation drops to the bank's target, but the dial and money printing are gone and minting falls 75%. Enacting any monetary law brings back your own currency, with inflation at 5% and ten years of Currency Reform; Command Economy ends it free |
| Wait: "It will pass. Everything passes." | Nothing; if inflation is still 50% or more in two years, the crisis fires again and takes another 5 |

## Monetary treaty articles

Five treaty articles tie money or debts across borders; all need the full Banking System (treaties in general: Diplomacy).

A pegged country is anchored. It has no dial and prints no money: its policy rate is the anchor's plus a spread, its exchange rate is the anchor's, and it gets up to 80% of the anchor's credibility. It keeps its own inflation and banking cycle; when its inflation runs ahead of the anchor's, the difference shows as overvaluation, which drains Peg Confidence. While overvaluation stays at 5 points or less, confidence recovers on the same schedule as a gold peg's: +3, +2 or +1 a month by band.

| Article | Unlocked by | What it does |
|---|---|---|
| Currency Peg | International Exchange Standards | The weaker party pegs to the stronger at the anchor's rate plus half a point, and its credit standing improves |
| Swap Line | Macroeconomics | Lowers the recipient's risk premium and steadies its peg; in a financial crisis it borrows from the provider, repaid interest-free |
| Lender of Last Resort | Intergovernmental Organizations | Halves the ward's debt-load premium; if the ward defaults, the guarantor pays out or reneges at a cost in infamy and prestige |
| Imposed Currency Peg | International Exchange Standards | A peg at the anchor's rate plus a point, with no credit benefit and −10 legitimacy, that the pegger cannot break |
| Debt Receivership | International Exchange Standards | The debtor pays the creditor 0.1% of GDP a month, with −10 legitimacy, until its debt stays low for 12 months |

The last two are hostile: a larger power can demand them only of a country in default. If the stronger party leads a power bloc, each article also gives it leverage generation over the weaker, from 150 to 500 by article, which builds the Leverage Advantage it needs to invite that country into its bloc (see Other power bloc changes). A country may receive only one swap line at a time. A treaty pegger at Peg Confidence 20 gets The Peg Under Siege: impose capital controls for a year (+40 confidence), break the peg by leaving the treaty, or re-peg lower. AI countries look for a swap line when their peg wobbles and a guarantee when heavily in debt, and are reluctant to peg their own currency.

Each side signs for its own reasons:

| Article | Why the weaker side takes it | Why the stronger side agrees |
|---|---|---|
| Currency Peg | A small economy that trades mostly inside a bigger country's market, or a fiat currency whose bank commands little trust (a rate set by hand, State-Owned Banking, inflation running high). It borrows more cheaply and takes on the anchor's credibility, but gives up its dial and money printing, and suffers when its inflation outruns the anchor's. | A great or major power, or a market leader, drawing trading partners into its monetary orbit at no risk to itself: +2% prestige, better relations, and cheaper borrowing as more of the world's GDP pegs to it. A bloc leader also gains 200 leverage generation over the pegger. It pays the article's upkeep. |
| Swap Line | A country with high borrowing costs, a shaky gold peg or a habit of financial crises: −1 point of risk premium at once, +2 Peg Confidence a month, and credit of up to 2% of its GDP in a crisis, repaid afterwards. | A larger economy backing a partner cheaply: +1% prestige and better relations, and as a bloc leader 150 leverage generation over the partner, for extra risk premium in proportion to the partner's size (a sliver when a great power backs a minor) and loans in a crisis. It pays the upkeep. |
| Lender of Last Resort | A heavily indebted country: its debt-load premium halves, imported panics start shallower, and a default brings a payout unless the guarantor refuses. Each payout makes the guarantee worth less. | A rich power sure of its ward: +2% prestige and better relations, and as a bloc leader 300 leverage generation over the ward, for extra risk premium in proportion to the ward's size. On a default it pays 5% to 15% of the ward's GDP, or refuses at a cost in infamy, prestige, relations and its own borrowing costs. |

| Article | Why the weaker side takes it | Why the stronger side agrees |
|---|---|---|
| Imposed Currency Peg | Nothing it would choose: it is forced on a country in default. | A major power or better that wants a defaulted country's money tied to its own for good; a bloc leader also gains 400 leverage generation over it. |
| Debt Receivership | A defaulted country that needs cheaper credit more than it needs legitimacy, though it is usually forced. | A larger financial power facing a defaulted country: it collects 0.1% of the debtor's GDP a month until the debtor recovers, and takes on a little of its risk. A bloc leader also gains 500 leverage generation over the debtor. |

### The Monetary Union principle group

A power bloc can bind its members' money together through the Monetary Union principles. Each tier includes the ones before, and all need Central Banking.

| Tier | Principle | Effect |
|---|---|---|
| I | Monetary Cooperation | Every member gets half a swap line from the leader |
| II | Common Currency | Members may adopt the leader's money: its rate with no spread, its exchange rate and +5% export and import advantage |
| III | Fiscal Backstop | The leader is lender of last resort to every adopter |
| IV | Banking Union | Adopters get −25% crash likelihood and volatility and an intervention point; needs Keynesian Economics |
| V | Reserve Currency | The leader borrows more cheaply and earns on the world's balances; adopters gain more trade advantage; needs Globalization |

Adopting needs Fiat Money or Digital Currency with a national bank, inflation within 3 points of the leader's and debt under half your credit limit. Leaving worsens your credit standing for ten years, costs investment pool, relations and bloc cohesion, and bars you for ten years. A leader can Press for Convergence at 50 influence per holdout, which waives the debt condition and puts The Question of the Common Currency to each holdout; one that refuses costs the bloc cohesion and gains Monetary Independence. Under the Monetary Union row a leader sees each member: adopted, holding out, or holding out and set to refuse, with what backing it costs. The Backstops row lists each treaty partner the same way, received and extended. Power blocs in general are covered in Power bloc principles and identities.

### Currency boards for subjects

A puppet, vassal, colony or crown land whose overlord has a dial runs a currency board: the overlord's rate plus a quarter point, its exchange rate, and half the subject's minting goes to the overlord. Six months at Very Tight or Very Loose bring The Overlord's Rate, which raises liberty desire. More autonomy ends the board.

### How the AI runs monetary policy

AI countries always delegate. They run Peg Defense on a convertible gold standard, Growth at war or with debt at half their credit limit, and Price Stability otherwise. Once Fiat Money is open to them they stop favoring gold and lean toward fiat, more so in later eras, in deflation and while convertibility is suspended; at the end of a suspension they keep the paper more often the later the era and the deeper the slump. They print money only at war with heavy debt and stop gradually afterwards, and a gold-standard AI tops up its vault when it is under a quarter full and the treasury is flush.

## 4.6 The Simplified banking setting

*Simplified* keeps the cycle and everything tied to it: crashes, contagion, rescue appeals, the Great Depression, the seventeen cycle tools and the financial regulation laws. The five additional monetary interventions in External & Currency are unavailable; Export Credit Facility and Capital Controls (Outflows) remain. It drops the monetary layer: no rate target or mandates, inflation, exchange rate, gold reserve, monetary treaty articles, Monetary Union principles or currency boards.

You still borrow at a rate of your own: the Reference Rate, a point more without a national bank, plus credit standing and risk premium, so a Panic still raises your interest. Open-Market Operations needs only Keynesian Economics, four points and a law that allows it, and The Gold Window Closes, a gold-standard crisis event, appears only under this setting. With the rule on *Disabled* interest works the same way, minus the cycle's premium.

## 4.7 Restructuring the public debt

Once your debt passes a quarter of your credit limit, or you are in default, the decision Restructure the Public Debt opens talks with your creditors. It needs a financial cycle that has started. In Restructuring Talks you choose how much of the debt they write off:

| Option | Written off | What it costs |
|---|---|---|
| Reprofile: write off a tenth and stretch the rest | 10% | Reprofiled Debt: +1.5 points of risk premium and −5% prestige, fading over five years; −10 relations with your investors |
| Negotiate a haircut of a quarter | 25% | Partial Default: +4 points of risk premium and −15% prestige, fading over ten years; Wary Lenders: −10% building cash reserves, fading over five years; −25 relations with your investors |
| Make them take half (in default only) | 50% | Deep Default: +6 points of risk premium and −25% prestige, fading over ten years; Shut Out of the Markets: −25% building cash reserves, fading over five years; −50 relations with your investors |
| Break off the talks | Nothing | Nothing; the talks wait a year |

Your investors are the countries that own more than 5% of your GDP. Every haircut also radicalizes the upper strata who held the bonds, more for a larger one, and costs the banks some cycle value. Your credit limit is your buildings' cash reserves plus a base set by your GDP, so the cash-reserve cuts shrink it for a few years: that is the markets shutting you out. After a reprofiling your creditors will not talk again for five years, after a haircut for ten.

A write-off buys room under your credit limit, not cheaper credit. The debt-load part of your risk premium falls by a point and a third for a quarter of a debt at the limit, while the haircut's own premium adds four. Half is for a government that has already stopped paying: the middle way between paying and the base game's Declare Bankruptcy, which erases every loan at a far heavier price. The Sovereign Debt Scare's partial default is the same quarter haircut, with the same costs. The AI opens talks in default, or with debt at three quarters of its credit limit and still growing.

## 4.8 Bailouts and the Great Depression

One country can ask another to rescue its banks, a wave of crashes can become a worldwide depression, and the United Nations can lend to a member caught in one.

### Appeals for a banking rescue

A rescue comes through the random banking events of a possible rescuer: a country that has researched Keynesian Economics, holds cycle value 40 or more and is not in default. Its event can pick a smaller trading partner in a Downturn or Panic, not at war with it, where one of the two markets relies on the other for at least 5% of its trade. That partner gets Appeal for a Rescue?; asking costs 5% prestige for a while (Hat in Hand). The rescuer then answers International Bailout Request:

| Answer | Effect |
|---|---|
| Extend a generous rescue package | A sum scaled to the rescuer's GDP, capped at a fifth of the recipient's; +30 relations; five years of Restored Banking Confidence for the recipient |
| Offer a smaller emergency grant | An eighth of the package; +10 relations; two and a half years of Restored Banking Confidence |

| Answer | Effect |
|---|---|
| They must solve their own problems | −20 relations, and the largest of the three hits to the rescuer's own cycle |
| Lend, on conditions | The rescue package's sum, offered as a loan on conditions; +10 relations. Nothing is paid unless the recipient signs |

Restored Banking Confidence lowers crash likelihood by 10%, raises company throughput by 5% and blocks another appeal while it lasts.

A loan on conditions reaches the recipient as The Rescue's Terms. Refusing, the default, costs nothing. Signing pays a third of the sum at once and puts you under two years of Rescue Program (−1 point of risk premium, −5 legitimacy) and Restored Banking Confidence; Trade Unions disapprove and Industrialists approve. The other two thirds come at yearly reviews, each only if your budget took loans in no more than 6 of the 12 months before it. Miss a review and the program is suspended: no more payments, −20 relations with the rescuer and +2 points of risk premium fading over five years. Both sides get a notice at each review. While a program still owes you money, no one else is asked to rescue you.

### The Great Depression

Every crash that starts at home begins a new crisis wave, replacing any wave still running, and each contagion crash adds to it. If ten or more countries crash in one wave and together hold half of world GDP, every country whose financial cycle has started gets The Great Depression three days later. It happens at most once per game.

| Your situation | Modifier |
|---|---|
| Crashed in the wave | Great Depression for ten years: −0.4 momentum a month, +3 points of risk premium, −5% prestige; radicals |
| Not crashed, under Isolationism or Command Economy | Global Depression (Sheltered) for five years: −0.1 momentum a month |
| Not crashed, any other country | Global Depression for ten years: −0.2 momentum a month, +1 point of risk premium |

### The UN emergency loan

A United Nations member whose financial cycle has started and that is in a Panic or in default can be offered An Emergency Lending Facility through the UN's docket (see The UN docket), once a crisis wave has reached it or while any country suffers the Great Depression. The offer comes at most once in ten years, and never while you are still repaying a UN loan or your representation is suspended.

| Option | Effect |
|---|---|
| Accept the facility and its conditions | 2% of your GDP at once, repaid as 110% over five years through the weekly UN Emergency Loan Repayments expense. For the same five years you carry UN Emergency Loan Conditions (−5 legitimacy, −50 Authority, the government resource rather than UN Authority) and Restored Banking Confidence. Cycle value +10. Trade Unions and Rural Folk disapprove, Industrialists approve. The loan adds to UN Authority's delivery pillar |
| We will weather this alone | Nothing |

The loan is capped at half a year of the dues members pay together, so a Moribund UN, which collects no dues, has nothing to lend.

## 4.9 Banking events

The cycle also draws random events: railway bond fever, bank runs, derivatives scandals, mortgage-backed securities, shadow banking and many more, with their own versions for command economies and cooperatives. Which ones can fire depends on the phase, the bubble, your technology and your economic system. After one, no random banking event fires for at least 18 months, and none in a crash month. Their options move

the cycle's readings, and the tooltip shows by how much. The Sovereign Debt Scare, drawn only while your debt is past a quarter of your credit limit, can end in a partial default, the quarter haircut of Restructuring the public debt at the same price. Its central bank option, where your bank may monetize the deficit under the full Banking System, has the bank buy government bonds: Monetize Deficit rises one level for 12 months, with that level's inflation and risk premium, then steps back. A few tie into the banking tools: The Bank Holiday appears only when you could declare one, and its first option does.

CHAPTER 5

# Taxation (experimental)

Under the Legislated Tax Code game rule, taxation stops being a taxation law plus a tax level. Each country has one tax code instead: a set of rates written into law. You change it by drafting a reform, introducing it as a bill, winning enough interest groups over and choosing the month it takes effect. The rule is experimental and off by default. It has four settings: *Tax Code Disabled* keeps base-game taxes, *Tax Code Enabled* legislates domestic taxes, *Tax Code and Customs Enabled* also legislates tariffs and subsidies, and *Light Tax Code* keeps the taxation laws and lets you set each tax within your law's range, at once and without bills (see the Introduction). Choose it when you set up the campaign; it can't be switched later.

Everything happens in the Tax Code journal entry, in the Domestic Affairs group, and in the Budget panel's Tax Code tab. Both show the same panels. Under the two legislated settings AI countries legislate through the same bills and the same rules (see How AI countries legislate taxes). The next section covers the light setting; the rest of the chapter covers the legislated ones.

## 5.1 Setting taxes within your taxation law

Under *Light Tax Code* the five taxation laws stay, and you enact them as in the base game. Your law sets a range for every tax it levies: from no tax at all up to its rate at a Very High tax level. Within that range you set each tax yourself on the Tax Code tab of the Budget panel, and a change applies at once. There are no drafts, bills, debate, offers or promises. At the start of the game every tax sits at your law's rate for your tax level, so you collect what the base game would.

| Taxation law | Wage Tax | Dividend Tax | Rural Assessment | Head Tax | Consumption Tax |
|---|---|---|---|---|---|
| Consumption-Based Taxation | not levied | not levied | not levied | not levied | 0–35% |
| Land-Based Taxation | not levied | not levied | 0–1.0 | not levied | 0–35% |
| Per-Capita Taxation | 0–15% | not levied | 0–0.5 | 0–1.0 | 0–35% |
| Proportional Taxation | 0–30% | 0–20% | not levied | not levied | 0–35% |
| Graduated Taxation | 0–20% | 0–30% | not levied | not levied | 0–35% |

Wage and dividend taxes move in 2.5% steps, the consumption tax in 5% steps, the rural assessment in steps of 0.025 and the head tax in steps of 0.05. Click – or + on a tax's row to move it a step; shift-click sends it to zero or to the highest rate your law allows. You still choose which goods carry consumption tax, and tariffs and subsidies work as in the base game.

### The tax level under the light code

Each rate sits on a step: the lowest tax level whose rate, under your law, is at or above it. A tax cut below your law's Very Low rate, to zero if you like, sits on the Very Low step. The Step column shows it. Your tax level is the highest step among your taxes, and the overview shows which tax sets it. The tax level buttons on the Budget panel are grayed, because your rates set the level, and the level's legitimacy, radicalism and expected standard of living effects apply as in the base game. A tax below that step is still collected at its own rate.

So your heaviest tax sets the political price. Under Proportional Taxation, with wages taxed at 25% and dividends and consumption at their Medium rates (10% and 25%), your tax level is High and you pay High's effects. Lowering one tax costs revenue, and lowers the tax level only when no other tax shares its step. A button's tooltip says when a step would raise or lower your tax level.

### How interest groups react to rate changes

Each change draws a reaction from every interest group, by how much of that tax its members pay and by its views on taxation: a group that favors progressive taxation welcomes a higher wage or dividend tax, one that opposes it a higher consumption, rural or head tax. A reaction that pleases a group counts half as much as one that angers it, so raising a tax and cutting it back again still costs you. Reactions add up, reach at most ±10 approval and fade by about a third each month; they show as Resents Recent Tax Changes or Welcomes Recent Tax Changes. Raising one tax across its whole range costs a group that pays all of it about 10 approval before its views count.

Groups also judge whom you spare. When the taxes a group's members pay sit below your tax level, by about one base-game tax level in all, the group Endorses the Tax Code (+1 approval), and at about two it Strongly Endorses it (+2), for as long as the relief lasts. Each group still judges your taxation law itself as in the base game.

### Changing taxation law under the light code

When you enact another taxation law, every rate moves into the new law's range. A tax the new law doesn't levy stops, a tax you had cut to zero stays at zero, and a tax it adds starts at the new law's rate for your current tax level, as it would in the base game. No tax ends up above the new law's rate for your current tax level, so a change of law never raises your tax level: a 20% wage tax under Proportional Taxation at Medium becomes 15% under Graduated Taxation. The law change's own approval changes are the base game's; the move adds no reaction of its own.

AI countries keep choosing their tax level as in the base game, and their taxes follow it: every tax at their law's rate for that level. Their interest groups react to each change as yours do. A country released from yours, a rebel country and a newly formed one start from their own law and tax level; when a revolution ends, the winner keeps its rates.

## 5.2 The Legislated Tax Code game rule

At the start of the game each country's taxation law and tax level are carried over into its code, rate for rate. A country that was taxing grain keeps grain on its list of taxed goods. The code starts out collecting what the base game would have collected. Decentralized countries have no tax code.

### What the tax code replaces

The five taxation laws, from Consumption-Based Taxation to Graduated Taxation, can't be enacted under the rule. Every country holds the Legislated Tax Code law in the Taxation group instead, and each rate is an amendment to it, such as Wage Tax 10%. The Taxation Level stays at Medium, because the code's rates are the whole tax. The base-game controls that would change a rate are grayed out: the tax level buttons, the add-consumption-tax menu, and the Tax and Untax toggles in the goods panel and the goods right-click menus. Under *Tax Code and Customs Enabled* the tariff and subsidy buttons are grayed too, the Market panel's All Goods rows among them.

On the 1st of every month the code puts its rates, taxed goods and relief back in place. A change made any other way, by an event or by an AI country's own tax decisions, lasts at most until then.

No interest group takes a stance on the Legislated Tax Code law itself. Each group judges your code as the taxation law it comes closest to:

| The code counts as | When it |
|---|---|
| Graduated Taxation | taxes wages at 10% or more and dividends at least as heavily |

| The code counts as | When it |
|---|---|
| Proportional Taxation | taxes wages at 10% or more and dividends at all |
| Per-Capita Taxation | levies a wage tax and a head tax |
| Land-Based Taxation | levies a rural assessment |
| Consumption-Based Taxation | does none of the above |

A group's approval then counts as it would for that law, and its approval breakdown says so: Endorses the Tax Code, Strongly Opposes the Tax Code and so on. The judgment is renewed on the 1st of every month. The same mapping decides three base-game journal entries that ask for a taxation law: Great Reforms: Bureaucratic Reform, Imperialism of Promise and Our Fortunate Regeneration complete when your code counts as the law they ask for.

Two things don't carry over. The Consumption-Based Taxation law's bureaucracy and authority discounts go with the law. The base game's Short-Term Tax Cuts amendment can't be added under the rule; a bill with an expiry date replaces it. The religious tax of the Millet System and People of the Book and the land tax of State Redemption Payments stay outside the code, collected at the Medium level.

### Taxes the code sets

| Provision | Steps and range | Who pays | Budget line |
|---|---|---|---|
| Wage Tax | 2.5% steps, 0% to 50% | pops on their wages | Income Taxes |
| Dividend Tax | 2.5% steps, 0% to 50% | pops who own private buildings, on their dividends | Dividends Taxes |
| Rural Assessment | steps of 0.025, up to 1.2 | peasants and farmers in agricultural buildings, who then pay no head tax | Land Taxes, part of Poll Taxes |
| Head Tax | steps of 0.05, up to 1.5 | working adults who pay no rural assessment | Per Capita Taxes, part of Poll Taxes |
| Consumption Tax | 5% steps, 0% to 60% | pops buying a good on the taxed goods list | Consumption Taxes |
| Taxed Goods | any good pops buy, on or off | as Consumption Tax | Consumption Taxes |
| Agricultural Relief | 25% or 50% | halves (or cuts by a quarter) the wage tax of workers in agricultural buildings | Income Taxes |
| Regional Relief | 25% or 50% | every tax collected from pops in up to three named states | every line |

Rural Assessment and Head Tax are flat amounts, shown with a money icon. The Consumption Tax rate applies to every good on the list, so the list decides what a consumption tax collects and the rate decides how much. Under *Tax Code and Customs Enabled* the code also holds an import and an export level for every tradeable good (see Customs under the tax code).

## 5.3 The Tax Code tab

The Tax Code tab sits beside the base game's Budget tabs. It opens once your taxes are under the code, at the start of the game, and its tooltip lists what is missing while it is grayed. The journal entry has the same panels and an Open Budget button.

The overview's boxes say what needs you. Tax Code gives your code's version and its next change. Draft Reform reads Blocked while a change waits to be accepted, Too soon while a bill introduced now couldn't pass in time, and Ready once it can be introduced. The bill's box reads Can pass, Short of support or Month has come, with the committed clout or the offers waiting. Passed Bills reads Held for a bill held back, and Promises counts the ones failing. Hover a box for the checklist or the dates behind it.

Below it, the sections you watch come first: the Bill Under Debate, the Passed Bills and your Promises, each shown only while it has something in it. Then:

- Enacted Code lists every rate in force, with when it came into force on the rate's hover and, for a temporary rate, when it expires and what replaces it. The taxed goods show as icons, and regional relief lists its states.
- History lists the last eight changes: bills passed and taking effect, rates expiring, promises kept or broken, held bills.

The "?" at the right of the tab's header, and at the top of the journal entry, opens the Help page, How Tax Legislation Works. It repeats the rules of this chapter with the exact figures your game uses; under the Light code it explains the Light code instead.

Another country's National Info window has its Tax Code tab too: its enacted code, its passed bills, the bill it is debating with its support bar, and its history. Its draft, its promises, each interest group's stance and offers, and every button stay hidden.

## 5.4 Drafting a tax reform

A reform starts as a draft, and a draft changes nothing. Draft a Reform opens one. If a bill is already under debate the button reads Draft a Revision, and the draft starts from that bill.

### The draft reform workbench

The Draft Reform section groups the provisions under Income, Land and Head, Consumption, Relief and Dates, with a Goods Catalog for the taxed goods, sorted into staples, industrial goods, luxuries and military goods. Each row shows three columns:

| Column | Shows |
|---|---|
| Law | the rate existing law will have in the month the draft takes effect: today's rate, unless a temporary rate expires or a passed bill takes effect before then |
| Draft | the rate the draft sets |
| Change | the difference |

Click – or + to move a rate one step, shift-click to set it to zero or its highest rate, and right-click to take it out of the draft, leaving it as existing law has it. In the Goods Catalog, Tax and Exempt change a good and Undo takes it out of the draft. Discard closes the draft.

If a passed bill, or a change made outside legislation, alters a tax after you changed it in the draft, Changed Since Drafted appears at the top of Draft Reform with an Accept button for each such change, and the draft summary counts them. Accept the new baseline before you introduce the draft; your own value stays.

### Commencement and temporary tax rates

Takes Effect sets the month the reform comes into force, on the 1st. It can be next month or up to 60 months ahead, and starts three months ahead. The bill must pass before that month comes.

Each changed rate can be Permanent or expire after 6, 12, 24, 36 or 60 months. When it expires, the rate beneath it returns: the rate that was law before this bill, or the rate an earlier temporary change was due to return to. A later bill that changes the same tax replaces the expiry. A permanent change clears it, and a temporary one keeps the rate that was due to return. A temporary rise is how you pay for a war without raising taxes for good.

### Agricultural and regional tax relief

Both kinds of relief move in bands of 25%, up to 50%.

Agricultural relief cuts the wage tax of workers in agricultural buildings. It is worth nothing to interest groups while you levy no wage tax.

Regional relief reduces every tax collected from pops in up to three of your incorporated states. Choose States lists them, with each state's population and the taxes it pays a week. Name adds a state and Drop removes

one. A named state shows Regional Tax Relief 25% or 50% among its modifiers. A state you lose drops out of the relief.

### Reviewing a tax draft

Review opens the comparison of your draft with existing law. Each changed tax shows Existing Law and With This Bill, with the expiry and what returns after it, and a revenue estimate (see Revenue estimates). The review also lists the passed bills waiting to take effect and any customs changes. The draft summary's Revenue line gives the whole draft's change in weekly taxes.

Bill Class says whether the bill is minor or major. A minor bill moves at most two taxes by at most two steps each, and touches no goods, relief or customs. It needs 15 days of debate instead of 30.

## 5.5 Passing a tax bill

Introduce makes your draft a bill and opens debate. The draft stays open, so you can work on a revision while the bill is debated.

### Conditions for passage

The Bill Under Debate section lists the Conditions for Passage. Pass is available once all of them hold:

| Condition | Requirement |
|---|---|
| Committed Clout | committed groups hold more than 50% of the clout of all groups that aren't marginal |
| Legitimacy | at least 25 |
| Debate | 30 days since the last revision, 15 for a minor bill |
| Takes Effect | the month the bill takes effect is still ahead |
| Waiting Bills | fewer than two passed bills waiting to take effect |
| Outside Changes | nothing outside legislation has changed a tax the bill changes since you introduced it |

Pass, Revise Bill and Withdraw sit right under the conditions. On the Budget panel's Tax Code tab, while no draft is open, the bill's committed clout, its debate and its buttons stay pinned at the bottom of the tab.

When a bill passes, each group that opposed it and did not commit disapproves of the government: −3 approval, fading over 180 days. If the bill's month arrives while the bill is still in debate, Move to Next Month moves it a month later without losing its commitments.

### How interest groups judge a tax bill

Each interest group gets a card with a score from −100 to +100, made of:

| Reason | What it counts |
|---|---|
| Cost to Members | the taxes, taxed goods and relief the bill changes, weighted by what the group's members pay |
| Views on Taxation | whether the bill makes the code more progressive (wage and dividend taxes) or more regressive (rural assessment, head and consumption taxes, taxed staples), against the group's views of the taxation laws |
| State of the Budget | with a deficit on the 1st of the month, a bill that raises revenue scores better; with a surplus, one that cuts taxes does |
| In Government | +10 for a group in government |
| Promises | promises made to the group under this bill |
| Trust | the group's record of your earlier promises |

A group scoring 20 or more is Committed: it commits to the bill's current revision and stays committed whatever happens to its score. A group at −40 or below draws a Red Line. Between the two a group is Persuadable if its score is 0 or more and Opposed below that. A Marginal group counts for nothing until it grows, but it can still hold a commitment that counts once it does. The cards are refreshed on the 1st of every month and whenever the bill changes. A card lists the reasons only for a group you still have to win over,

and only those that aren't 0; a Committed or Marginal group shows its line alone, with every reason on the score's hover.

### Offers from interest groups

A group that has not committed may make one offer per revision:

- Soften the rise it minds most by one step, never below existing law.
- Add a band of agricultural relief.
- Leave a staple it would pay for untaxed.
- A promise (see Promises to interest groups).

A group at its red line offers only promises. The card shows what the group asks for, what accepting gives, and for a tax change the revenue you give up. Accept carries the offer out. Any acceptance makes a new revision, so debate starts again and every other group reconsiders. A group that was Persuadable when you accepted commits at once. An Opposed or Red Line group only opposes the bill less, and commits only if its new score reaches 20.

| Offer | You give up | Worth it when |
|---|---|---|
| Soften a rise | part of the revenue the bill raises | the group's clout carries the bill over 50% and the rest of the rise still meets your need |
| Agricultural relief | wage tax from farm workers | Rural Folk or Landowners hold the clout you need and few of your wage earners farm |
| Leave a staple untaxed | that good's consumption tax | the bill raises other taxes enough without it |
| A promise | nothing now; an obligation for years | you can deliver it, see below |

A group never gains the same change twice in one bill, and no group can be promised what is already pending or in force.

### Revising, withdrawing and forcing a tax bill

To change a bill under debate, open Draft a Revision, edit it, and click Revise Bill. A revision releases every commitment, withdraws every offer and restarts the debate. Promises you accepted under the bill stay on the new revision; a revision from the draft drops them. Withdraw closes the bill and leaves the code as it was. Introducing and withdrawing a bill costs nothing.

Force Through passes a bill that is short of a majority, and its button shows only while the bill is. It needs Legislative Override Capacity of at least 2, committed groups holding at least 35% of the clout, enough authority to pay for it, and every other condition for passage. It costs authority and 5 legitimacy, angers the opposition and spends override capacity, as forcing a law through does, with the effects fading over five years. It also lowers Judicial Independence by 10.

## 5.6 Promises to interest groups

A promise is something the government undertakes in return for a group's commitment. It does nothing by itself: it records what you must bring about, and the game checks on the 1st of every month whether you did. You can have at most four promises at once.

| Promise | Deliver within | Then keep for |
|---|---|---|
| Ministry of Education or Ministry of Health one level higher | 12 months per level to gain, plus 6 | 24 months |
| No bureaucracy deficit | 12 months | 24 months |
| Fixed income above fixed expenses, six months running | 24 months | 12 months |

A promise starts as Pending: it binds only if the revision it was made under passes, and a revision or a withdrawn bill releases it. Once the bill passes it binds when the bill takes effect, and its clock starts then. A

promise of something you already have reads Maintain instead of Reach; it wins less support and earns no trust when kept.

The institution counted is the level reached, not the level it is growing to. A bureaucracy deficit never breaks an institution promise: each month of deficit moves its deadline a month later, since the institution can't grow meanwhile. During its term an institution level must hold every month. A bureaucracy or budget promise breaks only after three failing months in a row; the budget and the bureaucracy count as they stood on the 1st.

| Outcome | Group's approval | Trust |
|---|---|---|
| Kept Tax Promise | +3 | +1, not for a promise that only maintained |
| Broken Tax Promise | −5 | −1 |
| Renegotiated Tax Promise | −2 | −1 |

Each reaction fades over 180 days. Renegotiate ends a promise that is in force before it breaks. Trust shows on the group's card, 10 points per step, and fades a step back toward neutral after 24 months without a kept, broken or renegotiated promise. Breaking a promise never repeals a tax bill.

Promise what you were going to do anyway. A ministry you planned to expand, or a deficit your new taxes will close, buys commitment for nothing extra.

## 5.7 Passed tax bills

A passed bill waits in Passed Bills until its month, then takes effect on the 1st, all at once. Collections change then, not before. If another sync of the code already ran that day, the new rates start collecting on the 2nd.

### When passed bills overlap

At most two passed bills can wait at once. A bill passed later that takes effect no later than an earlier one replaces whatever the two change in common. A bill passed later that takes effect after it simply follows it.

### Held tax bills

A passed bill is held back, whole and changing nothing, in two cases:

- A change made outside legislation reached a provision it changes before it took effect. Such a bill can only be dropped; Drop frees its slot.
- Its month went by before it could take effect. If it is at most three months late and no later bill has changed what it changes, Move to Next Month puts it back on the calendar. Otherwise it can only be dropped.

Promises bound to a dropped bill lapse without penalty.

## 5.8 Revenue estimates

The review and the offers show a Revenue Estimate for each Budget line a change touches: Income Taxes, Dividends Taxes, Poll Taxes and Consumption Taxes. It scales the line's current weekly receipts by the change in rate, against the Economy Snapshot taken on the 1st of the month. It is a static estimate: pops, wages, prices and purchases as they are now. Poll Taxes shows a range when both the rural assessment and the head tax collect, because today's receipts don't say how they divide.

The review gives it as a table, each line's weekly receipts under existing law and with the bill and the change, and the draft summary's Revenue line adds the changes up.

A tax you don't collect now has no current base, and its line says so. Taxed goods, agricultural relief and customs get no figure. Regional relief shows its estimate on each named state. An estimate whose snapshot is older than the code or the bill it describes is marked out of date.

## 5.9 Customs under the tax code

Under *Tax Code and Customs Enabled* the code also holds an import and an export level for every tradeable good, from Max subsidy through None to Max tariff, and a bill changes them like any tax. The workbench's

Customs section lists the goods as Staples, Industrial Goods, Luxuries and Military Goods; click a step, or right-click to take a level out of the draft. A customs change always makes a bill major.

Only the country that owns a market sets its levels. A country trading in another country's market sees them as set by that market's owner, and its bills can't change them. A country that comes to own its market takes the market's levels into its code the next month.

On the 1st of every month the code sets each level back. A level that doesn't take, because a treaty forbids it or the tariff cooldown refuses it, is tried again every month. After four months in a row the code adopts the market's level instead, and the review marks the good as blocked. Losing your market keeps your levels on record, drops the customs changes from your draft and bill (withdrawing a bill that changed nothing else), and holds back a passed bill that changes customs.

Interest groups weigh customs like taxes: an import tariff on staples pleases Rural Folk and Landowners and costs Trade Unions, one on industrial goods pleases Industrialists, and subsidies work the other way.

## 5.10 The tax code in revolutions and new countries

When a revolution breaks out, both sides keep the code, the passed bills with their dates and the promises in force. Each side collects from its own territory and can pass its own bills. When the war ends, the winner's code, including its wartime bills, applies across the reunited country; History shows The tax code was restored after a civil war. Bills in debate and open drafts are closed. Promises whose bill no longer waits, or whose group no longer exists, lapse without penalty (see also After a revolution).

A country released from yours starts with your enacted code, without your bills or promises. A country released from one without a code, and a country formed by unification, carries over its own taxation law as it would at the start of the game.

## 5.11 How AI countries legislate taxes

AI countries draft, introduce and pass bills through the same conditions as you. They get no taxes for free, and no bill passes for them without the committed clout, legitimacy and debate it needs.

An AI country counts deficits and surpluses from its budget on the 1st of each month, and reads its income, debt and gold reserves when it acts:

- After three months of fixed deficit, with income at most 90% of expenses (110% if it has significant debt) and its gold reserves under a tenth of their limit, it introduces a bill raising one or two taxes by a step. It picks the taxes its interest groups mind least. When raising the consumption rate is the cheapest choice and it taxes fewer than four goods, it taxes up to two luxury goods instead.
- At war it prefers a two-step rise of a single tax that expires after 24 months.
- After six months of surplus, with income at least 125% of expenses (150% if it has significant debt) and its gold reserves at least a tenth of their limit, it cuts the tax its groups mind most by a step.
- In a financial emergency (default, heavy debt, or bankruptcy within about 30 weeks) it introduces a minor bill at once, with 15 days of debate, and forces it through if enough groups back it and it has the override capacity and the authority. After an emergency bill passes it waits about three months before the next, so the first can show in its budget; after one fails it waits six.

It looks at new bills about once every three months, and waits a year after a bill takes effect before starting another. It accepts interest-group offers until the bill can pass, and accepts a promise only when it can keep it. When it is about to miss a Ministry of Education or Ministry of Health promise it raises the institution itself. It renegotiates a bureaucracy or budget promise rather than break it. A bill that can't pass is withdrawn, and the country waits six months before trying again. Held bills are moved or dropped.

Under *Tax Code and Customs Enabled* AI countries pass no customs bills yet. Their tariffs follow the base game's trade decisions, and the code adopts a level an AI market owner keeps choosing after four months.

## 5.12 How the tax code connects to other systems

The State of the Budget reason and the budget promise read your fixed income and expenses, so the banking cycle and your borrowing costs (see Banking and monetary policy) feed into how groups judge a tax bill. Traditionalism forbids wage and dividend taxes: under it a draft can't be introduced while it levies either.

CHAPTER 6

# Government, laws and characters

The mod adds 164 laws, most of them in 39 new law groups, with the institutions, amendments, government types, ideologies, parties and movements that go with them. The new law groups exist from 1836: every country starts with a law in each, and most of the others unlock with technology. The chapter also covers three systems for rulers and states: Heir Education and aptitude traits (the Heir Education and Universal Aptitude Traits rules, both disabled by default), the custom religion creator (the Custom Religion Allowed rule, not allowed by default) and State Collapse, which has no rule.

## 6.1 The mod's new laws

The new law groups sit beside the base game's in the laws panel. Sixteen of them are ministries, covered in Ministries. The other twenty-three are grouped here by theme, and fifteen further laws join base-game law groups.

Many laws need another law in force, and some can't stand beside one. Anarchy needs Cooperative Ownership, for example, and Cooperative Ownership needs a Council Republic, Corporate State or Collective Governance. The base game checks this only when you enact a law. The mod checks again whenever a law changes and once a month: a law left without what it needs, or beside a law that rules it out, is replaced at once by the nearest law in its group that you can hold. Each replacement can set off more, but the law you enacted stays. Its tooltip doesn't list these replacements, so before you change your Governance Principles or Economic System, check which of your laws depend on them. An anarchist Council Republic that enacts a Presidential Republic loses Cooperative Ownership, and with it Anarchy, Collectivized Agriculture and Possession by Use.

### Rights and society laws

| Law group | Laws | What it governs |
|---|---|---|
| Minority Rights | Violent Hostility, Ghettoization, Discrimination, Cultural Assimilation, Indifference, Protection, Affirmative Action | How the state treats cultural and religious minorities. |
| LGBTQ+ Rights | Active Persecution, Legal Limbo, Basic Protections, Anti-Discrimination Laws, Full Equality and Protection | Legal status of LGBTQ+ people. |
| Criminal Justice | Penal Labor Camps, Punishment-Focused Criminal Justice, Restorative Justice, Rehabilitation-Focused Criminal Justice | Prisons, sentencing and reintegration; who polices is the base-game Policing group (see Who polices). |
| Privacy Rights | Intrusive Surveillance System, Minimal Privacy Protection, Moderate Data Privacy, Strong Privacy Rights | Surveillance against technology spread and tourism. Intrusive Surveillance System needs Social Media or Automated Surveillance. |
| Family & Reproductive Policy | Traditional Family Structure, Pro-Natalist Policy, State-Sponsored Family Planning, Population Control Measures, Communal Child-Rearing | Birth rates, dependents and the working-age share. |

| Law group | Laws | What it governs |
|---|---|---|
| Language Policy | Local Vernacular, Civic Monolingualism, Multilingual Federalism, Linguistic Purity, State-Led Language Reform, Ubiquitous Translation | Assimilation, separatism and acceptance by language. |
| Human Augmentation | No Augmentation, Human Purity, Medical Augmentation Only, Unrestricted Augmentation, Regulated Augmentation Market, Mandatory Augmentation | Cybernetic and genetic implants, from Brain-Computer Interfaces on. |
| Bodily Autonomy | Natural Form, Licensed Modification, Morphological Freedom, Clade Citizenship | Changing how your body looks (skin, sex characteristics, fur, tails), from Morphological Freedom (era 12) on, and from Self-Directed Evolution (era 13) whether someone who leaves the human body plan behind is still a citizen. |

The mod renames the base game's six Citizenship laws and changes their effects. This guide, like the game, uses the new names:

| Base-game name | Name in the mod |
|---|---|
| Subjecthood | Sovereign Subjecthood |
| Ethnostate | Ancestral Citizenship |
| National Supremacy | Cultural Citizenship |
| Racial Segregation | Racialized Citizenship |
| Cultural Exclusion | Assimilatory Citizenship |
| Multiculturalism | Universal Citizenship |

Minority Rights pairs with Citizenship: most rungs need a compatible Citizenship law. Violent Hostility needs Ancestral, Cultural or Racialized Citizenship, and Affirmative Action needs Assimilatory or Universal Citizenship and the Civil Rights Movement technology. Violent Hostility turns on violence against minorities, and the harsh end lets primary-culture homelands form more easily but wrecks colonial stability. Affirmative Action raises minimum local acceptance by 20, pulls migrants and steadies colonies, but institutions cost 20% more bureaucracy.

Universal Citizenship needs the Ministry of Refugee Affairs, which in turn rules out Closed Borders. Adopting Closed Borders abolishes the ministry at once, and Universal Citizenship falls to Assimilatory Citizenship the same day.

State-Led Language Reform starts a chain of choices. On enactment an event asks whether to revive a classical language, build a constructed one or simplify the existing tongue. Revival offers the classical language that fits your primary culture or state religion (Latin for Romance speakers, Sanskrit for Indo-Aryan ones and so on, 18 in all), each with a small bonus of its own. Later events ask whether an academy or an advisory panel governs the standard, and whether minority languages are protected. Each answer becomes an amendment on the law; repeal one and its event can fire again.

Three Human Augmentation laws need another law as well: Medical Augmentation Only needs Charity Hospitals, Private Health Insurance or Public Health Insurance; Regulated Augmentation Market needs a Ministry of Consumer Protection; Mandatory Augmentation needs a Ministry of War. The group's laws also drive a movement journal entry, described in The augmentation debate.

Bodily Autonomy is about how a body looks; what it can do is Human Augmentation's. Every country holds Natural Form until Morphological Freedom opens the next two laws. They raise the acceptance of pops whose culture shares no heritage trait with your primary cultures, Licensed Modification by 5 and Morphological

Freedom by 10, on top of your Citizenship law, and pull more migrants. The Devout and the Rural Folk oppose both, and the Intelligentsia support them. The group drives the morphological freedom debate.

Clade Citizenship, with Self-Directed Evolution (era 13), keeps every right of people who have left the human body plan behind, and takes heritage out of acceptance. Each month it raises the acceptance from a shared heritage trait, a shared heritage trait group and no shared heritage to the highest of the three, so your Citizenship law's heritage figures rank no pop above another (language, religion and homeland still count). Every pop gets at least what a shared heritage trait gives, 60 under Racialized Citizenship, and pops with no shared heritage at least Morphological Freedom's 10 more than your Citizenship law gives them. It also pulls 15% more migrants. The Devout, the Rural Folk and most groups that only tolerate Licensed Modification strongly oppose it. The groups that strongly support Morphological Freedom are content with either law, so the push from Morphological Freedom to Clade Citizenship comes from transhumanist leaders and the Transhumanist movement.

## Laws on state power and information

| Law group | Laws | What it governs |
|---|---|---|
| State Power | Traditional Vassalage, Unitary State, Federal System, Devolved Administration | Centralization: authority, incorporation, decree costs, separatism. |
| Judiciary | Executive Tribunals, Religious Courts, Statutory Courts, Constitutional Review, Judicial Supremacy, Algorithmic Jurisprudence | Who checks the government: authority and override capacity against legitimacy and credit. Stronger courts earn their benefits through Judicial Independence, which grows over decades. |
| Inheritance Laws | Primogeniture & Entail, Customary Inheritance, Forced Heirship, Freedom of Testation, State as Universal Heir, Possession by Use | Who inherits an estate: the concentration of wealth, and birth rates and migration on the land. See Who inherits. |
| Electoral Finance | No Campaign Finance Laws, Unregulated Donations, Donation Limits, Publicly Funded Elections | Money in elections. The first is only for countries without a voting franchise; the other three need one. |
| Right to Information | State Secrets, Informal Government Secrecy, Freedom of Information, Open Government | Authority against legitimacy and cultural pull. |
| Rules of War | Total War, Traditional Rules of War, War Crimes Forbidden, Humanitarian Regulations, Limited War | How you fight and what you promise the world: combat strength, occupation, war support, diplomatic standing and nuclear strikes. See The rules of war. |
| Internet Governance | No Internet Policy, Unregulated Internet, State-Controlled Internet, Net Neutrality | From the World Wide Web on: technology spread, radicalism, covert defense. |
| Frontier AI Control | No Frontier AI Policy, Private Labs, National Licensing, State Monopoly, International Oversight, Open Weights | Who builds the most capable AI, from AI Agents on: innovation against covert defense, authority and intelligence capacity. See Who builds frontier AI. |

In State Power, Unitary State gives 200 authority and faster incorporation. Federal System makes decrees 25% cheaper and calms movements but lowers tax capacity by 10%. Devolved Administration cuts the bureaucracy cost of institutions by 20%, slows assimilation and resists separatism and coups. Traditional Vassalage strengthens the aristocrats and costs authority.

In Judiciary, each law sets a ceiling for **Judicial Independence**, a score from 0 to 100: 10 for Executive Tribunals, 40 for Religious Courts, 60 for Algorithmic Jurisprudence, 70 for Statutory Courts and 100 for Constitutional Review and Judicial Supremacy, halved under Autocracy, Single-Party State and Algorithmic Governance. The score rises 4 a year toward the ceiling and falls 10 a year above it. At 100 it gives +10 legitimacy, half a point off credit standing, 10% more Capitalist investment and calmer losers of law changes.

Executive Tribunals give authority and override capacity. Statutory Courts give an elected government legitimacy from votes and an unelected one authority. Constitutional Review and Judicial Supremacy cost override capacity and lower the enactment success of the constitutional law groups by 10% and 15%, and Judicial Supremacy that of the rights laws too; each law's tooltip lists what it does. Cutting the courts to a ceiling 20 or more below the score raises your risk premium and costs legitimacy for five years, and from a gap of 40 lawyers take to the streets. Religious Courts need an established religion. The Nation tab shows the score heading for its ceiling, and what the courts give you now.

Under Constitutional Review and Judicial Supremacy a law you force through goes to the courts once it passes (see Forceful Legislation), and they rule six months later. They strike it down if Judicial Independence is at least 90 when you only expedited it, 60 when you pushed it forward decisively, or 30 when you forced it through, and otherwise uphold it. Against a ruling you can comply, which puts the old law back and earns legitimacy and 5 points of the score; defy the court, which costs 20 points and raises your risk premium for five years; or pack it, which costs 15 points and authority, and leaves a court that challenges nothing for ten years. A year after packing you may turn the courts into Statutory Courts.

Seven amendments change what the courts do (see Amendments to the mod's laws), and landmark cases come once each when the country's situation calls for them: a ruling on slavery while abolitionists organize, a foreign power demanding an execution, a labor law struck down under Judicial Supremacy, an election voided, a new regime asking its courts to call it lawful, and two paths out of Religious Courts toward Statutory Courts.

## Who builds frontier AI

Every country holds No Frontier AI Policy until it researches AI Agents, and then moves to Private Labs on its own. The other four laws need AI Agents too. Economic Covert Defense raises the Detection Risk of a rival's Industrial Espionage, the operation that steals production technologies (see Covert defense), so the group trades your research against what you keep from rivals and how firmly the state holds the technology.

| Law | Gives | Costs |
|---|---|---|
| Private Labs | +10% innovation; the Industrialists gain political strength | −10 Economic Covert Defense |
| National Licensing | +5% innovation, +5 Economic Covert Defense, +1 intelligence capacity | |
| State Monopoly | 100 authority, +3 intelligence capacity, +10 Economic Covert Defense; the Armed Forces gain political strength | −5% innovation |
| International Oversight | +5% innovation, +2 diplomatic reputation, +0.5 UN Authority Target | Needs membership of the United Nations |
| Open Weights | +10% technology spread; more Global Technological Development for every country; the Intelligentsia attract more pops | −10 Economic Covert Defense |

Leaving the United Nations, or its dissolution, drops International Oversight to National Licensing at once. While you are a major power or greater, Open Weights adds 100 technology spread to Global Technological Development for every country, up to 300 from all the countries that hold it, so it helps the world catch up on what you and the other leaders have researched. The Industrialists back Private Labs, the Armed Forces State Monopoly and the Intelligentsia Open Weights; most other interest groups prefer National Licensing.

## The rules of war

Each Rules of War law is best at one thing, so the choice follows what you are doing rather than your research. Every country starts on Traditional Rules of War.

| Law | From | Best for | What you give up |
|---|---|---|---|
| Total War | Chemical Warfare | A war you must win. Kill rate +0.1, double devastation (it drains the enemy's war support but ruins land you mean to keep), casualties cost | 10 diplomatic reputation, a stronger anti-war movement, 25% faster escalation in every play you are |

| Law | From | Best for | What you give up |
|---|---|---|---|
| | | half the usual war support, and armies mobilize 25% faster. | part of, one level of Ministry of Foreign Affairs |
| Traditional Rules of War | Start | A free hand. Armies lose 5% less morale and gain experience 25% faster, and nothing is forbidden. | What the other four offer |
| War Crimes Forbidden | Wargaming | Conquest you mean to keep. 25% fewer radicals in the states you conquer and 25% less devastation. | Kill rate −0.05 |
| Humanitarian Regulations | Nuclear Weapons | Allies and treaties. +6 diplomatic reputation, the most of any law, so other countries sign treaties with you and back your plays more readily; 10% more of your wounded survive; half the devastation. | Kill rate −0.05; no strategic nuclear strikes |
| Limited War | Intercontinental Ballistic Missiles | Small wars for limited aims. You gain 25% less infamy, plays escalate 30% slower, battles move war support 20% less either way, the anti-war movement is weaker and devastation falls by three quarters. | War goals in plays you start cost twice the maneuvers, so each play takes fewer states; kill rate −0.05; no nuclear strikes at all |

Patriotic, Fascist and Vanguardist groups and Jingoist leaders favor Total War, Traditionalists and Theocrats Traditional Rules of War, Reformers and the Pious War Crimes Forbidden, Liberals, Market Liberals and Humanitarians both Humanitarian Regulations and Limited War, and Pacifists and Anarchists Limited War.

Humanitarian Regulations and Limited War show their ban in the law's tooltip, as Strategic Nuclear Strikes Forbidden and Tactical Nuclear Strikes Forbidden. The ban lifts once you or a country you protect has been struck, or when an enemy's war goal would annex or subjugate you; see Strategic and tactical strike actions. The laws also count toward UN Authority: Total War −2, War Crimes Forbidden +0.5, Humanitarian Regulations +1 and Limited War +1.5 on the UN Authority Target. A positive figure counts only while you are a UN member not undermining it; see What counts toward Policy.

Each law also carries an infamy standing, charged once when the law takes effect: Total War +25, Traditional Rules of War 0, War Crimes Forbidden −10, Humanitarian Regulations −15 and Limited War −20. Moving to a looser law adds the full difference between the two standings to your infamy. Moving to a stricter law takes off 90% of it, rounded down. So moving from Traditional Rules of War to Total War adds 25 infamy, moving back takes 22 off, and moving from Total War to Limited War takes 40 off. A move there and back costs you 1 to 5 infamy. Infamy can't fall below 0, so a cut made while you have little infamy is partly lost. The enactment preview shows the change.

A change of law takes the usual enactment time, so moving to Total War when a war breaks out costs months and holds up your other laws. The AI moves to Total War in a war against a major or great power, or one whose war goals would annex or subjugate it, and back to another law once it is at peace.

## Economic law groups

| Law group | Laws | What it governs |
|---|---|---|
| Antitrust & Market Structure | Guilds and Chartered Monopolies, Freedom of Contract, Antitrust Enforcement, Regulated Utilities, Dirigiste Policy, Command/Cooperative Economy | Companies and monopolies. |
| Intellectual Property | No IP Protection, Creative Commons, Traditional IP Protection, Strict IP Protection, Open Source Innovation, State Intellectual Property | Innovation and technology spread against capitalist investment. |

| Law group | Laws | What it governs |
|---|---|---|
| Genetic Rights | Traditional Heredity, Ban on Genetic Modification, Corporate Genetic Licensing, Open-Source Genetics, State Eugenics Program | Biotechnology output and companies (Companies added by the mod), birth rate, authority and minority acceptance. |
| Monetary Policy | Commodity Money, Gold Standard, Fiat Money, Digital Currency, Decentralized Cryptocurrency | Currency regime. |
| Financial Regulation | Unregulated Banking, Free & Mutual Banking, Universal Banking (Light Prudence), Prudential / Narrow Banking, Directed Credit & Development Banks, State-Owned Banking, Central Bank Independence | How banks are supervised. |
| Resource Transition | Unrestricted Extraction, Fossil Expansion Moratorium, Managed Fossil Phaseout | Whether coal mines, oil rigs and power plants may be built, and retiring them (The fossil transition). |
| Automation Policy | Unrestricted Automation, Robot Tax, Human Labor Quotas, Automation Dividend | Robots and AI doing the work of people, from Humanoid Robotics on (Automation policy). |

Monetary Policy and Financial Regulation are explained in Banking and monetary policy.

## Automation policy

Every country starts with Unrestricted Automation. Humanoid Robotics unlocks the other three laws. Two of them scale with your automated share: the part of your building levels that run a robotic or AI-managed production method, out of all levels of the buildings that have one. The tooltips of the law group and of those two laws show the share, counted once a month. The methods are factory automation from robotic assembly up and the AI-managed tier, AI-managed power plants and networks, Fully Automated Content Creation, Automated Labs, the automated Deep-Sea Mine, robot and drone services in Urban Centers, and self-driving trains, highways and harvesters.

| Law | At an automated share of 100% | You pay |
|---|---|---|
| Unrestricted Automation | Nothing changes. | Nothing. |
| Robot Tax | +10% taxes collected from pops | Capitalists invest 5% less of their profits |
| Human Labor Quotas | The methods above can't be used, so the buildings on them keep their workers | The output those methods add |
| Automation Dividend | +2 standard of living for the lower strata and +1 for the middle strata, half as much again with Universal Basic Income or Post-Scarcity Economy | Capitalists invest 10% less of their profits |

At a share of 40%, Robot Tax gives +4% taxes. Its effect and the Dividend's follow the share each month and start at the first count after the law passes. Under the quotas, buildings on a withdrawn method change to one still allowed. The Trade Unions back Human Labor Quotas and the Industrialists Unrestricted Automation.

## New laws in base-game law groups

| Law | Law group | Unlocked by | What it does |
|---|---|---|---|
| Universal Basic Income | Welfare | Universal Basic Income | Large welfare payments and higher living-standard expectations. |

| Law | Law group | Unlocked by | What it does |
|---|---|---|---|
| Post-Scarcity Economy | Welfare | Post-Scarcity Economy | The largest welfare payments; expectations rise further. |
| Protected Class | Rights of Women | Sexual Revolution | Gender discrimination is illegal: more workers and political participation, lower birth rate. |
| Neocameralism | Governance Principles | Marketing Research, with Laissez-Faire or Interventionism | The state run as a corporation: industrialists and wealth hold power. |
| Collective Governance | Governance Principles | Political Agitation | No single head of state; see Collective Governance. |
| Contracted Administration | Bureaucracy | Supply Chain Management, with Laissez-Faire or Interventionism | Private firms run the administration under contract; see Who staffs the administration. |
| Communal Administration | Bureaucracy | Council Republic, Anarchy or Collective Governance | Local assemblies and recallable delegates run the administration; see Who staffs the administration. |
| Automated Bureaucracy | Bureaucracy | AI Agents | Machines run the administration under your government: institutions cost less bureaucracy and change size faster. |
| Algorithmic Governance | Distribution of Power | Artificial General Intelligence | Machines hold supreme authority and run the administration too. |
| Liquid Democracy | Distribution of Power | Liquid Democracy | Everyone votes on every question or lends the vote to someone they trust, and can take it back at any time. A voting law with Universal Suffrage's franchise: ideological incoherence costs less legitimacy and enacted laws stir their opponents less, but laws pass more slowly. With Neural Lace you also gain the Continuous Vote, checked once a year: more enactment success and fewer radicals from movements, but the groups in government draw less support. Women's Suffrage, Elected Bureaucrats, the campaign finance laws and the Tradition of Free Elections work with it as with any voting law; First Past the Post and the voting-age amendments don't. |
| Neocolonialism | Colonization | Decolonization | See Colonial empires and decolonization. |

| Law | Law group | Unlocked by | What it does |
|---|---|---|---|
| Private Policing | Policing | Computer Networks, with Laissez-Faire or Interventionism | Licensed private firms do most police work for whoever pays them; see Who polices. |
| Private Military Contractors | Army Model | Guided Missiles | Deniable force: 20% less infamy, faster infamy decay, fewer radicals from conquest and +0.25 war support a month, but casualties cost 30% more war support; −10% authority, 30% less conscription, 20% dearer military goods and −10% unit offense and defense. |
| Littoral Defense | Navy Model | Naval Convoy Defense | A coastal navy: ships operate only half as far from port, torpedo craft build faster and capital ships slower, navy goods cost 20% less, and Naval Fortifications get two more levels and resist invasion better. Coastal countries only. |
| Auxiliary Fleet | Navy Model | Predictive Logistics | A navy that carries the army: ships reach 50% farther from port, naval invasions go better, ships carry 30% more marines and supply ships 30% more supply, supply ships build faster and capital ships slower. Coastal countries only. |
| Legislated Tax Code | Taxation | The Legislated Tax Code game rule (off by default), except under its Light Tax Code setting | Replaces the five taxation laws: every rate is set by a tax bill. See Taxation (experimental). |

## Who staffs the administration

Three of the mod's Bureaucracy laws decide who does the work of your Government Administrations: Contracted Administration, Communal Administration and Automated Bureaucracy. Each replaces Professional Bureaucrats there with its own staffing, and each moves political strength away from bureaucrats. Automated Bureaucracy is covered further in Machine administration and machine rule.

| | Contracted Administration | Communal Administration | Automated Bureaucracy |
|---|---|---|---|
| Available | From Supply Chain Management, with Laissez-Faire or Interventionism; not in a Council Republic | In a Council Republic, under Anarchy or under Collective Governance; not with Technocracy or Elder Council | From AI Agents |
| Staffing | Service Contractors: 250 bureaucrats a level, buying 20 Services | Recallable Delegates: 250 bureaucrats and 250 clerks a level | Automated Casework: 150 bureaucrats and 100 engineers a level, buying Digital Access and Software |
| You gain | Institutions change size 25% faster; capitalists +15% political strength | Movements 5% less radical; Trade Unions and Rural Folk +10% political strength | Institutions cost 20% less bureaucracy and change size 50% faster; engineers +15% political |

| | Contracted Administration | Communal Administration | Automated Bureaucracy |
|---|---|---|---|
| | | | strength; Legislative Override Capacity +1 |
| You pay | Institutions cost 5% more bureaucracy | Tax capacity −10%; Legislative Override Capacity −2 | Nothing beyond the staffing |
| Bureaucrats | −25% political strength | −25% political strength | −25% political strength |

Contracted Administration suits a market economy with high wages. Each Government Administration level employs half the bureaucrats and buys Services from its state in their place, so the switch saves money where wages are high, and the bureaucrats it frees can take other jobs. Services are a local good: in a state whose Urban Centers make few of them, the price rises and the saving shrinks or disappears, and a shortage cuts the building's output. Contracts also run over budget, so every institution costs more bureaucracy.

Communal Administration is the bureaucracy for a council republic or an anarchist society. Delegates from local assemblies do half the work as clerks, who earn less than bureaucrats and qualify at lower literacy, so Government Administrations cost less and fill faster. The assemblies meet local needs first, which costs tax capacity, and delegates answer to the assembly that can recall them, which costs override capacity. If a Council Republic or a country under Collective Governance loses Elected Bureaucrats because its Distribution of Power stops being a voting law, it moves to Communal Administration rather than Appointed Bureaucrats. A country that can no longer hold Contracted Administration moves to Appointed Bureaucrats (Automated Bureaucracy under Algorithmic Governance), and one that can no longer hold Communal Administration moves to Elected Bureaucrats where it can, otherwise to Appointed.

## Machine administration and machine rule

Automated Bureaucracy and Algorithmic Governance are two steps. With Automated Bureaucracy, machines run your offices while your government keeps the authority, so any Distribution of Power can hold it. With Algorithmic Governance, the machines rule as well.

Under Automated Bureaucracy your institutions cost 20% less bureaucracy and change size 50% faster, and Legislative Override Capacity rises by 1 (see Forceful Legislation). Bureaucrats lose a quarter of their political strength and engineers gain 15%. Government Administrations staff Automated Casework in place of Professional Bureaucrats: 150 bureaucrats and 100 engineers a level instead of 500 bureaucrats, buying Digital Access and Software. Bureaucrats earn the higher wage, so the switch saves money where wages are high and costs money where they are low, and a shortage of either good cuts the buildings' output.

Algorithmic Governance enacts Automated Bureaucracy, and while it holds you can enact no other Bureaucracy law. It dissolves your parties and makes you an Algorithmic Directorate, or an Algorithmic Commons under Collective Governance. Laws pass 50% faster, research runs 10% faster, decrees cost a quarter less and you can run two more companies. It has no clout or votes to draw legitimacy from, so it gives 70 legitimacy flat and the rest comes from results: the Algorithmic Mandate adds legitimacy while living standards run ahead of what people expect and takes it away while they fall behind, up to 40 either way, and it shrinks as expectations catch up. Taxes, your head of state's interest group and timed modifiers count as usual, so with a steady standard of living you sit near 70 plus those, a boom takes you to or near 100 even under heavier taxes, and a deep slump drops you toward Unstable Government. Radicals and loyalists from changes in living standards double as well, so a falling standard of living turns on you fast.

## Who polices

The base-game Policing group decides who provides ordinary law enforcement, who pays for it and whom it answers to. Local Police Force is town and county constables under local notables, Dedicated Police Force a national professional force, and Militarized Police Force a national police run along military lines. The mod adds Private Policing, where licensed firms do most of the work, hired and directed by property owners, businesses and residents' associations. All four police laws keep the Ministry of Public Safety, so switching between them keeps its investment.

| | Private Policing |
|---|---|
| Available | From Computer Networks, with Laissez-Faire or Interventionism; not in a Council Republic |
| Ministry of Public Safety | Costs 40% less bureaucracy, as base-game Private Schools and Private Health Insurance do for theirs |
| Each Ministry level | Upper strata +0.25 standard of living, middle strata +0.1; the Ministry's turmoil reduction is halved, to 5% a level; under the Full Demographics rule deaths from accidents and violence −2% a level, against a Dedicated Police Force's 3%; Services output −1% |
| Also | Trade Unions −10% political strength; Labor Movement pop attraction +15%; Legislative Override Capacity −0.5 |

Protection follows money. In every state the wealthy districts are guarded and the poor ones are not, so the upper and middle strata live better while turmoil, which comes from the poor, is held back only half as well. Owners pay for their guards: each Ministry level takes 1% of the country's Services output, so Services cost more for the Urban Centers' customers, mostly the better-off. Guards break strikes, which costs the Trade Unions strength and feeds the Labor Movement, and a force that answers to its clients is harder to turn on the legislature (see Forceful Legislation).

Only Corpocrat leaders, and a custom religion's Corporate Rule, favor Private Policing. Market liberals and plutocrats are indifferent. Socialists, social democrats, radicals, humanitarians, egalitarians, paternalists, patriots, jingoists, fascists, integralists and authoritarians oppose it, and anarchists and communists strongly oppose it. The AI enacts it only under a Corpocrat ruler. A country that loses its market economy or becomes a Council Republic moves to Dedicated Police Force, keeping its Ministry's investment.

Two amendments adjust a police law without replacing it (see Amendments to the mod's laws). Corporate Security Powers gives companies police powers in their own mines, mills and company towns beside a public force, from Corporate Charters on. Interest groups take the same stance on it as on Private Policing, so until Corpocrats appear (Mutual Funds) it depends on the Industrialists, who sponsor it whatever their leader; the Guards at the Mill Gate event offers it during the debate. Civilian Oversight sets up review boards for complaints against police officers and licensed firms, from the Civil Rights Movement technology on, and interest groups judge it as they judge Guaranteed Liberties; the Who Watches the Watchmen event offers it during the debate. It pays most where many pops are Second-Rate Citizens or face Open Prejudice.

## Who inherits

The Inheritance group decides who inherits an estate. Interest groups judge the inheritance law in force for as long as it stands, so a group that dislikes yours loses approval the whole time. Every reform movement (liberal, radical, socialist and land reform) opposes Primogeniture & Entail and grows more radical while it stands, and the traditionalist movements defend it.

| Law | Available | Wealth Concentration target | Also |
|---|---|---|---|
| Primogeniture & Entail | From the start | +30 | In farming states, birth rate (outside the Demographics rule's Full setting) and migration up; tax capacity −10%; local administration costs 10% less bureaucracy |
| Customary Inheritance | From the start | 0 | Tax capacity −5%; local administration costs 5% less bureaucracy |
| Forced Heirship | Egalitarianism | −25 | In farming states, birth rate and migration down; tax capacity +10%, Rural Folk +10% political strength |

| Law | Available | Wealth Concentration target | Also |
|---|---|---|---|
| Freedom of Testation | From the start | +10 | Capitalists invest 10% more |
| State as Universal Heir | Command Economy | −50 | +200 authority, Bureaucrats +10% political strength; the upper and middle strata invest less |
| Possession by Use | Cooperative Ownership | −50 | Rural Folk and Trade Unions +10% political strength; the upper and middle strata invest less |

France starts with Forced Heirship, Britain and Japan with Primogeniture & Entail, the United States with Freedom of Testation and every other country with Customary Inheritance. A country that leaves Command Economy loses State as Universal Heir and moves to Possession by Use if it now has Cooperative Ownership, else to Forced Heirship if you have Egalitarianism, else to Customary Inheritance; Possession by Use falls the same way.

**Wealth Concentration** measures, from 0 to 100, how much of your wealth sits in great family fortunes. Each state keeps its own score, which moves 3% of the way to its own target each year, and your country's figure is the states' average, weighted by where property is held. A state's target starts at 50, and the law (the table above) and its amendments move it, together at most 50 either way. Land tenure, who owns the state's capital, how unequal incomes are, taxes on wealth, your economic laws, whether health care, schooling and ideas are provided or sold, and your slavery law move it from there. Laissez-Faire (+10), Extraction Economy (+5), Freedom of Contract and Guilds and Chartered Monopolies (+5) raise it; Interventionism, Antitrust Enforcement and Regulated Utilities (−5 each) lower it. Private Health Insurance, Private Schools, Terakoya and Traditional IP Protection (+5 each) and Strict IP Protection (+10) raise it; Public Health Insurance and Public Schools (−5 each) lower it. Colonial Slavery, Legacy Slavery and Debt Slavery (+10 each) and Slave Trade (+5) raise it too, whatever the number of slaves; an open trade keeps slaves cheap and ownership wide. Each inheritance, land, tax, economic, health, school, IP and slavery law that moves it lists its part among its modifiers, and so does each inheritance amendment. Once your banking cycle has started, the return on capital against growth moves it too: twice the gap between your policy rate less inflation and your economy's growth over about five years, at most 5 either way. War losses, a lost war that cost you a war goal, a winning revolution of the left, a banking crash, a hyperinflation crisis and land reform knock scores down at once. The hyperinflation crisis takes 5 each time it fires and 5 more if you reform the currency. Land reform is Peasant Proprietorship or Collectivized Agriculture replacing Serfdom or Tenant Farmers, and takes up to 10 from a farming state. A banking cycle's Frenzy raises every score by 5, once a cycle. When the state-owned share of a state's capital rises 10 points or more in a year, whether the state takes buildings over or builds faster than private owners do, the state loses half a point for each point, up to 10; when the share falls as far, it gains the same. Devastation lowers a state's score every year it lasts. A new law therefore takes about a generation to show: a country that abolishes entail still has great houses twenty years later.

Your country's figure sets the political weight of great fortunes and how fast speculation builds. Above 50, Great Family Fortunes strengthens the Aristocrats and Capitalists and adds bubble pressure to the banking cycle, by 20% and 0.3 a month at 80; below 50, Dispersed Wealth weakens both and takes bubble pressure off, by 20% and 0.4 a month at 20. Concentrated wealth grows faster but crashes more often; dispersed wealth grows slower but steadier. Each state's own score does the rest in that state:

| A state's score | Modifier | Effect at 80 (or 20) |
|---|---|---|
| Above 50 | Concentrated Property | Aristocrats and Capitalists invest 20% more; political strength from wealth +20%; qualifications −20%; education access −10% |

| A state's score | Modifier | Effect at 80 (or 20) |
|---|---|---|
| Above 50, in farming states | Land Hunger | Migration +30%, Land Reform movement support +25%, radicals +10%, scaled by the state's share of Peasants and Farmers |
| Below 50 | Dispersed Property | Political strength from wealth −10%; qualifications +10%; education access +10%; radicals −10% |

Every effect reaches its full size at 80 or 20, so a state at 65 gets half of it, and keeps growing beyond: the concentrated side to two thirds more at 100, the dispersed side to a third more at 0. A state's modifiers update with its yearly count, so a war loss or an event that moves the scores reaches them within a year. A capital full of Financial Districts can carry Concentrated Property while your country as a whole sits below 50. The Inheritance group's description shows the score and where it is heading, and the Wealth section of the Demographics tab shows what sets the target. A state's Wealth Concentration tile, on the state panel, shows its own score and the modifiers it carries from it and from the inheritance law. With the census off (the rule's Disabled setting) the tab is gone, but Wealth Concentration and all its effects still run and the group's description still shows the score.

The farming effects follow the share of Peasants and Farmers in each state. Under Forced Heirship, smallholders have fewer children so the farm is not split again, and stay on their plots; under Primogeniture & Entail families are large and the younger sons leave the land. Industrial states are hardly touched, so the effect fades as you urbanize. Under the Demographics rule's Full setting the census carries the birth effects instead: Forced Heirship's through family limitation and Primogeniture's through late marriage (see Reading the figures). States update once a year, so a new law takes up to a year to reach them.

Four amendments adjust the inheritance laws (see Amendments to the mod's laws). Estate Duties and the Birthright Endowment tax estates and pull the score down; Perpetual Dynastic Trusts push it up; Undivided Farm Succession lets one heir take the family farm whole, which switches off Forced Heirship's farming effects and sends the other siblings off the land. A law's amendments leave with it. When you enact another law with private estates, the debate events offer to carry an estate duty, the Endowment or the farm rule over.

Once you have held a law for 20 years, its generation event can come, at most once every 25 years: The Cadet Sons under Primogeniture & Entail, The Last Division under Forced Heirship (one answer attaches Undivided Farm Succession) The Founder's Will under Freedom of Testation, The Hidden Estates under State as Universal Heir and The Empty Holding under Possession by Use. The Great Estates for Sale comes after 15 years of an estate duty, and The First Cohort once, five years after you attach the Birthright Endowment. Their answers move the score, a pop group or a modifier for up to ten years.

With Mind Backups or Biological Immortality, Posthumous Title asks whether a person can outlive their estate. Recognizing continuity of title freezes Wealth Concentration where it stands, at a cost of 10% innovation and 10% qualifications, and five years later the young who will never inherit organize. Declaring legal death lets estates pass and strengthens the Transhumanist movement for ten years.

Traditionalists, paternalists and the patriarchs favor Primogeniture & Entail. Liberals and individualists favor Freedom of Testation. Reformers, social democrats and agrarians favor Customary Inheritance and Forced Heirship, and Bonapartists strongly favor Forced Heirship. Communists favor State as Universal Heir, and anarchists Possession by Use.

### Law enactment events

The mod's laws have their own events at the enactment checkpoints, such as a banking scare that helps financial reform or a new ministry struggling to find staff. Several offer an amendment as the price of passage (see Amendments to the mod's laws).

The checkpoint decides what the choices can do. At an advance, every choice helps the law. At a debate, one choice pushes the law and another gives ground to its critics: the law loses 5% enactment success, and you gain something in return, such as an interest group's approval or a few loyalists. Turning an amendment down

is that kind of choice. At a stall, every choice slows the law, and you pick between a setback and a longer drag on its success.

### Forceful Legislation

When you start enacting any law, the **Forceful Legislation** event offers to bypass the legislature, if your authority income covers the cheapest option. What it allows depends on your **Legislative Override Capacity**, summed from your laws, your institutions and a few technologies and power bloc principles. The Nation tab shows it with every source. Among the laws, Autocracy and Single-Party State give 3, Algorithmic Governance 2 and Unitary State 1; Universal Suffrage, Liquid Democracy, Anarchy and Traditional Vassalage take away 3, and Guaranteed Liberties, Devolved Administration and Communal Administration 2. Among the courts, Executive Tribunals give 1 and Algorithmic Jurisprudence 2 under Algorithmic Governance; Constitutional Review takes away 2 and Judicial Supremacy 3. Establishing the Ministry of Thought Control adds 2, and the ministries of Propaganda and of Intelligence and Security 1 each. The Policing laws count too, from −0.5 for No Police and Private Policing to +1 for a Militarized Police Force, and the Civilian Oversight amendment takes away 0.5. Each level of investment adds a fraction more: 0.4 in Thought Control, 0.2 in Propaganda and 0.1 in the Ministry of Public Safety, the base game's Law Enforcement institution under its new name. Below 0, the event never appears, and each stronger option also needs authority income above its own cost.

| Option | Needs capacity | Authority cost | Effect on the law | Lasting cost (fades over 5 years) |
|---|---|---|---|---|
| Expedite the process with minimal intervention | 0 | 250 | +10 amenability, +5% enactment success, a quarter fewer stalls | Interest groups in opposition −1 approval, capacity −1 |
| Push the law forward decisively | 2 | 500 | +25 amenability, +15% success, half the stalls, one more setback allowed, 20% faster | Opposition −3 approval, −5 legitimacy, movements radicalize more, capacity −2 |
| Force the law through, consequences be damned | 4 | 750 | +50 amenability, +25% success, three quarters fewer stalls, two more setbacks, 35% faster, jumps one enactment phase | Opposition −5 approval, −10 legitimacy, more radicals, some upper-strata radicals at once, capacity −3 |

Each option also lowers Judicial Independence by 5, 10 or 20. That loss doesn't fade: the score climbs back at its usual 4 a year. Under Constitutional Review or Judicial Supremacy the law then goes to the courts, which may strike it down (see Laws on state power and information).

The authority cost fades out over five years too. More capacity makes it cheaper: a few percent off from 2 (about 9% at 4), then 20% at 5 and half at 8. Each use lowers your capacity for a while, so a country exactly on a tier's threshold can use that tier once and must then wait.

## 6.2 Ministries

Sixteen law groups each decide whether a ministry exists. A ministry is an institution you invest in like schools or police. Its law sets a starting cap of one to three levels, and later-era technologies raise the cap, most of them by one level. The mod raises the ceiling on every institution, the base game's included, from 5 levels to 9.

| Ministry | Technology | Also needs | Each level of investment |
|---|---|---|---|
| Ministry of Foreign Affairs | International Relations | | +100 influence and +10% influence, faster relations changes, more leverage |
| Ministry of War | Wargaming | | More army experience and military research, +2% offense and defense, dearer military wages and goods |

| Ministry | Technology | Also needs | Each level of investment |
|---|---|---|---|
| Ministry of Commerce | International Exchange Standards | | +5% trade advantage and trade capacity |
| National Bank | Central Banking | not Traditionalism | Lower borrowing premium, larger gold reserves, better investment efficiency; see Financial regulation laws |
| Ministry of Culture | Mass Propaganda | | +10% assimilation and conversion, cultural pull, cultural program funding; see Cultural programs |
| Ministry of Labor | Public Works Programs (pro-labor) or Marketing Research (pro-capital) | | Both raise the workforce ratio, pro-capital twice as much. Pro-labor weakens the political power of wealth and adds company workforce dividends; pro-capital strengthens it, lowers minimum wages and radicalizes movements slightly |
| Ministry of the Environment | Pollution Control | | −5% pollution and emissions, larger national parks, slightly lower extraction output |
| Ministry of Intelligence and Security | Mass Surveillance | | +4 intelligence capacity and one more covert operation slot |
| Ministry of Refugee Affairs | Civil Rights Movement | Any migration law except Closed Borders | Faster cultural acceptance, more mass migration |
| Ministry of Propaganda | Mass Media | Single-Party State, Secret Police or Outlawed Dissent | +100 authority and primary-culture fervor; the law also steadies war support through battles |
| Ministry of Science | Nuclear Weapons | | +5% research speed, academics' political strength |
| Ministry of Thought Control | Television Broadcasting | Single-Party State, Secret Police or Outlawed Dissent | Loyalists every month; low electoral confidence hurts less |
| Ministry of Consumer Protection | Consumer Credit | Regulatory Bodies or Workers' Protections | +0.5 standard of living, lower mortality, production research, −10% opium plantation throughput a level |
| Ministry of Urban Planning | Urbanization | | Urban centers produce more and need less urbanization per level; +10% migration crowding tolerance; faster recovery from devastation |
| Ministry of Religion | Mass Media | State Religion or Freedom of Conscience | +20% conversion, clergy political strength, acceptance of the state religion |
| Ministry of International Aid | Intergovernmental Organizations | | Strengthens the Ministry of Foreign Affairs, which also costs more; +0.15 UN Authority Target a level (see What counts toward Policy) |

The Environmental Movement radicalizes while your Ministry of the Environment is missing or funded below its cap, and the Anti-War Movement does the same over the foreign affairs, international aid and refugee ministries (see New political movements).

The mod also adds one institution that isn't a ministry: the base game's Migration Controls law now sets up a Migration Controls institution in place of its flat restriction. Each level raises migration restrictiveness, the acceptance a pop needs to move to your country, by 10, so six levels match the base law. Each level also cuts mass migration attraction by 10% and adds 5% resistance to foreign leverage. The institution costs a tenth of the usual bureaucracy, and countries that start with the law begin at level 6.

## 6.3 Amendments to the mod's laws

Interest groups sponsor amendments to the mod's laws as they do for base-game laws, and every one of them also comes with an event or another way into the game: most are offered by enactment events as the price of passage, and most can also be asked for later, on a law already in force (see Amendment petitions). Four amendments attach to base-game laws: the Spoils System, Corporate Security Powers, Civilian Oversight and First Past the Post. Under the Legislated Tax Code rule's legislated settings the tax rates are amendments to the Legislated Tax Code law, and only tax bills change them (see Taxation (experimental)).

| Amendment | Attaches to | Effect |
|---|---|---|
| Minority Script Preservation | Civic Monolingualism, State-Led Language Reform | Slower assimilation, less separatism |
| Language Requirement for Naturalization | Cultural, Assimilatory or Universal Citizenship, alongside Civic Monolingualism, State-Led Language Reform or Linguistic Purity | Faster assimilation, tighter migration. The A Test of the Tongue event offers it while one of those citizenship laws is enacted, sponsored by the Petite Bourgeoisie (or the Rural Folk) |
| Official Bilingualism | Multilingual Federalism, State-Led Language Reform | Weaker separatism, higher bureaucracy cost |
| National Champion Exemption | Antitrust Enforcement, Regulated Utilities | Company output up, general output slightly down |
| Worker Cooperative Preference | Antitrust Enforcement, Regulated Utilities | Trade Unions gain political strength, Industrialists lose it |
| Mandatory Minimum Sentences | Restorative or Rehabilitation-Focused Criminal Justice | Petite Bourgeoisie and Devout approval, fewer workers |
| Community Service Alternative | Punishment-Focused Criminal Justice | Intelligentsia approval, more workers |
| Corporate Data Access Exemption | Moderate Data Privacy, Strong Privacy Rights | Industrialists gain strength, technology spreads faster |
| Whistleblower Protection Act | Moderate Data Privacy, Strong Privacy Rights | +5 legitimacy, less authority |
| No-First-Strike Pledge | Humanitarian Regulations, Limited War | +5 diplomatic reputation; holds your nuclear doctrine at No First Use |
| Military Necessity Clause | War Crimes Forbidden, Humanitarian Regulations | Armed Forces approval, −5 diplomatic reputation |
| Small Donor Matching | Donation Limits, Publicly Funded Elections | +5 legitimacy, higher bureaucracy cost |

| Amendment | Attaches to | Effect |
|---|---|---|
| Religious Exemption Clause | Any LGBTQ+ protection law | Devout approval |
| Environmental Grandfather Clause | Ministry of the Environment Established | Industrialist approval, 5% more pollution and emissions |
| Inflation Targeting Act | Central Bank Independence | Sets the central bank's mandate to Inflation Targeting, 12 months after it passes. The What the Bank Is For event asks for it or the Dual Mandate Act while the law has neither; the Industrialists sponsor it. See Rate target, delegation and mandates |
| Dual Mandate Act | Central Bank Independence | Sets the mandate to the Dual Mandate, 12 months after it passes. Offered by the same event and by the Central Bank Independence Debate; the Trade Unions sponsor it |
| Corporate Security Powers | Local, Dedicated or Militarized Police Force | Ministry of Public Safety costs 10% less bureaucracy; Trade Unions −10% political strength; Labor Movement pop attraction +15%; the sponsoring group gains 10% political strength. Not under Command Economy, Cooperative Ownership or Industry Banned. The Industrialists sponsor it, and so does any group that approves of Private Policing; the Guards at the Mill Gate event offers it |
| Civilian Oversight | Local or Dedicated Police Force, Private Policing | Ministry of Public Safety costs 10% more bureaucracy; Legislative Override Capacity −0.5; radicalism rises 10% slower for Second-Rate Citizen and Open Prejudice pops; −5% turmoil penalties; the sponsoring group gains 10% political strength. The Intelligentsia sponsor it, and so does any group that approves of Guaranteed Liberties; the Who Watches the Watchmen event offers it |
| Spoils System | Appointed or Elected Bureaucrats | The sponsoring group gains 15% political strength; tax capacity −10%. Groups in government sponsor it, and so does any group led by a Jacksonian Democrat. The To the Victor Belong the Spoils event offers it while either law is enacted, sponsored by the strongest group in government |
| First Past the Post | Landed, Wealth or Census Voting, Universal Suffrage, Anarchy | Interest groups join the two largest parties far more readily than any other (see Two-party politics); 25% less legitimacy lost to disagreement inside the government; −5 legitimacy. Members of the two largest parties sponsor it once those parties hold most of the clout, and on the same condition The Winner Takes All event offers it while a voting law is enacted |
| Estate Duties | Primogeniture & Entail, Customary Inheritance, Forced Heirship, Freedom of Testation | Wealth Concentration target −30; no standing revenue. From Political Agitation; not with the Birthright Endowment. Interest groups judge it as they judge Graduated Taxation; the Death Duties event offers it |
| Perpetual Dynastic Trusts | Freedom of Testation | Wealth Concentration target +30; Capitalists invest 10% more. From Globalization. Judged as Laissez-Faire; the Industrialists sponsor it, and The Trust That Never Ends event offers it |
| Birthright Endowment | Primogeniture & Entail, Customary Inheritance, Forced Heirship, Freedom of Testation | Wealth Concentration target −20, since its duty pays the stakes; qualifications +10%, education access +5%, Laborers, Farmers and Shopkeepers invest 10% more; institutions cost 5% more bureaucracy. From Universal Basic Income. It levies the estate duty itself, so it replaces Estate Duties. Judged as Old Age Pension; the A Stake at Twenty-One event offers it |
| Undivided Farm Succession | Forced Heirship, Customary Inheritance | Wealth Concentration target +10; ends Forced Heirship's farming effects, and in farming states the siblings who don't inherit leave the |

| Amendment | Attaches to | Effect |
|---|---|---|
| | | land. Judged as Peasant Proprietorship; the Rural Folk sponsor it, and the One Heir to the Farm event offers it |
| Suspension of Habeas Corpus | Statutory Courts, Constitutional Review | Suppression 25% cheaper; Judicial Independence ceiling −20%; −5 legitimacy. Only at war, and it lapses when the war ends. Judged as Secret Police; the Armed Forces ask for it |
| Judicial Elections | Statutory Courts, Constitutional Review | +5 legitimacy; Judicial Independence ceiling −20. Under an elected government. Judged as Universal Suffrage; offered while either law is enacted |
| Mixed Commercial Courts | Religious Courts, Executive Tribunals | Half a point off credit standing; ceiling +10; Devout −5% political strength. Judged as Free Trade; the Industrialists or Petite Bourgeoisie ask for it |
| Jury Trials | Statutory Courts, Constitutional Review | Ceiling +10; +3 legitimacy; suppression 10% dearer. Judged as Guaranteed Liberties; offered while either law is enacted |
| Military Tribunals for Civilians | Executive Tribunals, Statutory Courts | Suppression 15% cheaper; Armed Forces +10% political strength; ceiling −20%. Judged as Outlawed Dissent; the Armed Forces ask for it in unrest |
| People's Tribunals | Executive Tribunals under a Single-Party State | Suppression 20% cheaper; movements give 10% more loyalists; half a point on credit standing. It lapses without the Single-Party State. Judged as Single-Party State |
| Human-in-the-Loop Appeals | Algorithmic Jurisprudence | +5 legitimacy; ceiling +10; local administration costs 5% more; ends the override a captured court gives under Algorithmic Governance. Judged as Strong Privacy Rights; offered while the law is enacted |

The mod also changes one base-game amendment. Anarchy holds elections, so the Tradition of Free Elections attaches to it as to the voting laws. Keeping elections free under Anarchy builds toward the tradition, enacting Anarchy keeps one you already hold, and the tradition no longer makes Anarchy harder to enact.

## Amendment petitions

Once a law has passed, an interest group can still ask you to amend it when your situation gives it a case: the Armed Forces want the Military Necessity Clause in wartime, the Industrialists the National Champion Exemption in a recession. Attaching the amendment pleases the petitioners and angers its opponents, and you can't repeal it for five years; refusing angers the petitioners. Each petition has a yearly chance of 2.5% to 25% while you qualify, and after any petition no other comes for five years. Amendment petition list gives each one, what brings it on and its chance.

## Sunset clauses

Three enactment events can attach an amendment for a fixed term. When the term runs out, the amendment lapses and a follow-up event asks what to do next.

| Amendment | Offered while enacting | Term | Expiry event |
|---|---|---|---|
| National Champion Exemption | Antitrust Enforcement or Regulated Utilities ("The Trust Question") | 10 years | The Champion's Decade Ends |
| Corporate Data Access Exemption | Moderate Data Privacy or Strong Privacy Rights ("The Privacy Bargain") | 3 years | The Grace Period Closes |
| Environmental Grandfather Clause | Ministry of the Environment Established ("The Grandfather Clause") | 10 years | The Phase-In Ends |

At the end of the term you make the clause permanent, costing legitimacy for a few years and radicalizing some shopkeepers (for the National Champion Exemption) or academics (for the other two), or let it lapse, angering the Industrialists. While the Environmental Grandfather Clause applies, its extra pollution and emissions cancel one level of the new ministry's reductions.

## 6.4 Collective Governance

Collective Governance is a governance principle in which nobody holds supreme executive power: a body of equals governs, and its chair is first among equals. It unlocks with Political Agitation and needs one of these Distribution of Power laws: Landed Voting, Wealth Voting, Census Suffrage, Universal Suffrage, Liquid Democracy, Single-Party State, Technocracy, Oligarchy, Organic Regulation, Anarchy or Algorithmic Governance. Cooperative Ownership, which Anarchy needs, is open to it as to a Council Republic. Under it you have 15% less authority and laws pass 10% more slowly, but broad coalitions earn more legitimacy and ideological incoherence costs less. The chair's interest group gets no head-of-state legitimacy bonus.

What the body is depends on your Distribution of Power, expressed as an amendment the game attaches and swaps whenever that law changes. You can't repeal it, and the law's tooltip names it before you enact.

| Distribution of Power | Amendment | Effect |
|---|---|---|
| Any voting franchise | Direct Democracy | −10% enactment success; movements draw more support and are 50% more active, but enacting a law stirs its opponents and calms its supporters 75% less; wealth brings 25% less political strength; votes add legitimacy; one more agitator slot |
| Single-Party State | Collective Leadership | +25% coup resistance, which its modifier list doesn't show |
| Technocracy | Collegial Administration | Institutions change size 50% faster; decrees cost 25% more |
| Anarchy | Free Federation | −10% enactment success; movements draw 50% more support; passing a law a movement backs turns 50% more of its supporters loyal; bolstering a movement costs half |
| Oligarchy or Organic Regulation | Patrician Council | Aristocrats and capitalists gain 15% political strength |
| Algorithmic Governance | Algorithmic Commons | Wealth brings 25% less political strength; political movements are 10% less active |

Under Direct Democracy and Free Federation, a law needs a political movement behind it to pass at a normal pace. A movement adds its support to the enactment success chance of the laws it backs, so a law a large movement carries more than makes up the 10%, and a law no movement cares about passes slowly. If it has little support among the interest groups in government, it can't start at all.

## 6.5 New government types

These follow from the governance laws above. Each Collective Governance amendment brings a government type of the same name, from a Direct Democracy led by a Speaker to an Algorithmic Commons led by a Steward. Neocameralism makes you a Corporate Democracy, Corporate Syndicalism or Corporate Autocracy, led by a Director, and Algorithmic Governance on its own makes you an Algorithmic Directorate. Government type list gives every type with its conditions and ruler title.

## 6.6 Ideologies, parties and movements

The mod adds ideologies, three parties and four political movements, and gives many base-game ideologies stances on the new law groups. Every ideology with a view on a law group also states one on each law the mod adds to that group, so a group that dislikes the current law doesn't back a new law it has no view on.

### New ideologies

| Ideology | Held by | Appears |
|---|---|---|
| Environmentalist | Leaders | From Pollution Control; strongly wants a Ministry of the Environment |

| Ideology | Held by | Appears |
|---|---|---|
| Multiculturalist | Leaders; also added to the Intelligentsia | Leaders once a Civil Rights Movement exists; the group ideology with the Civil Rights Movement technology |
| Inclusive Multiculturalist | Leaders | As Multiculturalist, after the Sexual Revolution; adds LGBTQ+ rights |
| Anti-Colonialist | Leaders; also added to the Trade Unions and Intelligentsia | With the Decolonization technology; opposes colonial laws |
| Optimist Transhumanist | Leaders | From Brain-Computer Interfaces; favors augmentation, open genetics, Automated Bureaucracy, Algorithmic Governance and Liquid Democracy |
| Corpocrat | Leaders | From Mutual Funds, mostly Industrialists; favors Neocameralism, Contracted Administration and Private Policing |
| Islamic Inheritance | The Devout in Sunni and Shiite countries | From the start; favors Customary Inheritance and Forced Heirship |

Interest-group leaders, commanders and politicians are more often women under Protected Class, the higher LGBTQ+ Rights laws and related technologies.

## The three new parties

| Party | Technology | Draws |
|---|---|---|
| Green Party | Environmental Movement | Intelligentsia and Rural Folk, environmentalist, pacifist and reformist leaders; not the Industrialists or the far right |
| Populist Party | Social Media | Rural Folk, Petite Bourgeoisie and Devout, nationalist and reactionary leaders; not the left |
| Technocratic Party | Machine Learning | Industrialists and Intelligentsia, Corpocrat, transhumanist and market-liberal leaders |

All three need a voting franchise and take national names in some countries, such as Die Grünen or Movimento 5 Stelle.

## Two-party politics

The First Past the Post amendment gives each seat to whoever tops the poll, so a vote for a small party is wasted. While it is on your voting law, the two parties whose members hold the most clout are big tents: +10 affiliation, less 1 for each group joining ahead, and the base game's party-size limit (about three groups to a party) doesn't apply to them. Every other party gets −20. So every interest group joins one of the two unless it dislikes both, taking the one it fits better, and groups with no preference split between them. A third party forms only around a group that fits it strongly. The party affiliation tooltips show these lines. The two are re-ranked every month, so a party that gains clout can take a major party's place at the next campaign. Until two parties exist the amendment does nothing to parties.

Each major party is a broad coalition, so the government loses 25% less legitimacy to disagreement inside it, and 5 legitimacy for the voters left without a winning party. The United States starts with the amendment, and the Confederacy gains it if the Civil War breaks out; it replaces the base game's fixed two-party rule for American countries.

A law's amendments leave with it, so while you hold first past the post, a change of voting law brings the Counting the Votes event, asking whether to keep it. Keeping it puts it on the new law; declining adopts proportional representation, and a later change of voting law won't bring it back. If you move to a law without elections, the question comes when elections return, and the same holds for a revolution that wins on a new voting law. Once two parties hold most of the clout, other countries can adopt it through The Winner Takes All event while a voting law is enacted, or the Two Parties Are Enough petition. You can repeal it when legitimacy is high enough, as with other amendments; the AI keeps it.

### New political movements

| Movement | Forms when | Wants | Notes |
|---|---|---|---|
| Environmental Movement | Pollution Control researched and no Ministry of the Environment | A fully funded Ministry of the Environment | See The Environmental Movement |
| Civil Rights Movement | Civil Rights Movement technology, or Cultural Unity at tier III or higher in your power bloc | Universal Citizenship and Affirmative Action | Strongest among discriminated pops |
| Anti-War Movement | Anti-War Movement technology | Pacifist policy; its leaders may be Anti-Colonialist | Grows with casualties and while at war; never starts a revolution or secession, and never disbands |
| Transhumanist Movement | Brain-Computer Interfaces researched, without Unrestricted Augmentation | Unrestricted Augmentation | Radicalized by Human Purity and bans on genetic modification; never disbands |

The Anti-War and Transhumanist Movements bring events of their own. The Anti-War Movement's come mostly in wartime: Draft Resistance, Veterans Speak Against the War, Peace Rally Fills the Capital, and War Profiteering Exposed, which needs a military industry building of level 3 or more. The Transhumanist Movement's arrive with augmentation technology: Neural Implant Human Trials, The Augmentation Divide, The Biohacker Underground and The Digital Consciousness Debate.

The social-movement journal entries (civil rights, digital rights and others) are separate systems, described in Movement journal entries at a glance.

## 6.7 Modern election events

During an election campaign, countries with elections draw from 35 extra events, from radio broadcasts and war heroes through televised debates to deepfakes, microtargeting and space-colony voters. Each needs its own technology, from Mass Propaganda for the earliest through Television Broadcasting and Social Media to Generative AI and Space Colonization for the last. Single-party states never get them. They share the base game's pacing (one election event every three months at most), and each fires at most once every 20 years. Most options shift momentum between the governing and opposition parties or trade loyalists, radicals or a temporary modifier. Your laws, such as an unregulated internet or strong privacy rights, decide which option the AI favors.

## 6.8 Heir Education

With the Heir Education rule on, a monarchy with an heir gets the Heir Education journal entry. Each month every focus you set has a 3% chance of advancing its track, and when the heir comes of age the tracks become traits, an ideology and an interest group.

### Heir Education focuses

You can run one focus from each track at once, and each running focus costs 100 authority.

| Track | Choices | Who approves and who objects |
|---|---|---|
| Skills | Administrative Focus, Diplomatic Focus, Military Focus | Industrialists, Intelligentsia or Armed Forces approve their own focus; another group objects |
| Ideology | Progressive Tutors, Conservative Tutors | Intelligentsia and Trade Unions against Devout and Landowners |
| Faction | Radical Mentors, Moderate Mentors, Regressive Mentors | Radical: Intelligentsia, Rural Folk and Trade Unions against Landowners and Devout; Regressive: Landowners, Devout and Rural Folk |

| Track | Choices | Who approves and who objects |
|---|---|---|
| | | against Intelligentsia and Trade Unions; Moderate: Industrialists, Petite Bourgeoisie and Armed Forces, and nobody objects |

Each successful skill lesson also lowers innovation by 10% for 30 days. Every heir has a hidden intelligence from 1 to 5: a bright heir sometimes gains two points from one lesson, a slow one sometimes nothing, and the status line says "a quick study" or "struggles to grasp concepts". An heir of 15 or older starts with one to three points in each skill. Three events can interrupt: A Promising Pupil, A Difficult Student and Foreign Correspondence, which needs an ally.

## How the education resolves

The education ends when the heir is an adult and at least a year has passed since the journal entry began, or after 18 years. If the heir dies or changes, the tracks reset at the next monthly check, and the focuses switch off along with their Authority cost; set them again for the new heir.

| Points in a skill | Terrible | Poor | Average | Skilled | Exceptional |
|---|---|---|---|---|---|
| 0 | 35% | 30% | 25% | 8% | 2% |
| 1–2 | 15% | 25% | 35% | 20% | 5% |
| 3–4 | 5% | 15% | 30% | 35% | 15% |
| 5–7 | 2% | 5% | 18% | 45% | 30% |
| 8 or more | 0% | 2% | 10% | 38% | 50% |

The tutors' net lean sets the heir's ideology.

| Tutors' net lean | Possible ideologies, likeliest first |
|---|---|
| Strong progressive (4 points or more) | Reformer, Liberal, Social Democrat, Humanitarian, Radical, Communist or Vanguardist; rarely Moderate |
| Light progressive | Reformer, Liberal, Humanitarian or Social Democrat; often Moderate |
| None | Usually Moderate, sometimes Reformer, Traditionalist or Liberal |
| Light conservative | Royalist, Traditionalist, Authoritarian or Theocrat; often Moderate |
| Strong conservative (4 points or more) | Royalist, Traditionalist, Authoritarian, Theocrat, Ethno-Nationalist or Jingoist; rarely Moderate |

Strong progressive tutoring can raise a Radical, Communist or Vanguardist, an heir who may want to abolish the monarchy they inherit. Each heir has an 8% chance to rebel and take the opposite lean.

Only ideologies your country can have are rolled. Reformer needs Egalitarianism and is no longer possible once you have Feminism; Radical also needs Egalitarianism, Liberal needs Democracy, Social Democrat and Communist need Socialism, Vanguardist needs Political Agitation, Humanitarian needs Feminism and Ethno-Nationalist needs Pan-nationalism. France never gets a Royalist or Traditionalist heir, and countries with a strong republican tradition never get a Royalist. Until you have Democracy or one of those other technologies, progressive tutoring can only produce a Moderate, at the full Authority cost. Conservative tutoring always has outcomes, since Authoritarian, Theocrat and Jingoist need nothing.

The strongest mentor faction picks the heir's interest group.

| Strongest mentors | Interest group |
|---|---|
| Radical | Trade Unions, Intelligentsia or Rural Folk |
| Moderate | Industrialists, Petite Bourgeoisie or Armed Forces |
| Regressive | Landowners, Devout or Rural Folk |
| None | Random |

The first group listed is the likeliest, at 35–40%; the other two are about 30% each.

### Aptitude traits

Aptitude traits rate a character's administration, diplomacy and command on five tiers each. A good administrator as ruler raises bureaucracy, tax capacity and legitimacy, a good diplomat prestige, influence and reputation, a good commander army morale, experience and Armed Forces support; a poor one does the reverse. Every tier except the middle one also changes the character's popularity. The middle tier has no effect.

| Tier | Administration | Diplomacy | Command |
|---|---|---|---|
| 1 | Terrible Administrator | Terrible Diplomat | Terrible Commander |
| 2 | Poor Administrator | Inept Diplomat | Inept Commander |
| 3 | Average Administrator | Average Diplomat | Adequate Commander |
| 4 | Skilled Administrator | Skilled Diplomat | Skilled Commander |
| 5 | Exceptional Administrator | Master Diplomat | Military Genius |

With Heir Education on, rulers and heirs have aptitude traits. Heirs in education earn theirs; any other ruler or heir without them is rolled once a year, usually at the middle tier or below, with generals and admirals rolled higher for command. A ruler who leaves power loses the traits and gets the same ones back on returning. The Universal Aptitude Traits rule gives them to every adult character instead, with or without Heir Education.

## 6.9 Custom religion creator

With the Custom Religion Allowed rule, a player country (never the AI) gets the Create New Religion journal entry. It walks through four screens with Next and Back buttons.

| Screen | You choose |
|---|---|
| 1 | Three doctrines, at most one per category (economy, society, governance, outlook), such as Divine Wealth, Imperial Cult or Pacifism. |
| 2 | Three traits for the religion's interest group, one each for the loyal, happy and unhappy approval levels (the group holds one trait per level, so picking one rules out the rest of that level). Only traits matching one of your doctrines can be picked. |
| 3 | A name, from 20 such as Neohellenism or Church of the Sacred Machine. |
| 4 | The religion's heritage: Christian, Muslim, Judaism, Eastern, Animist, Buddhist or Custom. |

Create New Religion founds it at once. It becomes your state religion, your ruler and heir convert, and a fifth of your capital's population and a tenth of every other incorporated state convert with them. The Devout take the religion's name, your doctrines as their ideologies and your chosen traits, and every current Devout character is exiled.

## 6.10 State Collapse

The State Collapse journal entry appears when your average standard of living falls below 5 and starts counting 52 weeks below 4. If the count runs out, the state fails:

- Every state loses its barracks, naval administrations, naval fortifications, naval logistics centers, military bases, ports, railways, universities and government administrations.
- Every institution and ministry is abolished.
- You get Failed State for ten years, fading as it goes: no resistance to foreign leverage, the governing interest groups lose half their attraction, and subsistence farms produce twice as much.

The count then starts again. A second collapse restarts Failed State's ten years; it does not add a second one. Raising the average standard of living to 4 or more stops the count, and it restarts from zero if living standards fall again. Decentralized countries are exempt.

## 6.11 New decrees

The mod adds 16 decrees, one of them only under the Demographics rule's Full setting. They cost authority like the base game's, 100 unless noted, scaled by your rank.

| Decree | Unlocked by | Effect in the state |
|---|---|---|
| War Propaganda | War Propaganda | +5 war support a month |
| Political Patronage | Mass Media | Pops' political strength doubled |
| Bureaucratic Reform | Intergovernmental Organizations | Government administrations +25% throughput |
| Greenest Grass Campaign | Freedom of Movement power bloc principle, tier IV or higher | Large migration pull |
| Cultural Emigration Initiative | Cultural Unity power bloc principle | Violence against minorities |
| Pollution Control | Pollution Control | −50% pollution |
| Cultural Integration | | +5 cultural acceptance a year |
| Natalism Initiative | Mass Media | +15% birth rate |
| Medical Outreach (Demographics, Full only) | Medical Degrees | New medicine reaches the state faster; see States and population |
| Tax Breaks (50) | Keynesian Economics | −50% taxes, fewer radicals, more migration |
| Trade Reform | Containerization | +25% trade advantage and capacity, lower tariffs |
| Anti-Terrorism Campaign | Terrorism and Antiterrorism | Fewer radicals from movements, less turmoil |
| Promote Tourism | | Tourism industry +50% throughput |
| Encourage Emigration | Social Media | −50% migration pull |
| Subsidize Immigration | Public Works Programs | +50% migration pull, welfare payments |
| Resettlement Recruitment Drive | A Settlement Authority | Resettlement draws settlers here first; see States and population |

War Propaganda works only while you are at war, and its +5 is multiplied by the state's share of your population: in a state holding a fifth of your people it adds 1 war support a month. Issue it in your most populous state.

## 6.12 After a revolution

When a revolution wins, the winner carries on as the same nation and keeps what the old government built. UN membership, seat and programs, the central bank's gold and monetary settings, the banking cycle's position, nuclear arsenals, space-race rewards and milestone progress, the arable-land bonuses from farming breakthroughs, and under the Legislated Tax Code rule the tax code with its passed bills and promises (under the Light Tax Code, its rates), all carry over. So do UN sanctions on the nation, which return within a month. A Civil Rights Movement, Human Augmentation Debate, Morphological Freedom Debate or Mental Health Crisis journal entry you already finished does not start over. Some things are still lost: policies switched on from journal-entry buttons, such as banking tools and climate policies, have to be switched on again, and history charts start empty. The colonial empire is the exception: the winner keeps the colonial programs you were running while the Colonial Empire entry is open, Colonial Empire Solidified and the path rewards of a finished empire, and the Imperial Federation and Mandate System modifiers (see Colonial empires and decolonization).

CHAPTER 7

# Social movements

The later eras bring social questions your government must answer: equal rights for discriminated minorities, human augmentation, the freedom to change how your body looks, digital privacy, mental health and life after work. Each has a journal entry that opens with a technology and ends when you settle the question, mostly by passing a law, or when time runs out. Other movements arrive as events without a journal entry, and religious revival events push back against secularization. The Social Movements game rule (on by default) controls the six journal entries and their events; the movements carried by events run either way.

## 7.1 Movement journal entries at a glance

All six sit in the journal's Domestic Affairs group, except the Post-Scarcity Transition (Development). Decentralized countries never get them. The laws are described in The mod's new laws.

| Journal entry | Appears with | Succeeds when | Fails when | Time limit |
|---|---|---|---|---|
| Civil Rights Movement | Civil Rights Movement (era 7), while an incorporated state holds pops below Provisional Acceptance | Movement Support reaches 100, or you enact Affirmative Action, or Universal Citizenship together with Protection | Movement Support falls to 0, or the Civil Rights political movement is gone | None |
| Human Augmentation Debate | Human Augmentation or Brain-Computer Interfaces (era 11) | Regulated Augmentation Market, Unrestricted Augmentation or Mandatory Augmentation | Human Purity | 20 years |
| Morphological Freedom Debate | Morphological Freedom (era 12) | Morphological Freedom or Clade Citizenship | Natural Form still in force when time runs out | 20 years |
| Digital Rights & Surveillance | Automated Surveillance or Cybersecurity (era 9) | Strong Privacy Rights | Intrusive Surveillance System and Ministry of Intelligence and Security Established, both at once | 15 years |
| Mental Health Crisis | Mental Health Awareness (era 10) | Rehabilitation-Focused Criminal Justice and the Social Security institution at level 4 or higher | Punishment-Focused Criminal Justice or Penal Labor Camps still in force 10 years after the entry opens | 20 years |
| Post-Scarcity Transition | Post-Work Society (era 12) | Post-Scarcity Economy law | Never | 30 years |

## 7.2 The Civil Rights Movement journal entry

The Civil Rights Movement has no timer. A support bar rises or falls, you set your stance with policy buttons, and how long you held each stance decides which of ten endings you get.

### When the civil rights struggle begins

The entry appears once you have researched Civil Rights Movement and an incorporated state holds pops with acceptance below Provisional Acceptance (under 60: Open Prejudice or worse). It becomes active when the Civil Rights political movement, which the technology brings, has formed. If every such pop rises to Provisional Acceptance, the entry closes with no outcome and can return with the grievance. Once it succeeds or fails, it is over for the campaign. A revolution's winner carries the struggle on, keeping the bar, the policies and any reward (see After a revolution).

### The Movement Support bar

Movement Support starts at 30. At 100 the movement wins; at 0 it collapses. The bar drifts down on its own, but an ignored movement hovers around 5: only a crackdown or a Violent Hostility or Ghettoization law can drive it to 0. It rises with Social Justice Movements (era 9), with Protection or Affirmative Action in force, with the number of incorporated states holding poorly accepted pops, with widespread radicals and with the pro-movement policies below. With Social Justice Movements, Protection and all three pro-movement policies, support gains about a point a month, so the struggle takes some six years; without the technology, a decade or more. Enacting Affirmative Action, or Universal Citizenship with Protection, skips the bar and completes the entry at once.

As support climbs, the bar's phase adds turmoil penalties and radicals from political movements, and from 90 it also raises your chance of enacting laws. The bar's tooltip lists every line. The full list of what moves the bar, with the phase effects, is in Appendix: social movement details.

### Civil rights policy buttons

The entry has six policies, each switched on and off with a pair of buttons. The three pro-movement policies stack, and so do the two crackdowns, but the two groups exclude each other. Cooptation is limited only by its law requirement.

| Policy | Requires | What it does |
|---|---|---|
| Grassroots Organizing | No crackdown | Raises support; helps enact laws but brings more radicals |
| Federal Protection | Indifference, Protection or Affirmative Action; Minor Power rank or above; no crackdown | Raises support the most of the three; raises acceptance, costs bureaucracy and authority |
| Gradualist Accommodation | No crackdown | Raises support a little; eases turmoil, raises acceptance slightly |
| Cooptation / Token Reform | Discrimination, Indifference or Cultural Assimilation | Raises support for 12 months only; eases turmoil, costs legitimacy through ideological incoherence |
| Suppression / Crackdown | No pro-movement policy | Lowers support; adds authority and turmoil |
| Segregationist Consolidation | A discriminatory minority law; no pro-movement policy | Lowers support fast; adds authority, but lowers acceptance and brings more radicals |

The discriminatory minority laws are Violent Hostility, Ghettoization, Cultural Assimilation and Discrimination. Cooptation stops pushing the bar after 12 months in total for the whole struggle; switching it off and on doesn't restart the count, and its other effects stay. Each button's tooltip lists the modifier it applies, its monthly push on the bar and the ending its months count toward; the figures are also in Civil rights policy effects.

## Civil rights events

Four milestone events fire once each, as support first reaches 25, 50, 75 and 90; your minority law and your policies decide which version you get. A random event also comes about one month in seven. Sympathetic options please the Intelligentsia, anger the Devout, Rural Folk and Armed Forces, and make discriminated pops loyal; repressive options do the reverse. The events are listed in Civil rights event list.

While you lack Protection, Affirmative Action and Universal Citizenship (or keep Active Persecution or Legal Guardianship), a great power that has one of those three and has researched Civil Rights Movement may be asked whether to condemn you. Only a great power with a stake in you is asked: one strategically adjacent to you, a rival (either way), a member of your power bloc, or your overlord or subject. If it condemns you, you get International Pressure on Human Rights: reject it (−30 relations with the critic), promise reform (+20 relations, a small acceptance bonus) or counter-accuse (−50 relations and a diplomatic incident). As such a great power, you get the same question (A Stain on Their Record) about countries you have a stake in, at most once every two years.

## How the civil rights struggle ends

The ending depends on how the struggle finished and which policy ran longest; a policy must run more than 18 months to count. Results decay over ten years.

| Ending | When | Result |
|---|---|---|
| Victory on Paper | Support reached 100 while a discriminatory minority law is in force | Radicals, and either Civil Rights Movement Betrayed (−15% prestige, more turmoil and radicals) or Civil Disobedience Crackdown (+50 Authority, +15% turmoil penalties) |
| The Civil Rights Mandate is Signed | Federal Protection ran longest | +20% prestige, +20 acceptance, −5% Authority |
| The Movement Carries the Day | Grassroots Organizing ran longest | +15% prestige, +15 acceptance, +10% political movement attraction |
| A Settlement, At Last | Gradualist Accommodation ran longest | +10% prestige, +10 acceptance, −5% turmoil penalties |
| Reform on Paper | Cooptation, if no pro-movement policy passed 18 months | +5% prestige, +8 acceptance, more legitimacy penalty from incoherence |
| A New Era of Equality | No pro-movement or cooptation policy passed 18 months | +15% prestige, +15 acceptance, +10 acceptance for cultures sharing no heritage trait with yours |
| The Movement Broken (failure) | You used the crackdowns, or a discriminatory minority law is in force | A choice set by your minority law, from Order Restored to Minority Expulsion Program; all raise Authority and please the Devout, Rural Folk and Armed Forces |
| The Movement Demobilizes (failure) | Cooptation ran over 18 months | Radicals among Academics |
| The Long Drift (failure) | Gradualist Accommodation ran over 18 months | Radicals among lower-strata pops |
| The Movement Fades (failure) | Otherwise | Radicals among lower-strata pops |

Ties between pro-movement policies go to Federal Protection, then Grassroots. The entry's status lists the months each policy has run and names the ending the movement would reach if it won or collapsed now.

## How the AI handles civil rights

AI countries pick policies mostly by their minority law: Grassroots Organizing under Indifference or better, Federal Protection under Protection or Affirmative Action, Cooptation under Discrimination, the crackdowns under Violent Hostility or Ghettoization, and Gradualist Accommodation anywhere. The Intelligentsia in

government also draw them to Grassroots Organizing, and the Landowners to Suppression / Crackdown. They never switch a policy off.

## 7.3 Debates settled by law

The other five journal entries share a simpler shape. Each applies a modifier while it runs and has up to a one-in-five chance each month (one in nine for Digital Rights) of one of its five events (six for Morphological Freedom), whose options run from paying for reform to a hard-line response. It ends when you enact the settling law, or times out with a penalty. Outcome modifiers decay over ten years.

| Journal entry | While active | Success | Failure | Time-out |
|---|---|---|---|---|
| Human Augmentation Debate | +5% innovation, +5% turmoil penalties | +10% innovation, +5% prestige | −10% innovation and −2% prestige, or +15% turmoil penalties and −5% prestige (your choice) | +15% turmoil penalties |
| Morphological Freedom Debate | +5% Cultural Pull from Art, +5% turmoil penalties | +10% Cultural Pull from Art, +5% prestige | −10% Cultural Pull from Art and −5% migration pull, or +10% turmoil penalties and −2% prestige (your choice) | +10% turmoil penalties, under Licensed Modification |
| Digital Rights & Surveillance | +50 Authority, +5% turmoil penalties | +5% prestige, +5% innovation | +150 Authority, −5% innovation, −5% prestige; or radicals | −25 prestige, +5% turmoil penalties |
| Mental Health Crisis | +3% mortality | −3% mortality, +5% prestige | +5% mortality, −5% prestige; or radicals | +5% mortality, −25 prestige |
| Post-Scarcity Transition | +10% turmoil penalties | +15% prestige, +10% innovation | Can't fail | +15% turmoil penalties, −50 prestige |

### The augmentation debate

Every country starts on No Augmentation, and staying on it or Medical Augmentation Only runs the clock out. Regulated Augmentation Market needs Human Augmentation and the Ministry of Consumer Protection Established law, Unrestricted Augmentation needs Brain-Computer Interfaces, and Mandatory Augmentation needs Bioenhanced Soldiers and the Ministry of War Established law. The events cover a society divided by augmentation, black-market implants, workplace discrimination, augmented crime and religious condemnation.

### The morphological freedom debate

Every country starts on Natural Form, and the Morphological Freedom technology (era 12) unlocks Licensed Modification and Morphological Freedom. Enacting Morphological Freedom settles the debate, as does Clade Citizenship if Self-Directed Evolution (era 13) arrives while it runs. Licensed Modification is a step on the way: held to the end, it leaves the question unsettled. If Natural Form is still in force when the twenty years run out, the debate is lost, and an event asks whether to enforce the law against people already changed or look away. The Devout and the Rural Folk oppose both freer laws and the Intelligentsia support them; the laws themselves are in Rights and society laws. The events bring the first clinics, a television host's transformation, sport and eligibility, a teenager's case in court, a backlash from the pulpits and a fashion that sweeps the young, no two of them within two years of each other.

### The digital rights debate

The entry opens in era 9, but Strong Privacy Rights needs Universal Digital Identity from era 11 and the Guaranteed Liberties law, and you have 15 years. The events bring a whistleblower, a data breach, hackers, predictive policing and foreign cyber-espionage. The first three are leaks, and no two of them come within two years of each other. With Covert Warfare on, the last follows only when your counterintelligence exposes a foreign espionage operation (see What the target learns).

### The mental health debate

Rehabilitation-Focused Criminal Justice unlocks with the technology that opens the crisis, so raising Social Security to level 4 is usually the slow half. You have ten years to move off Punishment-Focused Criminal Justice, the law every country starts with. If it is still in force when the decade ends, the crisis fails, and Penal Labor Camps fails it the same way. Reforming in time leaves the other ten years to raise Social Security. The events deal with burnout, youth mental health, addiction, veterans' PTSD and a care-home scandal.

### The post-scarcity debate

This entry can't fail: it succeeds or times out. It opens with Post-Work Society in era 12, while the Post-Scarcity Economy law needs the era 13 technology of that name, which requires Post-Work Society: the 30 years are a race into the last era (see The extended timeline). The events bring unemployment protests, a crisis of meaning, AI replacing bureaucrats, neo-Luddite terrorism and an art renaissance.

## 7.4 Movements carried by events

Second-wave feminism, LGBTQ+ rights, secularization and environmentalism have no journal entry. They come as random events once you have the technology, and the Social Movements rule doesn't stop them.

| Movement | Needs | Events |
|---|---|---|
| Second-wave feminism | Second Wave Feminism (era 7) | The Second Shift and Half the Salary, once each, over women's work and equal pay |
| LGBTQ+ rights | LGBTQ+ Rights Movement (era 9) | Pride and Protest, at most once in ten years until you enact Full Equality and Protection; The Marriage Equality Question, once, under Basic Protections or Anti-Discrimination Laws |
| Secularization | One of several technologies, such as Sexual Revolution or Social Media, or the Total Separation law | Religious Revival Sweeps the Nation and Faith Against Modernity, each at most once in ten years and never under State Atheism |
| Environmentalism | The Global Warming journal entry | Climate events as world temperature crosses thresholds, under the Global Warming rule; see Climate and pollution |

Their options are in Details of the movements carried by events.

The Anti-War and Transhumanist political movements also bring events of their own, described with the movements in New political movements.

## 7.5 Religious revival events

Seven religious revival events push back against the Devout interest group's decline. Each is tied to a technology of eras 7 to 9 and fires at most once per campaign, at least five years apart. They need a Devout group that is weak but present (clout above 2% and below 10%, or 15% for three of them), a church law other than State Religion or its Millet System and People of the Book variants, and neither Autocracy nor Oligarchy. State Atheism blocks all but two of them.

Each event offers three choices. Embracing the revival grants a modifier for 20 years that raises Devout political strength and attraction, some with a side effect such as lost Authority or a little standard of living. A moderate choice gives a weaker modifier that fades over 20 years, and a secular choice radicalizes Devout pops. The embrace modifiers stack, so taking several can make the Devout a leading interest group again. AI countries embrace more readily with a powerful Devout group or a Ministry of Religion. The seven events and their modifiers are in Religious revival event list.

Once you have Decline of Organized Religion and any embrace modifier, the Secularization Campaign decision appears. It removes all seven embrace modifiers at once, radicalizes Devout pops, and leaves the moderate modifiers in place.

## 7.6 What the Social Movements rule turns off

With Social Movements Disabled, none of the five journal entries opens, so their buttons, events and endings never happen. The technologies and laws stay, the Civil Rights political movement still forms with its technology, and the feminism, LGBTQ+, faith, religious revival and political-movement events still fire. Environmental events follow the Global Warming rule.

CHAPTER 8

# States and population

The mod changes how people spread across your country and how its states develop. Crowded states draw fewer migrants, homelands follow the cultures that actually live in a region, a state's tourism depends on what it offers visitors, a government can move settlers onto an empty frontier, and a yearly census counts your people by age and sex. The state panel gains tiles that show most of these. Everything in this chapter is always on except internal resettlement, which the Internal Resettlement game rule controls (see the Introduction), and the census, which is experimental and has a game rule of its own (see Demographics).

## 8.1 State panel additions

The mod adds tiles to the status grid of the state panel, and above the grid a tourism card and, in a state with a Grand Monument, a Grand Monument card. Each tile is titled with a game concept you can hover for an explanation, and its tooltip breaks the numbers down.

| Tile | Shown | What it tells you |
|---|---|---|
| Homeland Dynamics | Where a culture could gain or lose a homeland, even while changes are paused | Which cultures are gaining or losing homeland status here, with a progress bar for each, or why progress is paused. |
| Arable Land | Always | Total arable land, how much technologies and other regional effects add to the geographic base, and the arable-land multiplier. |
| Migration Crowding | While the state is over its crowding threshold | Population, crowding threshold, crowding ratio with a bar, and the migration pull penalty. |
| Local Climate | Always | The climate where most of the region's people live: its hottest days, summer humidity, winter temperature, water and malaria, and Comfort, what the climate costs or adds to the state's migration pull. |
| Wealth Concentration | While the state carries a modifier from its wealth concentration or the inheritance law | The state's score heading for its target, and the modifiers it carries from them (Concentrated or Dispersed Property, Land Hunger, and the inheritance law's farming effect), which the state's modifier icons leave out; see Who inherits. |
| Solar Collector | In a state with a Solar Collector or a Solar Receiver | Your country's receiver slots available and generated (3 per staffed level); a receiver's build tooltip also gives the free slots. A megaproject; see The extended timeline. |
| Antimatter Facility | In a state with the Facility, an engine or a warhead plant | Your country's engine and warhead-plant slots available and generated (5 per staffed level). |
| Seabed Licenses | On a coastal state, once your country has had its first share of seabed licenses | The state's ocean basin, the licenses its Deep-Sea Mine levels hold and how many are long-haul, your unused claims in that basin and the claim weight the state adds there; see Seabed licenses. |
| Ocean Basin | On a coastal state without the Seabed Licenses tile (another country's, or one that never shared licenses), once | The ocean basin the state faces, whose license pool its mines and ports would count in. |

*Figure 9: A state panel with the tourism card above the status grid and the Homeland Dynamics, Arable Land and Migration Crowding tiles at the bottom.*

| Tile | Shown | What it tells you |
|---|---|---|
| | you have researched Abyssal Plain Mining | |
| UN Mission | While a UN mission works in the state | The mission's type, months, strength and progress; see The United Nations. |
| Tourism card | Always | Tourism output and throughput from each source, with a bar toward each source's cap. |
| Grand Monument card | While a Grand Monument stands in the state, yours or another country's | The monument's status, name and dedication, its grandeur with a bar toward its next step, its levels and age once finished, and what it gives the state: tourism output and its dedication's local effect. A contested monument also names its Old Supporters and the group it is Resented By. On your own state, while six monuments stand there, the least grand has Clear It Away, and the arrow at its corner opens the Grand |

| Tile | Shown | What it tells you |
|---|---|---|
| | | Monuments tab of the Timeline Extended window; see The extended timeline. |
| Grand Monument (Under Construction) card | While a Grand Monument is being built in the state | The levels built, the grandeur they come to, what a level finished now adds, whether levels are still being built, and on your own state the Dedicate button. |

In every version of the mod, the birth, death and workforce effects of laws, technologies and institutions appear in the birth rate, mortality and workforce tooltips of a state and its pops as Laws and Institutions lines, one for each kind of effect rather than one for each law. The workforce line also holds the workforce effects of amendments, interest groups, Woman Soldiers and pollution. Each law's or technology's own tooltip still lists its effect.

## 8.2 Migration crowding

Every state has a crowding threshold of 10,000 people for each unit of its geographic arable land, the land the state starts with before technologies add to it, plus 1,000 people for each point of Urban Engineering. Once the population passes the threshold, the state carries a Population Pressure modifier that cuts its migration pull. The modifier is recalculated once a year.

Urban Engineering takes the place of the base game's Construction Sector Max Level, which the mod's construction has no use for, on the same technologies: Urbanization gives 10 points, and Urban Planning, Modern Sewerage, Steel-Frame Buildings and Elevators 5 each. With all five, every state houses 30,000 more people before crowding. That barely moves a large farming province, but it doubles the threshold of a state with three units of arable land, so a small state can still hold a large city.

Public Housing, a government building unlocked by Urban Planning, adds Urban Engineering to its own state, for each staffed level:

| Method | Needs | Urban Engineering per level |
|---|---|---|
| Model Dwellings | | +100 |
| Sanitary Tenements | Modern Sewerage | +150 |
| Steel-Frame Apartments | Steel-Frame Buildings | +200 |
| High-Rise Apartments | Elevators | +300 |
| Prefabricated Tower Blocks | Prefabricated Construction | +450 |
| Arcologies | Advanced Structural Engineering | +700 |

Technology caps how many levels a state can hold: 12 with Urban Planning, then +3 with Paved Roads, +5 with Modern Skyscrapers, +10 with Computer-Aided Design and +5 each with Advanced Nanofabrication and Programmable Matter, for 40 in all. On the newest method you can build, a state at the cap houses 1.2 million people with Model Dwellings, 3.6 million with High-Rise Apartments, 13.5 million with Prefabricated Tower Blocks and 21 to 28 million with Arcologies, before tolerance. With the tolerance your technologies give, that is between half and nearly all of the world's largest rich city of the time.

Its upkeep comes out of your budget, so build it where crowding bites: in a state below its threshold it does nothing.

The penalty starts gently and then climbs fast. Up to ten times the threshold it grows with the square of the excess; beyond that it adds 10 percentage points for each further multiple.

| Population (× threshold) | Migration pull penalty |
|---|---|
| 1× or less | none |
| 4× | 5% |
| 7× | 20% |
| 10× | 45% |
| 15× | 95% |
| 20× | 145% |

### Raising crowding tolerance

The Migration Crowding Tolerance modifier multiplies the threshold, Urban Engineering's share included: +50% tolerance lets a state hold half as many people again before the penalty starts. The tile's tooltip breaks the threshold into arable land, Urban Engineering and tolerance; hover the Urban Engineering and tolerance figures for their sources.

| Source | Tolerance |
|---|---|
| Ministry of Urban Planning (institution) | +10% per level |
| Highway on the Civil Highway, Electric Civil Highway or Autonomous Highway method | +10% per level, scaled by staffing |
| Modern Skyscrapers, Wireless Internet (Wi-Fi), Autonomous Vehicles, Post-Scarcity Economy | +15% each |
| Modern Urban Planning, Advanced Structural Engineering | +20% each |
| Urban Planning power bloc principle | +25% at tiers I to III, +50% at IV, +100% at V, plus +5% per Ministry of Urban Planning level (+10% at IV, +20% at V) |
| National Park building on National Park, National Forest or National Wildlife Refuge | −5%, −7.5% or −10% per level, scaled by staffing |
| Metropolitan Crowding, under the Demographics rule's Full setting (see The urban pattern and a planned capital) | Up to −10%: it starts when more than 40% of the state's urban people live in its largest city and is full at 60% |
| Green Belt or Urban Sprawl, from the Overspill event (see Demographic events) | −10% or +20%, for twenty years |

Ministries are explained in Ministries. The same tolerance multiplies the densities that open and close a resettlement frontier (see The Settlement Authority).

## 8.3 Local climate

Every state region has a climate record, taken where most of its people live: the hottest day of a typical year and how hard such days are to bear, the warmest month's humidity, the coldest month's temperature, how much water there is, and how common malaria was in 1836. The Local Climate tile shows it, and every part of a split region shares it.

Migrants weigh the climate, so a harsh one cuts a state's migration pull, and through it the pull that draws mass migrations. Five things count:

| Part | When |
|---|---|
| Heat | The hottest days are hot and humid enough to strain the body; each degree costs more than the last |
| Winter | Measured against a winter like London's: a harsher one costs, a milder one adds a little unless the summer is hot |
| Frost | The coldest month averages below freezing, on top of Winter |

| Part | When |
|---|---|
| Dry land | Water is limited (semi-arid country) or scarce (desert) |
| Malaria | It was seasonal or endemic in 1836 |

Heat counts a region's worst days rather than its average summer, so a steady warm island such as Hawaii costs little and a city whose summers spike, such as Paris, a little more. Humid heat counts for more than dry, since damp air stops sweat cooling the body.

The tile's Comfort row gives the total as a percentage of the state's migration pull, worked out again once a year, and hovering a reading shows what it costs before and after technology. In 1836 a hot, humid, malarial state such as Georgia loses about 27%, Bengal about 50% and Yakutsk about 64%; London loses nothing and Paris 4%. A mild winter with little else against it adds pull, at most about 7%: Madeira gains 7%, Hawaii 5% and Cape Town 5%. A hot summer cancels that, so Bengal's or Singapore's warm winter adds nothing. Climate never costs more than 80%.

Technologies ease each cost, for the country that researches them:

- Frost eases early, with steam heating and concrete.
- Winter's pull, harsh against mild, shrinks more slowly with heating and insulation, and goes only with the last eras' technologies.
- Malaria goes with Quinine and then Malaria Prevention.
- Water follows mechanized farming, welding and chemistry.
- Heat waits for the 20th century's appliances and air conditioning.

By the last era none of the five costs or adds anything. On the way, air conditioning eases the South's heat sooner than heating closes the gap between winters, so once central air arrives Georgia costs less than Massachusetts.

The record is today's climate. With global warming at 0 the world runs about a degree colder than that, and winters more toward the poles. As it warms, every state's hot days and winters rise, winters most near the poles: hot states lose more to heat, and cold ones less to cold.

A mass migration's target needs no arable land, so a city-state or a small partial state can draw one. Deserts are kept back by their climate instead, until the technologies catch up.

## 8.4 Dynamic homelands

With the mod, homelands follow population. A primary culture of yours that makes up enough of a state's people can gain a homeland there, and a culture that is not primary and has dwindled can lose one. Each change fills its own progress bar over years, with no random rolls. A completed change applies to the whole state region, and you get a notification.

### Enabling homeland changes

Nothing changes until your country enables homeland changes with one of these:

- the Ancestral Citizenship law (Citizenship), Violent Hostility (Minority Rights) or Linguistic Purity (Language Policy);
- tier II or higher of the Cultural Unity principle in your power bloc;
- the Mass Media technology.

You must also own the whole state region. Losing either condition pauses every project in the state, and the Homeland Dynamics tile says why. Progress resumes where it stopped.

### Homeland thresholds

A primary culture without a homeland in the state can start forming one once it makes up at least the creation threshold of the state's population, 60% by default. Laws and principles move the creation threshold, but never below 10% or above 90%.

A culture with a homeland in the state that is not one of your primary cultures starts losing it once its share falls below the removal threshold. The default removal threshold is 0%, so no homeland can be removed until a law or principle raises it.

A culture's progress resets to zero if the state changes owner or the culture stops qualifying; the tile marks such a project before it is cleared.

### Speed of homeland change

A project gains 10 percentage points a year by default, so a homeland forms or disappears in ten years. The Homeland Change Speed modifier scales that rate. Turmoil of 25% or more multiplies it by 1.25, and 50% or more by 1.5. Legitimacy below 25 halves it, and legitimacy of 75 or more multiplies it by 1.25. The rate stays between 1 and 95 points a year. The Promote National Values decree adds +300% speed in its state, which on its own brings a change down to 30 months. In the states of a party to the UN Convention on Cultural Diversity, a homeland is lost 30% more slowly at the Established tier, scaled by the UN's enforcement (see UN conventions and agencies); forming one is not affected.

Laws and principles that move the three numbers:

| Law or source | Creation threshold | Removal threshold | Change speed |
|---|---|---|---|
| Ancestral Citizenship | −20% | +10% | +50% |
| Violent Hostility | −20% | +10% | +25% |
| Ghettoization | −15% | none | none |
| Linguistic Purity | −10% | +10% | +50% |
| Cultural Assimilation | −10% | +5% | +25% |
| Racialized Citizenship | −10% | +5% | +25% |
| Protection | +10% | −5% | −15% |
| Affirmative Action | +20% | −10% | −25% |
| Universal Citizenship | +20% | −5% | −25% |
| Cultural Unity principle, tier II / III / IV / V | −5 / −15 / −25 / −35% | +5 / +10 / +20 / +40% | +5 / +10 / +20 / +30% |

Other Citizenship, Minority Rights and Language Policy laws, a few technologies and the language-reform amendments move them too; the tile's tooltip lists every source. Under Ancestral Citizenship alone, a primary culture needs 40% of a state to start a homeland, a minority's homeland goes once it falls below 10%, and each change takes under seven years.

## 8.5 Cultural acceptance and minorities

When a new cultural community appears in a state, the base game gives it a temporary local acceptance penalty that wears off over time. The mod makes the penalty harsher and much slower to fade: a new community starts 40 points of acceptance down instead of 30 and recovers 1 point a year instead of 3, so the penalty lasts about 40 years rather than 10. The mod also adds two state modifiers that act on the penalty:

- Annual Cultural Acceptance closes the penalty faster, adding its value in acceptance over a year to every culture that still carries one. The Cultural Integration decree gives +5, the Ministry of Refugee Affairs +0.2 per level, and tiers III to V of the Cultural Plurality principle +0.25 to +0.5.
- Minimum Local Acceptance limits how deep the penalty can go: no deeper than −40 plus the modifier's value. Indifference gives +5, Protection +10 and Affirmative Action +20 (all Minority Rights laws), and Cultural Plurality tiers III to V give +5 to +10. Without either modifier there is no floor.

These matter most where migration keeps bringing new communities into a state.

The Violent Hostility law and the Cultural Emigration Initiative decree give a state Violence Against Minorities. Each year, every culture there with acceptance below 60 risks an outbreak, more likely the lower its acceptance. A severe outbreak kills 10% of that culture in the state, radicalizes the survivors, devastates the

region and sets off a mass migration; a lesser one kills 2% and does the same on a smaller scale. Minority laws are in Rights and society laws.

## 8.6 State tourism

Tourism is a luxury good in the mod, made by the Tourism Industry building (unlocked by Romanticism, see The extended timeline) and bought by wealthy pops. Two state modifiers, refreshed monthly and shown on the tourism card, set how much a Tourism Industry makes: Tourism Output and Tourism Throughput.

### Tourism output

Three sources on the card set a state's output bonus:

| Source | Bonus |
|---|---|
| Base Appeal | A score for each state region, from its scenery, climate and townscape. It runs up to +100%; Wyoming scores 90%, Nebraska 20%, and any region without its own score gets 20%. A region's historical monuments count under Monuments and in its Grand Monuments instead, so their share goes with them: Attica's Acropolis takes 40 of its 100, and Base Appeal reads +60% while the Parthenon gives it back; Lower Egypt's pyramids take 50 and the Pyramids of Giza give it back. Ruins still buried, such as Teotihuacan, take theirs until they are dug up. The tooltip says how much has moved. |
| Cities | +100% for the largest city in the world, 5% less for each place below it, nothing below 20th. See World city rankings. |
| Monuments | +10% for each monument. The wonders that stood in 1836 give more: the Taj Mahal +40%, Angkor Wat +30%, and Hagia Sophia, the Vatican, the Forbidden City, Saint Basil's Cathedral and the White House +20%; the Great Mosque of Djenné gives +20% once rebuilt in Eastern Mali. |

The card's "All modifiers" line adds everything else that changes output. A Grand Monument adds +10% for each step of its grandeur (see Grand monuments). An Airport or Spaceport adds +15% per level, scaled by staffing. A Highway adds +5% on the Civil, Electric Civil and Autonomous Highway methods and +2.5% on the Industrial and Electric Industrial ones. Laws add between −30% (Secret Police, Outlawed Dissent) and +15% (Universal Citizenship). Pollution, turmoil and obstinance cut output in proportion to how bad they are, at their worst by 200%, 100% and 50%, so heavy pollution can shut a state's tourism off entirely.

### Tourism throughput

Four sources set a state's throughput bonus, and each capped one shows a bar toward its cap:

| Source | Bonus | Cap |
|---|---|---|
| Excess Infrastructure | The infrastructure the state produces less what its buildings use: +25% at 10,000, +50% at 30,000, +75% at 90,000. Each point adds less than the one before. | +100%, never quite reached |
| Art | +1% per Creative Industry level up to 20, +0.25% up to 100, +0.05% up to 500, +0.01% up to 1,500 | +70% |
| Parks | +25% with a National Park, rising with Base Appeal to +37.5% | +37.5% |
| Skyscraper | +10% | none |

Railways and Highways raise Excess Infrastructure by supplying infrastructure. Each building you add uses some, so a crowded industrial state has little to spare for visitors.

The "All modifiers" line for throughput also carries the Promote Tourism decree (+50%) and bonuses from events.

## 8.7 World city rankings

Each month the game ranks every state in the world by city size: the services it produces (only with an Urban Center), weighted up by its average standard of living, ×1.5 for a capital and ×1.5 for a great power's states,

halved for an unrecognized country. The tourism card shows the rank. The top 20 get the Cities bonus to tourism output, from +100% for first to +5% for twentieth; nothing else uses the ranking.

## 8.8 Demographics

The census is experimental. The Demographics game rule controls it, set to Full by default: choose Display only or Disabled when you start a new game. A save from an earlier version keeps the setting it was started with: off, unless you chose otherwise. With it off, some laws, technologies and decrees still list lines only the census reads, such as Contraception or Health Care Access; their names end in (Demographics off), since they do nothing. The rest of this section describes the census with the rule on.

With it on, every state keeps a census of its people by single year of age and by sex. It is taken when the game starts and again once a year, and it opens a Demographics tab in the Population panel, beside the base game's four, and a Demographics subtab on the state panel's Population section. The tab is open from the first day.

Under the Demographics rule's Full setting, the census sets each state's births and deaths from its age structure, children per woman and causes of death, described below. From its first step in a state, the Laws and Institutions lines (see State panel additions) go. The effects the census models become part of its own rates: those of the child labor and women's rights laws, the health laws, the Contraceptive Pill, State-Sponsored Family Planning, vaccines and antibiotics, mild starvation, Primogeniture's larger farm families, which late marriage replaces, and Forced Heirship's smaller ones, which Family Limitation replaces. The late medical technologies' cuts to deaths become part of its rates too (see Life extension). The rest, such as Communal Child-Rearing, the State Eugenics Program, the Sexual Revolution and the family-policy laws, it applies at its own rate, as it does an effect on the whole state, such as pollution, devastation or a decree. Either way, a new law or technology takes effect at the state's next census step, up to a year later. An effect on some of a state's people, such as severe starvation, applies to their rate in the base game. Private Health Insurance's cut for the wealthy works at the census's rate, and no pop's deaths fall below half of it. A state's birth rate and mortality tooltips show the census's part as Census: Births and Census: Deaths. The Birth Rate and Mortality lines it takes over, on most laws and technologies, end in (Demographics).

Under Full the census also sets each state's workforce, from who is of working age and how many of them work. The working ages run from 15 (children of 10 to 14 work at a share each child labor law lists among its modifiers) to the pension age, or with no pension law for as long as people's health allows. Men of working age mostly work; how many women work depends on the women's rights laws, the technologies that free women's time and the state's lower strata, whose women work under any law. This replaces the laws' and technologies' own flat workforce lines: in 1836 a state works about as in the base game, but a rich modern country has a sixth to a quarter fewer workers than the flat lines would give it. Each yearly count sets a state's workforce to the census's figure, shown as Census: Workforce in its workforce tooltip, and the game's own count moves toward it over the years that follow. Under Full, the laws' and technologies' Workforce Ratio lines say what the census does with them: the ones it counts toward how many women work read Women's Work, and the rest read Workforce Ratio (replaced by the census).

With Old Age Pension, you choose the pension age with five buttons on the Demographics tab. A setting holds for five years once chosen, and the AI picks its own with the same considerations: a heavy bill or debt toward Raised or Linked to Average, high unemployment toward Early.

| Setting | Pension age | What you gain and what you pay | Interest groups |
|---|---|---|---|
| Early Retirement | 60 | More of the old retire and jobs open for the young; the largest bill and the smallest workforce. | Trade Unions +2, Industrialists −1 |
| Standard | 65 | The default. It grows more generous as lives lengthen, since more people live to draw it, and for longer. | None |

| Setting | Pension age | What you gain and what you pay | Interest groups |
|---|---|---|---|
| Raised Pension Age | 70 | A smaller bill and more old people at work. | Trade Unions −2, Rural Folk −1, Industrialists +1 |
| Linked to Average Longevity | Five years below where old age begins for your people, at least 65 | Follows lengthening lives, so the bill stays in check, a year at a time. | Trade Unions −1, Rural Folk −1, Industrialists +1 |
| Linked to Common Longevity | Five years below where old age begins for those who still age, at least 65 | The same until longevity treatment exists; after that it keeps the pension within reach of the untreated, at a larger bill. | Trade Unions +1, Rural Folk +1, Industrialists −1 |

The pension age changes at once, but a linked setting moves at most a year a year; the bill and the workforce follow at the next census. Under Full, Old Age Pension costs money for its pensioners: each year's census sets a weekly expense worth 6% of your output per person for each pensioner, per level of Social Security: Census: Pension Bill, which the Budget panel's Breakdown tab lists under Old-Age Pensions. A rich country at Standard with a fifth of its people pensioned pays about a fifth of its revenue. Universal Basic Income and the Post-Scarcity Economy pay everyone already and have no pension bill. An older people also needs more care: the Ministry of Health costs more bureaucracy as the census's health need rises above 1836's, and 5% more for each of seven chronic-care technologies (Modern Pharmaceuticals, Telemedicine, Personalized Medicine, Genetic Engineering, Rejuvenation Therapies, Molecular Assemblers, Biological Immortality): about two fifths more for a rich country today. Outside Full none of this changes anything, and the buttons are grayed.

The census also counts each state's men of fighting age, 20 to 39, its young adults and its men per 100 women aged 20 to 59, and takes war dead from the men of fighting age. Under Full each count sets a state modifier, refreshed at the state's yearly census:

- **Census: Fighting-Age Men** scales the state's conscription by its share of men of fighting age against 1836's: a state short of young men raises fewer conscripts, and one rich in them proportionally more, with no upper limit. An aging country loses conscripts as its people grow older. This one stays out of the state's modifier list; the conscription tooltip names it and the Fighting-Age Men row on the Demographics tab shows the figure.
- **Census: Youth Bulge** appears where young adults aged 20 to 29 are more than 18% of the people while work is short. At full strength (24% young adults and a fifth of the workforce without work) radicals from movements rise by 25% and the effects of turmoil by 15%. Past that each factor keeps adding at half the pace, up to twice the full size.
- **Census: Fewer Men** appears below 90 men per 100 women, as after a war with many dead. Fewer women marry, so children per woman fall with the balance (by at most a fifth), more women work, and support for women's rights movements rises, by 25% at 80. **Census: More Men** appears from 110 men per 100 women and raises radicals from movements and the effects of turmoil, by 15% each at 130. Past those marks both keep growing at half the pace, up to twice their size at 60 or 170.

Under Full the state's Gini sets one more: **Census: Inequality** appears where it is above 0.45 and raises radicals from movements, and support for the socialist and labor movements, by 5% for each 0.1 above that, twice as fast above 0.70, with no cap.

The census books migration as whole families once a state gains or loses many people in a year, so migration alone seldom carries a state's balance past 90 or 110. Internal resettlement is the exception: settlers arrive as their program recruits them, so a penal colony fills with young men and a Land Grants frontier with families. The American West of 1836, from Michigan and Wisconsin to Missouri and Arkansas, also starts with the surplus of men the 1840 census found there, and so do the Old Southwest and Florida.

Display only keeps the tabs and changes nothing: births, deaths and the workforce follow the base game's rules, and the tab reports what they produce. Disabled also removes the census and both tabs. Wealth Concentration and its effects on each state (see Who inherits) keep working under all three.

## What the tab shows

The tab covers your country as a whole. Most figures have a tooltip that explains them, and some break the figure down for your capital.

| Section | What it shows |
|---|---|
| Overview | Eight readings, each with an icon and a figure beneath: natural change (births less deaths per 1,000 people), net migration, life expectancy (over 10,000 past 10,000 years; the share of infants who die), the leading cause of death and its share of deaths, children per woman, median age (the share aged 65 and over), the working-age share (the dependency ratio) and the Gini coefficient (Wealth Concentration and its target). Hover a reading for more. |
| Causes of Death | Last year's deaths per 1,000 people and a pie of those deaths by cause; see below. |
| Population pyramid | Men per 100 women, then people by five-year age band (wider rows past 85 where many live past 100), with an outline twenty years ahead; once longevity care exists, a button shows the people in care at their biological ages. |
| Work and Retirement | The workforce as the working-age share times the share of them at work, plus those still working past the working ages; old age from (the age with fifteen years of life left; None when no age has); under Old Age Pension the pension age with its five buttons, the pensioners and, under Full, the pension bill; then health need against 1836's, the men of fighting age with the conscription they give, and young adults where they are many while work is short, with up to ten of the states where they are, strongest first. |
| Longevity | Once longevity care exists: Living to 100, the mean biological age beside the calendar one, the share of people within the care's reach, with the health system's part and the wealth from which the market brings it, and the longevity gap. |
| Family Policy | Under Full, your Family & Reproductive Policy law and the measures it allows, each with its running cost, its term and a button; the measures only other laws allow, dimmed. See Family policy measures. |
| History | Eight charts by year, over 5, 20 or 100 years: births and deaths per 1,000 people, net migration, life expectancy, children per woman, median age, deaths by cause, the Gini coefficient and Wealth Concentration. A mark over a year shows war dead, a famine or a demographic event. |
| Wealth | The Gini coefficient, the average income of each stratum, and what sets the Wealth Concentration target, with your most and least concentrated states; the arrow beside each opens it. |
| Where people live | Urban share, the largest city's share of urban people, the effective number of cities, the urban pattern, how many of your states are Metropolises with up to ten of them listed, and your three largest cities. Under Full, the urban pattern's tooltip shows its effects. |
| States | Your ten largest states, with people, life expectancy, median age, children per woman and Gini; hover a row for its age shares, men per 100 women and Wealth Concentration, and click it to open the state. Show All lists up to 40. |

The "?" at the right of the tab's header opens the Help page, How Demographics Works, which explains the method, topic by topic, from the census to work and retirement and longevity. A state's Demographics subtab has the same "?" at its top right.

Every state the tab lists, in States, Where people live and Work and Retirement, opens on its Demographics subtab when you click it.

The subtab shows one state's overview, causes of death, pyramid, work and retirement (without the buttons), longevity, wealth and modifiers, in the tab's order, for any state you open, and says when the state is next counted. Its Modifiers section lists the standing modifiers the census gives the state, each with its tooltip: Census: Inequality, Youth Bulge, Fewer Men and More Men, the Longevity Gap, Metropolis, Dispersed Towns,

Primate City and Provincial Neglect. The state's own modifier list leaves them out, because the yearly count works each one out again; the Planned Capital, and the modifiers your choices in a demographic event give, stay in the list with their time left. The first count of a state comes from the age structure its own birth and death rates would settle into, so 1836 opens with a settled population, not a baby boom or a gap.

National Info, the base game's Census Data window renamed, which the Population panel's button opens for your country and which shows any other country's pops too, has the census as well (Introduction). Its Demographics tab, beside Pops, the base game's list, holds the same sections as the Population panel's tab for whichever country the window shows, in three columns. Its Wealth section leaves out the strata's average incomes, which only the Population panel shows. For another country it leaves out what only a played country has: the buttons, the pyramid's twenty-year outline, and, for a country that is not a major power, the history, whose empty charts say so. The tab shows while the Demographics rule is on.

## Reading the pyramid

The pyramid has a row for each five years of age, from 0 to 4 at the bottom to 85 and over at the top, with men on the left and women on the right. A wide base means many children; a column that narrows slowly means few births and long lives. Hover a row for the numbers.

Where people aged 100 and over make up at least 1% of a state or country, its top row splits into wider rows: 85 to 99, 100 to 124, 125 to 149, 150 to 199, 200 to 249, 250 to 299, and 300 and over. A row shows while the people of its age and older are at least 1% of the people, and the highest row shown also holds everyone older. A row wider than five years is drawn per five years of age, so its bar compares with the rows below it; hover it for the people in it. These rows have no outline. The biological view splits the same way by biological age. The pale outline behind the bars carries your people twenty years forward at the census's own birth and death rates for this year, five years at a time. It leaves out migration and holds the rates where they are, so it shows where today's course leads, not a forecast. Only your own country gets it.

## Reading the figures

Each year, the people of every age lose their dead at that age's rate, the newborn enter at age 0, and the war dead, people killed by events and migrants are taken from or added to the ages they belong to. The war dead are people aged 18 to 40, nineteen in twenty of them men. Children per woman follows wealth and women's legal standing, and two customs lower it. Under late marriage, strongest at the start in north-western and central Europe, women marry in their mid-twenties and many never marry. Under birth spacing, strongest in Japan and slight in China and Korea, couples space and stop births by custom. Modern contraception wears both away, and health care wears away birth spacing; women's paid work and farms that pass whole to one heir make late marriage more common. Both customs also spread slowly from neighboring states. The figure then falls with literacy, child survival and city life, and family policy moves it (see Family policy measures). Under Full, the figure shown is what births came to that year, so decrees, famine and other effects on births count in it; its tooltip lists them as births added on top of the census. The means to plan a family grow with literacy and with contraception: Vulcanization, Feminism, the Contraceptive Pill and Modern Pharmaceuticals each add some, in any order, and State-Sponsored Family Planning adds more, whoever can read. Their tooltips show each line. Under Full, family limitation adds to them too: the share of couples who stop at the family size they want by older methods, which France starts with at 60%. It spreads from neighboring countries that practice it, grows under Forced Heirship in farming countries and in nearly universally literate ones, less under a state religion, and never falls back, though modern contraception takes over its part of the means as it arrives. It shows as Family Limitation among a country's modifiers. Under Full, too, where a cultural hegemon's culture outweighs yours (the Foreign Cultural Benchmark), the size of family its people want pulls yours a little toward it: a few percent at most, shown as the cultural hegemon's example in the children-per-woman tooltip. Life expectancy follows five causes of death, which standard of living, medicine, health laws and institutions bring down. Medicine saves lives only as far as health care reaches. Markets and charity give some care everywhere; a health law adds more with each level of the Ministry of Health, in incorporated states only. So a Public Health Insurance law does little before antibiotics, and medicine without a health system reaches less than half the people. The medical technologies and the health, policing, criminal justice, pension and labor laws show what they do to each cause in their tooltips. Deaths from accidents and violence

fall with each level of the Ministry of Public Safety in incorporated states: 3% a level under a Dedicated or Militarized Police Force, 2% under Local Police or Private Policing, nothing under No Police. Restorative Justice lowers them 4% and Rehabilitation-Focused Criminal Justice 8%; Penal Labor Camps raise deaths at work. Later technologies make work and everyday life safer whatever the laws: Prefabricated Construction, Industrial Robotics, Containerization and Knowledge Economy together prevent nine in ten deaths at work, and nineteen in twenty with Self-Directed Evolution's bodies built for the work. Modern Urban Planning, Computer-Aided Design and Cellular Networks prevent two in five deaths from accidents and violence, half once Autonomous Vehicles is researched. Medicine cuts adults' deaths from infection by more than children's, since it cures the tuberculosis and pneumonia that killed adults in 1836, so in a rich country adults die of infection about as rarely as in the rich world today; Personalized Medicine and Advanced Nanofabrication cut deaths from infection to about a sixth of that. Genetic Engineering and Rejuvenation Therapies let medicine prevent nine in ten deaths from chronic disease where it reaches, and Molecular Assemblers and Biological Immortality nearly every one. Life expectancy and infant mortality are the census's own figures, before famine, war and disasters.

Medicine also takes time to arrive. A new medical technology, health law or Ministry of Health level moves a state's medicine part of the way at each yearly count: a tenth of the gap at first, and up to just over a third once the Electric Telegraph, Modern Nursing, Radio, the World Wide Web and Telemedicine are researched. A party to the Global Pandemic Response, the convention that founds the World Health Organization, closes a further share, larger as the UN's enforcement rises. Under Full, the Medical Outreach decree closes a further fifth of the gap at each count in one state. It can be issued only in a state whose next count would leave its medicine short of the targets, and it ends once the state has caught up. Turmoil raises deaths from accidents and violence, by half at full turmoil. Where the map marks a state with malaria (most of tropical Africa, parts of Indonesia and the Pacific islands), deaths from infection run higher, more so with severe malaria, until Malaria Prevention; Quinine protected colonists, not the people who lived there, so it changes nothing here.

The Causes of Death section, on the tab for your country and on the subtab for one state, splits last year's deaths in a pie: infection and hunger, chronic illness and old age, accidents and violence, work, childbirth, people killed by events, and war and devastation. The census counts deaths, not causes, so the five causes share out its own deaths by each cause's rate in each age band, at the body's age for people in longevity care; the rest are counted as they are. Under Full the game also adds deaths on top of the census's own (famine, pollution, devastation, dangerous work, decrees and events), and each counts under its own cause: severe starvation and epidemics under infection and hunger, pollution and the climate's heat under chronic illness, dangerous workplaces under work, and devastation's dead, people of every age, beside the war dead. Each of those effects' tooltips names the cause it counts under. Hover a cause for its rate against a poor country's without modern medicine, what makes it up, where medicine is heading and, under Full, how many of its deaths the game added on top. Below the pie is health care access. Under Full the Deaths row's tooltip gives the census's own death rate against the game's. The Landowners' Famine Reserves is the exception: under Full the census sets the farm workers' saved lives against the rest of the state, so it moves deaths within a state without lowering them.

An arrow beside a figure shows whether it rose or fell since the last census. Life expectancy rising is green; the other figures are neutral. Arrows appear only after the census has been brought forward a full year twice, so in a new 1836 game the first ones come with the census at the end of 1838. Births and deaths per 1,000 people appear from the first full year.

## The urban pattern and a planned capital

The urban pattern counts each state as one city and its urban people as everyone who does not work a farm, plantation, ranch, mine or other rural workplace. The pattern is Primate when the largest city holds 40% or more of all urban people, Dominant from 25%, Balanced from 10% and Dispersed below that.

Under the Full setting a Dominant or Primate pattern has effects; Balanced and Dispersed are the baseline. Your largest city gets Primate City, with more Urban Center and University throughput and migration attraction. Every other state gets Provincial Neglect, with less migration attraction, higher penalties from turmoil and, where the state has a culture of its own, more support for its Cultural Minority Movement. Both grow with

the largest city's share of urban people, from nothing at 25% to their full size at 50%: Urban Center throughput +20%, University throughput +10% and migration attraction +25% for the city, migration attraction −15%, turmoil penalties +15% and cultural minority support +15% for the rest. Past 50% they keep growing at half the pace, to twice their full size when one city holds all the urban people. A country of fewer than three states gets neither. Each state takes the new figure at its yearly count, and the Urban Pattern tooltip names your largest city and shows the current figures. Under Display only it says what Full would do.

Within a state, the metropolitan share is the share of the state's urban people who live in its largest city. It reads Metropolis at 60% or more, Leading City from 40%, Several Cities from 20% and Market Towns below that, on the Settlement row under Urban People on the state's subtab (a state with fewer than 5,000 urban people has none). The share moves a twentieth of the way to its target each year. The national capital, a World Market Hub and a Financial District raise the target; a large region, a farming and mining economy, infrastructure to spare over what the state uses, Migration Crowding and urban planning (the Ministry of Urban Planning, Planned Towns and a Planned Capital) lower it, and a country's urbanization raises it most at half its people in towns and lowers it a little below a tenth or past nine tenths. Hover the row for each term as a bar. Under Full a share above 40% has effects, full at 60%: the state crowds 10% sooner, its people give 20% more infrastructure, its Urban Center works at 10% more throughput, turmoil penalties rise 10% and families are a little smaller. Past 60% all but the crowding keep growing at half the pace, to nearly twice their size at the highest shares. Below 20% its towns give up to 20% less infrastructure. Where People Live counts your Metropolises.

The decision Plan a New Capital, also under Full, moves your seat of government away from your largest city. It appears once you research Modern Sewerage and can be taken with Steel-Frame Buildings, a Dominant or Primate pattern, 100 authority to spare, at peace and outside any diplomatic play, when you have not moved your capital recently, and once in 50 years. It offers up to three sites: your largest incorporated states that are a homeland of one of your primary cultures, hold at least 1% of your people, and are neither the capital nor the largest city. The capital moves at once.

| You pay | You get |
|---|---|
| 3% of a year's GDP, at once | The capital in the chosen state |
| 100 authority for five years (Building the New Capital) | Planned Capital on the new capital for 20 years: migration attraction +25%, state construction efficiency +20%, infrastructure +10 and Government Structures throughput +10% |

Planned Capital ends if the capital moves away again. Not now spends nothing, and the decision stays open; it is also what happens if you let the choice lapse. A site you lose, or a war that starts, while the choice is open takes that site off the list. A planned capital draws migrants and building to a region your largest city leaves behind, so over the years it can lower that city's share and with it Provincial Neglect. The AI plans one only with a Primate pattern of 42.5% or more, eight states or more, a largest city crowded to Migration Crowding's ceiling (ten times the reference density), and the money in its treasury.

## Life extension

From mRNA Therapeutics on, medicine slows aging itself as well as treating illness, and the census tracks each person's body apart from the calendar. People in longevity care age nine-tenths of a year each year with mRNA Therapeutics, seven-tenths once Personalized Medicine, Genetic Engineering and Synthetic Biology follow, half a year with Rejuvenation Therapies, and not at all with Biological Immortality, under which bodies older than 35 also wind back toward 35. Children are never treated: people join at 25.

Who is in care depends on your laws and your people's wealth. Each law below lists its part among its modifiers.

| Law | Longevity care |
|---|---|
| Public Health Insurance | Everyone the health system reaches. |
| Charity Hospitals | A quarter of those it reaches. |

| Law | Longevity care |
|---|---|
| Private Health Insurance, No Health System | The market alone; Private Health Insurance brings its price down twice as fast. |
| Mandatory Augmentation | At least nine in ten people. |
| Unrestricted Augmentation | The market reaches 20 levels of wealth further down. |
| Regulated Augmentation Market | The market's price falls half again as fast; with Private Health Insurance, two and a half times as fast. |
| Human Purity | None. |

The RejoovenEsense company also brings the market 10 levels of wealth further down while its prosperity bonus is active.

The market reaches the rich first: from your first aging technology, people from wealth 80 can buy care, and the line falls about a level a year, faster with the technologies that spread medicine. Care rolls out as fast as new medicine spreads, and a ban winds it down the same way; people who lose care keep the bodies they have and age normally from there. The Longevity section shows the share in reach and the gap between those in care and those not, as the chance of living to 100.

Under Full, people in care die at their bodies' rates. Where the market reaches the rich first they die less, so your wealthy pops grow against the poor. The health system's share of the care costs your Ministry of Health more bureaucracy. Women in care can have children they put off, and births nobody planned go on as long as their bodies allow, even past 50 by the calendar; State Youth Centers on Ectogenesis carry a share of planned births with no risk in childbirth. A society that stops aging turns a slow decline into slow growth, about 1% a year at 1.6 children per woman: the brakes are the means to plan a family and the size of family people want. Old Age From rises with longer lives and reads None once no age has fifteen years left; Linked to Average Longevity follows it, Linked to Common Longevity follows those not in care, and the AI moves to Linked to Common when the gap is wide. Under Full a wide gap also costs you politically: the Trade Unions and the Rural Folk lose up to 3 approval, more the wider the gap and the fewer of your people are in care, and the labor movement gains support in the states where the gap is widest. Mental Health Awareness cuts deaths from accidents and violence, and Mind Backups halves deaths from accidents, violence and work among the people in care, whose minds it restores to new bodies: a state with half its people in care loses a quarter of those deaths. Outside Full these technologies cut deaths by a flat share, and nothing else of this changes the game.

With Mind Backups and a Mind Upload Nexus, under Full, people in care who face death from illness or old age upload instead and go on as minds, as far as your Nexuses have room: a million minds a level, shared across the country. Their bodies die and leave your states' people, but Living to 100 counts them as living, deaths per 1,000 leave them out, and the Longevity section's Uploaded Minds row shows how many you host. Each year a thousandth of the minds choose to end. A Nexus makes only what its minds produce: an empty one makes nothing, and a full level about five times what a Nexus makes outside Full, most of it software, with its research bonus following the same share. The fill is fastest in the decades after Biological Immortality, while bodies that aged before it still die of old age; once they have wound back, few in care die of illness and a Nexus fills slowly. If you lose Nexus levels, the minds that no longer fit are archived, frozen, and come back before anyone new uploads once you build again.

## Family policy measures

Under the Full setting, the tab's Family Policy section lets you act within your Family & Reproductive Policy law. It shows the law in force, a row for each measure that law allows, with its running cost, its term and a button, and, dimmed, the measures only another law allows; hover one for the laws that allow it. Under Display only the rows show what each measure would do and the buttons are grayed.

A measure takes effect as soon as you start it and costs its upkeep while it runs: a weekly share of GDP for the ones paid in money, recalculated each year, authority, or a costlier Ministry of Education. A measure acts on every state at once, so the ones that need enforcement cost more authority than a decree does for one state, and one child with the Crash Campaign takes 700. As with a decree, a measure paid in authority can't start

until you have its upkeep free, and Birth Limit can't tighten to one child without the further 100; ending a measure needs no authority. Each also moves the approval of the interest groups its tooltip lists. Once started, a measure runs for at least five years, and its button can't end it before then. A change to a law that doesn't allow a measure ends it at once, with a notice, and taking it up again under another law starts a new term. Under Communal Child-Rearing your form of government adds measures: a Council Republic or Anarchy also allows Sex Education, and a Single-Party State or Autocracy that has outlawed dissent also allows Honors for Large Families and Restricted Contraception. A change of government ends what the new laws don't allow. Birth Limit's button moves from two children to one child, then ends the limit. The Crash Campaign ends itself after five years and can't return for ten.

| Measure | Laws that allow it | What it aims at | Upkeep |
|---|---|---|---|
| Child Allowance | Pro-Natalist Policy, Communal Child-Rearing | Families want more children | 0.5% of GDP a year, more where children are many |
| Family Housing | Pro-Natalist Policy, State-Sponsored Family Planning, Communal Child-Rearing | Offsets half of city life's pull toward smaller families | 0.25% of GDP a year, by the share of people in cities |
| Honors for Large Families | Traditional Family Structure, Pro-Natalist Policy; Communal Child-Rearing under a Single-Party State or Autocracy with Outlawed Dissent | Families want a few more children | 50 authority |
| Restricted Contraception | Pro-Natalist Policy; Communal Child-Rearing under a Single-Party State or Autocracy with Outlawed Dissent | Families have less means to plan their children; more mothers die; the feminist movement gains support | 200 authority |
| Parental Leave | State-Sponsored Family Planning, Communal Child-Rearing | Families want more children, and more women work | 0.3% of GDP a year if every woman of working age worked, less as fewer do |
| Free Contraception | State-Sponsored Family Planning, Population Control Measures, Communal Child-Rearing | Families have more means to plan their children | 0.1% of GDP a year |
| Sex Education | State-Sponsored Family Planning, Population Control Measures; Communal Child-Rearing under a Council Republic or Anarchy | Knowing how to plan a family reaches people as if literacy were 20 points higher | The Ministry of Education costs 10% more bureaucracy |
| Later, Longer, Fewer | State-Sponsored Family Planning, Population Control Measures | Families want fewer children | 150 authority |
| Birth Limit | Population Control Measures | Families want far fewer children, fewer still at one child; once Microprocessor is researched, more boys are born than girls | 300 authority, 400 at one child |
| Rural Exemption | Population Control Measures, with Birth Limit | Halves Birth Limit for farming families, by the state's share of Peasants and Farmers | None |

| Measure | Laws that allow it | What it aims at | Upkeep |
|---|---|---|---|
| Crash Campaign | Population Control Measures | Families want fewer children still; legitimacy −10 and more radicals from movements | 300 authority |

The census reads a measure at each state's next yearly count: how many children families want, city life's pull toward smaller families, the means to plan a family, mothers' deaths in childbirth and, once Microprocessor is researched, how many boys are born per 100 girls. Births follow over the year after. Under Full, Pro-Natalist Policy itself makes families want a few more children, and the law's own boost to births and Population Control Measures' cut no longer apply: the measures carry those laws' force. Pro-Natalist Policy with Child Allowance, Family Housing and Honors adds about 0.3 children per woman in a rich country that has few, for 0.4–0.5% of GDP a year. Restricted Contraception, Birth Limit and the Crash Campaign move births further, at their political price. The Children per Woman tooltips list family policy beside the other terms. Parental Leave's women's work counts toward the workforce as the women's rights laws' does.

The AI starts the measures that raise births when its population shrinks or many of its people are old, and the ones that lower births in a famine, when half its people live in states crowded to Migration Crowding's ceiling, or with a youth bulge, as its treasury and authority allow, at most one measure paid in money a year. It takes Restricted Contraception, Birth Limit's one child and the Crash Campaign only when the need is pressing and its legitimacy is high.

### Demographic events

Under the Full setting, ten events mark the moments the census records: the Demographic Dividend (the working-age share a tenth above 1836's), the Baby Boom (after a long war with heavy losses), the Lost Generation (war dead past a twentieth of the men of fighting age), the Graying of the Nation (pensioners at a fifth of the people), the Youth Bulge (a fifth of the people in their twenties while work is short), Missing Girls (110 boys born per 100 girls under Birth Limit), Bare Branches (110 men per 100 women), the Empty Villages (an old, shrinking farm state), the Overspill (a Metropolis state crowded to Migration Crowding's ceiling) and the Last Generation? (Biological Immortality). Each comes at most once in 25 years, and the Last Generation? once a game; at most one of the yearly ones arrives each year. The Lost Generation and the Empty Villages offer only harms: each choice spares one and pays its own cost. Some choices use the census's own levers: the pension age, Birth Limit and, for the Overspill, your Settlement Authorities' Planned Towns. Every event and choice is listed in Demographic event list.

## 8.9 Internal resettlement

Internal resettlement is government-run settlement of an empty frontier. You build a Settlement Authority in a thinly populated state. Each month it recruits settlers across your country under its program and moves them there. You never choose a culture or religion to move: programs select by occupation, strata, acceptance and radicalism.

The Internal Resettlement game rule has three settings. Enabled (the default) gives everyone every program their laws allow. AI Voluntary Only is the same for players, but each month it switches AI countries off Penal Transportation, Special Settlements and Rustication. Disabled removes the system. Moving people between countries by treaty is a different mechanic, the Population Transfer article in Diplomacy.

### The Settlement Authority

The Settlement Authority is a government building available from the start of the game, with a low construction cost. It can only be built on an open frontier, measured over the whole state region whoever owns each part:

- a new Authority needs fewer than 2 people per km²;
- an existing one keeps running, and can add levels, until the region reaches 10 people per km².

*Figure 10: A Settlement Authority's building panel. Its three production method groups sit at the bottom left, and the tooltip shows the settlers who arrived last month.*

These figures hold at a world population of one billion, about the world's population at the 1836 start, and grow in proportion to it: at two billion people they are 4 and 20. A frontier is thin compared with the rest of the world, so it closes only when its region fills faster than the world grows.

Migration Crowding Tolerance multiplies both densities as it does the crowding threshold: at +50% tolerance and one billion people they are 3 and 15. The tolerance that counts is that of the state the Authority stands in, although the density is the whole region's, so two owners of one region can face different limits. The building's requirements show the region's current density against each limit, with the two multipliers. In an 1836 start the open frontiers include the American West, Siberia, the Kazakh steppe, Hokkaido and the Argentine pampas.

An Authority can have up to 5 levels. Nationalism, Civilizing Mission, Mass Propaganda, Keynesian Economics and Civil Rights Movement each add 5, to a maximum of 30. Capacity and costs scale with level.

When the region reaches the closing density, the Authority is wound up and you get the event "The Frontier Is Closed". To keep settling, build one on the next frontier.

### Resettlement programs

The Resettlement Program method decides who is recruited and how many move each month. Half the capacity comes with each level and half depends on staffing, so a new Authority moves settlers from the first month and speeds up as it fills its jobs. The state's modifiers show the total as Settlers Moved per Month. The table gives each program's settlers a month for every fully staffed level.

| Program | Needs | Recruits | Settlers a month per level | Die in transit |
|---|---|---|---|---|
| Land Grants | nothing | Unemployed and peasants of the lower strata who are at least second-class citizens; no peasants under Serfdom | 300 | none |
| Military Colonies | Standing Army | As Land Grants, but fully accepted only | 250 | none |
| Penal Transportation | Law Enforcement; not under Guaranteed Liberties, Restorative Justice or Rehabilitation-Focused Criminal Justice | Lower-strata pops in which at least a fifth are radicals | 100 | 5% |
| Organized Colonization | Railways | Unemployed, peasants and laborers of the lower strata who are at least second-class citizens | 500 | none |
| Special Settlements | Mass Propaganda and Collectivized Agriculture; not under Guaranteed Liberties, Protected Speech or Right of Assembly | Farmers | 1,000 | 15% |
| Development Program | Keynesian Economics; not under Laissez-Faire | As Organized Colonization, plus machinists, engineers and clerks, all at least second-class citizens | 800 | none |
| Rustication | Mass Media and Single-Party State; not under Guaranteed Liberties, Protected Speech or Right of Assembly | Laborers and clerks who do not work in farming, plantations, ranching or subsistence | 800 | 1% |
| Managed Retreat | Environmental Movement | Everyone except slaves, from coastal states while you suffer Coastal Flooding or Coastal Population Relocation, and from states hit by a nuclear | 600 | none |

| Program | Needs | Recruits | Settlers a month per level | Die in transit |
|---|---|---|---|---|
| | | strike or a weapons accident | | |

Programs cost bureaucracy, paper and staff. The voluntary programs and Managed Retreat also use services; Military Colonies and the coercive programs (Penal Transportation, Special Settlements and Rustication) hire soldiers and use small arms instead. Slaves are never recruited. If a law change retires the running program, the Authority falls back to Land Grants. Managed Retreat's damage comes from Climate events and What a nuclear strike does.

Each program also speeds up the destination's incorporation and its growth as a colony, by 10% to 25%. Military Colonies also reduce the effect of turmoil there by 25%, and Penal Transportation and Special Settlements raise mortality there. These bonuses grow with the month's arrivals and reach full strength at 2,000 settlers a month, so a small or idle Authority buys little of them.

## Settlement plans and transport

The Settlement Plan decides what settlers build. Any plan works with any program, and every plan adds +1 migration pull per level to help the frontier hold its settlers.

| Settlement plan | Needs | Effect on the destination |
|---|---|---|
| Homesteads | nothing | Virgin Soil: +10% farm and ranch throughput and more subsistence output. No arable land is added, and the bonus ends with the Authority. |
| Work Settlements | nothing | The Authority employs 100 laborers per level and uses tools; +10% mining and logging throughput, more infrastructure. |
| Planned Towns | Modern Urban Planning | +10% construction, more infrastructure and migration pull; employs engineers and uses services. |

The settlement bonuses scale with arrivals in the same way as the program's.

Transport adds capacity for every fully staffed level and costs transportation goods:

| Transport | Needs | Extra settlers a month |
|---|---|---|
| Overland | nothing | none |
| Rail and Steamship | Railways | +200 |
| Motor Transport | Combustion Engine | +400 |
| Airlift | Commercial Aviation | +600 |

## How settlers are recruited

Each month an Authority works down your other states, most eligible people first, until its capacity is met; it skips other Authorities' states. A state gives at most 2% of its eligible people a month across all your Authorities, and any draw of fewer than 100 people from a pop is skipped, so a pop smaller than 5,000 is never recruited. The Authority's Modifiers tab shows "Settlers arrived last month" and "Died in transit last month", and each source state's modifiers show how many it gave.

The Resettlement Recruitment Drive decree steers recruitment. Every Authority recruits from a drive state before any other, and may take 4% of its eligible people a month instead of 2%, which reaches pops of 2,500 or more. The decree lowers the state's migration pull by 10% and needs at least one Authority elsewhere.

## Costs and consequences of resettlement

Coercive programs lose some recruits in transit. Special Settlements radicalize the farmers left behind, and Rustication the laborers and clerks.

Settlers press on the people already there: each month, destination pops below second-class citizens gain radicals in proportion to the arrivals.

Each program has one political modifier for your country, however many Authorities run it. It reaches full strength when the program settles 0.25% of your population a year, and fades over several years after it stops (a third is left after one year).

| Program | Interest group approval at full strength |
|---|---|
| Land Grants | Rural Folk +3, Landowners −3 |
| Military Colonies | Armed Forces +3, Rural Folk −3 |
| Penal Transportation | Intelligentsia −3 |
| Organized Colonization | Rural Folk +3 |
| Special Settlements | Rural Folk −10, Intelligentsia −5 |
| Rustication | Intelligentsia −10, Petite Bourgeoisie −5 |
| Development Program, Managed Retreat | none |

If you are a party to the Universal Declaration of Human Rights, a UN convention (UN conventions and agencies), running Penal Transportation, Special Settlements or Rustication gives you Violating the Declaration: up to −10% prestige as the programs grow. It fades over several years after they stop. The first month you run a coercive program as a party, an event explains the penalty and offers to switch every coercive program to your best voluntary one.

Settlers change the culture shares at the destination. Over time that can create or remove a homeland there, under the rules in Dynamic homelands.

## Resettlement events

While an Authority is receiving settlers, you have a 10% chance each month of a resettlement event, and at most one every eighteen months however many Authorities you run. The events follow the program. Voluntary programs bring land rushes and speculators, Homesteads bring dust storms after ten years, and settling among people below second-class citizens brings land disputes. The coercive programs draw the harshest ones: reform campaigns, famine in Special Settlements, and after five years a petition from the settlers to go home. Most events offer a choice between paying (money, authority or fewer settlers) and letting the settlers or their neighbors bear it (deaths, radicals, lost throughput or prestige). Every event and option is listed in Resettlement event list.

## Resettlement and the AI

The AI values a Settlement Authority on an open frontier when one of its states has more than 10% of its workforce out of work or migration crowding, and on any frontier emptier than 0.5 people per km² (at one billion people, scaled with world population like the frontier densities). It stops founding new ones once it has three. It issues the Recruitment Drive in states with more than 10% of the workforce out of work or a crowding penalty above 10%. The mod gives the coercive programs a low AI weight, but only the AI Voluntary Only rule guarantees that AI countries stay off them.

CHAPTER 9

# Diplomacy

The mod adds 41 treaty articles, extra escalation in diplomatic plays, events between rival powers, a reunification system for divided nations, two more tiers of power bloc principles, ten late-game formable countries, and 22 city-states you can release. None of this has a game rule or journal entry of its own: it works through the base game's treaty screen, diplomatic plays, power bloc panel, formation panel and releasable countries list, though some pieces depend on other systems' rules. Much of it opens up in era 6, with Decolonization and Intergovernmental Organizations.

## 9.1 Treaty articles

The new articles appear in the treaty draft beside the base-game ones. Most have a technology gate (listed in the tables), a power bloc principle (see principles that unlock articles and actions), or another system's game rule. A directed article binds one side, the conceding country, and gives something to the other. An enforceable article can also be demanded as a war goal. The draft shows each article's influence cost and who pays it. A demand is paid for by the country making it, so Seize Company, Disband Company, Return Cultural Property, Surrender a Monument and the Lease Seabed Claims articles cost the conceding country no influence.

### Choosing a company or state for an article

Nine articles ask you to name something. Seize Company and Disband Company take a company. Minority Protection, Free Port Concession, Religious Mission Rights, Demilitarized Zone, Surrender a Monument, Excavation Concession and Excavation Permit take a state. When you add one to a draft, the article shows a button with its current choice; click it to pick from a list.

The state or company always belongs to the conceding country, so only a country that owns at least one company can be asked for Seize Company or Disband Company. Free Port Concession needs a coastal state, and a demilitarized zone can't be the capital. A country can concede each of the four state articles to the same partner only once at a time, across all their treaties, so a second demilitarized zone against the same country waits until the first ends. The company articles can be repeated for different companies.

The four state articles act on the state itself, and only while the conceding country owns it under a treaty in force. They end with the treaty, and they leave the state at once if it passes to another country, which includes the conceding country being annexed. If the other party is annexed instead, they end within a month. A state taken by rebels in a civil war keeps them, and one that returns to the conceding country while the treaty still stands gets them back.

Excavation Concession and Excavation Permit leave nothing on the state. They let the other side dig there only while the treaty is in force and the conceding country owns the state, so a dig under one stops as soon as either ends.

The four Grand Monument articles appear in a draft only where they have something to act on. Surrender a Monument is listed for a conceding country that owns a monument it could give up, Return Cultural Property for one that holds a trophy from the other side's lands, and the two excavation articles for one with a buried site open to the other side.

*Figure 11: Choosing the company for a Disband Company article. The list holds the conceding country's companies.*

## Treaty articles by purpose

The corporate articles let a winner, or a patron, reach into another country's economy.

| Article | Unlocked by | Enforceable | What it does |
|---|---|---|---|
| Seize Company | Always available | Yes | The conceding country hands one of its companies to the other, unless the receiver already has a company of that type. |
| Disband Company | Always available | Yes | The conceding country dissolves one of its companies. |
| Enforce Privatization | Stock Exchange | Yes | The conceding country must privatize its government-owned buildings while the treaty stands. The demanding country needs Laissez-Faire or Interventionism, and the conceding country Cooperative Ownership, Agrarianism or Traditionalism. |

| Article | Unlocked by | Enforceable | What it does |
|---|---|---|---|
| Free Port Concession | International Trade | Yes | The chosen state drops its tariffs (within a year) and gets +50% trade capacity, +25% trade advantage and more migration pull, but collects half the tax and assimilates more slowly. Blocked by Isolationism on either side. |

The humanitarian and cultural articles move people, faiths and cultures.

| Article | Unlocked by | Enforceable | What it does |
|---|---|---|---|
| Minority Protection | International Relations | Yes | Halves assimilation and conversion in the chosen state. The conceder loses legitimacy and pays authority; the other side gains prestige. The protecting side must be in a Cultural Commonwealth or Religious Convocation bloc. |
| Religious Mission Rights | Colonization | Yes | Each year 5% of the chosen state's pops of other faiths convert to the other side's religion. The side sending the missionaries must be in a Religious Convocation bloc. Neither country can have State Atheism, and the treaty freezes if the conceding country adopts it later. |
| Cultural Exchange Program | Pan-nationalism | No | Mutual: +1 yearly cultural acceptance, +2% prestige and +1 cultural pull for both, better relations, −5 ideological covert defense. |
| Population Transfer | Pan-nationalism | Yes | Moves pops of the receiving country's primary cultures out of the conceding country, once. See population transfers by treaty. |
| Return Cultural Property | The Grand Monuments rule | Yes | The conceding country returns every Grand Monument trophy taken from the other's lands, once. Each is rebuilt where it stood. |
| Surrender a Monument | The Grand Monuments rule | Yes | The conceding country gives up the Grand Monument in the chosen state as a trophy, once: the whole of it, a quarter or a tenth, by its form. Never a Grand Shrine. |
| Excavation Concession | The Grand Monuments rule; the digging side needs Romanticism | Yes | The other side may dig the buried sites around the chosen state, and half of what each dig finds goes to its capital as a trophy. It pays 75 influence. |
| Excavation Permit | The Grand Monuments rule; the digging side needs Empiricism | No | The other side may dig the buried sites around the chosen state; the finds stay, and it gains Foreign Scholarship. It pays 40 influence. |

The military and security articles cover disarmament and cooperation between armed forces.

| Article | Unlocked by | Enforceable | What it does |
|---|---|---|---|
| Demilitarized Zone | International Relations | Yes | No conscription, Barracks, Naval Fortifications or Military Bases in the chosen state; an existing Military Base is dismantled. While it stands the conceding country uses 25 authority and loses 5 prestige. |
| Forced Disarmament | Intergovernmental Organizations | Yes | Military wages −25%, conscription halved, military industry throughput −25% and −10 prestige. Every Arms Industry, Artillery Foundry, Munition Plant, Naval Administration, Naval Fortification, Naval Logistics Center and Military Base in the conceding country is dismantled; Shipyards and Military Shipyards are spared. |
| Intelligence Sharing Pact | Intergovernmental Organizations | No | Mutual, between two major powers or greater that have both established a Ministry of Intelligence and Security: +3 intelligence capacity, a stronger but dearer intelligence ministry, and a defense shield. Costs each side 1 infamy. |
| Joint Military Exercises | Combined Arms | No | Mutual: +25% army experience gain and admiral rank impact, military goods 5% dearer for each treaty that carries it. Needs an alliance or defensive pact, already in force or in the same treaty, a subject relation between the two, or a shared Military Treaty bloc. Each country can hold it in at most three treaties. |

The requirements on Minority Protection, Religious Mission Rights and Joint Military Exercises apply when a treaty is signed or renegotiated. An article already in force stays if they stop being met, but renegotiating its treaty means meeting them again or dropping it.

Arms control between nuclear powers belongs to the nuclear system: Nuclear Arms Limitation holds two countries to the same ceiling on warheads, and leaving it weakens the nuclear taboo unless another such treaty still binds you as tightly (see The Nuclear Arms Limitation treaty).

The aid and influence articles are how power blocs extend their reach. All but Request Influence and Reduce Subject Liberty need a principle in your bloc. In the aid articles the donor must outrank the recipient, pays a running cost and gains leverage over it, and relations improve.

| Article | Offered by | What it does |
|---|---|---|
| Request Influence | A country outside any bloc, to a bloc leader | The leader gains leverage over you very fast, so it can soon invite you into its bloc. It is the way for a player to join a bloc: an AI leader accepts readily, but an AI country essentially never agrees to be the one asking. |
| Extend Influence | Bloc leader | The leader gains leverage over the other country. |
| Crisis Resolution | Bloc leader | The recipient gets +40 minimum legitimacy, lower liberty desire and half the radicalism from enacting unpopular laws. |
| Education Aid | A more literate member | +10% education access for a recipient with a weak Ministry of Education or none. |
| Healthcare Aid | A member with a Ministry of Health at level 3+ | −5% mortality and +1 standard of living for a recipient with a weak health system. |
| Security Aid | A member with a Ministry of Public Safety at level 3+ | −20% turmoil effects and lower liberty desire for a recipient with weak policing. |

| Article | Offered by | What it does |
|---|---|---|
| Development Assistance | A member | +2 standard of living for the lower strata of a recipient whose primary cultures average below 15. |
| Modernization Aid | A member with more technologies and a Ministry of Science at level 2+ | The recipient gains technology spread; the donor loses weekly innovation. |
| Extensive Modernization Aid | A member with a Ministry of Science at level 4+ | Four times Modernization Aid, which it requires. |
| Reduce Subject Liberty | A subject, to its overlord | Lowers the subject's liberty desire. The overlord pays its influence. |

The remaining articles belong to systems with chapters of their own.

| Article | Game rule | Chapter |
|---|---|---|
| Currency Peg, Imposed Currency Peg, Swap Line, Lender of Last Resort, Debt Receivership | Banking System set to Enabled | Banking and monetary policy |
| Nuclear Disarmament, Nuclear Program Freeze, Nuclear Program Aid, Nuclear Guarantee, Nuclear Security Assistance, Nuclear Arms Limitation | Nuclear Weapons | Nuclear weapons |
| Require UN Membership | United Nations | The United Nations |
| Enforce Emissions Reduction | Global Warming | Climate and pollution |
| Lease Pacific, Atlantic, Indian, Arctic and Southern Seabed Claims | Global Commons | The extended timeline |
| Cede Colony | Space Race set to Simple Colonies or Full Colonies | The space race |
| Grid Interconnection Agreement | None | Economy and construction |

## Treaty article events

Some articles have events that can fire in any year the treaty stands. Intelligence Sharing Pact, Minority Protection and Cultural Exchange Program each have a good and a bad one, such as a foiled plot or a nationalist backlash. Joint Military Exercises, Education Aid, Security Aid, Nuclear Disarmament and Nuclear Program Freeze have one each, and Population Transfer has one that can fire while its disruption lasts.

## Dynamic treaty names

A treaty containing one of the mod's articles gets a fitting name instead of "Treaty of" plus the signing city: a Nationalization Accord for a seized company, a Convention on Minority Rights, a Demilitarization Agreement, and so on for about half of the new articles and for money transfers. The draft's randomize button draws from the names that fit.

## The Intelligence Sharing Pact's defense shield

Once a year, each partner in an Intelligence Sharing Pact compares its own covert defense with its partners'. If a partner's is higher, you gain half the gap between your combined economic, military and ideological defense and your strongest partner's, spread evenly over the three: a sixth of the gap on each. The stronger partner gains nothing from the shield, so the pact lets a great power cover a weaker partner against covert operations, and partners can't run covert operations against each other. The AI signs readily with a country that shares one of its rivals, and almost never with a rival.

## Population transfers by treaty

Population Transfer takes effect once, on ratification. Every pop in the conceding country whose culture is primary for the receiving country, and not for the conceding one, moves to the receiving country, mostly to

its most populous states. Both countries then take a five-year Population Transfer Disruption to bureaucracy and legitimacy, with more radicals from conquest. It fades over time, and its strength grows with the share of each country's population that moved, never falling below a quarter.

The draft blocks the article when the receiving (demanding) country has Universal Citizenship. It also needs at least one community of the receiving country's primary cultures inside the conceding country with acceptance below 60. The transfer then moves every such pop, well-accepted communities included.

The AI demands a Population Transfer, in a treaty or as a war goal, only when it would move at least 100,000 people. You can still demand smaller ones.

## 9.2 Diplomatic play escalation

In the base game a play escalates by one point a day. The mod adds extra escalation about once a week from the countries in the play, each for its own role as initiator, target or committed participant. No country adds more than 10 a week, each country's weekly figure is rounded to a whole point, and nothing slows a play below the base rate: laws and seats that slow plays only cancel extra escalation.

| Source | Effect |
|---|---|
| Fourteen technologies of eras 6–12, mostly military, from Bombing Aircraft and Mass Media to Orbital Weapon Platforms | +0.5 to +2 a week each in plays you start; three also add 10% |
| Total War (Rules of War law) | +25% in plays you start and plays against you |
| War Crimes Forbidden, Humanitarian Regulations, Limited War (Rules of War laws) | −10%, −10%, −30% in plays you start and plays against you |
| Vassalization V and Aggressive Coordination V principles | +0.5 and +1 a week in plays you start |
| Defensive Cooperation V principle | −20% in plays against you |
| UN membership, a Security Council seat, permanent membership | Small reductions; see The United Nations |
| Rising Global Tensions and Home Front Strain | +1 and +2 a week in plays you start, in the run-up to and during a world war; see Ideological tension |

A country with all of these technologies reaches the cap, so plays it starts escalate about two and a half times as fast as in the base game, leaving the defender little time to gather support. The mod adds no escalation events of its own; Nuclear Brinkmanship, below, fires during plays between nuclear-armed countries.

## 9.3 International relations events

Major and great powers roll monthly for a pool of Cold War events. The more of them a country qualifies for, the likelier one fires, up to about one chance in ten a month. Most offer a move against a rival of major power rank or greater (a buildup, a proxy insurgency, propaganda, an embargo or a planted agent) or restraint. The others are a great-power summit, a space milestone to celebrate, and Nuclear Brinkmanship when both sides of a play hold nuclear weapons. With Covert Warfare on, its operations replace the proxy war, propaganda and espionage events (see The covert operations). Each event, what it needs and its choices are in International relations event list.

### Rival-choice event chains

When an event says a rival has acted against you, the rival chose to. Plans on the Table, The War of Words and An Agent in Place go to the rival first, and you hear only if it acts; your answer goes back to it. The chains are in Rival-choice chain list.

## 9.4 Irredentism and reunification

After Decolonization, countries that share a people but not a flag can press for reunion, by war or by consent. The AI does this through its Irredentist Pressure strategy.

### Reunification candidates and drivers

A country is a candidate for you when it shares one of your primary cultures, holds no homeland of a culture you lack, owns a homeland state of yours that you claim and where your people still live, and either borders you or has a coast while you have one too. You can't both be subjects of the same overlord.

Your government must also give the cause a voice: Ancestral Citizenship, Cultural Citizenship, Universal Citizenship or Council Republic, a governing interest group whose leader's ideology takes up the cause (jingoist, fascist, communist, market liberal, humanitarian, pacifist and several more), or the base game's Warlord Era journal entry. The ideology sets whether the AI prefers war or union; it never locks either path.

The Reunify the Nation war goal annexes a candidate outright. It needs enough irredentist pressure, which comes mostly from claims on the target's homeland states: several of them, or fewer when your government leans hard toward war or union, and more when you share a bloc with the target. Its infamy is what the base game charges for annexing that country: it grows with the size and wealth of the states taken, and it falls for states that are homeland to your culture, states you hold claims on and states you border. It isn't offered against countries with the base game's German, Italian or Chinese unification entries.

### Irredentist events

Two events come with a reunification candidate. "The Diaspora Calls" offers a claim, cultural ties that make a later union likelier, or a demand that the target cede a state or face your diplomatic play. "An Opportunity for Union" comes when relations are warm. Their options and costs are in Irredentist event list.

### A bloc leader's blessing

A bloc member that doesn't lead its bloc can't use the reunification war goal without its leader's blessing. Ask with the Seek Blessing for Reunification action, taken on the country you would fight. A blessing lasts five years and covers that one target. A refusal costs 10 relations with the leader and stirs radicals at home for five years, during which you can't ask again. A war on the bloc leader can't be blessed. AI members with warlike governments ask on their own.

### Voluntary union

Voluntary Union is a diplomatic action that merges one of your reunification candidates into your country. You need Decolonization, amicable relations, no truce and a higher rank than the target, which can't be another country's subject. Each side must be outside a bloc, lead one, sit in a bloc the other leads or in a Diplomatic Framework, or have its leader's blessing. The AI starts from a strong no and wants friendly relations, cultural ties or a powerful kinsman's protection before it accepts.

Integrate Decentralized Power, also from Decolonization, absorbs a decentralized nation without consent, if it borders you or you both have a coast. Until Globalization it works only on your own peoples; after it, on any. The AI absorbs at most one such nation a year.

## 9.5 Power bloc principles and identities

The mod adds one identity, fifteen principle groups, and tiers IV and V to every base-game group. A bloc can hold five mandates instead of three.

Each tier's effects are the complete list for that tier: tier V replaces tier IV rather than adding to it. Compare tiers by their tooltips, not by summing them.

### Principle slots

A bloc can hold up to eight principles instead of four. As in the base game it starts with two slots, gains a third at rank 5 or better with five members and a fourth at rank 3 or better with ten. On top of those, each of four technologies the leader has researched adds a slot: Intergovernmental Organizations (era 6), Containerization (era 8), Globalization (era 9) and Universal Digital Identity (era 11). The slots are counted, not tied to positions, so a small bloc whose leader has two of the technologies fills its third and fourth slots with them. The panel shows all eight slots from the start and locks the ones above the bloc's count. Hover a locked slot to see the count it needs, the count the bloc has, and a tick or a cross for each rank condition and each slot technology.

The Power Bloc Principle Slots game rule is Up to Eight by default; set to Four, the technologies add nothing, a bloc keeps the base game's two to four slots and the panel shows four.

A filled slot stays filled if the count drops, when the bloc loses rank or a leader without the technologies takes over, but the bloc can't add a principle until it is back under its count. Upgrading a principle, or changing a slot to another principle, never needs a free slot. The leader can remove a principle with the cross on its slot; the mandates spent on it are not refunded. AI leaders keep to the same count.

## The Diplomatic Framework identity

Diplomatic Framework is a bloc built on collective security rather than control. Members get +10% influence; the leader gets +50% pact leverage generation and +25% infamy generation. Every member adds mandate progress, great powers most. Cohesion rests on the leader's legitimacy, the worst relations inside the bloc, and the infamy of the leader and the worst member. The leader can use Diplomatic Alignment, a pact that slowly aligns a country's relations with third parties to its own, and a Diplomatic Framework never blocks a member's voluntary union.

## New principle groups

Foreign Service and Multilateral Institutions are open only to a Diplomatic Framework, and Artistic Expression only to a Cultural Commonwealth. Vassalization, Colonial Offices and Sacred Civics are closed to a Diplomatic Framework. Urban Planning and Rural Development exclude each other.

| Group | First tiers need | What it does |
|---|---|---|
| Foreign Service | Nothing | Influence, faster relations, cheaper sway. |
| Multilateral Institutions | Intergovernmental Organizations | Diplomatic reputation and UN alignment; at tiers IV and V members can't fight each other. |
| Global Security | Nuclear Weapons | Less radicalism and turmoil in members, more maneuvers for the leader. |
| Education | Nothing | Cheaper schools, education access, technology spread. |
| Healthcare | Pharmaceuticals | Cheaper health ministries, recovery and birth rate. |
| Welfare | Nothing | Welfare payments, lower living-standard expectations. |
| Cultural Unity | Nothing | Faster assimilation and homeland change. |
| Cultural Plurality | Nothing | Cultural acceptance, migration, technology spread. |
| Artistic Expression | Nothing | Fine art, arts services, academics' clout. |
| Military Training | Nothing | Army experience, mobilization speed. |
| Engineering and Logistics | Nothing | Supply, entrenchment, convoy protection. |
| Environmental Sustainability | Environmental Movement | Less coal and oil use, fewer emissions. |
| Urban Planning | Urbanization | More room in crowded cities, better urban services. |
| Rural Development | Nothing | Arable land, farm throughput. |
| Monetary Union | Central Banking | A shared currency; see Banking and monetary policy. |

## Principle tiers IV and V

Every base-game group, from Construction to Maritime Supremacy, gains tiers IV and V, and the new groups' last tiers work the same way. Each needs the bloc leader to have researched a particular technology, all but one of them from the mod's eras. Mass Media opens tier IV of ten groups, Combined Arms of the four military ones, Globalization tier V of six. The principle's tooltip names its technology.

## Principles that unlock articles and actions

Several principles give your bloc an article, action or decree.

*Figure 12: The Change or Upgrade Companies V window: every principle group laid out in tiers I to V, with Companies V selected and five mandates available.*

| Principle | Unlocks |
|---|---|
| Global Security, any tier | Crisis Resolution |
| Multilateral Institutions I+ | Require UN Membership |
| Multilateral Institutions III+ | Development Assistance |
| Foreign Service IV and V | Extend Influence |
| Education IV and V | Education Aid |
| Healthcare IV and V | Healthcare Aid |
| Police Coordination IV and V | Security Aid |
| Advanced Research IV and V | Modernization Aid; V adds Extensive Modernization Aid |
| Welfare IV and V | Humanitarian Aid, a pact that sends part of your income to another country's poor in return for leverage and better relations |
| Vassalization IV and V | Peaceful Integration: annex a puppet or colony with under 25 liberty desire, cordial relations and under a tenth of your GDP, without a play |
| Cultural Unity II+ | Enforce Cultural Acceptance: a member of five years or more adds your primary cultures to its own, if the bloc has 50% cohesion and you have three times its prestige |
| Cultural Unity V | Enforce Cultural Adoption: a subject replaces its primary cultures with yours |
| Cultural Unity, any tier | Cultural Emigration Initiative decree |
| Freedom of Movement IV and V | Greenest Grass Campaign decree |

## Other power bloc changes

A bloc holding unspent mandates gains cohesion from a Mandate Reserve, set once a year: +3 per unspent mandate, up to +12. Example new bloc names include the Global Accord for a Diplomatic Framework, the Anglosphere for a cultural bloc and the Concert of Arms for a Military Treaty. Any bloc can take the name of a historical league or standing body, such as the Congress System or the Central Commission, or a name from its founder's capital city, such as the Berlin Pact or the League of Vienna. An AI country never gives its bloc one of the base game's names that fit any country, such as the Grand Coalition or the United Front; it

picks from these, its bloc identity's names and those tied to its own region, culture or government. You can still choose those names for your own bloc, but a bloc that passes to an AI leader loses one when it is renamed for its new leader.

Joining a bloc gets easier with technology. In the base game a leader needs a Leverage Advantage of 200 over a country to invite it, and the same to accept its request to join. The mod lowers that with the bloc leader's technology: to 100 with Intergovernmental Organizations and to 50 with Globalization. The Request Influence article and the mod's monetary articles (see Monetary treaty articles) build leverage toward it.

With Intergovernmental Organizations, a bloc leader can build one Power Bloc Headquarters, a government building with a very high construction cost. What it does depends on the bloc's identity: authority and cheaper decrees for a Sovereign Empire, military industry throughput and faster training for a Military Treaty bloc, trade advantage and capacity for a Trade League, and so on. Its production method lists the effects.

Subjugate, the leader's action that makes a bloc member a protectorate, is open to every identity. It needs a Subjugation Strength of at least 1.0, which means roughly that your prestige is twenty times the target's: ten times if your bloc is a Sovereign Empire, and about twelve times when your identity's bond applies (shared religion, governance principle or culture, economic dependence, or an army five times the target's). Its infamy grows with the target's population, up to 25; a Sovereign Empire pays half as much, so it reaches that cap only against far more populous targets.

## 9.6 Formable countries

The mod adds ten formable countries. All need Decolonization and form through the base game's leadership and unification plays, with up to three candidates. A candidate needs major power rank and, except for United Earth, a capital in the region. The regional unions also need a world that pushes them together: a top-three great power outside the region, every major power of the region allied to you, in a defensive pact or customs union with you, in your bloc, or your subject, and no rivalry or war with any of them.

| Country | States needed | Also needs |
|---|---|---|
| African Union | 50% | Pan-nationalism |
| East African Federation | 65% | |
| European Union | 50% | Pan-nationalism |
| Intermarium | 65% | |
| North American Union | 65% | |
| Dar-Al-Islam | 50% | Pan-nationalism; a Sunni, Shiite or Ibadi state religion, without Total Separation or State Atheism |
| United Earth | 75% | Quantum Communications and Space Colonization; the United Nations at the Supranational tier (with the United Nations game rule off, no great power outside your bloc instead); you lead a bloc holding every great power but at most one, which must be aligned with you; no rivalry or war with any great power |
| India | 65% of its core | |
| Indonesia | 65% of its core | |
| China | 65% | |

India, Indonesia and China are nations the base game can already form, but there one country forms each by holding enough of its states. From Decolonization on, the mod's versions are major formations like the unions: major powers with a capital in the region can be candidates and compete through leadership and unification plays. They need none of the unions' world conditions, so in a world of former colonies these three come together far more often. Once Decolonization is researched, the base game's India and Indonesia formations leave the list and these replace them. While a base-game India or Indonesia exists, the new one can't be unified.

### Government-flavored country names

Each formable renames itself with its government. The African Union becomes the Pan-African Socialist Federation under communism, the African Imperium under fascism, the African Empire as a monarchy, and a theocracy or technate otherwise. Some names follow the ruler's culture or the country that formed it: the Empire of the Great Qing for a China with a Manchu monarch, the Ottoman or Safavid Caliphate, or Pax Americana for a United Earth formed by the United States.

### India, Indonesia and Operation Polo

Forming India claims the Pakistani, Bangladeshi, Himalayan and Ceylonese periphery; forming Indonesia claims Malaya, northern Borneo and eastern New Guinea. A Muslim-ruled state inside India's core isn't swept up in unification. While an independent, Muslim-ruled insignificant power borders India, "An Island of Resistance" can fire and lets India annex it by police action (20 infamy), pressure it into accession as a major power or greater (8 infamy), or leave it be.

## 9.7 Releasable city-states

The mod adds 22 city-states you can release, from New York and Singapore to Constantinople, Lagos and Guayaquil. Each is a port city carved out of a larger state. They work like the base game's free cities, such as Hamburg and Danzig: releasing one hands over only the city's own province (New York also takes Long Island), so the state splits and you keep the rest of the region as a smaller state. Whoever holds the city releases it from the Releasable Countries list on the Diplomacy panel, and another country can demand it in a diplomatic play with a Liberate Country war goal. The AI never releases one on its own.

Most of these cities sit on one of their state's hub provinces. A split state shares the region's arable land and resources out by province, with hub provinces counting for more, so a city-state takes a larger share than its one province suggests.

Some start in a colonial power's hands: Britain holds Singapore, Portugal holds Macau and Goa, and France holds Pondicherry, each as a small holding of its own, so they can be released from the first day. Hong Kong is the province a treaty port in Eastern Guangdong takes, so a power that wins it as a treaty port can later release it. Constantinople holds the European end of the Bosphorus strait. San Francisco appears only while the Releasable Nations game rule is set to All, the default, as with the base game's less plausible releasables.

The full list, with the state each city-state comes from and who holds it at the start, is in the appendix.

## 9.8 Country ranks and other diplomatic actions

The mod doesn't change rank thresholds. Great powers gain +25% cultural pull, two covert operation slots and +10 intelligence capacity; major powers +10%, one slot and +5. Each rank also sets part of your credit standing, and so what you pay to borrow (see What your government pays to borrow).

Colonial Culture Change lets an overlord with three times a colonial subject's prestige replace the subject's primary cultures and religion with its own, at +20 liberty desire. Administer Seabed Claims lets an overlord take its subject's unused seabed claims while it stands (see Leasing seabed claims). Covert operations are in Cultural hegemony and covert warfare, nuclear warnings and strikes in Nuclear weapons, and UN lobbying in The United Nations.

CHAPTER 10

# The United Nations

One great power founds the United Nations; the rest of the world joins, ignores or works against it. The journal entry appears once you research Intergovernmental Organizations, an era 6 society technology, and for every country once a UN exists. It shows UN Authority (how seriously the world takes the organization), the Security Council of five permanent members with a veto, the conventions in force, missions in individual states and the dues members pay. It also holds the General Assembly, where you vote and table resolutions. The same panels appear as a UN tab in the Diplomacy panel (see The UN panels). The United Nations game rule turns the system off.

## 10.1 Founding the United Nations

A great power that has researched Intergovernmental Organizations and is not a subject can press Found the United Nations, under the overview's Membership cell, while no UN exists. Founding uses 500 influence, fading over five years, and needs that much available. The founder becomes the first member, a founding member, the headquarters host and the first permanent member of the Security Council, and UN Authority starts at 50.

Every other country that conducts its own foreign policy then receives The United Nations Charter:

| Option | Result |
|---|---|
| Sign the Charter | You join as a founding member (UN Founding Member: prestige and leverage generation). A signer without Intergovernmental Organizations is a founding member too, from the moment its UN journal entry opens. |
| We shall observe, but not yet commit | UN Observer Status for ten years (+5% relations improvement speed). You can join later. |
| This undermines our sovereignty | UN Rejectionist for five years (−10% relations improvement speed, +50 Authority, the government resource rather than UN Authority). |

Signing seats nobody on the Security Council; seats go by prestige (see The UN Security Council). The host may build one United Nations Headquarters, which raises influence, prestige and society research speed and adds +1 to the UN Authority Target (see What counts toward Policy), but adds 10% infamy generation. While the UN is above Moribund, the host's covert networks in other members grow 25% faster. If the host leaves or loses its representation, the headquarters passes to another member and the old building is demolished.

### Joining and leaving the UN

Any country that conducts its own foreign policy may press Join the United Nations, under the overview's Membership cell: an independent country, or a dominion, protectorate or tributary. Puppets, vassals, colonies, personal unions, crown lands, chartered companies and decentralized nations cannot. No technology is needed.

Members get UN Membership Privileges at once: +10% relations improvement speed, +5 leverage generation, +3% prestige, +3% influence and slower escalation when defending in a diplomatic play. From the Contested tier they also get UN Membership Benefits, scaled by authority ÷ 50 (at 50: +10% relations improvement speed,

+1% research speed, +3% prestige). Joining binds you, without a choice, to every convention already in force, and makes you pay UN dues.

Leave the UN, which asks you to confirm, ends your membership, seat, programs and conventions. It costs 5 standing and gives UN Withdrawal Consequences for five years (−25% relations improvement speed, −5% prestige), during which you cannot rejoin. Sanctions against you stay in force, and so do unpaid dues. Sanctions you enforce lapse the month after you leave. A permanent member walking out knocks authority down at once (4 points for a typical great power, up to 8).

When a revolution wins, the mod carries the nation's membership, seat, conventions and any UN sanctions on it over to the winner and rebuilds them in the following months (see After a revolution).

### Subjects and suspended representation

A member that stops conducting its own foreign policy, for example by being made a puppet, keeps its membership but has its **representation suspended**. It casts no vote, tables nothing, holds no permanent seat, cannot host the headquarters and runs no UN programs. It pays no dues itself: if its direct overlord is a represented member, the overlord is assessed on the subject's GDP too, and while the overlord pays, the subject keeps its membership benefits. Representation returns as soon as the country conducts its own foreign policy again, and Conventions Passed in Our Absence then offers it the conventions adopted meanwhile, all or none (the nuclear ban only to a member holding no warheads). Countries that cannot join are never treated as pariahs, and an annexed member's seat simply ends.

### The Require UN Membership treaty article

A power bloc leader that is a UN member, and whose bloc holds a Multilateral Institutions principle, can add Require UN Membership to a treaty with a non-member that has researched Intergovernmental Organizations. The target joins at its next monthly update if it conducts its own foreign policy. The AI resists it under Isolationism and refuses it under Total War. See Diplomacy.

## 10.2 UN Authority

UN Authority runs from 0 to 100 and is the bar at the top of the UN panels. Each month it closes a 48th of the gap to a target, at most 1 point, so a change in the world shows over years. The target is the sum of eight pillars:

| Pillar | Range | What moves it |
|---|---|---|
| Base | 15 | Constant. |
| Participation | 0 to +25 | The share of world prestige held by members. |
| Commitment | −25 to +25 | Members championing (+1) or undermining (−1) the order, weighted by their share of world power. Members of a bloc with Multilateral Institutions count a little in favor. |
| Credibility | −25 to +25 | Resolutions carried (aid, peacekeeping and observer requests among them) or failing, vetoes, members backing or defying the UN in events, lifted sanctions, abused mandates, World Court rulings obeyed or defied, ceasefire orders and referendums defied. |
| Funding | −10 to +10 | The power-weighted share of major and great power members running peacekeeping, human rights and arms control programs, plus the money members give the World Development Fund (0.7% of the members' GDP a year counts in full), minus up to 15 for dues withheld. |
| Peace and order | −20 to 0 | Members at war with fellow members, nuclear use, and aggressors fighting on under a ceasefire order. |
| Delivery | −10 to +10 | Aid and peacekeepers delivered, emergency loans, mandates discharged, missions accomplished less missions failed, ceasefires kept, referendums held and grain the World Food Reserve sends: 0 to +10. While the World Food Reserve stands, hunger among members adds −10 to +3. |

| Pillar | Range | What moves it |
|---|---|---|
| Policy | −25 to +25 | The UN Authority Target modifier carried by laws and institutions, with each country's total scaled by its weight. A member that is not undermining the UN counts for or against; any other country counts only against. A negative total pulls authority down. See What counts toward Policy. |

Credibility, delivery and the nuclear half of peace and order are kept as ledgers: each act adds or subtracts points. Credibility records votes and whether the UN is obeyed; delivery records what it gets done, so a failed mission takes delivery away. Credibility entries halve every ten years; delivery entries and nuclear use halve every four. The delivery ledger holds no more than its pillar shows, 0 to 10, so points past +10 are not banked for later. Acts by powerful countries count for more. Each entry is multiplied by the actor's **weight**, its share of world prestige against a typical great power's 10% (×1), up to ×5; the Policy row scales each country's modifier the same way. A permanent member walking out and a nuclear first strike also knock authority down directly. A new UN starts at 50, but with empty ledgers, no champions and no programs its target sits well below that, so expect authority to fall in its first years unless great powers champion it and run programs.

Policy counts a country's UN Authority Target in full only while the country is a member and is not undermining the UN. A non-member, or a member undermining it, counts only when its total is negative: its harmful policies pull authority down, and its good ones can only offset them. Leaving the UN, or pressing Undermine the Order, stops a positive total from counting. Hover the Policy row to see your own figure, your weight and what you count for.

## What counts toward Policy

These carry the UN Authority Target modifier. Your total is multiplied by your weight, so for a typical great power each point is one point of target.

| Source | UN Authority Target |
|---|---|
| Rules of War law | Total War −2, Traditional Rules of War 0, War Crimes Forbidden +0.5, Humanitarian Regulations +1, Limited War +1.5 |
| Isolationism (trade policy) | −1.5 |
| International Oversight (Frontier AI Control law, members only; see Laws on state power and information) | +0.5 |
| Ministry of International Aid | +0.15 for each level of investment (+1.35 at the cap of 9) |
| Nuclear doctrine | No First Use +0.5, Existential Deterrence 0, Flexible First Use −0.5, Nuclear Compellence −1, Nuclear Warfighting −1.5 |
| The Burden of the Bomb | Up to −1, in step with your arsenal's burden |
| United Nations Headquarters (the host) | +1 |
| Peace Palace | +0.5 |
| Palais des Nations | +0.5 |
| Party to the Antarctic Treaty | +0.25 |

The positive entries count only while you are a member and are not undermining the UN; the negative ones count whatever you do. A member great power under Limited War, with a funded Ministry of International Aid and a No First Use doctrine, holds about +2.5 to +3.4. A great power under Total War and Isolationism holds −3.5 before its nuclear posture. The AI does not weigh this modifier when it chooses laws or doctrine.

## UN authority tiers

Authority sets the tier, and the tier sets the UN's **enforcement**: the multiplier on every penalty the Assembly imposes and on every convention's effects. A tier is entered at its floor and left 4 points below it.

*Figure 13: The Why UN Authority Is Moving section in the UN tab of the Diplomacy panel. Authority is 68.1 and rising toward a target of 72.4, the sum of the pillars; below come the tier and the charter and the champions and underminers, with Recent Entries collapsed at the foot.*

| Tier | Authority | Enforcement | Dues (GDP a year) | What changes |
|---|---|---|---|---|
| Moribund | below 20 | ×0 | none | Resolutions are recommendations, vetoes cost only relations, members lose UN Membership Benefits, and power blocs gain cohesion and leverage (Vacuum of World Order). |
| Contested | 20–45 | ×0.5 | 0.1% | Membership benefits are paid. |
| Established | 45–70 | ×1 | 0.2% | Countries that could join but stay out become International Pariahs. Each war goal a play's initiator adds costs 2 extra infamy, except in the play a mandate is used in. Peacekeeping requests send full deployments. |

| Tier | Authority | Enforcement | Dues (GDP a year) | What changes |
|---|---|---|---|---|
| Strong (needs Charter Reform I) | 70–85 | ×1.5 | 0.4% | Outsiders also lose trade advantage and leverage. Sanctions become embargoes, condemned countries are Shunned, the surcharge rises to 4, members share intelligence, and nationalist interest groups resent the UN. With the IAEA, members without the bomb are held to disarmament. |
| Supranational (needs Charter Reform II) | 85+ | ×2.5 | 1% | The surcharge rises to 10, condemned countries are also Restrained, and outsiders of major rank or with a nuclear program carry a standing case of 50, as does a member outside the Non-Proliferation Treaty with a nuclear program and no bomb, or a bomb built outside it. Forming United Earth requires this tier. |

From Strong, the surcharge doubles against a country hosting UN peacekeepers, your patriotic, jingoist, isolationist and sovereignist interest groups lose approval, and those led by humanitarians or pacifists gain it (twice as much at Supranational). If a resentful group is powerful and in government, or two are powerful, a yearly check can send Sovereignty First, at most once a decade: leave, buy their patience with concessions, or stay and anger them.

## UN charter reforms

The charter caps the target at 70, so the founding charter allows no more than Established. Charter Reform I, Standing Mandate Force and Compulsory Jurisdiction, raises the ceiling to 85. Charter Reform II, Veto Restraint and the UN Levy, raises it to 100 and restrains the veto: a binding resolution other than a charter reform that two thirds of the members with a vote carry is no longer stopped by a veto. Each reform also opens new business before the Assembly and strengthens the World Development Fund (see Business the reformed charter opens).

A reform is ripe once authority has held within 5 points of the ceiling for 24 months running; Why UN Authority Is Moving counts the months. A member of major-power rank may then table it, and the docket may offer it to a great power (The Charter Has Been Outgrown). It needs two thirds of all members with a vote, any permanent member can veto it outright, and the next reform cannot be tabled for five years.

## The UN crisis and dissolution

Below authority 10 the UN is in crisis until authority climbs above 20. Every great power then receives The United Nations in Crisis: members can stand by the organization (−10% influence for five years, credibility +2 × weight), outsiders can join to save it, and anyone can wait and see or let it go (+5% influence for five years, credibility −2 × weight). Why UN Authority Is Moving lists every great power that could lift the target, and by how much. If the target falls below 5 during the crisis, authority falls at least a quarter point a month, and at 0 the UN dissolves.

Dissolution ends every membership, seat, program, convention, agency, sanctions regime, mandate and mission, demolishes the headquarters and writes off unpaid dues. Every power bloc gains The UN Has Fallen (+20 cohesion, +20% leverage generation, fading over ten years). Twenty years later any great power with Intergovernmental Organizations can convene a founding conference with Convene a Conference, in the same place, at the founder's influence cost. The great power with the most prestige, if it is not the convener, receives A Founding Conference and may join, stay out or oppose; opposing wrecks it and blocks another for ten years. Otherwise the UN is refounded after twelve months at authority 25, under the founding charter and with no agencies.

## 10.3 The UN Security Council

The Security Council has five permanent seats. The founder takes one. The other four stay open for a twelve-month signing period, then go to the great-power members with the most prestige, never to whoever joined first. A seat that falls vacant later is filled the same way the next month. The overview shows the permanent members' flags; hover Security Council for the candidates in line.

A seat brings leverage generation, +250 Authority, prestige, influence and slower escalation in diplomatic plays. It is lost by leaving the UN, by ten continuous years below great-power rank, by a motion to expel, or by suspended representation. A member two years behind on its dues cannot be seated.

### Vetoes in the Security Council

A permanent member may veto one of the twelve binding topics, from the vote event or the chamber. The veto kills the binding form; if the Assembly still carried the resolution, a weaker form applies:

| Binding topic | If vetoed but carried |
|---|---|
| Condemnation of Aggression | A non-binding rebuke. |
| International Sanctions | Voluntary partial sanctions, with no enforcer. |
| Request a Peacekeeping Deployment | Observers only: five years of better relations and prestige for the requester, not a mission. |
| International Criminal Court | A symbolic censure of the vetoer; no court. |
| Charter Reform | Blocked outright. |
| Authorized Military Mandate | Blocked outright. |
| Refer a Dispute to the World Court | An advisory opinion: the court rules, but nothing changes hands; an opinion against the holder costs it 2 standing. |
| Arms Embargo | A voluntary embargo at half strength, for five years. |
| Establish the Standing UN Force | Blocked outright. |
| Binding Ceasefire | A call for a ceasefire: one drop in the aggressor's war support. |
| Supervised Self-Determination Referendum | Blocked outright. |
| Lift the Sanctions Against Us | Blocked outright. |

After Charter Reform II, a resolution that two thirds of the members with a vote carry overrides the veto and takes its full form. This covers every binding topic, military mandates included, except Charter Reform itself.

A veto costs credibility (1.5 × your weight), Diplomatic Isolation After Veto for five years, −5% influence for ten, 25 relations with the proposer, and 3 infamy when it blocks a condemnation, the ICC or a peacekeeping request. At Moribund it costs only the relations. While you carry the isolation, any member may table a Motion to Expel a Permanent Member against you: it needs two thirds of the members with a vote and cannot be vetoed. If it carries you lose the seat and 5 standing, stay a member, carry UN Withdrawal Consequences for ten years, and lose your standing benefits for five.

Each veto outside Moribund also adds 5 to your **veto record**, which fades by 5% a year. A motion to expel is weighed on its own case, not the case that decides condemnations and sanctions: your veto record, −10 for Exemplary standing or −5 for Respected (+5 for Poor, +10 for Disgraced), and −5 each for hosting the headquarters, running the development program, and running the peacekeeping program or serving in a mission (−10 at most). Every other member adds that case to its lean, and the habit of consensus does not apply. The seat goes to the great power with the most prestige that holds none, which can be the proposer; members lean 5 against a proposer that would take it. Once you have vetoed, Our Record shows your veto record and the case for expelling you.

## 10.4 Resolutions in the General Assembly

The floor takes one resolution at a time, and each is open for a year. The proposer's own vote counts in favor. Human members can vote in the chamber at once and receive the UN General Assembly Vote event after 30

days. Its "We will decide later." option closes the event without voting, and it comes back in the ninth and eleventh months of the session. AI members vote late: each votes in the ninth, tenth or eleventh month, drawn at random, by its lean at that time. An AI member that still has not voted (one that joined late, for example) votes in the last month of the session. After a year, General Assembly Vote Results applies the outcome.

Most topics pass by **Majority**: the votes in favor outnumber those against. Charter reforms, motions to expel and a suspension of credentials need **Two-Thirds**: two thirds of all members with a vote, so abstaining counts against them. A vote in favor gives +15 relations with the proposer, a vote against −15. On a resolution that accuses a country (a condemnation, sanctions, a mandate, a motion to expel, an arms embargo, a ceasefire, a suspension of credentials, a referral to the World Court or a referendum), voting in favor also costs 15 relations with the target and voting against gains 15. When a power bloc leader's resolution carries, its bloc gains 3 leverage in each of its other members that sits in the Assembly; when it falls, the bloc loses 3.

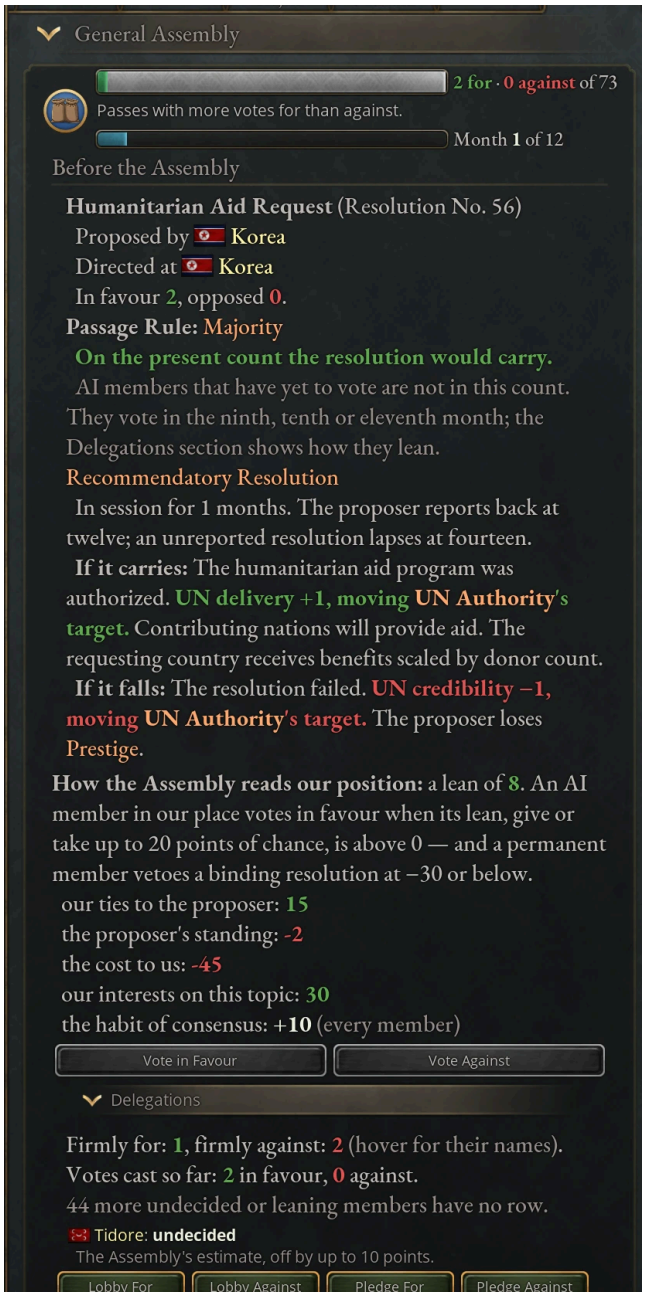


*Figure 14: The General Assembly with a Humanitarian Aid Request in session, in its first month with two votes for and none against: what happens if it carries or falls, how the Assembly reads our position term by term, the vote buttons and the start of Delegations.*

## General Assembly topics

There are thirty-four topics. Twelve are binding and can be vetoed: Condemnation of Aggression, International Sanctions, Lift the Sanctions Against Us, Request a Peacekeeping Deployment, the International Criminal Court, Charter Reform, Authorized Military Mandate, Refer a Dispute to the World Court, Arms Embargo, Establish the Standing UN Force, Binding Ceasefire and Supervised Self-Determination Referendum. The other twenty-two are recommendatory. Seventeen topics, the ICC among them, are conventions (see UN conventions and agencies). Nine open with the charter reforms (see Business the reformed charter opens). The other eight are:

| Topic | Who may table it | If it carries |
|---|---|---|
| Condemnation of Aggression | A member with a rival that started a war it is still fighting and has a case of 30+ | Condemned for ten years: prestige, relations improvement speed and infamy decay, × enforcement. |
| International Sanctions | A major power, against a rival with a case of 50+ | Sanctioned: trade advantage, influence and prestige, × enforcement. The proposer enforces them until it presses Lift Sanctions or the Assembly lifts them; if the target voted for them, none apply and nobody enforces them. Sanctions whose enforcer leaves the UN or is gone lapse the next month. |
| Lift the Sanctions Against Us | A member under sanctions, for itself | The sanctions end, with the embargoes the Assembly imposed, and the enforcer may table sanctions again. |
| Authorized Military Mandate | See UN military mandates | A mandate for the proposer. |
| Request a Peacekeeping Deployment | A member at war or with a devastated state, for its own territory | A peacekeeping mission. Major-power members that voted and take part (a yes vote, or a no vote they then accept) contribute and pay for it, and cannot leave the Peacekeeping program for ten years; nobody else is asked. |
| Request Humanitarian Aid | A member with a state below 8 standard of living or devastated | An aid mission; major-power members that voted and take part pay for ten years. |
| Charter Reform | A major power, once the charter is ripe | The next reform. |
| Motion to Expel a Permanent Member | Any member, against a permanent member that vetoed within five years and is not its ally, overlord or subject | The seat is stripped. |

From Strong, sanctions also become embargoes: each major-power member with diplomatic relevance to the target that voted for them (at Supranational, every such member) loses 30 relations with it and, if relations are then Poor or worse and the pact can be made, embargoes it at its own influence cost. Dropping such an embargo by choice once its first year is out counts as busting the sanctions. A condemnation adds Shunned by the United Nations (−6 diplomatic reputation × enforcement) from Strong, and Restrained by the United Nations (fewer play maneuvers, more infamy) at Supranational.

Members weigh a petition to lift sanctions by how far your case strength has fallen below the 50 that supports them, and are slow to lift sanctions that have stood less than five years. The members embargoing you vote against it, and so does the enforcer while it is still your rival or its relations with you are Cold or worse; only then will an AI permanent member veto it. Your allies and bloc partners lean for it and your rivals against. The Propose a Resolution row names the enforcer and how many months the sanctions have stood. A failed petition bars your next motion as any other does, so wait until your record has cleared.

## Grounds for UN censure

Punitive topics need grounds. Every country has a **case strength** from 0 to 100 built from its record:

| Record | Case |
|---|---|
| War begun, other than a mandate's own | 10 to 30 (more against a greater power) |
| Binding resolution refused | 10 |
| Sanctions busted | 8 |
| Court ruling defied | 6 |
| Mandate abused | 20 |
| Severe covert operation exposed | 15 |
| Nuclear first strike | 40 |
| Tactical nuclear strike | 20 |
| Nuclear retaliation | 10 |
| Nuclear program referred under the Non-Proliferation Treaty | 20, or 30 for a signatory |
| Destructive anti-satellite strike | 15, or 30 for a party to the Moratorium on Destructive Anti-Satellite Tests |
| Hostile act in space (see Space fleets) | A colony destroyed 20, a free colony attacked 10 (twice each for a party to the Convention on the Settlement of Outer Space); an interdiction begun 10 |
| Unilateral Licensing as a party to a Law of the Sea that governs the seabed | 10 |
| Antarctic Treaty denounced | 10 |
| Infamy | 0.8 per point, up to 50 |

Each entry halves every five years. A condemnation needs 30, an arms embargo or a binding ceasefire 40, sanctions or a suspension of credentials 50, and a military mandate 60, so a clean record cannot be censured. The Our Record section shows your own case, each part of it, and whether it would support a condemnation, sanctions or a mandate against you today.

## How members decide their UN votes

Every ballot comes from one number, the member's **lean** on that resolution. The General Assembly prints your lean term by term under the resolution, and the Recorded Ballot shows every voter's lean and why the members voted as they did.

| Term | Value |
|---|---|
| The habit of consensus | +10 (0 on a motion to expel) |
| The case against the target | Half of how far the target's case clears the threshold, −20 to +30; on a motion to expel, the case for expulsion, whole |
| It is our own censure | −50 (−30 from Strong) |
| Our ties to the proposer | Alliance +20, same bloc +20, rivalry −30, diplomatic relevance +5, relations up to ±10; −5 on a motion to expel that would seat the proposer |
| Our ties to the target, on accusing topics | Alliance −60, same bloc −35, rivalry +30, relations up to ±10 |
| Our own record (glass houses) | −10 with a case of 30, −20 at 50 |
| The proposer's standing | Exemplary +5, Respected +2, Poor −2, Disgraced −5 |
| A pledged vote | +100 for, −100 against |
| The target accepted the verdict | +15 |
| The cost to us, on aid and peacekeeping requests | −15 per earlier contribution, up to three; more for aid to a great, major or richer country; −30 for a major power in default or near its debt ceiling, which a carried request would bill |

| Term | Value |
|---|---|
| Our interests on this topic | Laws, technologies and what the convention's terms would do to us. On the World Development Fund: +30 below its line and up to +30 more for the grant it would pay us; the richest −5 and laissez-faire governments −5 under the founding charter, when it takes nothing from the budget; once a charter reform gives it a share, the richest −25, those above the average −10, laissez-faire −15 |
| Lobbying campaigns on us (AI members only) | 3 a month per campaign, up to 15, more from a lobbyist with UN Lobbying Effectiveness; at most 20 each way |

An AI member votes in favor when its lean, plus a random −20 to +20, is above 0. An AI permanent member vetoes a binding resolution at a lean of −30 or below (−50 if it vetoed recently, −10 at Moribund), leaving any pledge out of the count, and never when it pledged to vote for. A pledge against therefore never makes a member veto, and never stops one that would have. Lobbying campaigns do count, so a campaign can push a permanent member toward a veto or away from one.

## Reading how the members lean

The Delegations list under the General Assembly shows each AI member's lean as a band, not a number:

| Band | Lean |
|---|---|
| Firmly for | +30 or more |
| Leaning for | +10 to +29 |
| Undecided | −9 to +9 |
| Leaning against | −10 to −29 |
| Firmly against | −30 or less |

The band is the Assembly's estimate. Each member's reading is off by −10, 0 or +10, fixed for the whole session, so a member shown as undecided may lean either way. While you run a campaign on a member, you see its true band. Votes still carry the random −20 to +20, so a close vote can turn on the day it is cast.

## Complying with a UN resolution

When a resolution carries, every member that voted against it may accept it or refuse. Refusing a convention (Refuse to ratify it) keeps you outside it, with neither its party terms nor its benefits, though any term that reaches every country still reaches you. Refusing an aid or peacekeeping request (Refuse to take part) means you contribute nothing. Refusing anything else (Denounce the decision) is a statement on the record that changes nothing. Refusing costs 20 relations with the proposer and 3 infamy, and for a binding resolution also 5 standing and 10 case strength. Accepting a binding resolution you opposed earns 2 standing, except a peacekeeping request, whose contribution is its own reward. A resolution that a veto blocked or softened binds no one, so accepting or refusing it moves no standing or case strength. A referral to the World Court and a supervised referendum bind only the country they name, which answers in its own event, and a petition to lift sanctions asks nothing of the members, so accepting or refusing these moves no standing or case strength. A member that voted in favor, or did not vote, ratifies a carried convention without being asked, except that a member holding warheads that did not vote stays outside the nuclear ban.

## Tabling UN business

You table the eight topics above, the nine the reformed charter opens and any convention from the Propose a Resolution rows, which sit under the General Assembly while no resolution is in session. Its heading says how many topics are ready, and its first line when your next motion may come. The topics with a case come first, under Ready to Table; the rest are under No Case Now, collapsed; a convention in force or an institution already founded is not listed, only counted. Each row gives the topic, whether it is a Binding Resolution (which can be vetoed) or a Recommendatory one, its passage rule and its target. "Target: None" means no country gives grounds right now; hover it for why. On a topic aimed at a country, Nearest names the candidate with the strongest case and the case strength it needs. Hover a topic's name for everything it would do, to its target, its proposer, the members and other systems, and for a convention the agency it founds and when the docket

raises it. On the resolution in session, Its Terms after "If it carries" shows the same. A grayed Propose button lists in its tooltip what stops you.

Tabling on your own motion uses influence for two years, and you need that much available to table:

| Topics | Influence |
|---|---|
| International Sanctions, Authorized Military Mandate, Motion to Expel a Permanent Member, Refer a Dispute to the World Court, Arms Embargo, Suspension of Credentials | 300 |
| Every other topic and convention | 150 |
| Request a Peacekeeping Deployment, Request Humanitarian Aid, Request Electoral Observers | None |

Tabling on your own motion also puts UN Request Cooldown on you, during which you can table nothing else yourself. It lasts ten years for an ordinary member. A great power waits half as long, and standing stretches or shortens the wait: 15% shorter at Respected, 30% shorter at Exemplary, 25% longer at Poor and 50% longer at Disgraced (never under one year). Suspended standing benefits count as Neutral. The sponsor mark that also bars a second proposal shortens the same way, from five years. A convention or charter reform the docket offers you is free, so the docket's offer is the cheap way to bring one to the floor. A topic cannot return to the floor for five years after a resolution on it closes (ten for a motion to expel), except that an appeal over a nuclear strike can table a condemnation despite that cooldown (see The UN docket), and mandates have a five-year cooldown per proposer. After every vote the floor is in recess for three months, when only human members may table, and you are told when it reopens.

## The UN docket

Most business reaches the floor through the docket. Once a month the UN takes stock of the world and, at most once every three months, takes up the gravest situation and offers it to the countries it concerns. Roughly from gravest down:

| Situation | Offered to | Event |
|---|---|---|
| A nuclear strike | The struck country first | An Appeal to the Assembly |
| The end of a year-long war between great powers | A proposer | Declaration of Universal Human Rights, or International Criminal Court |
| A war of aggression on a member, six months or more, with heavy devastation | The attacked member first | An Appeal to the Assembly |
| A warming threshold (0.5, 1, 2 and 3 °C) | A proposer | Climate Change Resolution |
| A country's first nuclear bomb | A proposer | Nuclear Non-Proliferation Treaty |
| A severe covert operation exposed | The target first | An Appeal to the Assembly |
| A destructive anti-satellite strike | A proposer, never the striker | A Moratorium on Anti-Satellite Tests |
| A colony in space seized or destroyed | A proposer that holds a colony, never the country that did it | A Convention on the Settlement of Outer Space |
| A state collapse | Up to three peacekeeping powers | A State Has Collapsed |
| The charter outgrown | A great power | The Charter Has Been Outgrown |
| A famine | Up to three donor powers | Humanitarian Crisis Demands Response |
| A banking panic or default in a contagion wave | The member in trouble | An Emergency Lending Facility |
| A colonial empire's collapse | A proposer | Decolonization Resolution |
| The first Moon landing or colony | A proposer | International Space Cooperation |

| Situation | Offered to | Event |
|---|---|---|
| A country's first Antarctic research station | A proposer | The Antarctic Treaty |
| A trade embargo between members | The weaker party | International Trade Dispute |
| Nothing graver, at most once in 24 months (12 while a great power member is Respected or better) | A proposer | A convention no situation raises |

An appeal lets the wronged party table a condemnation or sanctions, ask for peacekeepers, take the accused to the World Court, or pass the matter to a member with a stake, never one on the accused's side. An appeal over a nuclear strike can table a condemnation even while that topic is on its five-year cooldown, as long as the striker's case gives grounds. That vote is the Assembly's verdict on the use: a condemnation that carries strengthens the nuclear taboo, and one that fails, or that a veto cuts to a rebuke, weakens it. Sanctions or a World Court case over the same strike give no verdict (see The United Nations and the taboo). Business with no wronged party goes first to a human member that qualifies. Every proposer event has "Leave it to another delegation", which passes the item on and earns nothing; refusing a convention outright costs credibility and bars you from tabling it for five years. A major power that refuses a famine appeal while UN authority is 40 or more loses prestige and relations improvement speed for five years. Sending aid to a famine, or a full peacekeeping force to a collapsed state, costs 0.5% of GDP a year, fading over five years; a country already running Contribute to Peacekeeping pays nothing extra for the force. The force earns leverage, prestige and +10% army experience gain, fading over ten years. Observers or a token gesture cost a quarter of that. The lending facility's loan and conditions are covered in The UN emergency loan.

A trade dispute goes to the weaker side of an embargo between two members. It can accept the UN's arbitration, settle outside the UN by offering the other side trade concessions, or impose unilateral tariffs on it. Tariffs lower the other side's trade advantage, fading over five years, and sour relations between the two. The other side never sees the event: a notification names the country and says what its choice costs or gains you. None of the three ends the embargo.

## Lobbying for UN votes

Any member can try to move an AI member's vote, for or against the resolution in session, in two ways: a campaign, which costs influence and builds up over months, or a pledge, which costs an obligation and works at once. Because AI members vote only in the ninth to eleventh month, you have most of the session to work on them.

A campaign is a diplomatic pact, Lobby For the Resolution or Lobby Against the Resolution, started from the diplomacy panel or the Delegations rows. It uses 100 influence while it runs. Each full month it runs moves the member's lean 3 points its way, up to 15, so it needs five months to reach its full effect. Campaigns on the same side stack to 20, and campaigns on opposite sides cancel out. A campaign ends when the member votes or the resolution closes, and you can stop it at any time, which frees the influence and loses what it had gained. Each Delegations row carries the member's flag and its band, marked (est.) for the Assembly's estimate or (true) for a member you lobby. While you lobby a member, its row shows Stop Lobbying in place of its two Lobby buttons, and a campaign you are running never starts over. Only AI members can be lobbied, never the resolution's target, and you need a vote yourself.

Lobby Top Members For and Lobby Top Members Against, above the Delegations rows, start campaigns in bulk. One press puts a campaign on each listed member the Assembly doesn't yet read as leaning your way, from the top of the list down, until your influence runs out. Members you already lobby are skipped. The tooltip says how many campaigns your influence covers. A button is grayed if you have no vote, less than 100 influence, or no member on the list left to lobby.

Seek a Vote Commitment: For and Seek a Vote Commitment: Against ask a member to pledge its vote. If it accepts, you owe it an obligation and its lean moves 100 points your way. A member pledges once per resolution, and the first pledge it accepts stands. You can ask each member once per resolution and collect two pledges per resolution, and you cannot ask the target or a member that has voted. From the Delegations

rows (Pledge For or Against) an AI member answers at once; a human member answers a request from the diplomacy panel. A kept pledge gives the member +1 standing and +10 relations with you; a broken one costs it 4 standing and 20 relations and cancels your obligation. A permanent member that pledged against and then vetoes has kept its word.

UN Lobbying Effectiveness makes your campaigns move a member further and faster (+25% turns 3 a month up to 15 into 3.75 up to 18.75), and a member weighing your request for a pledge counts it in your favor, +20 for every 25%. Its acceptance tooltip shows the line. The members the World Development Fund pays give it to those who fund it.

## 10.5 Business the reformed charter opens

Nine topics beyond the seven above come before the General Assembly. Seven wait for a charter reform; the World Food Reserve and the World Development Fund need none, and the Fund grows with each reform. The Propose a Resolution list shows the business of the next reform before it is adopted, and says which reform

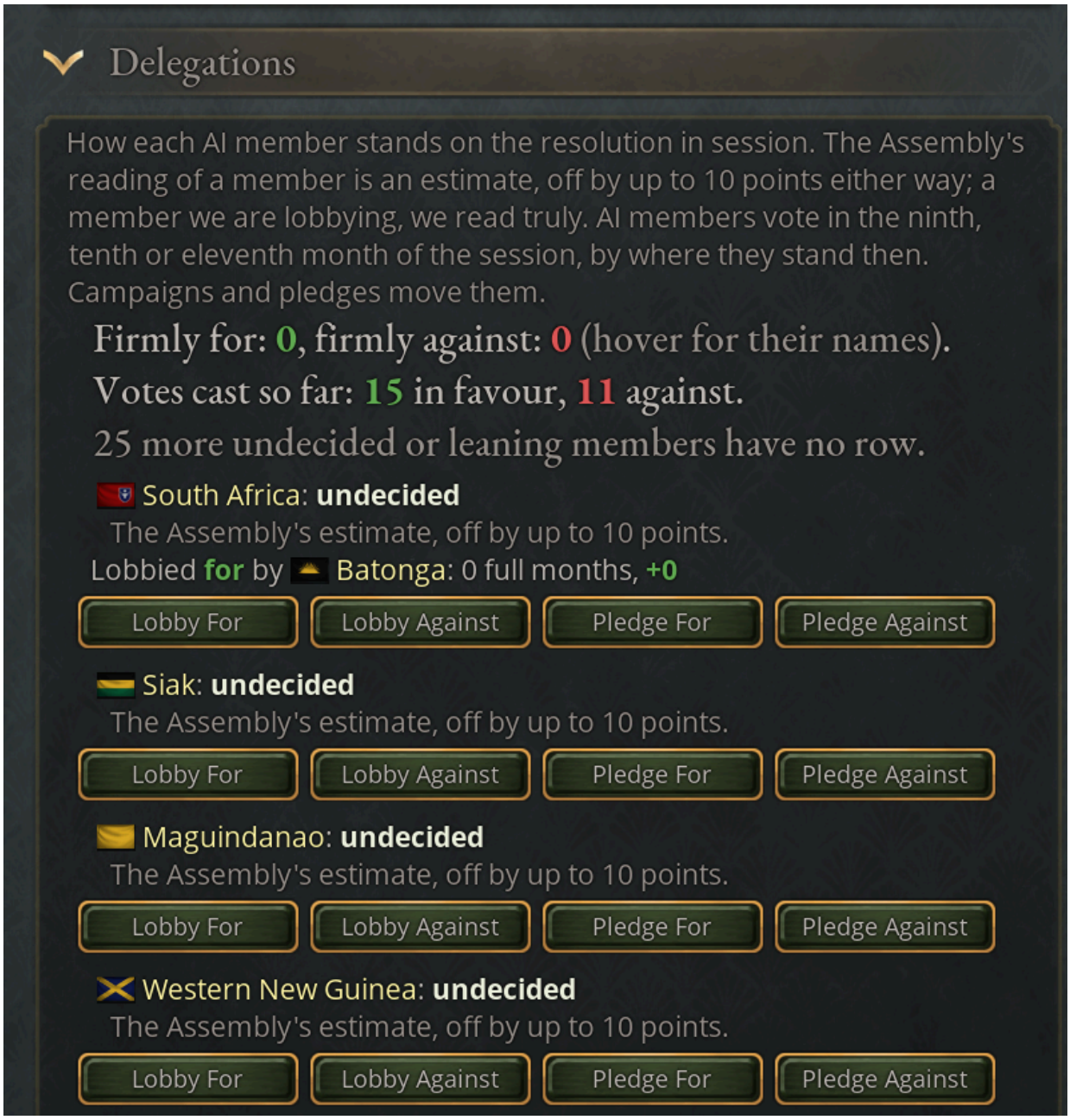


*Figure 15: The chamber's Delegations section. Each row shows a member's band as the Assembly estimates it and any campaign on it (Batonga is lobbying South Africa to vote for the resolution), with buttons to start a campaign or ask for a pledge.*

each row waits for. Charter Reform II's rows only appear once Reform I has carried. The AI tables these topics through the journal entry's buttons, as it does the older ones.

| Topic | Needs | Binding | Passes by | If it carries |
|---|---|---|---|---|
| Refer a Dispute to the World Court | Charter Reform I | Yes | Majority | The World Court rules on a state you claim. |
| Arms Embargo | Charter Reform I | Yes | Majority | A UN Arms Embargo for ten years. |
| Suspension of Credentials | Charter Reform I | No | Two-Thirds | The member's credentials are rejected for five years. |
| Establish the Standing UN Force | Charter Reform I | Yes | Majority | The Standing UN Force is founded. |
| Request Electoral Observers | Charter Reform I | No | Majority | An Electoral Observer Mission goes to your capital. |
| World Food Reserve | Nothing | No | Majority | The World Food Reserve is founded. |
| Binding Ceasefire | Charter Reform II | Yes | Majority | A ceasefire order against the aggressor for two years. |
| World Development Fund | Nothing (stronger after each reform) | No | Majority | The World Development Fund is founded. |
| Supervised Self-Determination Referendum | Charter Reform II and the Decolonization Resolution in force | Yes | Majority | The overlord must hold a referendum or refuse. |

Effects scale with the UN's enforcement like everything else the Assembly does, so a UN that slips back to Contested keeps this business but enforces it weakly. The charter only goes back to the founding one if the UN dissolves.

## Referring a dispute to the World Court

Any member with 20 standing or more can refer a state it claims to the World Court, if the state's holder is a member that is not its overlord, its subject or at war with it, and the state is not the holder's capital. The row picks your best case and shows the odds that the court finds for you, between 10% and 90%. Hover them for the terms:

| Term | Effect on your odds |
|---|---|
| Your primary cultures are half the state's people (a quarter) | +20 (+10) |
| The holder's primary cultures are half (a quarter) | −20 (−10) |
| The state is a homeland of yours, or of the holder's | +10, or −10 |
| The holder has incorporated it | −10 |
| Your standing against the holder's | ±0.3 per point of difference, up to ±15 |
| The holder's case strength, or yours | Up to +15, or up to −15 |

The court rules a month after the vote carries, on the odds as they stand then, so a court is less likely to hand a paragon's land to a pariah. If it finds for you, the holder chooses in The World Court Rules Against Us:

| Choice | Result |
|---|---|
| We abide by the ruling. | The state is yours. The holder gains 2 standing and the UN gains credibility. |
| The court has no say over our soil. | The holder keeps the state, loses 5 standing, takes a defiance record of 6 and International Court Defied, and the UN loses credibility. For ten years you may table a military mandate over that region against it without the case of 60. |

If the court finds for the holder, your claim on the region is withdrawn, and for twenty years each diplomatic play in which you take a war goal there costs you 10 × enforcement extra infamy, once per play. The case is

struck off as moot, and nothing changes hands, if by the time the court rules or the holder answers the holder no longer holds the state, has made it its capital, or either party no longer exists. A veto turns the ruling into an advisory opinion: the court still rules, nothing changes hands, and a holder it rules against loses 2 standing. The AI refers a case only when its odds are 60% or better.

## UN arms embargoes

An Arms Embargo can name a rival of yours, or a country at war with a member, whose case strength is 40 or more and that is not your ally, overlord or subject. If it carries, the target has a UN Arms Embargo for ten years: −30% trade advantage in small arms, ammunition, artillery, tanks, aeroplanes, man-o'-wars and ironclads, for imports and exports alike, and military goods 5% dearer, all × enforcement. It loses 6 standing. A vetoed embargo, or one the target voted for itself, or both, is a Voluntary UN Arms Embargo at half strength for five years and costs 3 standing. A country under an embargo cannot propose one.

## Suspension of credentials

The Assembly can reject the credentials of a member whose case strength is 50 or more and that breaks the Declaration of Human Rights' terms or has a defiance record of 10 or more, other than your ally, overlord or subject. A permanent member can be named only after Charter Reform II. It needs two thirds of the members with a vote, and no veto can stop it. For five years, or until its case strength falls below 30, the member is treated like a subject with suspended representation: no vote, no tabling, no benefits, no UN programs, no headquarters, and its permanent seat is gone for good. Unlike a subject, it still pays its own dues. It loses 8 standing, or 4 if it voted for its own suspension. If, when the vote closes, the member has left the UN or its case strength has fallen below 30, the motion is moot and nothing happens to it.

## Binding ceasefires

A Binding Ceasefire names a country with a case strength of 40 or more that started a war it is fighting against a member, and the member it attacked. If it carries, the aggressor is under a ceasefire order for 24 months. Nobody imposes a peace. Each month it fights on against a member:

- its war support in that war falls by 2.5 × enforcement;
- its aggression record grows;
- the UN's peace and order suffers, booked every six months.

After three months of fighting on it loses 5 standing (2 if it voted for its own censure). How the order ends decides what follows:

| Ending | Result |
|---|---|
| The war with the victim ends and the aggressor gained nothing from it | The order is kept. UN delivery rises, peacekeepers deploy to the victim's most devastated state, and an aggressor that stopped within three months gains standing for complying. |
| The war ends with the victim losing a state, becoming the aggressor's subject or losing its capital region to it | The order is defied. UN credibility falls and the aggressor earns nothing. |
| The order runs out with the war still on | The order is defied. UN credibility falls. |
| The victim stops being represented in the Assembly while the war goes on | The order has no effect and lapses quietly. |
| The victim is destroyed by someone else | The order ends with no penalty. |

A veto makes it a call for a ceasefire: one drop of 10 × enforcement in war support. Our Obligations shows an order against you, or against your enemy.

## Supervised referendums

With the Decolonization Resolution in force, a member can ask the Assembly to order a referendum in another member's direct subject whose liberty desire is 50 or more; a subject can ask for its own. A veto blocks it. If it carries, the overlord receives A Referendum Ordered:

| Choice | Result |
|---|---|
| Open the polling stations. | The subject becomes independent with a chance equal to its liberty desire, rounded down to 5, between 20% and 90%. Otherwise its liberty desire falls by 20. Either way, if the subject is still yours when you choose, you gain 4 standing and UN delivery rises. |
| We will not put our union to a vote. | The subject's liberty desire rises by 15. For five years you have Referendum Refused (less prestige, and every subject's liberty desire grows faster, × enforcement), you cannot ask for a referendum yourself, and you lose 5 standing and take a defiance record of 10. UN credibility falls. |

An overlord that voted for the referendum must hold it.

## The Standing UN Force

A major-power member can propose the Standing UN Force once. While it stands it serves in every peacekeeping and Stabilization mission as one more contributor, and those missions no longer lapse unstaffed or fail when the last contingent goes home. It costs a tenth of the UN's budget. At Supranational, a war goal a member adds against a country hosting such a mission, in a play it started, is struck as soon as it is added. Three kinds of goal are not struck but pay the doubled infamy surcharge instead: the goal the play opens with, another goal of that same kind, and a unification goal. Countries outside the UN, and members whose credentials are suspended, pay that surcharge on every such goal. Nothing is struck or charged in the play a mandate is used in.

## Electoral observers

A member that holds elections and is at peace can Request Electoral Observers. It is a request: no veto, and it shares the request cooldown. If it carries while you are still at peace and holding elections, an Electoral Observer Mission goes to your capital, paid from the UN's budget, so nobody sends a contingent. At strength 1 it gives you +5 legitimacy and makes Election Interference against you easier to catch, and its state −10% turmoil effects and −10% radicals from political movements. While it is there, unless the UN is Moribund, Election Interference cuts your electoral confidence by half as much. It succeeds when an election campaign ends while it is there and you still hold its state and your seat (UN delivery and 3 standing for you), fails if you are attacked or stop holding elections (which also costs 2 standing), and lapses after five years without an election.

## The World Development Fund's grants

Any member can propose the World Development Fund while the UN assesses dues. While it stands, it pays out every week to the members whose GDP per head is below a line, shared by population. What it pays and where the line sits grow with the charter:

| Charter | Paid out each week | Members paid |
|---|---|---|
| Founding charter | The donations alone | Below a tenth of the members' average GDP per head |
| Charter Reform I | 5% of the UN's budget, plus the donations | Below a quarter of the average |
| Charter Reform II | A quarter of the UN's budget, plus the donations | Below half the average |

The line never falls below the GDP per head of the poorest member that holds its seat and pays its dues. When no such member is under the charter's share of the average, the line rises to that member's figure, so the Fund always pays at least one member.

A reform strengthens a Fund that already stands; nobody votes on it again. The donations are the Development Programs contributions, which open only once the Fund stands, to any member that holds its seat, pays its dues and draws no grant. A contributor's money pays the grants, as long as it holds its seat and pays its dues. You receive a grant only with a seat in the Assembly and your own dues paid. It goes into your investment pool, or into your treasury if your laws allow no investment pool or your banking system has not started. The line and the grants are worked out each month. Our Obligations shows your grant, and the budget line shows what the Fund and the Standing UN Force take. The World Development Fund section of the UN panel lists

every member the Fund pays this month, largest grant first, with its weekly grant and GDP per head, under the cutoff, the members' average and what the Fund pays out a week. Hover the cutoff to see how it was set.

### The World Food Reserve and hunger

The World Food Reserve needs no charter reform. While it stands, a member with a Strategic Reserve Hub can press Pledge Grain, under the World Food Reserve line in Our Obligations, and Withdraw Our Pledge at any time. When a UN aid mission opens, the Reserve takes grain from the pledgers, other than the mission's host, holding at least 500: up to a quarter of each one's grain and no more than 2,500 units from any one, 5,000 in all. The mission's state gets up to +0.15 food security and −5% mortality for a year, scaled by the grain sent. Each pledger drawn on gains 2 standing, and UN delivery rises.

Founding the Reserve makes the UN answer for hunger. While it stands, the Delivery pillar loses up to 10 points as the share of the members' people living in famine-struck states rises to 5%, and gains 3 while that share stays below half a percent.

## 10.6 UN conventions and agencies

A convention is a standing regime. When it carries it founds its agency (the decolonization declaration founds none), and every member that ratifies it carries its member modifier. Its effects are multiplied by the UN's enforcement, never below ×0.01, so a Moribund UN leaves conventions dormant rather than lapsed. Most also name winners and losers among the parties. Our Obligations lists each convention you are party to, with your terms under it; hover a convention for all its terms, or a term for its modifier. An agency's hover in the overview says whether it is founded, with its convention's terms one hover further, on Its terms. Your terms start the day you ratify a convention or join, and are re-read when the tier changes and once a year.

| Convention (agency) | Can come to the floor with | Parties gain | Winners and losers |
|---|---|---|---|
| Universal Declaration of Human Rights (UNHRC) | Authority 40; a proposer with Human Rights | Acceptance of other cultures, prestige | Countries with Ancestral Citizenship, Outlawed Dissent, Penal Labor Camps or slavery lose legitimacy and prestige. |
| International Criminal Court (ICC), binding | Authority 40; the Declaration in force (a great-power war's end can raise it sooner); a major-power proposer with Human Rights | Less infamy generation, faster relations | The court indicts rulers. |
| Nuclear Non-Proliferation Treaty (IAEA) | Authority 40; a proposer with Nuclear Weapons | Faster infamy decay, prestige | Programs without a bomb run 25% slower; countries with neither a bomb nor a program defend better against strikes. |
| Climate Accord (UNEP) | Authority 30; Environmental Movement; the Global Warming rule | Environment ministry impact, prestige | Market leaders with 10%+ of world emissions cut emissions and heavy industry; low emitters get adaptation aid. |
| Global Pandemic Response (WHO) | Authority 20; a major-power proposer with Antibiotics | Cheaper health system, prestige, faster spread of medicine | None. |
| International Refugee Resolution (UNHCR) | Authority 20; a country at war or with a state below 6 standard of living | Refugee ministry impact, prestige, migration pull | The 20 richest countries per head take in migrants and turmoil; the poorest gain standard of living. |
| Cultural Heritage Program (UNESCO) | Authority 40 | Prestige, research speed | Countries with a monument (any building in the Monuments group, the Space Program included, but not a Grand Monument) gain cultural pull |

| Convention (agency) | Can come to the floor with | Parties gain | Winners and losers |
|---|---|---|---|
| | | | and tourism, and build slower in every state. |
| Decolonization Resolution | Authority 30; Decolonization; a country holding a subject | Binds every member | Colonial powers lose colonial stability; members' colonies gain liberty desire. |
| International Space Cooperation (UNOOSA) | Authority 40; a major-power proposer with Space Exploration | Science ministry impact, space race progress | The space race leader slows and laggards speed up; orbital battlestation holders lose prestige. |
| Convention on the Law of the Sea (ITLOS) | Authority 30; International Trade | Cheaper port connections, prestige | Great powers gain less prestige from their navies. Once anyone has Abyssal Plain Mining it also regulates the seabed (see The Law of the Sea and the seabed). |
| Convention on the Physical Protection of Nuclear Material (CPPNM) | Authority 30; warheads missing; a proposer with Nuclear Weapons; the Nuclear Weapons rule | Prestige | Half as many of the parties' warheads go missing, and they may recover lost ones. |
| Single Convention on Narcotic Drugs (INCB) | Authority 20; a member with Antibiotic Mass Production; a major-power proposer with Pharmaceuticals | Pharmaceutical Industries throughput +5%, prestige +3% | Opium Plantation throughput −15%; no additional exporter penalty. |
| Convention on Cultural Diversity | Charter Reform I; authority 40; UNESCO in force; a proposer with Mass Media that is not the cultural hegemon | Prestige; half the Foreign Cultural Benchmark (none at Supranational); minorities assimilate 20% more slowly and their homelands fade 30% more slowly | A cultural hegemon that is a party projects 10% less cultural pull. |
| Treaty on the Prohibition of Nuclear Weapons (TPNW) | Charter Reform II; authority 40; the NPT in force; a proposer with Nuclear Weapons and no warheads; the Nuclear Weapons rule | Prestige | An armed party is held to its arsenal at ratification, a fifth fewer every five years, none after 25; an unarmed party may build none. An armed member that cast no ballot on it, or joins later, is not bound. |
| Antarctic Treaty (ATS) | Authority 30; a country running an Antarctic research station; a proposer with Space Exploration; the Global Commons rule | Prestige +2%; +0.25 toward UN Authority's target, not scaled | Parties hold no new Southern seabed licenses. A party with a research station gains +3% research speed. |
| Moratorium on Destructive Anti-Satellite Tests (UNODA) | Authority 30; a destructive anti-satellite strike somewhere; a proposer with Space Exploration that has not struck in two years; the Space Race rule | Prestige +5% | A party's own strike costs twice the infamy and twice the case strength. A party running a debris-removal method counts toward the Delivery pillar each year. |
| Convention on the Settlement of Outer Space (COPUOS) | Authority 30; a colony held somewhere; a proposer that holds a colony and has not seized or destroyed one in two | Prestige +2% | A party that destroys a colony, or closes in on a free one, pays twice the infamy and twice the case strength. It forbids nothing. |

| Convention (agency) | Can come to the floor with | Parties gain | Winners and losers |
|---|---|---|---|
| | years; the Space Race rule's Full Colonies | | |

Four member modifiers also carry a cost: the NPT raises infamy generation and lowers your units' kill rate, the Climate Accord lowers bureaucracy, the ICC makes casualties cost more war support, and the Convention on Cultural Diversity slows the assimilation of your minorities. While in force, the NPT, the CPPNM and the TPNW also raise the target of the nuclear taboo, by up to 8, 4 and 6 points, in proportion to UN authority. While UNESCO stands, carrying off a Grand Monument or a trophy in a war costs every country twice the infamy, party or not, and the AI loots a quarter as often (see Monument trophies).

The narcotics convention shifts Drugs production toward medicine. Its figures above are at ×1 enforcement and scale with the UN's enforcement. Countries with Pharmaceutical Industries and no Opium Plantations favor it; governments with 10 or more plantation levels resist, and those with 30 or more resist more strongly. Public Health Insurance favors quotas; Laissez-Faire opposes them. A powerful Landowners group makes a country that voted against the convention more likely to refuse ratification.

The overview shows seventeen specialized agencies, each lit once founded. Sixteen come from the conventions in the table: WHO, UNESCO, UNHRC, IAEA, UNEP, UNHCR, UNOOSA, ITLOS, the ICC, the CPPNM, Cultural Diversity, the TPNW, the INCB, the Antarctic Treaty Secretariat, the Office for Disarmament Affairs and the Committee on the Peaceful Uses of Outer Space. The seventeenth, the International Court of Justice, is founded the first time a country accepts a World Court ruling against it, and has no effects of its own. An agency's hover leads to its convention's terms (Its terms). Once the International Criminal Court convention is in force, the court indicts the ruler of a country that ratified it (from Established) or of any country (at Supranational) for a nuclear first or tactical strike or an exposed regime-change operation, at most once a decade. Handing the ruler over sends them into exile, and your heir, if you have one, succeeds them. Defying the court costs 5 standing, 10 case strength and credibility, and brings International Court Defied (−25% infamy decay, −5% prestige, fading over five years, × enforcement); defying a World Court ruling brings the same modifier at full strength. A party to the Declaration that runs a coercive resettlement program is penalized (see Costs and consequences of resettlement).

## The Law of the Sea and the seabed

Under the Global Commons rule, the Convention on the Law of the Sea has seabed terms that sleep until some country researches Abyssal Plain Mining. From then on the seabed has one regime for the whole world, and the Timeline Extended window's Seabed tab and Our Obligations show it. It is read once a year, when the basins' licenses are shared out (see Seabed licenses for the basins, the claim weights and the seabed policies).

| Regime | When | How each basin's licenses are shared |
|---|---|---|
| Open | The convention is not in force, or the United Nations rule is off | By weight among all coastal countries. |
| Regulated | The convention is in force | Half of each basin, the reserved half, by weight among the parties whose GDP per head is below the members' average (the World Development Fund's yardstick), with or without Abyssal Plain Mining: the share of a party without it waits for it. The other half by weight among all coastal countries. While no such party has any claim weight in a basin, its reserved half is shared with the open half. |
| Scramble | Two or more great powers on Unilateral Licensing while Regulated | The whole basin among all coastal countries, every navy counting double, and Sponsor Contractors protects no one's claims. It is Regulated again two years after fewer than two great powers remain on Unilateral Licensing. |

While the seabed is Regulated, a party pays royalties each week: 4% of its Deep-Sea Mines' base output value, about 1,800 a level. The money goes to the World Development Fund while it stands, otherwise to the UN budget. A party that switches to Unilateral Licensing while the convention governs the seabed defects: 10 infamy once, 5 standing and 10 case strength. It loses its place in the reserved half and pays no royalties. It becomes a party again five years after it defected, once it has left Unilateral Licensing for Sponsor Contractors or Scientific Survey. A member that refused the convention, or a country outside the UN, claims in the open half only and pays nothing. Unilateral Licensing costs it nothing extra. The AI weighs the seabed terms when it votes on the convention: poorer coastal members lean for it, great powers with Deep-Sea Mines against.

### The Antarctic Treaty and research stations

Any country with Space Exploration can take the decision Open an Antarctic Research Station. A station gives +2% research speed and +25 innovation cap for a weekly upkeep of a tenth of a percent of GDP a year, never less than 1,000 a week. It also adds to your claim weight in the Southern basin, coastline or not. The decision Close the Antarctic Research Station ends both. The AI opens a station when its weekly surplus covers ten times the upkeep: great and major powers, and smaller powers with a Southern coastline or a place in the Antarctic Treaty.

A country's first station puts the Antarctic Treaty on the docket. Its parties hold no new Southern seabed licenses; levels built before they ratified keep theirs. A party running a station gains +3% research speed (scaled by enforcement), and every party adds 0.25 to UN Authority's target, times its weight. A party can take the decision Denounce the Antarctic Treaty, for 10 infamy, 5 standing and 10 case strength. It is then outside the treaty and free to claim the Southern seabed; rejoining the UN does not bring it back in, but a new treaty does. When two or more great powers outside the treaty hold Southern licenses, the treaty lapses for everyone and the Southern basin opens to all. The docket does not raise it again while that holds.

### The anti-satellite moratorium

A destructive anti-satellite strike (see The space race) puts the Moratorium on Destructive Anti-Satellite Tests on the docket, offered to a proposer other than the striker. A party's own strike then costs twice the infamy and twice the case strength. Parties gain prestige, and once a year each party running a debris-removal method on its Space Program adds 0.25 times its weight to the Delivery pillar, as aid delivered does. A country that struck in the last two years leans hard against it, and a space power leans for it.

### The outer-space convention

Under the Space Race rule's Full Colonies, where fleets can take and destroy colonies (see Space fleets), the Convention on the Settlement of Outer Space can come to the floor once any country holds a colony. The docket offers it to a member that holds a colony when a colony has been seized, destroyed or taken from its free settlers, never to the country that did it, and as Assembly business; a member holding a colony may also table it from the chamber. A party pays twice the infamy and twice the case strength for destroying a colony or closing in on a free one, and gains prestige. It forbids nothing, so seizing an enemy's colony at war and interdicting one cost a party what they cost anyone. A colony holder leans for it, and every member leans further for it while a seizure or destruction is on the docket; a power whose fleets have just seized or destroyed a colony leans hard against it, and one with a large fleet leans a little against it.

## 10.7 UN military mandates

A mandate licenses one member to recover one state region from one country. To move for one you need authority 40, no mandate in force and none proposed in the last five years, standing above Disgraced, and a target that is not a subject, has a case of 60 or more and holds a state in a region you claim. The button picks the case (a condemned or sanctioned target first, a rival before a stranger, a weaker country before a stronger). Mandates in Force previews it under Our Case, with a Propose the Mandate button, and so does the Propose a Resolution row.

A carried mandate lasts five years and makes the Authorized Restoration war goal available against that country, for that state, at no infamy, in one diplomatic play. Enforcing the goal adds delivery and 6 standing. Adding any other demand against that country in that play for territory, subjugation, regime change or

humiliation abuses the mandate, and so does backing down; reparations and similar demands are allowed at their normal price. Demands against that country in another play or war do not touch the mandate. The mandate covers its own war and no other: any other war you begin pays the surcharge, goes on your record and can come before the Assembly as usual. Abuse ends the mandate, brings a condemnation, costs 12 standing, suspends your standing benefits for ten years and adds 20 to your case. From Strong, a holder at war under the mandate loses less war support to casualties and defeats.

## 10.8 UN missions in the field

Missions put the UN's work in one state:

Hover the name of a mission (on the state panel, in Missions in the Field or in a tooltip) for its Peacekeeping Mission, Aid Mission, Stabilization Mission or Electoral Observer Mission entry, which sets out how that kind opens, what it does, what its progress means and when it succeeds, fails or lapses.

| Mission | Opened by | Effect on its state | Succeeds when |
|---|---|---|---|
| Peacekeeping | A peacekeeping request carried in full | Turmoil effects −20%, devastation recovery +50% | The host has 24 months of peace. |
| Aid | A carried aid request, or a full answer to a famine | Standard of living +1.5, food security +0.2, mortality −5% | After six months, no famine and, without the mission's help, a standard of living of 10 or 1 above where it started, whichever is higher. |
| Stabilization | A full answer to a state collapse | Turmoil effects −30%, standard of living +0.5 | About a year after the collapse ends, sooner if the mission served through it (it progresses slowly during the collapse). |
| Electoral observer | A carried request for electoral observers | Turmoil effects −10%, radicals from political movements −10%; the host also gains legitimacy and covert defense | An election campaign in the host ends while it is there. |

Effects scale with the UN's enforcement, fall by up to half when members withhold dues, and rise with each contributor up to three; the Standing UN Force counts as one in every peacekeeping and Stabilization mission, and an electoral mission's paid observers count as one. A mission fails if its host is attacked after it arrived, if every contributor leaves a peacekeeping or Stabilization mission while the Standing UN Force does not serve in it, if the host expels a Stabilization mission, or if an electoral mission's host stops holding elections. It lapses after six months if nobody has joined a peacekeeping or Stabilization mission (never while the Standing UN Force stands), after five years without an election for an electoral mission, or if its host or state is gone. A mission has no time limit: one that is not done goes on for as long as it keeps contributors, so a member that keeps paying for its contingent can keep it in the field. Time works on the AI instead, which grows less willing to send a contingent to a mission the longer it has run. Success adds delivery and gives each contributor 3 standing and 15 relations with the host; failure takes delivery away. The state panel shows a UN Mission tile, and contributors build covert networks in the host faster.

Each row in Missions in the Field shows the mission's progress as a bar, with its month and strength beside it, and its contributors' flags, the largest economies first. Any major-power member not under sanctions can press Send a Contingent on a mission's row, if it is not the host, not at war with it and has not left that mission before; a country that can send none anywhere sees one line saying why, and electoral missions take no contingents. Each mission joined this way costs a quarter of a percent of GDP a year, free for peacekeeping and Stabilization if you run the peacekeeping program. Withdraw Contingent costs 2 standing and 10 relations with the host, and a peacekeeping or Stabilization mission left empty fails.

*Figure 16: The UN Mission tile on a state panel: an aid mission three months in, working at ×1.31 strength, with no progress yet.*

## 10.9 UN programs and great-power stances

A major power's peacekeeping, human rights and arms control programs count toward the funding pillar while they run (two count in full); Development Programs counts by its money, whoever gives it. Every program earns standing after 24 months.

You join or leave a program with Join and Leave in its Ours column under Programs and Conventions, and fund Development Programs from the World Development Fund section.

| Program | Requires | Cost | Gives |
|---|---|---|---|
| Peacekeeping | Major power | 0.5% of GDP a year, 100 influence, +5% military goods cost; cannot be ended for ten years | Leverage generation, prestige and +10% army experience gain; covers peacekeeping contingents |
| Development Programs | The World Development Fund founded; you pay your dues and draw no grant | 0.0625% to 1% of GDP a year, by size | Faster infamy decay, prestige, lobbying with the Fund's recipients; the money pays the Fund's grants |
| Human rights champions | Universal Citizenship, Universal Suffrage, Liquid Democracy or Protected Speech | 100 influence | Prestige, refugee ministry impact |
| Arms control | War Crimes Forbidden, Humanitarian Regulations or Limited War | 100 influence, +5% infamy generation | Faster infamy decay, less devastation, cheaper military goods; lower kill rate |

Great powers can also Champion the Order or Undermine the Order, under the champions in Why UN Authority Is Moving, each using 500 influence while it holds (as much as leading a power bloc), which count +1 or −1 in the commitment pillar. Undermining also drains standing, and after 24 months it suspends your standing benefits until you stop. A program or stance that uses influence needs that much available to start.

### Development Programs sizes

Fund Development, at the top of the World Development Fund section, starts a Token contribution. Raise and Reduce move it one size at a time, and Withdraw ends it; changing size doesn't restart the 24 months before it earns standing.

| Size | Cost (share of GDP a year) | Infamy decay | Prestige | Standing a month after 24 months | Lobbying with the Fund's recipients |
|---|---|---|---|---|---|
| Token | 0.0625% | +2% | +1% | 0.075 | up to +5% |
| Small | 0.125% | +4% | +2% | 0.15 | up to +10% |

| Size | Cost (share of GDP a year) | Infamy decay | Prestige | Standing a month after 24 months | Lobbying with the Fund's recipients |
|---|---|---|---|---|---|
| Modest | 0.25% | +6% | +3% | 0.225 | up to +15% |
| Substantial | 0.5% | +8% | +4% | 0.3 | up to +20% |
| Generous | 1% | +10% | +5% | 0.375 | up to +25% |

Each size doubles the money but adds only a quarter of Substantial's benefits, so Token and Small buy them most cheaply. The larger sizes pay the Fund's recipients more and count for more in the funding pillar, which weighs the money, not the number of contributors.

The lobbying bonus is Grateful Aid Recipients on the UN journal entry: UN Lobbying Effectiveness with the members the Fund pays. It builds toward the figure for your size, about two thirds of the way in a year, and fades over a few years after you stop or cut back, so raising your contribution just before a vote buys little. The World Development Fund section shows your size, its weekly cost and the bonus first.

## 10.10 UN dues and Article 19

Every member that conducts its own foreign policy pays dues, the share of GDP set by the tier, as a weekly expense; that includes a member whose credentials the Assembly has rejected. Withhold Our Dues, under the dues in Our Obligations, saves the money but costs 3 standing, lowers the funding pillar and weakens every mission. Each month you withhold while dues are assessed is a month in arrears. After 24 months you lose your vote (Article 19): you cast no ballot, table nothing, cannot be given a permanent seat, and lose 5 more standing. Pay Our Dues settles everything owed at once and restores the vote. Arrears survive leaving and rejoining; only a dissolution writes them off.

The dues paid are the UN's budget. While they stand, the Standing UN Force takes a tenth of it and the World Development Fund 5% after Charter Reform I or a quarter after Charter Reform II, and the emergency lending facility lends only from what is left. Our Obligations shows the split as a table.

## 10.11 International standing

Standing is your own record in the organization, from 0 to 100, separate from UN Authority. It starts at 50 when you join.

| Tier | Standing | Effect |
|---|---|---|
| Exemplary | 80+ | +4 diplomatic reputation, faster infamy decay and relations |
| Respected | 60–79 | +2 diplomatic reputation, faster infamy decay |
| Neutral | 40–59 | None |
| Poor | 20–39 | −2 diplomatic reputation |
| Disgraced | below 20 | −4 diplomatic reputation, slower infamy decay; no mandates |

Standing also sways votes on your resolutions and sets how soon you may table business again (see Tabling UN business). When the docket offers a convention, it goes to the strongest member with a claim, ranked by power share and stretched the same way: a great power in good standing first, a member with a ruined record last. You earn it by delivering: programs kept for two years, aid and peacekeepers sent, missions accomplished, mandates discharged, binding resolutions accepted at a cost, World Court rulings obeyed, referendums held, ceasefire orders heeded within three months and grain sent through the World Food Reserve. Gains shrink as standing rises, and voting earns none apart from a kept vote pledge. Censure, sanctions, arms embargoes, suspended credentials, defiance (of the Assembly, the court, a ceasefire order or a referendum), abused mandates, leaving and withheld dues cost it. The World Court and the suspension of credentials need 20 standing to table.

*Figure 17: The UN tab of the Diplomacy panel: the overview (membership, tier, standing, the authority bar, the countries, GDP and population pies, the Security Council, the agencies and the headquarters) and Status.*

## 10.12 The UN panels

The United Nations journal entry and the UN tab in the Diplomacy panel show the same panels, and a change made in one shows in the other. The tab is grayed until the journal entry is active; hover it for what is still missing. Every action is a button beside the figure it acts on, in the panels; the journal entry's own buttons are the AI's. The tab adds the entry's status text, while a founding conference sits or after the UN has dissolved, and ends with an Open Journal Entry button.

The overview at the top is always shown. Its first rows are cells, two to a row: your membership (with Found, Join or Leave under it), the UN's tier and enforcement, the General Assembly (what is in session and how you voted, or that your vote is due), your motions (how many topics are ready, or how long until you may table again), your dues, your standing, the case against you and, while it runs, the crisis. Hover a cell for its detail. Below them are the authority bar, solid to the lower of authority and its target and pale to the higher (hover it for why the target is where it is), three pies of the members' share of the world's countries, GDP and

population with the Assembly's membership under them, then the Security Council's five flags (hover one for the country, click it to open the country), the sixteen agencies, lit once founded, and the headquarters.

The sections below are open by default when they change month to month, and collapsed when they are reference:

| Section | Starts | Shows |
|---|---|---|
| General Assembly | Open | The resolution in session: a tally bar (green for, gray not yet voted, red against, with a line at two thirds for the topics that need it) and how many of its twelve months have run, then your vote or the vote buttons and your lean (hover it for its terms), then the topic, the parties, what it does if it carries and the grounds. Delegations and Recorded Ballot sit under it while a resolution is in session; Propose a Resolution takes their place when none is. |
| Why UN Authority Is Moving | Open | A bar for each pillar, with its trend since last month (hover a pillar for how it is computed), then your weight, the tier and the authority cap, the champions and underminers with Champion the Order or Undermine the Order, and the newest ledger entries (collapsed). |
| Missions in the Field | Open | A row for each mission, with its progress bar, its contributors' flags and Send a Contingent or Withdraw Contingent; ended missions below, collapsed. |
| Programs and Conventions | Open | How many countries take part in each program and convention, whether you do, with Join or Leave for a program, and how many are under sanctions. |
| Mandates in Force | Open | The mandates in force, and the case a mandate of yours would take, with Propose the Mandate. |
| Our Obligations | Open | Your dues, with Withhold Our Dues or Pay Our Dues, beside the budget as a table (all members' dues and what the Standing UN Force, the World Development Fund and the lending facility take), your pledge to the World Food Reserve with Pledge Grain or Withdraw Our Pledge, any ceasefire order against you or your enemy, then each convention you are party to, with your terms under it, and from Strong the UN's reach (shared intelligence, your interest groups' reaction). |
| World Development Fund | Open | While the Fund stands: your position first (your grant, or your contribution with Fund Development, Raise, Reduce and Withdraw), then the cutoff GDP per head (hover it for how it was set), the members' average, how many members qualify (at or below the cutoff, seated and paying their dues) and what the Fund pays out a week, then each member it pays this month and how much, or why nobody is paid. |
| Our Record | Open | Your case strength and its parts, and whether it would support a condemnation, sanctions or a mandate. |
| UN Authority History | Open | A chart of authority over time. |
| Resolutions on the Record | Collapsed | A row for each closed resolution, its title and how it ended, with the rest of its card on hover; Voting Details adds how each member voted. |

The "?" at the right of the tab's header, and at the top of the journal entry, opens the Help page, How the UN Works. It explains authority, standing, missions, delegations, dues and the record.

## 10.13 How the AI uses the UN

AI countries play by the same rules and numbers you do:

- They vote by the same lean and veto rule, in the ninth, tenth or eleventh month of the session.
- A proposer lobbies for its resolution. On a resolution that accuses a country, that country, its allies and its bloc leader lobby against it. Each runs up to three campaigns at once on AI members close to the line,

starts at most one a month and only with influence to spare, and asks for pledges now and then, humans included. On a resolution that accuses no one, up to two AI members a side with a strong stake in the topic (interests of +35, or −35 or less) run one campaign each, when they have 300 influence to spare: a nuclear power against the prohibition treaty, poor members for a World Development Fund that would pay them well. They never run campaigns on human members.

- They table the seventeen non-convention topics through the journal entry's buttons when their situation calls for it, and reach conventions only through the docket, so a qualifying human is offered convention business first. They refer a dispute to the World Court only on odds of 60% or better, and petition to lift sanctions on themselves only once their case is below 50 and the sanctions have stood two years.
- They pledge grain to the World Food Reserve when they hold 2,000 units or more, and withdraw the pledge when famine strikes at home.
- They join or leave by what a seat is worth to them: their laws and rank, the dues against the grant the World Development Fund would pay them (both as a share of their GDP, so a poor country that would draw a large grant wants in most), what staying out costs at the UN's tier, the aid and peacekeepers only members can ask for, and the conventions they would have to keep. A country whose case is close to the line neither joins nor leaves, so it doesn't walk out and back. Below UN Authority 30 a seat is worth less with every point lost, so members leave a failing UN a few at a time as the crisis deepens, those with the weakest reasons first.
- They stop paying their dues when their treasury defaults or they undermine the order. Debt near the ceiling, isolationism and a high levy count against paying and can add up to it. A member the Fund pays more than its dues keeps paying even in default, and one that has lost its vote under Article 19 pays sooner. A healthy member doesn't withhold to save the levy and then pay again.
- Members whose GDP per head is above the members' average often fund Development Programs, usually at Small size. So do members with Humanitarian Regulations, champions of the order, members with notorious infamy, and members with Fund recipients among their subjects, bloc partners and allies. Only Humanitarian Regulations and championing make them give more, a size for each, and recipients among their own half a size. A member whose GDP per head is below the great powers' average, or that borrows, gives less and often stays at Token. Major powers contribute to peacekeeping for the same kinds of reasons, more when missions are in the field. They stop either one when their treasury fails or their laws turn against it, and stop peacekeeping at war; peacekeeping only once its ten years have run.
- They start a program or stance that uses influence only with 200 influence to spare beyond its cost, and give it up when their influence runs into deficit.
- Great powers champion or undermine the order for their own reasons: Humanitarian Regulations or Limited War, a permanent seat and their bloc's Multilateral Institutions principle pull toward championing; isolationism, Total War, conventions that cost them, UN sanctions and a heavy record toward undermining. UN Authority adds only a small pull (a weak UN draws champions, a strong one makes powers wary), never enough to turn a power on its own.
- They send contingents to missions hosted by allies, bloc partners and subjects, and bring them home when attacked or short of money. A country holding a mandate is steered toward the target and the authorized goal.

## 10.14 How the UN connects to other systems

Nuclear strikes lower peace and order and add to the striker's case, the NPT reaches threshold states, at authority 60 with the IAEA in place the Nuclear Program Aid treaty article is forbidden, the TPNW holds its armed parties to a falling arsenal ceiling, and the NPT, the CPPNM, the TPNW and the Assembly's verdict on a nuclear use move the nuclear taboo (Nuclear weapons). The Climate Accord exists only under the Global Warming rule (Climate and pollution). The decolonization declaration presses colonial powers and opens supervised referendums (Colonial empires and decolonization), and the space treaty slows the leader (The space race). Exposed covert operations feed appeals, cases and indictments, the headquarters host and mission contributors build networks faster, electoral observers halve Election Interference's cut to electoral confidence, and the Convention on Cultural Diversity eases the Foreign Cultural Benchmark (Cultural hegemony and covert warfare). The same convention slows homeland loss (States and population). The World Food Reserve

draws on Strategic Reserves, and the World Development Fund pays into the investment pool (Economy and construction). Power blocs gain cohesion when the UN is Moribund or gone, and the Multilateral Institutions principle strengthens membership benefits (Diplomacy).

CHAPTER 11

# Cultural hegemony and covert warfare

Two systems let countries compete for influence without going to war. Cultural Hegemony measures how much of the world's cultural influence each country holds and turns a large share into migrants, legitimacy pressure on trailing countries and pressure on foreign political movements. Covert Warfare gives you an intelligence agency that runs espionage, subversion and sabotage against other countries, at the risk of being caught. Each has its own game rule, Cultural Hegemony and Covert Warfare, and both are on by default.

## 11.1 The Cultural Hegemony journal entry

The game scores every country's cultural pull from the first day of the campaign, but the Cultural Hegemony journal entry and everything it does wait until some country in the world has researched Mass Media (an era 6 society technology) and you have Romanticism. Until then the scores run in the background and have no effect. The entry never completes. Before it opens, the entry shows only your tier and cultural share; after that, its panels show the rest (see The Cultural Hegemony panels).

### The Cultural Hegemony panels

The Cultural Hegemony journal entry and the Hegemony tab in the Society panel show the same panels, and a change made in one shows in the other. The tab is grayed until the journal entry opens; hover it for what is still missing. It ends with an Open Journal Entry button. Another country's National Info has a Cultural Hegemony tab with that country's tier, rank, share, programs and breakdown, and the same Top Cultural Powers, political models and history as yours; the funding and program buttons are only on your own.

The overview at the top is always shown. Its first row is icons with a word beneath: your influence tier (a gray sprig for Negligible, then a lyre that gains a laurel wreath and turns bronze, silver and gold up to Hegemon), your World Rank, the leading power's political model in its pie color, and a red Benchmark icon while you carry the Foreign Cultural Benchmark. Below them are a pie of your cultural share (amber for yours, gray for the rest of the world), with an arrow for the change since last month, and the flags of the three Leading Powers, the leader first. Hover a flag for the country and click it to open the country. Hover the other icons for the detail: the tier's full name and the share the next tier starts at, the model's share of world culture and its push abroad, the benchmark's strength. Last comes Programs, with the Ministry of Culture's icon: how many of the four programs are running and your funding level against its cap, or No Ministry in red while you lack the law. Hover it for each program and the funding cost.

The sections below all start open:

| Section | Starts | Shows |
|---|---|---|
| Top Cultural Powers | Open | A table of the ten largest cultural shares at the last recount: rank, flag, country, share, its change since that country's own share at the recount before (with an arrow up or down), and its political model as a colored swatch (hover it for the model). An arrow marks your row; hover a row for that country's pull components. When you are outside the top ten, Our Rank below the table gives your place. |
| Cultural Programs | Open | Program Funding with its – and + buttons (hover the level for where its cap comes from), then a row for each program with Running or Idle and |

*Figure 18: The Hegemony tab of the Society panel: the overview, Top Cultural Powers with each one's change since the last recount, Cultural Programs, and where our pull comes from.*

| Section | Starts | Shows |
|---|---|---|
| | | its button (International Cultural Outreach's row reads International Outreach). Hover a program's name for what it gives while it runs, and a button for what is missing and what it would do. While you lack Ministry of Culture Established, a line above the rows says so. |
| Where Our Pull Comes From | Open | A bar for each component of your pull, with zero in the middle: gains fill to the right in green, penalties to the left in red, each against that component's usual range. Hover a row for the working behind the number and the bar's range. Cultural Pull from Art sits under Art Production when it is not zero, the Pull Multiplier closes the components, and Raw Score stands under the rule. |
| Political Models of the World | Open | The pie of the world's political models and its legend, by short name. Hover a model for its full name and the laws that put a country in it. |

| Section | Starts | Shows |
|---|---|---|
| History | Open | Each political model's share of world cultural influence, stacked to 100% in one bar per year, over 5, 20 or 100 years, its legend by short name. Hover a bar for its date and all fifteen model shares. Every country sees the same history. Existing saves begin collecting at the next cultural recount; up to 100 annual snapshots are kept. |

The "?" at the right of the tab's header, and at the top of the journal entry, opens the Help page, How Cultural Hegemony Works. It explains cultural share and tiers, the programs, where pull comes from, the top cultural powers, political models and the Foreign Cultural Benchmark.

## Cultural share and influence tiers

Your **cultural share** is your raw cultural pull as a percentage of the world's total. Other countries' pull is recounted about twice a year and after every war, while yours counts as it stands, so your share can step a little without your own output changing. Your share sets your tier:

| Cultural share | Tier |
|---|---|
| Under 2% | Negligible Cultural Influence |
| 2% | Minor Cultural Influence |
| 5% | Moderate Cultural Influence |
| 10% | Significant Cultural Presence |
| 15% | Major Cultural Power |
| 25% | Cultural Hegemon |

The country with the highest share is the world's leading cultural power, the hegemon, whatever its tier.

## Where cultural pull comes from

Your raw score is the sum of the components below, multiplied by your pull multiplier. Where Our Pull Comes From, in the panel, shows each one for your country as a bar.

| Component | How it counts |
|---|---|
| Art Production | Your fine art output, capped at your percentage share of world production, with diminishing returns above a third of world output. Free Speech, Church and State, LGBTQ+ Rights and Bodily Autonomy laws, Romanticism, Realism and Film change it through the Cultural Pull from Art modifier. |
| Prestige | Your prestige divided by 5, capped at your percentage share of world prestige. Recognized countries only. |
| Standard of Living | Your average standard of living minus the world average (−5 to +20): in full for great powers, half for major powers, a quarter for minor powers, nothing below. |
| Tech Leadership | +2 each time you research a technology no other country has yet, up to 30, fading by a tenth at each recount. |
| Monuments | +3 per monument: the mod's wonders, the base game's monuments and the Space Program (power bloc statues don't count). Grand monuments add +1 for each step of their combined grandeur, counting only monuments that are not contested. The first step takes 10 grandeur, the second 20, the third 30 and so on, with no ceiling. |
| Megaprojects | +3 for each of the seven megaproject types you have completed. |
| Other Modifiers | Flat pull from program funding, events, tiers IV and V of the Education power bloc principle and the Cultural Exchange Program treaty article. |
| Infamy | −0.1 per point of infamy. |
| Instability | Up to −10 at full turmoil, and −20 during a civil war. |

The multiplier comes from your rank (great power +25%, major power +10%), laws (free trade, open borders and inclusive citizenship raise it; isolationist, exclusionary and repressive laws lower it), technologies such as Romanticism, Realism, Film and Television, the Ministry of Culture institution, programs and events. Wonders, megaprojects and grand monuments are described in The extended timeline.

## What cultural dominance gives

Once your share reaches 1%, you get a Cultural Hegemony modifier whose strength grows in step with your share:

| Effect | At 10% share | At 25% share |
|---|---|---|
| Mass migration attraction | +2.5% | +6% |
| Tourism output | +10% | +25% |
| Trade advantage for fine art | +10% | +25% |
| Loyalists from high legitimacy | +10% | +25% |
| Power bloc leverage generation | +10% | +25% |
| Diplomatic reputation | +1 | +2.5 |
| Infamy decay | +10% | +25% |
| Assimilation | +50% | +125% |

Tourism itself is covered in State tourism.

## Legitimacy pressure on trailing countries

Every country with the journal entry whose share trails the hegemon's by more than about three points carries the Foreign Cultural Benchmark modifier: its people measure their government against the leading power's, and legitimacy falls. The penalty grows with the gap. With the hegemon at 30% and you at 5%, it costs about 4 legitimacy. The hegemon never carries it. Tier V of the Education power bloc principle shrinks the penalty, and so does being a party to the UN Convention on Cultural Diversity: a quarter off at Contested, half at Established, three quarters at Strong and all of it at Supranational. A hegemon that ratifies the convention projects 10% less cultural pull, scaled the same way (see UN conventions and agencies). While you carry it, the overview shows a red Benchmark icon; hover it for the current strength. Under the Demographics rule's Full setting the same gap also pulls the family size your people want a little toward the hegemon's (see Reading the figures).

## Political models of the world

At each recount the game sorts every country into one of fifteen political models by its laws, from Liberal / Progressive Democratic and Constitutional Monarchist to Communist / Vanguardist, Technocratic, Developmentalist Junta and Mixed / Other. Some tests also ask who runs the government, meaning the strongest interest group in it. A Command Economy under a Single-Party State is Communist only when that group is Communist or Vanguardist, and any other left-wing lead makes it Socialist. Cultural Citizenship, a common starting law, makes a Single-Party State Fascist unless that group is left-wing, and makes Autocracy Fascist only when that group is Fascist or Ethno-Nationalist. A left-wing planned economy that keeps the law is still Communist or Socialist, and a monarch who takes absolute power over it is Absolutist Monarchist. The pie chart shows each model's share of world culture, weighted by the cultural pull of the countries running it rather than by their number. Hover a model in the legend for what qualifies. The tests run in a fixed order and a country counts as the first model it fits, so a Council Republic under Anarchy is Anarchist, not Communist, a monarchy with Universal Suffrage is Constitutional Monarchist, not Liberal, a monarchy under Technocracy is Technocratic, and a Single-Party State with Ancestral Citizenship and a Command Economy is Fascist, not Communist.

The hegemon's model pushes on every country that carries the Foreign Cultural Benchmark: the matching political movement there (a liberal movement under a liberal hegemon, for example, or a related one if it is absent) grows more active and attracts more pops. The push is zero while the model holds less than 15% of world culture and grows with its share after that; hover the model in the overview for its strength. A hegemon

with feminist, civil-rights, environmental, abolitionist, labor or land-reform laws also feeds those movements abroad, and one at peace without Mass Conscription feeds anti-war movements. Movements are covered in Government, laws and characters.

### Cultural programs

Every program needs the Ministry of Culture Established law (unlocked by Mass Propaganda). If you repeal it, funding drops to zero and all four programs stop at the next monthly update.

| Program | Controls | Effect | Cost |
|---|---|---|---|
| Program Funding | – and + | Each step adds +1 cultural pull and +5% pull. | A weekly expense per step that grows with your GDP, so the bill rises with every step. The tooltip shows the cost of one step and of your current level. |
| International Cultural Outreach | Begin / End | +10% pull, +5% prestige, +10% mass migration attraction. Needs Mass Media. | The weekly cost of one funding step. |
| Cultural Institutes | Fund / Defund | +10% pull. | 150 influence. |
| Global Media Campaign | Launch / End | +15% pull, +5% prestige. Needs Mass Media. | 200 influence. |
| Cultural Protectionism | Enact / End | +10% pull, +100 authority. | −5% society research speed, −10% mass migration attraction. |

The funding cap is one step per level of the Ministry of Culture institution, plus one each from Mass Media and Television; the panel shows it. The four programs run until you end them. Cultural Institutes and Global Media Campaign need their influence available to start. Global Media Campaign and Cultural Protectionism exclude each other: while one runs, the other's button is grayed out.

### Cultural hegemony events

Rising powers get opportunities as their share climbs past 8%, mostly trading authority or prestige for pull, research or migration, and two milestones fire once, at 10% and at 25% while ranked first. Trailing countries under the benchmark get events of their own. Most let you embrace the foreign influence (research, relations, often a push toward the hegemon's model) or resist it (authority and interest group approval, often at a cost in relations, legitimacy or pull). The harshest, Hegemonic Convergence, can press you to adopt one of the hegemon's laws once it holds 25% and your share is a tenth of its share or less. Four more events follow the political models. All of them are in Cultural hegemony event list.

### How the AI competes for culture

AI countries use the same programs. Those with a share of 10% or more begin outreach and launch media campaigns, a campaign or institutes only with 200 influence to spare beyond the cost, and they end either when their influence runs into deficit. An AI raises funding at a share of 10% or more, or below 5%, but only while it is out of default and its weekly surplus covers one more step; below 2% it is more likely to cut funding. Small countries under the benchmark enact Cultural Protectionism and end media campaigns, and drop protectionism once their share passes 10%.

## 11.2 The Covert Warfare journal entry

The Covert Warfare journal entry is where you run your intelligence agency. It becomes active when you hold one more covert operation slot than your rank grants for free (every country gets one, major powers two, great powers three). The first extra slots come from the Ministry of Intelligence and Security Established law (unlocked by Mass Surveillance), whose institution adds a slot per level, and from the era 7 technology Mainframe Computers. Once active, the entry stays open even if you later lose that extra slot. It never completes.

*Figure 19: The Covert tab of the Military panel: the overview (Fortress, Funding 2, Veteran, Spy Caught; three of nine operation slots in use; Capacity and Tradecraft), the Operations section with three operation rows, and the start of Funding.*

You launch operations from another country's diplomatic actions, listed as "Covert: ..." with the operation's name, which is also its name on your rows. Everything else is on the Covert tab of the Military panel, which shows the journal entry's own panels: a change made in one shows in the other. The tab is grayed until the journal entry is active; hover it for what is still missing. It ends with an Open Journal Entry button.

The overview at the top is always shown. Its first row is icons with a word beneath:

- Your intelligence standing, a shield from gold-rimmed (Fortress) to split in two (Vulnerable); see Intelligence capacity and operation slots.
- Your funding level ("Funding 3"), an envelope of banknotes, or an empty gray one at 0.
- Your Tradecraft tier, a fedora with up to four chevrons.
- Spy Caught, while you have exposed a foreign operation in the last ten years.

Below the icons, Operation Slots shows how many slots you hold and how many are in use, as case files lit while an operation fills them. Two bars follow: Capacity, your intelligence capacity as a share of the world's

best, with a line at each standing's threshold, and Tradecraft, with a line at each tier (hover a line for what it unlocks), the next tier's threshold beside it and an arrow for which way it last moved. Hover any of them for the detail: the standing's meaning, the funding level's name and cost, the capacity breakdown, this month's Tradecraft and why it last changed.

The sections below are open by default:

| Section | Starts | Shows |
|---|---|---|
| Operations | Open | A warning while funding is 0, or "No operations running" (hover it for how to launch one). Then the Detection Risk every operation shares (base risk, Funding Stealth, Efficiency Factor), and a row for each running operation: the target's flag, its icon, its name and target (hover it for what it does), its phase, its own Detection Risk (hover it for the breakdown), what its exposure would count as, the priority stepper and Stand down. |
| Funding | Open | A line while funded with nothing running, the Level stepper, and the Funding Levels table: each level's Stealth, Capacity and weekly cost, with your level's row shaded gold. |
| Agent Networks | Open, once you have a network | A table with a line for each network: the target's flag (hover it for the country, click it to open the country) and name, its strength with a bar and an arrow for growing or decaying, your operations there, the head start a new one would get, and what it reports (None, Partial or Full; hover it for the report). Hover the name for what the network adds to catching that country's operations. |
| Counterintelligence | Open | Unused slots redirected, your funding bonus and your economic, military and ideological defense, each with its breakdown on hover; then Last Caught while an exposure stands. |
| Known Foreign Operations | Open, while you are watching an operation | A row for each foreign operation you caught and let run: its sponsor's flag and name, the operation, the months you have watched it and an Expose button. |

The "?" at the right of the tab's header, and at the top of the journal entry, opens the Help page, How Covert Warfare Works. It explains operations, funding, detection, networks, Tradecraft, counterintelligence, watching and intelligence standing.

## Intelligence capacity and operation slots

Intelligence capacity is your agency's size: a base of 5, plus your rank (great power +25, major power +10), your literacy (up to 50 at full literacy), your share of world GDP (two points per percent, up to 50), and modifiers from technologies, laws such as Secret Police and Censorship, and the Ministry of Intelligence and Security (+4 per level). Tradecraft then multiplies the total. The overview's first icon rates your capacity against the world's best: Fortress at 80% of the best or more, Hardened from 60%, Defended from 40%, Exposed from 20% and Vulnerable below that.

Capacity cuts both ways. The stronger a target's capacity is against yours, the likelier your operations there are caught; the stronger yours, the likelier you catch operations against you. Rank and GDP share give a great power's service an edge that a small, literate country can't match: a small country's lightly funded operations against a great power are caught noticeably more often, though high funding still hides most of them. Capacity also gives separatism and coup resistance (the Intelligence Capacity modifier), growing up to a capacity of 125.

Each operation fills one slot for each level of its priority: one at priority 1, three at priority 3. Beyond the rank and ministry slots, Mainframe Computers, Computer Networks, Cyber Warfare and Quantum Computing

each add one, to a maximum of ten. You can run one operation of each type at first, and one more of each type with Mainframe Computers, with Cyber Warfare and with a Veteran agency.

## Covert funding levels

The Level stepper in the Funding section sets what the agency spends. Every operation needs funding of at least 1, so stepping down to 0 ends them all; the step's tooltip warns you first. The weekly cost grows with your GDP, the funding level and the number of operations (weighted by priority, plus a base share paid even with nothing running). The section's Funding Levels table shows what each level costs you a week at your current operations, with your level's row shaded gold.

Funding from level 2 up gives Funding Stealth, points off every operation's Detection Risk, and counterintelligence capacity:

| Level | Name | Funding Stealth | Counterintelligence |
|---|---|---|---|
| 0 | Dormant | Nothing can run | None |
| 1 | Operational | None | None |
| 2 | Professional | −3 points | +5 intelligence capacity |
| 3 | Covert Network | −8 points | +10 intelligence capacity |
| 4 | Black Budget | −13 points | +15 intelligence capacity |
| 5 | Deep State | −19 points | +20 intelligence capacity |

From level 3, idle networks also decay half as fast.

## The covert operations

There are seventeen operations: sixteen run against another country, and one, Counterintelligence Support, helps a friend. Every operation against another country needs a free slot, room under the per-type limit, no operation of the same type already running against that target, funding of at least 1, a target that is not decentralized, no truce with it, no promise from a nuclear crisis to run none against it for ten years (see Nuclear weapons), and no Intelligence Sharing Pact binding it (see The Intelligence Sharing Pact's defense shield). Good relations don't stop any of them: you can run an operation against a friend, and pay for it if it is exposed (see Detection and exposure).

A **friend** is a country with which you have cordial relations plus an alliance, a defensive pact, the same power bloc or an overlord or subject tie, or amicable relations on their own. Two members of a bloc who dislike each other are not friends.

Seven operations also need a technology, named first in the table's last column: Mass Media and Cryptography (era 6), Television Broadcasting, Pop Culture and Satellite Communications (era 7), Computer Networks (era 8) and Social Media (era 9). The operations that can bring down a great power come late in the tree, so a small country can't rush them before the great powers have agencies of their own.

| Operation | What it does at base strength | Also needs | Influence |
|---|---|---|---|
| Election Interference | Target −5 legitimacy; its electoral confidence falls at the end of each election campaign | Television Broadcasting; target has an elected legislature | 50 |
| Financial Subversion | Target +2 bubble pressure a month | Computer Networks | 50 |
| Infrastructure Sabotage | Target units lose morale 15% faster; each month a random target state gets −15% infrastructure and −10% throughput for three months | War or a diplomatic play against the target | None |
| Communications Disruption | Target −10% unit offense and defense | Cryptography; war or a diplomatic play against the target | None |

| Operation | What it does at base strength | Also needs | Influence |
|---|---|---|---|
| Industrial Espionage | You: stolen production technology. Target: −3 authority | Target has researched a production technology that you could research now | 25 |
| Military Espionage | You: stolen military technology. Target: −3 authority | Target has researched a military technology that you could research now | 25 |
| Influence Campaign | Target −15% leverage resistance | Mass Media; you lead a power bloc the target is not in | 50 |
| Ideological Subversion | Target −10% separatism resistance; movements there matching your political model grow more active and attract more pops | Pop Culture; target has a political movement | 100 |
| Destabilization | Target −15% separatism resistance, worse colonial stability; every movement there grows more active and attracts more pops | Social Media; target has a political movement | 100 |
| Regime Change | Target −5 legitimacy, −1 coup resistance, more radical movements; once fully operational, a coup under way there gains 5 progress a month | Satellite Communications; a rivalry; target not your subject; Seasoned Tradecraft | 100 |
| Nuclear Program Sabotage | Target's nuclear program 20% slower | Target runs a funded program; Seasoned Tradecraft | 100 |
| Space Program Espionage | You: space race milestones 10% faster and 10% less risky. Target: −3 authority | Target has completed a milestone you have not | 25 |
| Cultivate Assets | Nothing to either country; your network there grows half again as fast | Target not your subject | 25 |
| Secure Loose Material | Once established, a 3% monthly chance (6% fully operational) of finding and destroying one of the target's missing warheads | Target has warheads unaccounted for; Nuclear Weapons rule on | 25 |
| Incite Colonial Autonomy | Every colony the target holds gains 0.5 autonomy a month; a colony that reaches 100 breaks away | Target holds a colony; Simple or Full Colonies setting of the Space Race rule | 50 |
| Sabotage Colony | The target's most populous colony loses 1% of its people a month and gains 0.25 autonomy | Target holds a colony; Simple or Full Colonies; Seasoned Tradecraft | 100 |
| Survey Colony Defenses | Your fleets fire 20% harder on the defenses of the target's colonies, and your Fleets panel shows their Defenses, integrity and marines | Target holds a colony; Full Colonies setting | 50 |
| Counterintelligence Support | Target: +8 economic, military and ideological covert defense. You: nothing | Target is a friend and has no other protector | 25 |

An operation uses its influence for as long as it runs, more the worse the world would take its exposure (see Detection and exposure). As with any pact, a great power pays double and a major power half again as much, and you need the influence available to launch it.

Counterintelligence Support is open: it is never rolled for detection and never exposed, needs no technology, and an Intelligence Sharing Pact with the target doesn't block it. A country has one protector at a time, and the support ends if the friendship does. It uses a slot and upkeep like any operation but gives you nothing back. Its defense adds to the target's own (see Covert defense).

If you run several Space Program Espionage operations, their bonuses come from the strongest and don't add up. Stolen technology does add up: each Industrial or Military Espionage operation steals from its own target. Election Interference cuts electoral confidence by 5% once establishing and 10% once fully operational, half that against a country hosting UN electoral observers, which also catch it more often. Bubble pressure is explained in Banking and monetary policy, nuclear programs and loose warheads in Nuclear weapons, and milestones, colonies and their autonomy in The space race.

#### *Stealing technology*

From its sixth month, each Industrial Espionage operation steals production technology from its target every month, and each Military Espionage operation steals military technology. The progress goes to your current research if it is in that tree and the target has researched it. Otherwise it goes to a random technology in that tree that the target has and you could research now. Once the target has nothing left that you could take, the operation ends.

Each theft is 5% of the technology's era cost at base strength, 10% once fully operational, and 20% fully operational at priority 3. Progress that reaches a technology's cost completes it. Each operation's row shows how much it has stolen and the latest technology.

Hover over an operation before launching it: the tooltip previews its effects at base strength (established, priority 1), your network in that country and the head start it gives, the weekly upkeep the operation adds, and how its phase scales the effects, and the description says how the world will read it if it is exposed.

### Operation phases and priority

An operation does nothing for its first six months (Preparatory). From month 6 it is Establishing and has its base effect; from month 12 it is Fully Operational and its effects double. Each row shows the phase and the months to the next one; hover the phase for what it does. Ideological Subversion's push on movements is the exception: it doesn't double, and stays at its establishing strength.

Each row also has a priority stepper from 1 to 3; hover the word Priority for what the current level multiplies. Priority multiplies what the operation does, but costs and exposure rise faster, and each level fills another operation slot, so each step up needs a free one:

| Priority | Effect | Slots | Share of upkeep | Detection risk |
|---|---|---|---|---|
| 1 | ×1 | 1 | 1 | Base |
| 2 | ×1.65 | 2 | 1.6 | +2 points |
| 3 | ×2 | 3 | 2.4 | +4 points |

A slot spent on a second operation does more than one spent on priority, so priority pays when the per-type limit or the targets worth hitting run out. Stepping down frees the slot at once. If you lose slots, your operations keep their priorities, but none can rise until a slot is free again.

Priority 3 needs an Established agency; a fully operational operation at priority 3 works at four times base strength. A click changes detection risk at once, in either direction, but the effects only at the start of the next month. Raising priority raises upkeep at once; lowering it cuts upkeep only when the effects fall. Cultivate Assets stays at priority 1. Election Interference's cut to electoral confidence and Regime Change's push on a coup are fixed and don't scale with priority.

### Agent networks

Every country you run operations against keeps a network of yours, rated 0 to 100, that outlives the operations. It grows by about 2 a month with one operation there, faster with two or three and more slowly above 50, so one operation takes it to 50 in about two years (Cultivate Assets, in about 17 months). With nothing running it decays by 1.5 a month. An exposed operation costs its network 25. Counterintelligence Support plants no network, but an operation you watch grows your network inside its sponsor and keeps it from decaying.

A strong network gives new operations there a head start of up to five months (strength ÷ 20) and takes up to 5 points off their detection risk. It also adds up to 10 points (strength ÷ 10) to the detection risk of every operation

that country runs against you, because your agents inside its service hear of them; hover the network's name for how many points. From strength 50 it reports the target's intelligence capacity and technology count beside yours, and from 75 how many covert operations the target runs against you: the only way to learn of operations your counterintelligence has not caught, though it never says which. Each network's line shows its strength as a number and a bar, with an arrow for growing, decaying or at full strength, and under Report the word None, Partial or Full; hover the Report heading for what the report shows at 50 and at 75. In the space race, the same two tiers narrow your estimate of that country's progress on a milestone from ±25% of the goal to ±10% and ±5% (see The space milestone panel).

## Tradecraft

Tradecraft is your agency's experience, from 0 to 100. Each month every operation against another country past its preparatory phase adds 1 (a preparatory one half that, and Cultivate Assets always half), counting at most four operations, and gains slow above 50. An exposed operation costs 3 (routine espionage or wartime sabotage), 6 (interference) or 9 (an attempt on a government). With no such operation running, Tradecraft drifts down by a quarter point a month; Counterintelligence Support adds nothing and doesn't stop the drift. Standing an operation down costs nothing.

| Tier | Tradecraft | Unlocks |
|---|---|---|
| Untested | 0 | |
| Fledgling | 20 | |
| Established | 40 | Priority 3 |
| Seasoned | 60 | Regime Change, Nuclear Program Sabotage and Sabotage Colony |
| Veteran | 80 | One more operation of each type |

Each tier above Untested adds 5% to your intelligence capacity and makes your networks grow 15% faster. Falling below a tier never ends a running operation.

## Detection and exposure

Each operation against another country has its own monthly Detection Risk, shown on its row (hover it for how it is made up). It starts at 10%, falls with your Funding Stealth and your network in the target, and rises with priority, with the target's counterintelligence (its capacity plus its defense against that kind of operation, compared to yours; up to +20 points) and with any network the target holds inside your country (up to +10 points). Your Efficiency Factor, from the Covert Operation Efficiency modifier that technologies, laws and the Ministry of Intelligence and Security give, then cuts it by up to four fifths, within a range of 0.1% to 50%. The Operations section lists the base risk, your Funding Stealth and your Efficiency Factor above the rows. Each operation rolls separately, so three at 10% give about a 27% monthly chance of a catch, but at most one is caught a month.

A caught operation goes to its target first, which can expose it or let it run. An exposed operation ends in the Operation Compromised event, which also costs its network 25 and costs you Tradecraft. Recalling your operatives and apologizing takes the full infamy and half the relations damage. Denying everything takes half the infamy and the full relations damage, and the target gains Heightened Vigilance (+25 to all three of its covert defenses, decaying over five years). At the establishing phase the costs are:

| Exposure tier | Operations | Recall and apologize | Deny everything |
|---|---|---|---|
| Routine espionage | Industrial, Military and Space Program Espionage, Cultivate Assets, Secure Loose Material | +1 infamy, −5 relations | +0.5 infamy, −10 relations |
| Interference in internal affairs | Election Interference, Financial Subversion, Influence Campaign, Incite Colonial Autonomy, Survey Colony Defenses | +2 infamy, −10 relations | +1 infamy, −20 relations |

| Exposure tier | Operations | Recall and apologize | Deny everything |
|---|---|---|---|
| Attempt to bring down a government | Ideological Subversion, Destabilization, Regime Change, Nuclear Program Sabotage, Sabotage Colony | +4 infamy, −20 relations | +2 infamy, −40 relations |
| Wartime sabotage, exposed during the war | Infrastructure Sabotage, Communications Disruption | Nothing | Nothing |
| Wartime sabotage, exposed before the war | The same | +4 infamy, −10 relations | +2 infamy, −20 relations |

An operation exposed while preparatory costs half as much, and one exposed fully operational half as much again, which makes an operation its target has watched into that phase the costliest to have exposed. Exposing an attempt on a government also costs 10 relations with every country in the target's power bloc and every treaty ally of the target, and the United Nations records it (see The United Nations).

Spying on a friend costs more. The relations damage is half again as large, and at home the political lobby that favors that friend loses appeasement and your government loses legitimacy (Caught Spying on a Friend), fading over about two and a half years. Each is 3, 5 or 8 points by exposure tier (routine espionage, interference, attempt on a government; wartime sabotage exposed before the war pays the highest) if you recall your operatives and apologize, and 2, 3 or 4 if you deny everything. The operation's phase doesn't change them. With no lobby that favors the friend, only the legitimacy is lost. Hover the options for the figures.

## What the target learns

A country that catches your operation decides first whether to expose it. If it runs a Covert Warfare journal entry of its own, a player gets an event, A Foreign Operation Uncovered, and chooses between "Expose them" and "Let it run, and watch"; a country without the entry exposes at once, and the AI chooses by chance (see How the AI runs covert operations).

Exposing makes the operation public: you answer for it as described under Detection and exposure, and the operation ends. It also turns the exposing country's public against the interest groups at home that are friendly to you. Every group in a political lobby that favors your country loses 5, 10 or 15% political strength (Foreign Ties Discredited) by exposure tier, half as much again if you were its friend, fading over about two and a half years. That follows every exposure that costs anything, whoever chose it.

If the country lets the operation run, you are not told, and it keeps working against that country at full strength. In return its network inside yours stops decaying and grows each month, by 4 to 8 depending on the operation's priority (more with Tradecraft, more slowly above half strength), and the operation is not caught again. The country can expose it whenever it likes from Known Foreign Operations, a section of its Covert Warfare journal entry and Covert tab that lists each operation it watches with your flag and name and the months watched. The exposure goes out at once; while another exposure of your operations is still open, the button is grayed until you have answered for it.

A country that exposes your operation gets the Foreign Interference Detected event, which names you and the operation. It can shore up its counterintelligence (Counterintelligence Alert: +6 intelligence capacity and more separatism and coup resistance, fading over five years), retaliate in kind (a network of up to 25 inside your country, if it has a Covert Warfare journal entry of its own), or expel your diplomats (−25 relations; your network inside its country loses another 25, and any network it runs inside yours loses 25 when you answer in kind). The event fires at most once every two years per country, whether or not that country has a Covert Warfare journal entry.

When you are the target, the Spy Caught icon and the Last Caught line in the Counterintelligence section show the latest operation exposed against you, and who ran it, for ten years, and Known Foreign Operations names the sponsor of each operation you watch. Outside these two sections and the events A Foreign Operation Uncovered and Foreign Interference Detected, nothing names the country behind an operation. You can also

see an operation's effects among your own modifiers, get an unsigned Election Interference Detected notification with each confidence hit, and count the operations against you through a strong network of your own.

Watching pays when the operation does little harm to you and the sponsor's service is worth learning: one watched operation takes your network inside the sponsor past the 25 that retaliating in kind plants in about six months. Expose an operation that is hurting you.

## Covert defense

Each unused slot becomes Domestic Intelligence Focus: +2 intelligence capacity, more separatism and coup resistance, calmer movements and better colonial garrisons. You also have three covert defenses, economic, military and ideological, each added to your capacity against operations of its kind. Strict IP Protection, Central Bank Independence and State Monopoly (Frontier AI Control) raise economic defense and Publicly Funded Elections ideological defense, by 10 each; No IP Protection, Unregulated Banking, Private Labs, Open Weights and No Campaign Finance Laws lower them. A friend's Counterintelligence Support raises all three at once, by 8 each at base strength and up to 32 fully operational at priority 3. Against an agency of about 100 capacity, 10 points of defense add about 1.5 points to its Detection Risk. The Intelligence Sharing Pact treaty article adds intelligence capacity and lends the weaker partner part of the stronger one's covert defenses.

## When operations end

An operation runs until one of these happens:

- It is exposed.
- You press Stand down on its row, or break the pact in the diplomacy outliner. This frees the slot at once at no infamy or relations cost, but the months invested are lost.
- Your funding drops to 0, which ends every operation.
- A truce with the target begins (not for Counterintelligence Support).
- The target of Counterintelligence Support stops being a friend.
- A wartime operation's war ends, or its diplomatic play ends without one.
- Something it needs goes: the target's elections, its political movements, its lead in space, its last colony, the last technology you could steal from it, its missing warheads or its nuclear program (frozen by treaty, held at an arsenal ceiling or while it dismantles, or given up), your rivalry, or your power bloc lead over a target outside the bloc. The technologies named in the operations table and Tradecraft only gate the launch.

A war with the target does not end a peacetime operation.

With the Covert Warfare rule off, a few international-relations events such as The Shadow War and The Spy Who Was Caught stand in for the operations; see International relations events.

## How the AI runs covert operations

AI countries set their funding each month: none while idle, level 1 to keep operations alive when bankrupt or without rivals (and for a major power with a rival), level 3 for a great power with a rival or a major power at war, and level 5 for a great power at war. They launch rarely. Peacetime operations come mostly from major and great powers, aimed at rivals or countries they feel antagonistic or domineering toward; espionage goes to countries with technology to steal or ahead in space, and a funded AI at war or in a play uses the wartime operations. An AI that holds a colony or runs a space milestone incites autonomy in the colonies of countries that are not its friends, and sabotages them only when the colony's holder is its rival or at war with it. An AI with fleets of its own surveys the colony defenses of a rival or an enemy. Upkeep grows with the operator's GDP, not the target's, so an AI leaves a country with less than a tenth of its GDP alone unless its gold reserves are at least half full and its budget is in surplus. Espionage, Nuclear Program Sabotage, Secure Loose Material and the colony operations ignore the target's size, and so does a wartime operation against a country the AI is losing to. The AI raises priority to 2 against rivals and to 3 when a great power is at war with the target. Once launched, an operation stays until it is fully operational, unless it is exposed or something it needs goes. After that the AI calls it off only when its reason has gone: no rivalry either way, no hostile attitude and relations above poor, or no space milestone of its own under way (for the colony operations, no milestone and no colony of its own; for the defenses survey, no fleets of its own, or peace and no hostility), or when its influence runs into deficit. It launches an operation that uses influence only while it holds 300 or more. It

stops Cultivate Assets once its network reaches 75. It needs Seasoned Tradecraft for Regime Change, Nuclear Program Sabotage and Sabotage Colony, as you do, so it rarely launches them.

The AI doesn't start Election Interference, Financial Subversion, Infrastructure Sabotage, Communications Disruption, Ideological Subversion, Destabilization or any colony operation against a friend, and calls off Election Interference, Financial Subversion, Ideological Subversion and Destabilization once they are fully operational and the target has become a friend. Its other operations have no friend test: it still spies on friends, and pays when it is exposed. An AI with a Covert Warfare journal entry that catches an operation lets it run 6 times in 10 (15% for an attempt on a government; 20 points more when the sponsor is a friend), but only 1 in 10 once its network inside the sponsor reaches 75. It then exposes each operation it watches by a monthly chance that grows with that network: 25% once it reaches 75. An AI that funds its service and whose agency is stronger than that of a friend in a bloc it leads, or of its subject or ally, offers that friend Counterintelligence Support, one at a time.

CHAPTER 12

# Colonial empires and decolonization

Once the Decolonization technology is researched, holding colonies stops being free. A country with overseas colonies or colonial subjects gets the Colonial Empire journal entry, whose Colonial Stability bar drifts downward unless your rank, laws and programs hold it up. You can spend money, authority and bureaucracy to slow the drift, integrate colonies into the nation, or let them go on your own terms before they break away. The Decolonization game rule controls the system; with it off, the journal entry and its events never appear, and tiny countries never collapse.

Decolonization also ends your claims on decentralized nations. Within a month of researching it, you lose every claim you hold on a state of a decentralized nation, and a Colonial Claims Renounced notice tells you so. Stake Colonial Claim is no longer open to you, and a claim that reaches you some other way is dropped within the month. AI countries lose their claims and stop staking them too. Claims on other countries' states, such as the ones a conquest leaves you with or the ones behind reunification, stay.

## 12.1 The Colonial Empire journal entry

Decolonization is a society technology in era 6, the first of the mod's eras. After you research it, the Colonial Empire entry activates as soon as you hold either of these:

- An overseas colony: a state that is not a homeland of your cultures and has at least 100,000 people, which is separated from your capital, has a distinct people and is held down. It is separated when it isn't joined to your capital over land through your own states; a strait joins two states only when both are homelands of your primary cultures. It has a distinct people when a culture that shares no heritage trait with any of your primary cultures is more than a tenth of it. It is held down when it is unincorporated, when that people's average acceptance there is below 60, or when its average standard of living is below 80% of your national average. A state where more than half the people are of your primary cultures is settled land, not a colony.
- A colonial subject: a Colony or Chartered Company subject, or a Protectorate separated from your capital in the same way, none of whose primary cultures shares a heritage trait with yours.

This is the test the United Nations used in 1960. France's Algeria and its Tunisian and Moroccan protectorates, Japan's Korea and Taiwan, and Italy's Libya are colonies, across the sea from their capitals. Land empires are not: Russia's Central Asia and China's Tibet are joined to their capitals by land, and leave, if they leave, by the base game's secessions. Heritage traits are narrow, so a people you share one with doesn't make a state a colony: Ireland under Britain, Iceland under Denmark, Ryukyu under Japan, or a country's own islands.

A great power with a few well-integrated colonies and little international opposition can hold on; a smaller empire condemned by the big powers usually can't. Two era 9 technologies, Knowledge Economy and Globalization, add −0.75 and −1.5 a month to the drift, so holding on gets harder as the game goes on.

### The Colonial Empire panels

The journal entry opens with an overview, and the sections below it hold the detail. While the entry runs it has no status line, because the overview shows where you stand. An entry that isn't running says why: the empire is secured, it collapsed less than ten years ago, or you hold no colony or colonial subject.

The same panels appear on the Colonial Empire tab of the Timeline Extended window, which the button under the sidebar's Map List opens, and a change made in one shows in the other. It ends with an Open Journal Entry button. The tab opens once you have researched Decolonization, whether or not the journal entry runs. Without the entry it shows only Our Stance on Colonialism and The Colonial Powers, under a line saying why the entry isn't running; hover that line for what would start it. So the tab stays open after your empire is secured, collapses or lets its last colony go.

National Info shows another country's Colonial Empire tab with its holdings, stability, programs, pressure, stance and history, and no buttons. It counts that country's territories and colonial subjects and shows how much the great powers move its stability, but doesn't list any of them one by one, because the mod builds those lists only for the country you play. The Colonial Powers rows are the same as on your own tab, without their hover.

The overview starts with the Stability bar. Its solid part and a lighter stretch show today's value and where the bar will be in 12 months at this month's rate. The stretch is green while stability rises and red while it falls, so red is what the next year would cost you. A thin cream line marks where the next band begins; hover it for how many months away that is. Beside the bar are the value and this month's change, such as "62% (+0.42/mo)", and an arrow for its direction. Hover the bar for every term behind the change, and the value for the projection. Four boxes sit under it:

| Box | Shows |
|---|---|
| Band | Your band as an icon and a name, and where the next one begins, such as "Stable at 65". Hover it for the band's modifier. |
| Great Powers | Isolated or Consensus while one of the two escalations applies, otherwise No escalation, and what the great powers do to your stability a month. Hover it for how many condemn and support you. |
| Holdings | Your colonies and colonial subjects, what keeps the entry running. Hover it for what each one is. |
| Programs | How many of the three programs run, their icons lit while they do, and what they add a month. Hover it for each one's level. |

In the Solidified band, two more bars count toward winning: "Months at 100" against 60 and, for a great power, "Months Solidified" against 36, the requirement of the Imperial Federation Act and the Civilizing Mission Compact. Last comes a pie of great-power prestige, yours included: the condemning powers' share in red, the supporters' share in green, the rest in gray. Only the condemning share counts toward the two escalations; hover each line of the legend for what its share does.

Every section starts open:

| Section | Starts | Shows |
|---|---|---|
| Colonial Holdings | Open | Your wars of independence, your territories and your colonial subjects, with buttons beside each, then Planned Full Decolonization (see below). |
| Why Stability Is Moving | Open | A bar for each term that moves the bar: red to the left of the center line for a drain, green to the right for a gain, each on its own scale. Then Base decline, Monthly limit and Projected change. Hover a row for what it holds and its bar's scale. |
| International Pressure | Open | Every great power that condemns or supports you, with its prestige and what it does to your stability a month, then any escalation in force. |
| Colonial Programs | Open | The three programs, each with its level out of 3, what it adds a month, and a minus and a plus button to lower or raise it. Hover a row for its costs. |
| Our Stance on Colonialism | Open | Your stance, whether it is in force, when it can next change, and the Condemn, Stay Neutral and Support buttons. |
| The Colonial Powers | Open | The eight countries with the most people in their colonies and colonial subjects, yours among them if you are one: their people, their status and their stability (see below). |

| Section | Starts | Shows |
|---|---|---|
| History | Open | Charts of Colonial Stability and the Monthly Stability Change. |

The "?" at the right of the tab's header, and at the top of the journal entry, opens the Help page, How the Colonial Empire Works, which explains the system.

Colonial Holdings shows what keeps the entry running and what you can let go. Its Territories list holds every territory you could release, one row each, largest GDP first. A release takes the neighboring states where the same people are at home, so those states share one row: "Henan +1" is Henan and one more state that would leave with it. Each row gives the share of its people of your primary cultures (Ours), its standard of living (SoL) and its GDP, across all its states; hover the name for the states, the people who would form the new country and the row's weight on stability. The colony icon at the start of the row is lit when a state in it counts as a colony, which keeps the entry running and weighs on the bar through Imperial Overreach and Colonial Acceptance, and dim when none does. A red figure is the test that makes one a colony: a red Ours share means a people sharing no heritage with yours is more than a tenth of a state and accepted below 60, and a red SoL means a state's standard of living is below 80% of your national average while some of its people are of other cultures. An unincorporated state counts too; hover the icon for how many of the row's states are. Hover a figure for its threshold. A row with a dim icon can still be released, with the Release button at the end of the row.

While you fight a war of independence, a Wars of Independence list sits above the territories: the country fighting you, the months the war has run (red while your war support in it is 25 or below), the great powers arming it, and an Open Talks button.

The Colonial Subjects list shows your colonial subjects the same way: the flag, the name, the kind of subject (hover it for why it counts as colonial), its liberty desire and its GDP, then a Release and a Dominion button (see Freeing colonial subjects). Under The Whole Empire is Planned Full Decolonization. Both lists update monthly, and at once after a release.

## Colonial stability bands

The Colonial Stability bar runs from 0 to 100 and starts at 50. Once a month it moves by the sum of everything acting on it, but never by more than 1.67 up or down, so crossing the whole bar takes at least five years. Where the bar sits puts you in one of five bands. Each band applies a modifier to the journal entry, and the lower three also raise your colonial subjects' liberty desire.

| Band | Bar | While in this band |
|---|---|---|
| Solidified | 90 and above | +15% prestige, +15 acceptance for pops living outside their homeland, more loyalists from movements |
| Stable | 65–90 | +75 prestige, +5 acceptance for pops outside their homeland, more loyalists from movements |
| Strained | 40–65 | −25 prestige, a little bureaucracy lost; colonial subjects gain liberty desire |
| Crumbling | 20–40 | −50 prestige, −3% bureaucracy; subjects gain more liberty desire; civil rights movements radicalize |
| Collapsing | below 20 | −100 prestige, −6% bureaucracy, more radicals from movements; subjects gain severe liberty desire; civil rights movements radicalize further |

## What moves colonial stability

Why Stability Is Moving shows the projected change for the month and where it comes from, in the rows below. Hover the bar itself for every term with its current value.

| Row | What it contains |
|---|---|
| Base decline | −0.5 a month, always. |

| Row | What it contains |
|---|---|
| Laws and events | Your Colonization, Minority Rights, Citizenship, Distribution of Power, State Power, Free Speech and Internal Security laws; Knowledge Economy and Globalization; event modifiers such as Positive Colonial Development (+1 a month) and Colonial Crisis (−1 a month) while they last. |
| Interest Groups | Powerful Landowners +0.3 and Armed Forces +0.2; powerful Intelligentsia −0.4 and Trade Unions −0.3. |
| Rank | +0.3 as a great power, −0.5 as anything else. |
| Programs | Whatever your running colonial programs are worth. |
| War and Turmoil | −0.5 while at war, a further −0.5 for each war of independence (−0.25 once its accords are signed), −1 during a revolution, and a penalty that grows with national turmoil. |
| Imperial Overreach | A penalty once your colonial states outnumber your other states by more than 1.5 to 1. |
| Great Power pressure | Condemnation and support from great powers (see below). |
| Colonial Acceptance | −0.4 for each colony with a severely unaccepted population, +0.5 for each overseas state where everyone is well accepted. |
| Monthly limit | What the ±1.67 cap removed, if anything. |

Colonial Exploitation and Colonial Resettlement raise the bar; No Colonial Affairs and Neocolonialism lower it. The harshest minority-rights laws drain it heavily, even though they make a garrison far stronger. Being a nuclear power helps a little; the UN's Decolonization Resolution (UN conventions and agencies) and foreign destabilization (The covert operations) hurt.

## Great power stances on colonialism

Each country can hold an anti-colonial stance, a pro-colonial stance, or none, but only a great power's stance weighs on the colonial empires. The first time you are a great power that has researched Decolonization while another country holds an overseas colony, the event The Colonial Question asks you to declare one. That includes a power with no colonies of its own and one whose own Colonial Empire entry has ended. After that you change it from Our Stance on Colonialism, on the Colonial Empire tab, which stays open once your own entry ends:

| Stance | While you are a great power | Button |
|---|---|---|
| Condemns colonialism | Anti-Colonial Policy: +5% prestige, 150 influence. If you hold colonies yourself, −0.2 stability a month and weaker programs. | Condemn |
| Neutral | Nothing, and no cost. | Stay Neutral |
| Backs the colonial powers | Pro-Colonial Policy: +2% prestige, 150 influence. | Support |

A stance can change once every 24 months, whichever way it changes. Taking a side needs 150 influence available, unless you already pay for the other side. Condemning pleases the Intelligentsia and Trade Unions and angers the Landowners and Industrialists; backing the colonial powers does the reverse and radicalizes some academics. Below great-power rank a stance lies dormant: it costs nothing and moves no one until you are a great power again. Stay Neutral needs no rank, so you can always drop a dormant stance.

The AI is asked in The Colonial Question every three years instead. An AI power without colonies nearly always condemns; one with colonies leans toward backing them. It takes a side only with 150 influence and 100 more to spare, and drops its stance when its influence runs negative. When a Decolonization Resolution passes at the United Nations, every great power that voted for it, or voted against it and then accepted it, takes the anti-colonial stance automatically, and can't change it for 24 months.

Each condemning great power costs you about 0.6 a month and each supporter adds about 0.3, scaled by its prestige against yours: from a quarter as much, for a power with a quarter of your prestige, up to double. If the condemners together hold a third of the prestige of all great powers plus you, you lose a further 1 a month; if they hold two thirds, another 2. Two condemning peers that together hold a third already cost about 2.2 a

month; with the base decline, the rest of your terms must add more than +1 just to keep the bar off its fastest monthly fall. A dominant empire is hard to isolate; a small one reaches both thresholds as soon as a few large powers turn on it. The International Pressure section lists who condemns and who supports you, and what each one costs or gives you a month.

The Colonial Powers lists the eight countries with the most people in their colonies and colonial subjects, largest first, read each month and again when you change your stance. Each row gives the people of its colonies and subjects, its status and its stability to the nearest 5 while its entry runs. The status is Not begun before its Colonial Empire entry starts, its band while the entry runs, Secured once it has won, and Collapsed for ten years after its colonial system falls apart. Hover a row for what your stance does to its stability a month.

## 12.2 Colonial programs and decolonization decisions

The Colonial Empire panels are where you act: three programs you run at a level from 0 to 3, a button beside each territory and each colonial subject to let it go, and Planned Full Decolonization to let them all go. Two decisions offer the routes to a permanent empire.

### The three colonial programs

Each program runs at a level from 0 (off) to 3, set with the minus and plus buttons on its row. Level 3 is the full program and each level is a third of it: a third of its value to the bar every month, a third of its effects and a third of its running cost. The bar values below are the base at level 3; your laws change them, and the row shows what the program adds now; hover it for what one more level would add and for its costs.

| Program | Requirement | Bar, per month at level 3 | Costs and side effects at level 3 |
|---|---|---|---|
| Colonial Development Investment | none | about +0.5 | Colonial Development Spending, a weekly cost of 5% a year of your colonies' GDP for each level, counted as at least 1% of your own GDP. Each step up also costs a one-off half a year of one level's spending. Your lower and middle strata expect a higher standard of living. Industrialists and Petite Bourgeoisie disapprove when you start it. Unincorporated states attract more migrants. |
| Military Garrison | more than 100 authority for each step up | about +1.0 | 100 authority a level for as long as it runs, and 5 infamy when you start it. Infamy decays 20% slower. Small losses of bureaucracy and prestige from army projection, higher borrowing costs, weaker Intelligentsia and Trade Unions. |
| Cultural Assimilation Program | none | about +0.7 | Each step up costs a one-off quarter of a year of one Investment level's spending. −15% bureaucracy and 100 authority while running. Assimilation and conversion speed up, but movements radicalize more, and colonial states lose some qualifications while it runs. |

Investment's cost follows the size of what you develop, not of your whole economy. If your colonies produce a tenth of your GDP, each level costs about 0.5% of your GDP a year; a great power with a couple of small colonies pays little. Raise a program only while the bar can still use it: the bar's change is capped at about 1.67 a month, and the bar stops at 100, so a level that pushes the change past the cap, or adds to a full bar that is not falling, is paid for and does nothing. The Monthly limit row in Why Stability Is Moving shows how much of the change the cap is cutting off.

Laws pull the programs apart. Outlawed Dissent and the harshest minority-rights laws strengthen the garrison, while Protected Speech and the Protection and Affirmative Action minority-rights laws weaken it. Voting franchises strengthen investment. Assimilatory, Ancestral and Cultural Citizenship and the Cultural Assimilation minority-rights law strengthen assimilation, and Universal Citizenship weakens it. The program you run longest also decides how the empire ends (see How a colonial empire ends); a month at a lower level counts as a third of a month for each level.

## Releasing colonies

Every territory in Colonial Holdings has a Release button. It opens a popup about that territory alone, which names the people who would form the new country and how many of your states go with it. The new country is formed by the most numerous culture whose homeland the territory is or, if none of them lives there, by its most numerous culture that isn't one of yours, and it leaves with every connected state of yours that is a homeland of that culture. Choose how it goes, or reconsider. Reconsider releases nothing, and it is what the popup does if you leave it unanswered. A territory can be released if it isn't a homeland of your cultures, has at least 100,000 people, is separated from your capital, has a distinct people and isn't settled land (more than half its people of your primary cultures); it doesn't have to be held down. A land of your own heritage, such as an Australia you rule directly, leaves by the base game's release or as a dominion instead. Hover Release to see whether a Round Table is open to it.

| Choice | Needs | What you get |
|---|---|---|
| Grant it independence | nothing | +1 stability a month for 8 months; the new country starts slightly less friendly. |
| Round Table Conference | none of the territory's people severely unaccepted, and more than 200 authority; grayed with the reason otherwise | +1 stability a month for 12 months; Intelligentsia approve, Landowners disapprove; the new country starts friendlier. |
| Planned Full Decolonization, under The Whole Empire | nothing | Every territory goes at once, after a popup that lists them; the new countries start resentful. |

One of these popups can wait for an answer at a time; until you answer it, the other buttons are grayed. A release also grants Peaceful Decolonization, a decaying +15% prestige, +50 influence and −200 authority for five years (ten after Planned Full Decolonization). A territory or subject you free while it lasts doesn't add a second one. Each new country also shakes every other empire: all other countries running the Colonial Empire entry take a year of Colonial Crisis, −1 stability a month. An empire carries one such year at a time: new countries formed before it runs out add nothing to it.

## Freeing colonial subjects

Every colonial subject in Colonial Holdings has a Release and a Dominion button. Each opens a popup about that subject first, with Reconsider as its default. Both need it to be your own subject, not one of your subjects' subjects, and not your enemy in a diplomatic play.

| Button | Needs | What happens |
|---|---|---|
| Release | nothing more | The subject becomes independent, a former colony of yours. Relations rise by 30, the Landowners strongly disapprove, and you gain +1 stability a month for 8 months and ten years of Peaceful Decolonization. With the Neocolonialism law you can instead free it on your terms: it takes Neocolonial Dependency, you take Neocolonial Economic Concessions, and relations fall by 10. |
| Dominion | a Colony that can become a dominion, and no change of its | It becomes a Dominion, as with the base game's Increase Autonomy: its liberty desire falls by 30 and you take a five-year truce with it. A dominion isn't a colonial subject, so it stops keeping the entry running and your band no longer raises its liberty desire. It still joins your wars, and it may start diplomatic plays of its own. |

| Button | Needs | What happens |
|---|---|---|
| | autonomy by you lately | |

A chartered company or a protectorate has no looser kind of subject to become, so its way out is independence. Letting go of your last colony or colonial subject ends the entry (see How a colonial empire ends).

Once the UN charter carries Charter Reform II and the Decolonization Resolution is in force, the General Assembly can order a referendum in a direct subject whose liberty desire is 50 or more. You hold it, and the subject leaves with a chance equal to its liberty desire, or you refuse and pay in standing, prestige and the loyalty of every subject (see Supervised referendums).

### Integrating colonies instead

Common Bonds and two decisions keep colonies for good. The event Common Bonds fires when an overseas state's whole population is well accepted, and lets you make it a homeland of your primary cultures; it then stops counting as a colony.

The Imperial Federation Act and the Civilizing Mission Compact are decisions that complete the journal entry at once. Both need a great power that owns at least four overseas colonies itself (colonial subjects don't count), Colonial Exploitation or Colonial Resettlement, 36 consecutive months in the Solidified band, more than 300 authority and 100 prestige. The Act also wants repressive minority-rights, citizenship, speech and power laws; the Compact wants liberal ones and a voting franchise. The decision tooltip lists the accepted laws.

The Mandate System Decision suits a liberal empire in managed decline. It needs the bar at 65 or more, the Neocolonialism law, a voting franchise, no Outlawed Dissent and more than 100 authority. It costs 100 authority, easing over five years (Mandate System Transition), and can be taken only once. It gives a permanent diplomatic-reputation bonus at a small cost in acceptance, and a year of Positive Colonial Development.

## 12.3 Wars of independence

A colony can declare its independence, and you can let it go, negotiate, or fight to keep it. Fighting is hard to win, and winning buys years, not a settlement: the people are as unaccepted as they were.

### When a colony declares independence

Once your bar is below 40, a colony with severely unaccepted people (acceptance below 20) may declare independence in Declaration in (the territory), twice as often below 20. The one most likely to rise goes first: the colony with the largest share of severely unaccepted people, then the largest. Outside a collapse, you face at most one declaration every two years, and a territory that declared, or that you won back, waits ten years before it declares again. Below 40, Blood in the Colonies also turns its suppression into a war of independence; at 40 or above it stays a police action.

The popup names the territory, the people who would form the new country and, when your own people are a tenth or more of the territory, its settlers.

| Choice | What happens |
|---|---|
| Let them go | The country forms now, as in Blood in the Colonies' release: Colonial War Weariness, the Armed Forces disapprove, the Intelligentsia approve, five months of +1 stability. |
| Negotiate the handover | A Round Table release with the declared government as your partner, whatever its acceptance: +1 stability a month for 12 months and friendlier relations. It costs 100 authority, easing over a year. Under Neocolonialism it goes on your economic terms instead. |
| Send the army | The rebels form as a country with the territory and the neighboring states where its people are at home. The next day they open a diplomatic play against you, and about three weeks later you are at war. Until then you can still back down, which lets them go. You can fight three such wars at once; with three running the option is grayed. |

## Fighting a war of independence

The rebels start the war, a War of Independence. Their goal is Colonial Independence; yours, Reconquer, annexes them back. Neither costs infamy, and neither side can add goals. Holding their land doesn't end it: only a capitulation or a peace does, so a war can drag on long after your army holds every province.

The rebels fight like guerrillas. They lose less morale and fewer provinces after a defeat, defend their own provinces better, raise twice the conscripts and train them faster, and their war support gains "Fighting for independence" (+0.5 a week), or more with foreign arms. Yours loses "A war to keep a colony" (−0.25 a week) and "The world condemns it" (−0.25, or −0.5 once the condemning great powers hold two thirds of great-power prestige), and gains "Settlers to defend" (+0.25) when your own people are a tenth of the territory. Each war drains the bar by 0.5 a month on top of the at-war term. Every year of war, summed over all your wars of independence, costs 2 legitimacy, up to 10. Each year the Intelligentsia and the Trade Unions disapprove; the Armed Forces approve for two years, then turn on the government.

A colonial subject's independence war against you, and the secession of a colony, count as wars of independence too: they appear in the list, the subject or the secessionists fight with the same advantages, and they end the same way. Talks are only for a territory's rebels; you make peace with a subject or with secessionists from the peace screen.

## How a war of independence ends

| Ending | When | You get | The territory |
|---|---|---|---|
| Order Restored | The rebels capitulate or are annexed, or a subject that fought you is still yours when its war ends | 12 months of +1 stability, 12 months on Military Garrison's record (your ending leans toward The Iron Fist), the Armed Forces approve, the Intelligentsia disapprove | Yours again, with large radicals, and no declaration from it for ten years |
| The Flag Comes Down | The war ends and the rebels are free | The Flag Came Down: −15% prestige and −10 legitimacy, decaying over five years; the Armed Forces strongly disapprove for three years and The Veterans' Protest can follow; 12 months of −1 stability | A former colony with Bloody Decolonization's legacy, whatever program you ran: autocracy, outlawed dissent and relations −50 |
| The Accords | Talks dropped your claim and the war ended in a peace | The Accords: −15% prestige and −5 legitimacy, decaying over five years (no legitimacy under the general's mandate); the Armed Forces disapprove for a year | A former colony with your longest program's legacy and a Round Table's friendlier relations |

## Talks and the home front

Open Talks on a war's row asks for a popup first. Talks take six months while the fighting goes on. When the accords are signed, your claim is dropped and the rebels accept a peace: offer one from the peace screen, conceding their independence if they refuse a white peace. An AI opens talks after half a year of war support in the red, or after four years.

A war that has run four years, with your war support in it at 25 or below for a year running, breaks the home front. Without a voting franchise, junior officers rise in The Captains' Revolt: step aside, and talks open on every war, every territory you could release goes, and the country gets universal suffrage and a parliamentary republic (a monarchy keeps its crown); or crush the officers, at −20 legitimacy for two years, and face them again a year later. With one, it is The Republic in Danger: call the general, who brings a presidential republic, +100 authority for three years and a peace by the accords that costs no legitimacy; or hold the line at −15 legitimacy.

### The world takes sides

When a war of independence starts, every great power hears of it. One that condemns colonialism decides in War in (the territory): condemn it (relations −20), arm the rebels (they fight harder and keep their resolve, it pays them a monthly sum, and relations fall by 30), intervene, or stay out. Intervening joins the war on the rebels' side and is rare: it is open only to a great power at peace that is your rival with the larger army, and never to your ally, defensive-pact partner, power bloc partner, subject or overlord, or one under a truce with you. A great power that backs colonial rule may send you arms instead, for two years, but never joins the war.

## 12.4 How a colonial empire ends

The journal entry ends in one of four ways. The event you get when the empire holds or collapses depends on the program you ran longest, counting a month at level 1 or 2 as a third or two thirds of a month, and only a program with more than 24 months.

| Ending | Condition | Result |
|---|---|---|
| The empire holds | The bar stays at 100 for 60 consecutive months | Permanent Colonial Empire Solidified, plus a permanent modifier for your path if one program dominated. Overseas states where everyone is well accepted become homelands. After Quiet Assimilation every overseas state of 75,000 people or more does, whatever its acceptance; after The Iron Fist Holds none do, and you gain 25 infamy. You can never get the entry again. |
| Federation | You take the Imperial Federation Act or the Civilizing Mission Compact | Imperial Federation from the decision, plus a permanent The Iron Fist (Act) or The Commonwealth Path (Compact). Well-accepted overseas states become homelands. You can never get the entry again. |
| Collapse | The bar reaches 0 | Up to three of your territories declare independence, the one most likely to rise first, two months apart, and you can let each go, negotiate or fight; those that go inherit the legacy of your longest program (below). Every colonial subject gains Imperial Collapse Aftershock, a large liberty-desire increase for 20 years, and its relations with you drop by 50. The entry can't return for ten years; after that it starts again within two months if you still hold a colony or colonial subject. |
| Voluntary end | You no longer hold any colony or colonial subject while the bar is above 0 | A decaying boost to diplomatic reputation; the event's wording depends on whether you freed colonies by choice. There is no ten-year lockout: after a year, the entry starts again within two months of your holding a colony or colonial subject. |

| Longest program | If the empire holds | If it collapses | The countries you free |
|---|---|---|---|
| Military Garrison | The Iron Fist Holds | The Empire Crumbles | Bloody Decolonization Legacy; they start under autocracy with outlawed dissent, and their relations with you fall by 50 |
| Colonial Development Investment | The Commonwealth Path | Negotiated Withdrawal | Commonwealth Legacy; they start as parliamentary republics with protected speech, and relations rise by 30 |
| Cultural Assimilation Program | Quiet Assimilation | Negotiated Withdrawal | Cultural Assimilation Legacy; they start with Assimilatory Citizenship, and relations rise by 15 |
| none | The Empire Endures | The Empire Crumbles | no legacy |

The legacy applies to every country the system creates: those you release from Colonial Holdings, those an event frees as a new country, and those that go in a collapse. A colony that wins a war of independence takes Bloody Decolonization's legacy instead, whatever program you ran. On top of the legacy, a Round Table

adds 25 to relations, a single release subtracts 10 and Planned Full Decolonization subtracts 25. The Empire Crumbles also leaves you with Colonial Empire Collapsed, a decaying loss of prestige and standard of living; Negotiated Withdrawal instead gives the same decaying diplomatic-reputation boost as a voluntary end.

When a revolution wins, the new government carries the empire on. It keeps the programs you were running, Colonial Empire Solidified and the path rewards of a finished empire, the Imperial Federation and Mandate System modifiers, and any war of independence it is still fighting (see After a revolution).

## 12.5 Decolonization events

Most decolonization events fire for the colonial power while the journal entry runs: as a subject's liberty desire climbs, when great powers condemn or back you, when colonies resist, and as the cost of empire tells at home. Repressive options usually add Colonial Crackdown, which strengthens your garrison but costs prestige and bureaucracy and gives the Intelligentsia more clout; a war fought while it lasts can bring on the Conscription Crisis. Conciliatory options add Decolonization Negotiations or free the colony. With the Neocolonialism law, four events offer independence on your terms and give you Neocolonial Economic Concessions. Each event and what triggers it is in Colonial power event list.

### Former colonies

Every colony freed through this system, through an event, or through the base game's release actions is marked as a former colony of its old ruler. While that ruler still runs the Colonial Empire entry and is at least a major power, the former colony may propose closer ties or demand reparations of a tenth of the old ruler's yearly revenue, capped at a tenth of its own GDP. For its first 20 years it also faces the questions of a young state: development, borders, government, alignment and identity. A newly independent country in which one foreign country owns more than 5% of its GDP faces the Nationalization Question, and seizing everything can bring a great-power old ruler's fleet. The events are in Former colony event list.

## 12.6 Colonial collapse of tiny countries

Once any country has researched Decolonization, a tiny, poor AI country can collapse. The check runs once a year, and a country qualifies when its population and its average standard of living are both below one of these pairs:

| Population under | Standard of living under |
|---|---|
| 100,000 | 6 |
| 200,000 | 5 |
| 300,000 | 4 |
| 500,000 | 3 |
| 1,000,000 | 2 |

It must also be independent, at peace, outside any diplomatic play and not on the verge of a civil war, and it must hold no subjects, belong to no power bloc and be bound by no treaty. A country that became independent in the last 20 years, by a release, an independence war or a secession, is exempt, and so is every player's country.

A collapsing country is annexed by an AI neighbor of its most populous state whose primary cultures share its heritage, provided that neighbor is at peace and not a subject. A player's country never absorbs one. With no such neighbor, the country reverts to decentralized land, unless it owns a company or holds nuclear warheads, in which case nothing happens that year. Players with a state in the same strategic region, and player great powers, get a notification.

## 12.7 How the AI runs its colonies

The AI plays by the same rules and uses the same programs. It releases a territory through decisions that offer it the most populous candidates, by grant or at a Round Table, and frees subjects through events; the buttons

beside each territory and subject are yours. Once a month it reviews its programs. It lowers a program by a level when that level is wasted, or when it can no longer pay for it: in default, or with weekly income below its expenses, for Investment; short of authority or bureaucracy for the other two. A level is wasted when, without it, the bar's monthly change would still reach the monthly limit, or, with the bar full, would still be +0.5 or more. That margin keeps a full bar full through a new war. A program cut for cost stays down for a year. If it lowered nothing, the AI raises one program a level while the change is below the limit (below +0.5 with the bar full) and it can afford the step, preferring the garrison when several colonies are badly unaccepted.

At a declaration, an AI weighs its rank, the condemning great powers and its other wars. A great power with settlers in the territory, a powerful army lobby and no voting franchise fights about two times in three; a democratic great power with no settlers and two condemners about one time in four. A country that isn't a great power is much less likely to fight, and one in default or already at war with a great power never does.

An AI that isn't a great power leans strongly toward Planned Full Decolonization, more so when three great powers condemn it or three of its colonies are badly unaccepted. AI great powers are less willing, and no AI plans full decolonization while two of its colonies are well accepted. In events, AI empires lean toward negotiation and release, and an AI takes the Imperial Federation Act or Civilizing Mission Compact whenever it qualifies.

CHAPTER 13

# Military and war

The mod carries the army and navy through to era 13. It adds 25 battalion types, two new unit groups (Heavy Tanks and Aircraft) and 19 ship types, and it rewards armies that mix their arms. Military Bases fortify states against invasion and missiles, several of the mod's systems feed war support, and mobilization drives a sharp rise in ammunition demand. All of this is always on. The optional part is the World War journal entry, which needs the World War game rule (disabled by default): great powers of opposed ideologies drift into a global war and then settle the peace.

## 13.1 Combined arms bonuses

Combined arms rewards army formations that field several kinds of battalion. It starts when you research the Combined Arms technology (era 6, after Mobile Armor), which also unlocks Armored Infantry and Combined Arms Marines.

Each month, and whenever you create a formation, recruit a general or finish barracks, the game sorts each army formation's battalions into five groups: Infantry (marines count as infantry), Artillery, Cavalry (including Light Tanks and the mod's reconnaissance vehicles), Heavy Tanks and Aircraft. A group counts as present when it makes up at least 5% of the formation. With two or more groups present, the formation's general gets a trait for each group present, and a sixth trait when all five are. The traits apply to the troops the general commands.

| Trait | Earned by | Effect |
|---|---|---|
| Combined Arms: Infantry Screen | Infantry | +5% unit defense |
| Combined Arms: Fire Support | Artillery | +5% unit offense |
| Combined Arms: Reconnaissance | Cavalry | +1 maximum offensive battles |
| Combined Arms: Armored Spearhead | Heavy Tanks | +5% offense, +3% defense, −5% morale loss |
| Combined Arms: Air Superiority | Aircraft | +5% offense, +100% advancement speed |
| Combined Arms: Full Spectrum Dominance | All five groups | +5% offense, +5% defense, +1 maximum offensive battles |

The bar is low: a formation of 19 infantry and one artillery earns Infantry Screen and Fire Support. A formation of a single group earns nothing however large it is, and a formation without a general earns nothing at all.

## 13.2 New combat units

Every unit line runs past the base game's last tier, and the new units appear in the upgrade options of the older units in their line, so you can upgrade battalions in place. The base game's Heavy Tanks (Mobile Armor, era 5) move from the artillery group to the new Heavy Tanks group, with lower stats than in the base game, and aircraft form a group of their own. Each group gains three to five units, from Armored Infantry and Propeller Aircraft in era 6 to the Orbital Weapons Platforms of era 12 and the Utility Fog Phalanx of era 13, and later units come in smaller battalions, down to 100 for the orbital ones. Combat unit list gives each unit's technology, era and battalion size.

*Figure 20: A general's traits with all six Combined Arms traits. The formation fields all five groups, so the general has Full Spectrum Dominance as well as one trait for each group.*

Offense and defense climb steeply with each tier, but read them with the battalion size. A battle weighs each battalion by its manpower as well as its stats, so a 100-man orbital battalion carries less weight than its unit card suggests. The reconnaissance vehicles occupy territory fast (+100% to +200% occupation), and the new infantry and marines lose fewer provinces when pushed back.

Their upkeep runs to tanks, aeroplanes, radios, electronic components and industrial robotics, and launch capacity for the orbital units. Tanks, aircraft, and the new artillery, cavalry and marines after Combined Arms Marines can't be built under the Peasant Levies law.

## 13.3 Barracks training methods

The Barracks' training methods go on past Mobile Warfare Tactics. Two raise the training rate further, and three train more slowly in return for something else. Training rate only matters while a battalion is below strength, so a slower method costs little in peacetime. Switch back to a fast one before a long war.

| Method | Unlocked by | Training per level | What else it does |
|---|---|---|---|
| Officer Academies | Combined Arms (era 6) | 26 | 30% of the staff are officers. Academy Officer Corps raises the Armed Forces' political strength by up to 20% and their attraction by up to 10%, in proportion to the share of your regular battalions trained this way |
| Expeditionary Logistics | Predictive Logistics (era 8) | 10 | Cuts the base goods surcharge while the battalion is mobilized by a tenth. Mobilization options' own goods are unchanged |
| Neural Grafting | Bioenhanced Soldiers (era 12) and the Regulated Augmentation Market, Unrestricted Augmentation or Mandatory Augmentation law | 120 | Buys Drugs and Electronic Components, and raises its servicemen's mortality by 25% |

| Method | Unlocked by | Training per level | What else it does |
|---|---|---|---|
| Reserve Cadre | Public Works Programs (era 6) and the National Militia or Mass Conscription law | 12 | Each staffed level gives the state two reserve battalions. Each one raises the state's conscription rate and its conscription-center limit by enough for one more conscript battalion, so it still counts in a state already at the limit |
| Political Commissars | Mass Media (era 6) and the Single-Party State or Council Republic law | 20 | 5% of the staff are officers. Commissar Indoctrination raises attraction to the interest groups in government by up to 15%, in proportion to the share of your regular battalions trained this way |

The national bonuses and the reserve battalions update once a month. Expeditionary Logistics makes a short war you are winning a little cheaper, and its slow reinforcement tells in a long one. Reserve Cadre suits a militia: its conscripts cost nothing until you call them up. Few officers under Political Commissars also means a weaker Armed Forces. The AI mostly picks the method with the highest training rate it can use, so AI armies move to Officer Academies and later Neural Grafting. A Single-Party State or Council Republic AI prefers Political Commissars to Officer Academies. The AI never picks Expeditionary Logistics or Reserve Cadre.

## 13.4 Ships and ship modifications

Nineteen ship types extend the navy from era 6, with escorts, submarines, carriers, battle-line ships and troop transports in most eras: from the Fleet Carrier and ASW Destroyer of era 6 to the Orbital Support Mothership of era 12 and the Antimatter Battleship of era 13. Ship list gives each ship's class, role and technology, and the technology that makes it obsolete.

The Expeditionary Fast Transport is quick and carries little; the Expeditionary Sea Base is slow and carries four times as much. The base game's late ships go obsolete as their successors arrive: the Super Dreadnought with Bombing Aircraft, the Troopship with Combined Arms, the Destroyer and Light Cruiser with Sonar, the Submarine with Advanced Submarine Technology and the Aircraft Carrier with Nuclear Energy.

In the ship designer, nine of the new ships bring their own armor, guns, propulsion and range modifications in three tiers; the others reuse those of a related ship. Eight new utility modifications join the base game's, from the Sonar Suite (Sonar) to Composite Armor Plating (Modern Material Science); Ship utility modification list gives each one's technology and effect and the ships that mount it.

Shipyards gain a Production Focus group that trades civilian hulls for naval construction; its strongest setting, Wartime Mobilization, needs the Total War law.

## 13.5 Mobilization options

The mod adds 27 mobilization options, some in a new Training group. The new transport and medical options join the base game's exclusive sets: a formation picks one of Forced March, Truck Transport, Rail Transport, Air Transport and Space Transport, and one of First Aid, Field Hospitals and Medevac Helicopters. Many options can be chosen only while your market has the goods they consume for sale.

Beyond transport and medicine, the options run from logistics that cut attrition, through morale and combat gear such as Radar Equipment, Night Vision Gear and Exoskeleton Suits, to terrain training from your power bloc's principles and bioenhancement under the augmentation laws. Mobilization option list gives each option's unlock and effect.

Entrenchment, the three terrain trainings, Missile Defense System and Exoskeleton Suits add ammunition to a mobilized battalion's upkeep, on top of the fourfold rise that mobilization already brings (Wartime demand for munitions).

## 13.6 Military bases

The Military Base is a government building that fortifies a state. It unlocks with Trench Works and grows to five levels, one each from Trench Works, Defense in Depth, Concrete Fortifications, Guided Missiles and Precision Guided Munitions. Each level employs 200 soldiers and 50 officers, buys small arms and ammunition, and adds +5 infrastructure.

Three production method groups set what it does. The Fortification Type and Base Purpose methods decide the state's fortification level, 1 to 4 for each level of the base. The purpose can instead calm turmoil and add war support (Garrison Duty), or add infrastructure and cut your whole army's supply use (Logistics Hub). A Missile Defense method, from Missile Defense Systems, raises the state's chance of stopping a nuclear strike. Military base production method list gives each method's unlock and its effect per fully staffed level. The AI values bases far more at war or when committed to a diplomatic play.

### Fortification levels and battle conditions

A state's fortification level adds up across its base's production methods, so five levels of Hardened Positions with Territorial Defense give 20. When your general defends a state with a Military Base, one of four battle conditions can apply, set by that level. Each tier replaces the one below it.

| Battle condition | Fortification level | Effect for the defender |
|---|---|---|
| Fortified Position | Any base | +20% defense, −10% morale loss, −20% combat width |
| Fortified Stronghold | 5 or more | +35% defense, −20% morale loss, +10% kill rate, −40% combat width |
| Fortified Bastion | 10 or more | +50% defense, −30% morale loss, +15% kill rate, +5% recovery, −60% combat width |
| Impregnable Fortress | 15 or more | +70% defense, −40% morale loss, +20% kill rate, +10% recovery, −80% combat width |

The base game's Naval Fortification also gains two late production methods, Coastal Missile Battery (Guided Missiles) and Integrated Coastal Defense (Missile Defense Systems), which raise its resistance to invasion, bombardment and blockade.

### Bases under nuclear attack

A base's missile defense covers its own state, on top of the defense your technologies give every state. AI strike planners weigh target states by the chance a warhead gets through, so defended states draw fewer strikes.

A tactical nuclear strike halves every Naval Fortification and Military Base in the state, and what survives keeps its production methods; see What a nuclear strike does.

### Demilitarized zones and forced disarmament

Two treaty articles, Demilitarized Zone and Forced Disarmament, dismantle military buildings, Military Bases included, and both can be war goals; see Treaty articles by purpose.

## 13.7 War support from mod systems

The mod adds weekly lines to the war support breakdown.

| Line | When it applies | Per week |
|---|---|---|
| Home front strain (World War) | Your world war is 2 years old | −0.5 |
| Prolonged World War | Your world war is 4 years old (adds to the line above) | −0.5 |
| Condemned by the United Nations | The UN condemned you | −0.5 |
| Rebuked by the United Nations | The UN rebuked you and did not condemn you | −0.25 |
| Fighting a UN-condemned enemy | An enemy in this war is condemned | +0.25 |
| Enemy communications disruption | An enemy in this war runs a communications-disruption operation against you | −0.25 |

| Line | When it applies | Per week |
|---|---|---|
| Enemy nuclear arsenal | An enemy has nuclear weapons, you don't, and neither a nuclear-armed guarantor nor your overlord's nuclear umbrella covers you | −0.25 |
| Shelters and civil defense | The line above applies and you have Civil Defense, from a nuclear taboo event | +0.125 |

The United Nations, covert operation and nuclear lines come from systems with their own game rules, and appear only when those systems are on.

At war, states also add war support each month, scaled by their share of your population. The War Propaganda decree is worth 5 a month in its state, and a Military Base 0.25 per level on Garrison Duty (half that on Territorial Defense), each times that state's share. The Private Military Contractors law adds 0.25 across the country. **Fervor** is the religious kind, from the Sacred Civics power bloc principle at tier II and up: 5 a month times the Devout interest group's clout.

Laws and technologies also change how far battles move war support.

| Source | Victories | Defeats |
|---|---|---|
| Total War law | +25% | +25% |
| Limited War law | −20% | −20% |
| Ministry of Propaganda Established law | +15% | −15% |
| Unregulated Internet / Net Neutrality laws | +20% / +15% | +20% / +15% |
| State-Controlled Internet law | −10% | −20% |
| State Secrets / Freedom of Information / Open Government laws | −5% / +5% / +10% | the same |
| Television Broadcasting / Satellite Communications / Social Media technologies | +10% / +5% / +15% | +15% / +10% / +20% |

Total War and Limited War belong to the Rules of War law group; see The rules of war. Total War also halves the war support that casualties cost.

## 13.8 Ammunition and mobilization

Mobilization quadruples a battalion's ammunition use, even for a diplomatic play that never becomes a war; see Wartime demand for munitions.

## 13.9 The World War journal entry

The Gathering Storm is a journal entry for great powers that have researched Combined Arms, present only when the World War game rule is enabled. It follows the hostility between great powers of opposed ideologies through a leadup, a world war and three post-war years, and it fails if you drop below great power.

### Ideological camps

The entry sorts countries into camps by their laws, and its status shows your own camp on a second line (Non-Aligned when you are in none). Democratic opposes communist and fascist, and communist opposes fascist. Authoritarian countries, and those in no camp, oppose nobody, so their entry stays inactive.

| Camp | Laws |
|---|---|
| Democratic | Census Suffrage, Universal Suffrage, Liquid Democracy, Wealth Voting or Landed Voting; Presidential Republic, Parliamentary Republic or Monarchy; neither Single-Party State nor Autocracy |
| Communist | Council Republic, with Command Economy or Cooperative Ownership |
| Fascist | Corporate State with Single-Party State, or Autocracy with Single-Party State and Ancestral, Cultural or Racialized Citizenship |

| Camp | Laws |
|---|---|
| Authoritarian | Autocracy, Single-Party State or Oligarchy, when not communist or fascist |

## Ideological tension

Each great power has its own tension, from 0 to 100: +20 when one of your rivals is an opposed great power, +10 for each opposed great power, +15 if you are fascist, +10 if communist, +20 while you are at war, and −30 if you have nuclear weapons. The entry doesn't show the number; its status text names the band.

| Tension | Status text | What opens |
|---|---|---|
| 20 | The great powers watch each other warily | Pursue Appeasement |
| 30 | Tensions are rising | Begin Rearmament, the Ideological Demands event, and Rising Global Tensions (+25 maneuvers in your plays, faster escalation as aggressor, +5% military goods cost, +10% prestige from army power projection) |
| 40 | Tensions are rising | Provide Lend-Lease, the Border Incident event |
| 50 | High tension | The Diplomatic Crisis event |
| 70 | Crisis: war is imminent | The Brink of War event |

## Preparing for the world war

Three leadup policies sit on the entry as buttons, each with a matching button to end it.

| Button | Needs | Modifier while active |
|---|---|---|
| Begin Rearmament | Tension 30, not appeasing | +10% army offense and defense, +20% prestige from army power projection, +15% military goods cost. Armed Forces approve. |
| Pursue Appeasement | Democracy, tension 20, not rearming | +50 influence, +10% relations improvement speed, −5% military goods cost. Intelligentsia approve, Armed Forces disapprove. |
| Provide Lend-Lease | Major power, tension 40 or a world war under way, not at war | +50 influence, +15% prestige, +15% military goods cost. Industrialists approve. |

The AI rearms when it is fascist or authoritarian and tension is high; AI democracies appease between 30 and 70. An AI great power that stays out of a world war leans strongly toward providing Lend-Lease.

Leadup events come at random, each over an opposed great-power rival: demands that it renounce its system, a border incident, or a diplomatic crisis when it opens a play against a smaller country in your power bloc or under your protection. Their options are in World War event list.

## The brink of war

The Brink of War fires at tension 70 when one of your rivals is an opposed great power that holds the entry, at least three great powers exist, and no world war is running. Launching makes you the aggressor and your rival the defender, and opens a Cut Down to Size diplomatic play against it at escalation 90. Standing down angers the Armed Forces and keeps the event away for ten years. The AI usually launches.

At launch only those two powers are belligerents; allies who join the play fight beside them but take no side in the entry. If the play ends without a war, the crisis is defused and a later one can start another.

## Fighting the world war

Belligerents at war get three wartime policies, each with a button to end it. All three lift when the war ends.

| Button | Needs | Modifier while active |
|---|---|---|
| Total War Economy | Combined Arms | −15% military goods cost, +15% army offense and defense, +10% morale recovery, +25% conscription, +25% construction goods cost, −2 standard of living. Armed Forces approve, Trade Unions disapprove. |
| Launch War Propaganda | Mass Media | −10% morale loss, +100 authority, −30% radicalism of political movements. Intelligentsia disapprove. |
| Impose Rationing | Nothing extra | −3 standard of living, +10% agriculture throughput, −10% bureaucracy |

A long war wears the home front down. After two years Home Front Strain gives −10% bureaucracy and −0.5 war support a week, and plays you start escalate 2 a week faster. After four, Prolonged War Weariness adds −4 standard of living, +3% mortality, +100% radicals from political movements, +25% war support lost to casualties and a further −0.5 war support a week. The war support breakdown shows the two weekly losses as Home front strain (World War) and Prolonged World War.

Wartime events test the home front, from rallies and strategic bombing to resistance in occupied land, and from two years on they ask whether to push on or seek terms. They are listed in World War event list.

Great powers that stayed out receive The Hour of Decision. Entering joins the defender's side with Fresh Forces for five years (+20% army offense, +10% defense, +20% morale recovery), plus Arsenal of Democracy for a democracy. You can instead send material support (Lend-Lease Program for ten years) or stay neutral. AI democracies and communists join readily against a fascist aggressor.

A player great power outside the war need not wait for the event. While a great power still fights on each side, the entry's Enter the War button joins the defender's wars against the aggressor at once, with the same Fresh Forces and Arsenal of Democracy. Entering by either route, and every answer to The Hour of Decision, brings Recent Intervention Decision, which keeps the button unavailable for a year. A power that has already made its peace in the war can't enter again, and AI powers enter only through The Hour of Decision.

## Winning and losing the world war

The mod records a concession whenever a belligerent enforces a war goal on a belligerent of the other side, by capitulation or negotiated peace. You **won** if you fought, conceded nothing, and at least one enemy belligerent conceded; you **lost** if you conceded and enforced nothing. A white peace, or one that enforced goals both ways, is neither. Outcomes are per country, so one member of a coalition can win while an ally loses.

The Peace Conference fires when your war ends. A victor picks the peace, which decides what the powers it beat receive.

| Victor's choice | Victor gets | Each power it beat receives |
|---|---|---|
| Impose a harsh peace | Victor's Peace and Occupation Duties, 10 years | Regime Change: accept for Defeated Power and Imposed Regime Change, or resist for Defeated Power and heavy radicalism |
| Seek a just and lasting peace | Victor's Peace for 5 years, Post-War Reconstruction for 10 | Reconstruction Aid: accept for Post-War Reconstruction and a shorter Defeated Power, or refuse for Defeated Power for 10 years |
| Establish a new world order | Victor's Peace for 10 years, Liberation Hero for 5 | Nothing |

Victor's Peace gives +30% prestige, +200 influence, +500 authority and +10% leverage generation. Defeated Power gives −30% prestige, −200 influence, −3 standard of living and −25% army offense and defense. These modifiers fade over their duration. A power that lost or drew chooses between bringing the soldiers home and keeping the army ready.

### The post-war years

The entry stays open for 36 months after peace, and three events can reach great powers: a war crimes tribunal, a choice about the post-war order, and The New Rivalry, when a great power you fought beside now holds an opposed ideology. Either answer to The New Rivalry closes the entry at once. Otherwise it completes after the 36 months. The events are in World War event list.

CHAPTER 14

# Nuclear weapons

Once you research the Nuclear Weapons technology, a great power can fund a weapons program that turns research into warheads. Holding warheads brings a posture to choose (when you would use them, how ready they stand and who may launch), a standing bill, the risk of accidents, and crises in which you threaten others or are threatened. Strikes, civil wars that split an arsenal, and warheads that go missing follow from the same system. From the world's first warhead, the nuclear taboo, a single score for the whole world, sets what each nuclear act costs and can make holding the bomb a burden. The Nuclear Weapons game rule controls it and is on by default; with it off the technologies remain, but no program, stockpile, taboo or nuclear events exist.

## 14.1 The Nuclear Weapons journal entry

Everything happens in one journal entry, **Nuclear Weapons**. Before the world's first warhead, it is active for any country with a working program; a country with the technology but not the standing to build sees it inactive, with a status line saying what it lacks. Once the first warhead exists, the entry is active for every country except decentralized ones, whether or not it has the technology. An active entry has no status line: the overview at its top says the same at a glance.

The same panels appear as a Nuclear tab in the Military panel, and a change made in one shows in the other. The tab is grayed until the journal entry is active; hover it for what is still missing. There, the progress bar shows only while you have a program, and the tab ends with an Open Journal Entry button.

From the world's first warhead, another country's National Info has a Nuclear tab with what the world knows of it: its warheads as the world estimates them (none known before its first test), the nuclear guarantees it gives and holds, any crisis it is in, World Crises, the nuclear taboo's score and history, and Nuclear Powers. Its true count, program, posture, forces and reputation, and every button, are only on your own.

The overview is a grid of boxes, two to a row. Each box has an icon, a caption, the reading in a word and one figure beneath it; hover a box for the detail.

- While you are in a crisis, the first box is the crisis: its stage, and the weeks to its deadline, red from two weeks out.
- The program's state (Unfunded, Developing, Producing, Frozen, At Ceiling, Dismantling, No Program, Renounced or Disarmed; hover it for the program's status) and when its next warhead comes; your warheads, with the world's estimate beneath.
- While you hold warheads, your posture. Doctrine and launch authority each say when they can next change. Readiness says where it is moving or, when it is not, what your deterrent costs a week.
- Your credibility, once you have a record: Doubted, Steady or Trusted, and the figure out of 100.
- From the world's first warhead, the nuclear taboo: a bar, the score and its target with the monthly change ("62 → 68 (+0.40/mo)"), and an arrow for the direction. The solid bar runs to the lower of the score and its target, and a pale stretch to the higher. The stretch is pale rather than green or red because a stronger taboo cuts both ways: it weighs on an arsenal and on threats, and shields countries without one. Hover it for the target's breakdown.

The Program sits directly above the entry's progress bar, which is progress toward your next warhead. The other sections follow below the bar. Every section starts open. The "?" at the right of the tab's header, and at

*Figure 21: The Nuclear tab of the Military panel for a country holding 12 warheads: the overview (an Unfunded program, warheads, credibility, doctrine, readiness, launch authority and the taboo bar), The Program at funding step 0 with its progress bar, Nuclear Posture, and the start of Forces.*

the top of the journal entry, opens the Help page, How Nuclear Weapons Work, with one heading per topic. Doctrine, readiness and launch authority each open their choices from a Change button beside their row, and Disavow our strikes is a switch of its own.

| Section | Shown when | What it holds |
|---|---|---|
| The Program | You have a program | Program funding (a stepper), warhead production a month, the time to the next warhead, and the year of your first device and whether it was the world's first |
| Nuclear Crisis | You are in a crisis | A line naming your opponent, your side and the dispute; boxes for the stage, the weeks to the deadline, danger, pressure, whether the threat can be carried out and when the target is pressed next; your buttons under Our |

| Section | Shown when | What it holds |
|---|---|---|
| | | Moves; then Outcomes, what each ending would bring and do to your credibility |
| World Crises | Any crisis is running, anywhere | Every crisis, yours included and tinted, most dangerous first: the two countries' flags, the stage and the danger; hover a row for who threatened whom, the dispute, the deadline, any talks and whether it is public. Private warnings are listed too, since governments know of them even when the public doesn't |
| Nuclear Posture | You hold warheads | Doctrine, readiness (with the level it is moving to) and launch authority, each with a Change button that opens its choices; whether you disavow your strikes; your nuclear umbrella, warheads unaccounted for, upkeep and incident exposure |
| Forces | You hold warheads | Survivability, first-strike survivors against your main rival, your reach before Intercontinental Ballistic Missiles, command reliability, crew strain and the chance of a halt; safeguards and hardening under Investment; under Arsenal Size, warheads held beside the world's estimate, your arsenal ceiling and dismantling |
| At Home | You hold warheads | A small table for each interest group with a view of your posture |
| Reputation | You have a record | How your last crisis ended and your non-use pledges (your credibility is in the overview) |
| The Nuclear Taboo | From the world's first warhead | The score, where it is heading and its band, what nuclear acts cost now, your arsenal's burden, and The Taboo Over Time, its history in two charts |
| Delivery and Defense | Always | Your delivery capability and home defense |
| Nuclear Powers | From the world's first warhead | The ten largest arsenals as the world estimates them, each with its country's flag and public doctrine; your own row is tinted |

An armed power that loses its rank stops building but keeps its warheads, its posture, its upkeep and its accidents.

## 14.2 Building a nuclear arsenal

The program spends weekly innovation on warheads. It is slow to reach the first device and much faster afterwards.

### Who can run a weapons program

You need the Nuclear Weapons technology and one of these:

- Great Power rank.
- Major Power rank and the Intercontinental Ballistic Missiles technology.
- A Nuclear Program Aid treaty article in which a nuclear power helps you, at any rank.

A Nuclear Disarmament article ends the program, and so does renouncing the bomb by dismantling your arsenal (see Reducing or giving up an arsenal). A United Nations member without a bomb can't run one while the UN is at its Strong tier or higher and the IAEA exists (see UN authority tiers). A Nuclear Program Freeze holds funding at zero, and so does an arsenal ceiling, your own, a Nuclear Arms Limitation treaty's or the UN Prohibition Treaty's, while you hold at least that many warheads. Losing the rank that qualified you zeroes funding the same week.

### Program funding and warhead production

Program funding is a stepper in The Program. Each step costs 100 weekly innovation, taken out of your research, and you can add a step only while your innovation is above 100. A warhead needs 100 progress.

| Stage | Progress per funding step | Example |
|---|---|---|
| First device | 1 a month | 5 steps (500 innovation a week) finish it in 20 months |
| Series production | 10 a month | 5 steps then build a warhead every 2 months |

A Nuclear Program Aid article doubles the rate, covert sabotage slows it, United Nations inspections slow it (by a quarter at the UN's middle tier of enforcement, more at the higher tiers), and after any nuclear use every funded program gets +25%, fading over five years (The Bomb Has Been Used). Events can bring a laboratory accident, a discovery with civilian uses, or anti-nuclear protests. Most of the accident's and the discovery's choices cost 5 to 25 points of progress toward the first device, never taking it below zero.

### The first device and the world's reaction

The first country to finish a device gets "Dawn of the Atomic Age", the Nuclear Power modifier and a burst of loyalists, and its warhead brings the nuclear taboo into being at 20. Later powers get "Our Nuclear Arsenal is Complete" and the same modifier, which raises prestige (+30%), diplomatic play maneuvers, and leverage generation (+25%) and resistance. Once the taboo rises above 40, The Burden of the Bomb eats into that prestige and leverage generation (see What the nuclear taboo costs).

A later program is noticed at 75 progress on its first device: the great powers, its rivals and its neighbors get "Nuclear Proliferation Alert". A nuclear power can answer with a public ultimatum, or with a public denunciation where no crisis can be opened. A country with a program of its own can crash it or stay the course. Anyone can push for a non-proliferation treaty, accept the new reality, or, with the Covert Warfare rule off, sponsor sabotage. Once the Non-Proliferation Treaty is in force, a UN member pushing for it refers the program to the General Assembly instead: the first referral adds 20 to the proliferator's case strength, or 30 if it signed the treaty (see Grounds for UN censure). Several answers shave progress off the proliferator.

Accepting the new reality is an accommodation: +25 relations with the proliferator, and −15 with every country that opposes its program (its rivals, the countries hostile to it, and its opponent in a nuclear crisis). A nuclear power can instead extend its guarantee to the proliferator's neighbors, at −10 relations with the proliferator. Every neighbor without the bomb or a guarantee that the power could sign a Nuclear Guarantee with today is offered one, binding for ten years. A neighbor that accepts loses 10 relations with the proliferator. The power hears each answer, and an offer lapses if another power's guarantee was accepted first.

Other countries see your arsenal only as an estimate, re-observed yearly at between 60% and 150% of the true count, and exact after a test or a strike. The Nuclear Powers table and the AI use these estimates.

## 14.3 Keeping a nuclear arsenal

From the week your first warhead exists you pay for it, and a human player gets "A Standing Deterrent" to say so.

### Nuclear deterrent upkeep

Nuclear Deterrent Upkeep is a weekly expense that scales with GDP. The figures below are shares of GDP a year. The Upkeep row in Nuclear Posture shows the weekly total, and each readiness choice and investment row its own weekly cost.

| Item | Cost |
|---|---|
| Custody | About 0.01% per warhead, counting at most 50; half at Recessed |
| Heightened readiness | 0.08–0.2%, rising with arsenal size |
| High Alert | 0.25–0.6%, rising with arsenal size |
| Safeguards | About 0.1% per level |

| Item | Cost |
|---|---|
| Hardening | About 0.16% per level |
| Automatic Retaliation | About 0.16% |
| Cap | About 2% |

## Survivability, reliability and crew strain

The Forces section tracks three numbers on a 0–100 scale, updated monthly, and two investment steppers from 0 to 3, Safeguards and Hardening.

- Survivability is how much of your force would survive a first strike. It climbs toward a ceiling set by technology (25, plus 15 for Radar, 20 for Intercontinental Ballistic Missiles, 25 for Advanced Submarine Technology and 10 for Missile Defense Systems, at most 80) only while you pay for hardening, and erodes when you stop. Its bar shows where it is heading: a green stretch while it climbs, a red one while it erodes, and a cream line at the ceiling.
- Command reliability is how well the chain of command holds. Each safeguards level raises it by about 12; strain, buried incidents and enemy sabotage lower it.
- Crew strain rises 5 a month at High Alert, settles around 40 at Heightened, and falls 6 a month at Routine or Recessed.

Safeguards also make an unapproved launch likelier to be halted and cut the warheads lost when your arsenal changes hands. A crisis counts survivability at your present readiness: all of it at High Alert, nine tenths at Heightened, four fifths at Routine and half at Recessed. The Survivability tooltip shows the figure that counts now.

What decides how hard you are to coerce is how many warheads you would have left if your main rival spent its whole arsenal, as far as you know it, on yours: First-strike survivors, in the Forces section. Your main rival is your crisis opponent if it is armed, otherwise the armed rival, enemy or hostile country believed to hold the most warheads. Its hover shows how many of your warheads are within reach, what the rival could destroy, and the share your launch authority would get away first. Three survivors make a full answer, and a crisis weighs threats against you by them. Below 50, survivability also makes a crisis more dangerous and a launch less likely to be halted.

## Delivery capability and home defense

A strike lands with the attacker's delivery capability minus the target state's defense, never below 5% and never above 100%.

| Delivery | Bonus | Defense | Bonus |
|---|---|---|---|
| Nuclear Weapons | +100% | Military Aviation | +25% |
| Intercontinental Ballistic Missiles | +50% | Radar | +25% |
| Hypersonic Weapons | +50% | Missile Defense Systems | +50% |
| Orbital Weapon Platforms | +50% | Directed Energy Defenses | +50% |
| Heightened readiness | +5% | Orbital Weapon Platforms | +50% |
| High Alert | +10% | Non-Proliferation Treaty guarantee (members with no bomb and no program) | +5% at the UN's middle tier of enforcement, scaled by the tier |

Technology defends every state. A Military Base defends its own state only, by +2% per level with a Missile Defense Battery or +3% with Directed Energy Point Defense, and AI planners aim fewer strikes at defended states. Military bases in general are in Military and war.

## Range and Nuclear Basing

Until you research Intercontinental Ballistic Missiles, your weapons go only as far as your bombers fly. That is every country next to you by land or sea, next to one of your direct subjects or next to a country that hosts

your weapons, and every country with a state in a strategic region where you, a subject or a host holds one. Advanced Submarine Technology adds every coastal state. A war goal between you and a country puts it in reach, so range matters most in peacetime crises and against your enemies' distant allies. No strike, salvo or answer can hit a state out of reach, a Leadership salvo needs the capital in reach, and a threat against a country you can't reach is a bluff. The Forces section's Reach row counts the countries in reach and lists them on hover.

A Nuclear Basing treaty article stations your weapons in another country. Your reach then extends from its states as from your own, and your arsenal covers it as a ready guarantor does in a crisis, with the right to answer a strike on it. An armed country next to the host loses 10 survivability against you. About one in three of your bomber crashes and silo explosions happens in the host, which then chooses to order your weapons out (ending the whole treaty that holds the article, with no penalty for breaking it even while it binds), to protest (−30 relations) or to say nothing (+10 relations, and −3 legitimacy fading over two and a half years). A country struck from a host next door may have no reach back to you, and then no answer at all.

## 14.4 Nuclear posture

Posture is three separate choices. Nuclear Posture shows the ones in force, one row each; hover one for what it does now and when it can next change. Its Change button opens that choice's options beneath it, with the Adopt and Order buttons, the option in force tinted and a line above them saying when you can change it. A button you can't use says why, and each option's tooltip lists what it does.

### Nuclear doctrine

Doctrine says when your government may strike first against a country you are at war with. Retaliation is allowed under every doctrine. You can change doctrine once every two years, and the first choice is free.

| Doctrine | First use allowed when | Standing effect |
|---|---|---|
| No First Use | Never | +10% infamy decay and relations improvement, −10% play maneuvers, +0.5 UN Authority Target |
| Existential Deterrence (default) | The enemy's side means to annex or subjugate you, or holds goals on your incorporated states while you are losing | +5% leverage resistance |
| Flexible First Use | Also when you are losing to them, or they hold goals on any incorporated state | +10% leverage resistance, +5% maneuvers, −5% infamy decay, −0.5 UN Authority Target |
| Nuclear Compellence | Also when they defied your public ultimatum, or your threat against them has reached Confrontation | +15% maneuvers, +10% leverage generation, +10% infamy generation, −10% relations improvement, −1 UN Authority Target |
| Nuclear Warfighting | In any war | +20% maneuvers, +10% leverage generation, +20% infamy generation, −10% infamy decay, −20% relations improvement, −1.5 UN Authority Target |

Losing means a quarter of your land occupied, or under 35% of battles won after five significant battles. Adopting Compellence or Warfighting costs infamy equal to a tenth of the nuclear taboo (5 at a taboo of 50) and 10 relations with every rival. Leaving No First Use is a **repudiation**: −20 credibility, +10 infamy, a ten-year Broken Nuclear Pledge modifier, and your restraint-minded interest groups disapprove. While the No-First-Strike Pledge amendment is on your laws (see Amendments to the mod's laws), your doctrine is held at No First Use, and leaving it strikes the amendment. An AI country keeps the amendment unless a militarist government takes power, which may repeal it.

A launch your government never ordered (Launch on Warning, a commander cut off from the capital, or Automatic Retaliation misreading an accident) breaks the pledge too, but your doctrine stays No First Use and the amendment stays on your laws. Once the incident's own event is answered, "The Pledge After the Launch" lets you reaffirm No First Use in public (−10 legitimacy and −5% authority, fading over two and a half years, for +15 credibility at once), repudiate it at half the usual credibility and infamy cost with no two-year wait, or say nothing.

## Nuclear readiness levels

Readiness moves one step every two weeks toward the level you order.

| Readiness | What it means |
|---|---|
| Recessed | Warheads stored apart from their delivery systems. Half the custody cost, half the incidents of Routine, less dangerous crises. Nothing can be launched, not even in retaliation, until you are back at Routine, and your threats carry less weight |
| Routine | The default. Cheap and slow; crews recover |
| Heightened | +5% strike success; incidents four times as likely as at Routine |
| High Alert | +10% strike success and +5 play maneuvers; strain builds every month, and with it accidents and false warnings |

Higher readiness also makes your forces harder to catch in a first strike, so a crisis counts more of your survivability. The readiness options are a table: the deterrent's weekly upkeep and the survivability a crisis would count at each level.

Struck while Recessed, you can order the warheads mated; "Our Forces Are Ready" offers the answer when they reach Routine, if the war goes on. A stand-down agreed or conceded in a crisis holds you at Routine or below for two years, and dismantling your arsenal takes you down to Recessed and holds you there until it ends or you halt it.

## Launch authority

Launch authority decides who may act when a warning is uncertain. It can change once a year.

| Authority | Needs | Effect |
|---|---|---|
| Central Authorization | Nothing | Only the government orders a launch. A salvo on your capital can cut off your command for two to four weeks (see Salvos and the Strike section), and an issuer believed to hold more warheads than your command could absorb on the capital counts your survivors at three quarters |
| Conditional Delegation | Radar | A quarter of the warheads a first strike would destroy get away first, 5 points fewer per safeguards level. Cut-off commanders may fire, even at peace. In a war, a false warning can also end in a launch you never approved |
| Launch on Warning | Radar, Intercontinental Ballistic Missiles | 60% of the warheads a first strike would destroy get away first, 5 points fewer per safeguards level, and up to three of them may answer a Forces salvo before it lands. A false warning can become a launch under standing orders |
| Automatic Retaliation | Radar, Intercontinental Ballistic Missiles, Mainframe Computers | A strategic first strike on you is answered in full, up to three warheads, with no choice left to you. Extra upkeep; restraint-minded groups dislike it; an accident at home during a war or an acute crisis can set it off |

For an AI government, which sometimes gives the launch order on a false warning, Delegation with full safeguards is about as risky as Central Authorization with none, and makes it harder to coerce. Under Central Authorization your own country launches only when you give the order. Hardening is the slow, costly, safe way to the same end: survivability's ceiling is 80, and the last stretch of security comes from a faster launch authority.

The Forces section's Chance of a halt row is the chance that someone in the chain halts a launch begun under delegation or launch on warning, or an accident that Automatic Retaliation reads as an attack. Under Central Authorization it reads "not delegated". It starts from 53 plus half your command reliability and rises by 25 points at peace with no crisis and by 10 in a Warning or Confrontation, since a sudden attack makes little sense then; an Acute crisis or a war gives no such benefit of the doubt. It also rises 20 when the suspected

attacker's warheads are in storage and 10 while crisis talks are open, and falls 5 when the suspect is at High Alert and up to 10 when your own forces could not ride out a strike. Nothing makes it certain: it is at most 97% at peace, 92% in a Warning or Confrontation and 85% in an Acute crisis or a war. Automatic Retaliation never answers a retaliation, and answers a given attacker at most once in six months.

## How interest groups judge your posture

The At Home section has a small table for each interest group with a view. It names the group and how it judges you (militarist, restraint, officers or business, some of them mild, hawkish or restrained); hover the word for what that view likes and dislikes. Then comes the approval your posture adds, and a row for each term behind it: Doctrine, Readiness, Launch authority, Crew strain, The arsenal itself, and Alerts and crises. Hover a row for what the group objects to or wants. A group's approval moves by at most 2 either way (1 for a mild view), so when its rows add up to more, a last row, Held to the cap, shows what was cut (Doctrine +2 and Readiness +1 give an approval of +2, with −1 held to the cap). The approval appears in the group's breakdown as Nuclear Posture: Enthusiastic (+2), Approving (+1), Uneasy (−1) or Opposed (−2). Opinions are reviewed monthly, and doctrine and readiness count only once a doctrine has stood for six months.

| Group | Wants | Dislikes |
|---|---|---|
| Armed Forces (officers) | Flexible First Use at Heightened readiness | No First Use, Warfighting, Recessed, Routine when an armed enemy is plausible, High Alert with strain of 50 or more |
| Industrialists (business) | No doctrine preference | High Alert held three months or more; any crisis at Confrontation or beyond |
| Groups favoring Total War over Limited War | Compellence or Warfighting, High Alert | No First Use, Existential Deterrence, Routine or Recessed |
| Groups favoring Limited War over Total War | No First Use (Existential Deterrence less so), Recessed | Flexible First Use, Compellence, Warfighting, High Alert, Launch on Warning, Automatic Retaliation; the arsenal itself, once its burden reaches 33% (−1) or 67% (−2) |

A group that only approves of its preferred Rules of War law, rather than strongly, holds a mild view (±1). The Armed Forces and the Industrialists also lean with their leader: a jingoist-led army wants compellence. The objection to the arsenal itself, its own row in the group's table, comes from the nuclear taboo's burden and counts at once, whatever your doctrine's tenure.

## The nuclear umbrella and nuclear guarantees

Your direct subjects are under your **nuclear umbrella** while you hold Nuclear Power, unless you are at war with them or on opposite sides of a play. Anyone else needs a Nuclear Guarantee treaty article, or to host your weapons under Nuclear Basing. To sign or renegotiate one you need an alliance, a defensive pact or a Guarantee Independence of the country, already in force or in the same treaty, or a shared power bloc, and more warheads than the world believes it has (none, if it has never tested). A guarantee already in force stays if these lapse. Either cover spares the country the war-support drain of facing a nuclear-armed enemy without the bomb, weighs your arsenal in any crisis against it, lets you open a crisis against a country in a war or play with it, and lets you retaliate for a strike on it under any doctrine. You are asked to answer when it is threatened or struck (see Guarantors called in).

Withdraw Nuclear Umbrella is a diplomatic action on a direct subject: +10 liberty desire and −20 relations at once, then +0.1 liberty desire a week until you restore it. Subjects of your subjects are covered by their own overlord only, and power bloc members need a treaty.

Going Recessed while you protect anyone brings "Our Allies Are Alarmed" every time: −5 credibility, −10 relations with each country you protect, and +5 liberty desire in each covered subject. A protector in storage counts for half in a crisis.

## 14.5 Nuclear crises

A nuclear crisis is one pair of countries and one dispute, with a deadline. The issuer, the country that made the threat, runs it; the other is the target. A country can be in only one crisis at a time.

### Opening a nuclear crisis

Two diplomatic actions open one:

| Action | Cost | Deadline |
|---|---|---|
| Private Nuclear Warning | −5 relations | 10 weeks |
| Public Nuclear Ultimatum | Infamy equal to a tenth of the nuclear taboo, −20 relations, higher reputation stakes | 8 weeks |

Either action needs an arsenal of your own, both sides free of other crises, no non-use pledge between you, and one of the disputes below, taken in this order. After a crisis you can't threaten the same country again for 24 months, except in a war, or over their nuclear strike when that strike is what ended the crisis.

The threat reaches its target 14 to 21 days after it is made. Its answer, the deadline and the crisis's year all wait for it, though danger is reckoned from the start. Every other armed country hears of the threat at once: a public ultimatum as news, a private warning as an intelligence report. A Nuclear Guarantee for the target from an armed country lowers the pressure on it to concede.

| Dispute | What the target gives up if it concedes |
|---|---|
| A war between you | It loses 15 war support in the war. If it leads its side, the side drops its war goals against you, and any others of the same kind it holds in that war |
| A diplomatic play between you | It backs down, and the play should end in your favor |
| A play or war between the target and a country you cover, when you are not a party to it | The same, on your protégé's behalf |
| The target struck you, or a country you cover, with a nuclear weapon outside a war in the last 24 months, other than in answer to a strike of yours | It pays the struck country reparations, about a tenth of a year's income, and opens its arsenal to inspection for 20 years, taking Custody Inspections (−5 legitimacy, fading over ten years) |
| A weapons program run by a country the world believes unarmed that is your rival, or that you are antagonistic, belligerent or domineering toward | Its program is frozen for 10 years |
| The target's forces next to you: its Nuclear Basing article in, or nuclear guarantee over, a country next to yours by land or sea, or its missile defense in a state bordering yours | It leaves each such treaty (whole, so the treaty's other articles end too, with no penalty for breaking it), switches the missile defense off, and makes none of these next to you for 10 years |
| The target claims one of your incorporated states | It renounces those claims. A claim covers a whole state region, so one on a region you share goes for all of it |
| A covert operation of the target's against you that you have caught | It ends each one you have caught and runs no hostile operation against you for 10 years. Operations you haven't caught go on |
| The target is building an Orbital Battlestation (a site not left paused) | It tears down the construction site, losing the progress, and starts no new one for 10 years. Levels already built stay |
| An armed rival at Heightened readiness or higher | It stands down to Routine for 24 months |

Only conceding binds the target. Cutting its program's funding to zero, setting its own arsenal ceiling, standing its forces down, pulling a deployment back, ending an operation or tearing a battlestation site down without conceding doesn't end the crisis, because it could undo each of them a week later. A claim that disappears, struck down by the UN court or lost with the state, ends the crisis with no winner. The ten-year bars end if the two go to war. The AI threatens over forces next to it, claims and caught operations only against a rival,

and over a battlestation only as a great power against a country it is hostile to; over any of the four only when nothing weightier divides you and its doctrine makes the threat more than a bluff. An AI won't concede forces next to you when that would end a treaty holding its alliance or defensive pact.

## Crisis stages, danger and pressure

A private warning starts at Warning; a public ultimatum starts at Confrontation. A warning becomes a Confrontation when it goes public, is refused or answered with a counter-threat, or when either side holds an exercise. A Confrontation turns Acute when danger reaches 70 or the issuer is at High Alert while the target is at Heightened or higher. Any crisis turns Acute when the issuer holds firm, a play becomes a war, or a launch between the two is recalled at the last moment. A crisis can be settled at any stage, and it lapses after a year.

The Nuclear Crisis section shows its readings as boxes. Two of them are broken down in their tooltips: Danger, and Pressure on the target (Pressure on us when you are the target), each a word over the figure. Pressure reads Firm below 50, Wavering from 50, Yielding from 70 and Breaking from 85.

- Danger (0–100) rises with the stage, time, publicity, readiness, poor command reliability, counter-threats, exercises and a vulnerable force on either side (up to +10 at survivability 0), and falls with open talks. It is lower when the target has no arsenal and no armed protector. High danger makes incidents likelier.
- Pressure on the target (0–100) is what makes an AI target concede. The issuer's credibility, the danger, a threat that can be carried out, and the issuer's exercises and alerts raise it. The target's ability to answer in kind (up to −25, for three warheads that would survive the issuer's whole arsenal) and an armed protector behind it lower it, as does a threat from an issuer the armed target could disarm first (up to −15). So do the demands at stake: an annexation −30, a subjugation −20, each goal that takes a state or a subject −10 and each lesser goal −4, at most −40; the AI won't back with a nuclear threat a demand so large that it alone keeps an AI target from conceding. The course of the war, the target ruler's temperament and the nuclear taboo shift it either way: up to +8 where the taboo is near 0 and threats are believed, down to −8 near 100 where nobody believes them, with the doubt halved for a public ultimatum. When first threatened, an AI target concedes only rarely below 40. Pressed again later, it never concedes below 50, does so about half the time from 70, and two times in three from 85.

From Confrontation on, the target is pressed every six weeks unless talks are open. Talks are open while a proposal to stand down or to trade awaits an answer, and for six weeks after one side answers The Exercise They Mistook by sharing its schedule (the threatening side accepting, if the other shared it). Either side can still make a proposal during those six weeks.

## Moves during a crisis

Each side acts through events and the buttons under Our Moves in the Nuclear Crisis section.

| Move | Who | Effect |
|---|---|---|
| Concede | Target | The concession happens at once and the crisis ends |
| Refuse | Target | Confrontation. A refused public ultimatum counts as defiance, which lets a Compellence issuer strike you in a war |
| Counter-threat | Armed target | Confrontation, danger +10, readiness to Heightened |
| Propose a mutual stand-down | Either | Pressure pauses and danger falls 15 while the other side decides. If accepted, both stand down to Routine for 24 months and pledge non-use; the war or play goes on |
| Offer a trade | Either, in a crisis over forces deployed next to a country, claims, covert operations or a battlestation, when each side holds the same against the other and neither is under an earlier | Pressure pauses and danger falls 15 while the other side decides. If accepted, each side makes the concession the other demands of it (a base for a base, a claim for a claim), with the same ten-year bars, and the crisis ends with neither backing down |

| Move | Who | Effect |
|---|---|---|
| | crisis's ten-year bar of that kind | |
| Go public | Issuer | Infamy equal to a tenth of the nuclear taboo, −15 relations, a new 8-week deadline |
| Exercise | Either armed side | Money and 5 strain (and +3 credibility for the issuer); for six weeks danger +10, and pressure +10 if the issuer holds it. It can be mistaken for an attack |
| Back down | Issuer | The crisis ends in a climb-down |

At the deadline the issuer chooses: six more weeks (−3 credibility), hold firm (the crisis turns Acute, forces go to High Alert, six more weeks), offer a stand-down, offer a trade where one is open, let the deadline lapse (which counts as backing down), or, at war with the target and with every strike gate open, carry out the threat. A play under a public ultimatum that turns into a war brings "They Chose War": strike, fight conventionally (−10 credibility, militarists disapprove) or offer a stand-down.

## Credibility, bluffs and crisis outcomes

Doctrine is public, so the Nuclear Crisis section rates every threat in its "Can we carry it out?" box ("Can they carry it out?" when you are the target): Backed (+10 pressure; your doctrine permits a strike now, or you hold Compellence or Warfighting), Uncertain (Flexible First Use, or Existential Deterrence with something at stake), or A bluff (−25, or −35 under No First Use; your doctrine or Rules of War law forbids a strike, or your weapons can't reach them).

| Outcome | Issuer | Target |
|---|---|---|
| The target concedes | +10 credibility (+15 if public), Nuclear Brinkmanship Rewarded, hawks approve | −5 credibility, Yielded to Nuclear Coercion, hawks disapprove |
| The issuer backs down | −5 credibility if private; −15, Nuclear Climb-Down and angry hawks if public | +10 credibility, Faced Down a Nuclear Threat |
| Mutual stand-down | +5 credibility (−5 for trading away a public demand, unless the demand defended you or a country you cover: against a play the target started on you or a war on your core states, for a protégé, or over its nuclear strike, forces next to you, claims on your land or operations against you), Mutual Nuclear Restraint, restraint groups approve | +5 credibility, Mutual Nuclear Restraint, restraint groups approve |
| A trade | +5 credibility | +5 credibility |
| A public crisis runs out its year unsettled | −10 credibility | Nothing |

The Outcomes table at the foot of the Nuclear Crisis section shows, for your crisis, what conceding, backing down and a mutual stand-down would bring, and a trade in a crisis that can have one, and what each would do to your credibility. An AI target takes a trade once pressure on it reaches 40, an AI issuer once the danger reaches 40 and its threat is eight weeks old, and neither takes one that gives more than it gets or ends one of its alliances. An AI offers one only at its own deadline, and only when you would take it by the same rule.

A public bluff that ends in a climb-down or runs out its year costs 5 more credibility. The modifiers last five years. A crisis also ends when its war or play ends, when nuclear weapons are used between the two, when a treaty halts the program it is about, when the country on alert no longer holds warheads, or when no claim it is about is left. A non-use pledge from a stand-down blocks strikes on that partner until one side uses Repudiate Non-Use Pledge (−15 credibility, +5 infamy, −30 relations and a Broken Nuclear Pledge).

### Guarantors called in

When a country you guarantee or cover is warned or struck, you get "A Promise Called In" (not as the guest of a Nuclear Basing article, which covers the host in its crises but is not called in):

| Choice | Effect |
|---|---|
| Honor | +5 credibility. Under threat, you back the target. After a strike, you join its war, or open a public crisis against the attacker if there is no war to join |
| Retaliate (after a strike) | +10 credibility; you join the war and, if you are at war with the attacker a day later, strike back then, as retaliation, under any doctrine |
| Abandon | −15 credibility, Abandoned a Nuclear Guarantee for five years, −30 relations with the protégé and −10 with everyone else you protect; a subject gains 10 liberty desire |

After a nuclear strike outside a war on a country you cover, Honor opens a public crisis against the striker over it, if neither of you is in a crisis already.

An armed country fighting a war beside an unarmed one covers it against their common enemy for that war, with no treaty: it hears the same call, and its answer counts as a protector's. Nothing was promised, so walking away costs it only 5 credibility.

If your protégé used the bomb first and its victim answered, you get "Our Protégé Struck First" instead. Standing by it or answering in kind costs 5 infamy; declining costs no credibility, only 10 relations with the protégé.

## 14.6 Nuclear strikes

A single strike is a diplomatic action against a country you are at war with, aimed at a state you pick. A salvo of several warheads is ordered from the Strike section of the journal entry.

### Strategic and tactical strike actions

| | Nuclear Weapon Industrial Strike | Tactical Nuclear Strike |
|---|---|---|
| Needs | Nuclear Power and a warhead | The same, and the Tactical Nuclear Weapons technology |
| Target | Any enemy state in reach | A state in reach with a Barracks, Naval Administration, Naval Fortification, Naval Logistics Center, Conscription Center or Military Base |
| Rules of War that forbid it | Humanitarian Regulations, Limited War | Limited War |

A strike ordered through these actions or a crisis event also needs your doctrine to allow it, no non-use pledge with the target, and forces that are not Recessed. The law's tooltip lists its ban as Strategic Nuclear Strikes Forbidden or Tactical Nuclear Strikes Forbidden, and the ban lifts once you or a country you cover has been struck, or when an enemy's war goal would annex or subjugate you. Give the launch order, the government's answer to an unconfirmed early warning, passes the same tests; only launches nobody ordered skip them (see Nuclear incidents and accidents).

### What a nuclear strike does

A strategic strike that lands devastates the state, kills a tenth of its people and leaves Nuclear Strike Aftermath for two and a half years (decaying): −50% infrastructure, −3 standard of living, +25% mortality and −50% throughput. The victim loses 12.5 war support and 50 relations with you. A first use also costs you infamy equal to the nuclear taboo and relations with every other country; an answer to a strike on you or on a country you cover costs neither. The action's confirmation, and a salvo's Launch Order, name the figures. A strike that fails costs 5 infamy.

A tactical strike that lands kills half the soldiers and officers in the state and a few civilians, and stops unit training there for a while. As a first use it costs infamy equal to four tenths of the taboo, and relations with every other country; as an answer it costs neither. A tactical strike that fails costs no infamy. It halves every Naval Fortification and Military Base in the state, an odd level lost on a coin flip, so a level-1 site is destroyed half the time; what survives keeps its production methods. Barracks, conscription centers and the other naval buildings survive.

Both sides get a notification of the result. Every use ends any crisis between the two countries. One that lands also speeds up the world's funded programs and knocks the nuclear taboo down. The United Nations records it against you, and after a first strike or a tactical strike its court may indict your ruler (see Grounds for UN censure).

## Salvos and the Strike section

The Strike section appears below Nuclear Crisis while you hold warheads and are at war with a country your weapons reach, or in a nuclear exchange before war with it. Pick the target with the buttons beside its name, then a mission and the number of warheads (up to the warheads you hold, at most 25), and press Order. A Launch Order event asks you to confirm: Give the order fires the salvo, and Stand the crews down, its default, fires nothing. The order needs everything a single strike needs, and a command that is not cut off.

| Mission | Aimed at | What each warhead that lands does |
|---|---|---|
| Cities | the states a single strike would choose, a different one each | what a strategic strike does |
| Forces | states with a building a tactical strike can target (Barracks, Military Base and the rest), else the capital's region, else the capital, else any state in reach | a quarter of a city strike's deaths and devastation, and destroys part of the enemy's arsenal |
| Leadership | the capital, which must be in reach | what a strategic strike does, and may cut off the enemy's command |

Each Forces warhead that lands destroys one of the enemy's warheads, or one and a half with Intercontinental Ballistic Missiles or Precision Guided Munitions, and two with both. It can only reach the warheads the enemy's survivability leaves exposed. A share of those gets away first: a quarter under Conditional Delegation and 60% under Launch on Warning, each 5 points fewer per safeguards level. A fraction of a warhead is destroyed or not by chance. Under Launch on Warning up to three of the warheads that got away may answer at once, before your warheads land, if a launch is open to the enemy; the rest stay for its choice. A single warhead can't disarm a large arsenal, and a hardened one keeps most of its warheads whatever lands.

Warheads that land on the capital of a nuclear power under Central Authorization can cut its chain of command. The more warheads it holds, and with Satellite Communications and Mainframe Computers, the more redundant its command: a power with 30 warheads and Satellite Communications needs 7 on its capital before there is any chance, and a state with 3 warheads faces 21% from two. A country cut off can't order a launch, or change its launch authority, for two to four weeks. The salvo's event, "A Nuclear Salvo from ...", offers "Our command is cut off" in place of absorbing the blow or answering it, and so does any strike that lands before command returns. Once the command is restored, a "Command Restored" event asks how to answer, if the war or exchange is still on and the victim did not accept a claim that the salvo was unauthorized. The other launch authorities can't be cut off.

However many warheads it fires, a salvo is one use of nuclear weapons, recorded once. Its record at the United Nations, its infamy, the world's loss of goodwill and its blow to the nuclear taboo are a single strike's times the square root of the warheads that land: four count twice as much as one, nine three times. A salvo that lands nothing costs what a failed strike costs. A claim that the salvo was unauthorized counts every warhead that landed.

## Claiming a strike was unauthorized

When a strategic first strike lands, the launcher may tell the victim that the launch was unauthorized. After a launch nobody ordered (Launch on Warning acting on a false warning, a cut-off commander, Automatic

Retaliation misreading an accident), it says so, truthfully. Your own deliberate first strikes carry the same claim while Disavow our strikes is on, in the Nuclear Posture section; the AI makes it falsely about one time in seven after a single warhead on a country believed armed, and after a salvo only when one warhead landed (twice as often under an aggressive ruler, less under an honorable or direct one, more under a ruthless one or a grifter, half as often once the taboo is 50).

The struck country's event ("A City Erased", or "A Nuclear Salvo from …") shows the claim, with a choice to accept it: hold back the answer and demand reparations, which the launcher may pay or refuse. The choice comes once a strike, and not while Automatic Retaliation has the answer in hand. Paying settles the strike, so the victim can't also open a crisis over it. An AI believes a claim more readily when relations are good and only one warhead landed, less from an opponent in an Acute crisis, and much less from a country already caught lying. Six to eighteen months later (sooner with a covert network in the launcher) the truth comes out. A lie exposed costs the launcher 20 credibility, infamy and 50 relations with the victim, and makes its next claim far harder to believe. A true claim the victim refused to believe lets it make amends.

### Retaliation and Automatic Retaliation

The victim of a strategic strike gets "A City Erased" with its answer: condemn the attack, retaliate with one warhead, or launch a full retaliation of three if it holds more than two. Retaliation costs no infamy, counts for less with the United Nations than a first strike, knocks the nuclear taboo down half as far, and the attacker may answer it in turn. A Nuclear Weapon Industrial Strike you order in answer to one on you or on a country you cover counts the same way: no infamy, and retaliation for the taboo and the United Nations. A Tactical Nuclear Strike in answer costs no infamy and counts as an answer for the taboo, but the United Nations records it as it does any tactical strike. Under Automatic Retaliation the answer is made for you; while Recessed you can only mate the warheads and answer when they are ready.

A strike is an answer when its target struck you, or a country you cover, in your present war with it, or fights you in the same war as a country whose strike still leaves its aftermath on one of your states. A strike by a third country gives you no answer against your enemies, and a loose warhead's detonation, which nobody ordered, gives you none at all.

## 14.7 Nuclear incidents and accidents

Every armed country rolls once a week for a meaningful incident. The peacetime base rates are equivalent to 0.05% a month at Recessed, 0.1% at Routine, 0.4% at Heightened and 1% at High Alert, before command reliability, strain and doctrine. Flexible First Use raises exposure by 15%, Compellence by 35% and Warfighting by 60%. These are chances of an incident, not of nuclear use.

Crisis danger adds a separate weekly exposure to that base. At High Alert it adds 0.5% at danger 25–49, 3.5% at 50–74 and 8.5% at 75–100, before the same modifiers. A war with an enemy believed armed adds the same as danger 50–74, or as 75–100 when either side holds goals on the other's incorporated states, is losing, or has been struck; alongside a crisis the higher applies. Heightened takes six tenths of that addition and Routine one quarter. This crisis exposure only ever produces a misreading (a false warning, a mistaken exercise or a commander cut off), never a weapons accident, so it needs one of those to be possible: none is at Recessed, and at Routine only a cut-off commander under delegation or launch on warning is. Meaningful exposure is capped at 35% per country per week. The Nuclear Posture section's incident exposure reads Extreme in a dangerous crisis. Two High Alert countries at danger 75 or higher, with reliability 60, no strain and Existential Deterrence, have about a 28% chance of at least one meaningful incident over two weekly checks each.

No posture is safe. Over an eight-week Acute crisis between two AI powers at High Alert, the chance that one of them launches when nobody attacked runs from about 2% (Central Authorization with full safeguards) through about 5% (Delegation with full safeguards, or Central with none) to about 27% (Launch on Warning with none). A year of war between two AI powers under Central Authorization at High Alert carries about 9%, about 19% when the war threatens either side. Under Central Authorization those figures come from the AI governments' launch orders; a player's government launches only on its own order.

Routine mishaps have their own monthly roll: 0.2% at Recessed, 0.5% at Routine, 1% at Heightened and 2% at High Alert. After one, no other comes for five years. They do not replace a meaningful incident.

| Incident | Can happen when | What it is |
|---|---|---|
| A false warning ("Conflicting Indications") | Radar, Heightened or higher, and a plausible armed opponent | An early warning that no second source confirms. Launch on Warning, or delegation in a war with the suspected attacker, can execute it without another order; otherwise the government decides |
| The Exercise They Mistook | A crisis at Confrontation or beyond, Heightened or higher | Your opponent reads your exercise as cover for an attack |
| Silence from the Capital | Delegation or Launch on Warning, assembled weapons, and war, an Acute crisis or a heightened force facing a plausible attacker | A commander cut off from the capital; another officer refuses, or the unit fires. Slightly likelier in the incident draw in a war or an Acute crisis |
| A Weapons Accident | Any armed posture; six times more weight after six months at High Alert or strain of 60 or more | Crashes, silo explosions, lost weapons, a contaminated state or failed maintenance. With Nuclear Basing, a third of crashes and silo explosions happen in a host, and can't be classified |
| A Routine Mishap | Separate monthly roll | Base alarms, dropped training weapons and storms; cannot launch |

Most incidents cost money, readiness, reliability or reputation. Safeguards and professional skepticism make launch sequences easier to halt. A buried incident has a 3% chance each month of coming out (Nuclear Cover-Up Exposed: −10 legitimacy, −10% authority, −5% prestige).

When the government decides on a false warning, Give the launch order needs a war with the suspected attacker or an Acute crisis against it, and every test for a deliberate strike except the doctrine, which only No First Use blocks. If the warning was false, the launch is a first use. Central Authorization cannot launch without this decision. Launch on Warning and isolated delegated commanders can fire without it, including at peace. Automatic Retaliation can misread a bomber crash or silo explosion as an attack during war, an Acute crisis or High Alert against a plausible attacker. Nothing launches from Recessed.

## A nuclear exchange before war

A mistaken launch can strike a country you are at peace with. The victim can answer with nuclear weapons under any doctrine, including No First Use, while that exchange remains open. This permission belongs to the two countries in the exchange; it does not open ordinary peacetime strike actions. The normal stockpile, disarmament, pledge and assembled-force requirements still apply to an ordered response. An unauthorized first use breaches your pledges and pays the usual consequences, but it does not end No First Use or strike the No-First-Strike Pledge amendment: your government never chose to. It decides afterwards (see Nuclear doctrine).

After at least seven days without another launch, the next weekly review offers both governments "After the Exchange". If both stand down, the exchange ends without a conventional war and a five-year truce binds the two countries. If either refuses, a conventional war starts after both have answered. The refusing country initiates it, and either side may still answer the strike with nuclear weapons until the war is under way. If no war has begun two weeks after the refusal, the exchange ends and both governments are told. Another launch cancels earlier votes and starts the quiet period again. An old offer cannot settle a renewed exchange. A recalled order is now a fallback for an invalid launch context, rather than the automatic result of being at peace.

An AI government that launched by mistake usually stands down. One that was struck refuses more often: more so if it took the blow without answering, for every state of its that was hit, and if its army is half again the launcher's. An AI whose enemy's army is half again its own leans to standing down.

### The Monopoly Window

A Compellence or Warfighting power at war with an enemy that has no arsenal, no armed protector and no armed ally in that war, and whose doctrine allows a strike on it, has an 8% chance a month, at most once a year, of "The Monopoly Window": its general staff proposes using the bomb while nobody can answer. You can win conventionally, issue a public ultimatum, or authorize the strike. The strike obeys the Rules of War like any other: under Limited War or Humanitarian Regulations it is offered only once their block has lifted (see Strategic and tactical strike actions).

## 14.8 Arsenals in civil wars and annexations

Warheads outlive the government that built them. When an armed country is annexed by war, diplomacy or a formable nation, its arsenal passes to whoever holds the most populous state in its old capital's region, and a human recipient gets "The Arsenal Changes Hands". Every such transfer loses some warheads on the way: 7% with no safeguards, 2 points less per safeguards level (1% at level 3). Conditional Delegation or Launch on Warning adds 3 points, and the loss doubles if the holder was at war. **Secured custody** halves the loss. You have it while a foreign custodian helps you in a civil war, for twenty years after opening your depots to inspectors over a stolen warhead, while a Nuclear Security Assistance article is in force, or as a party to the United Nations' Convention on the Physical Protection of Nuclear Material. Missing warheads become loose warheads.

### Who Holds the Button?

When a revolution or secession breaks out in an armed country, the government chooses in "Who Holds the Button?", whose tooltips show the exact numbers:

- Hold the line: the rebels seize a share matching their share of the population, half again more under delegated authority and 20% less per safeguards level, and some of those go missing.
- Bring them in: readiness goes to Recessed and stays locked there until every civil war of yours is over, so you can't strike, answer a strike or raise readiness. Rebels take half their share under delegated authority, none under central control.
- Take them apart: your stockpile goes to zero and you lose Nuclear Power, nothing is seized or lost, and every country's relations with you rise by 10.

Striking your own civil war, from either side, brings Nuclear Weapons Used on Our Own People (−50 legitimacy, decaying over twenty years) and a wave of radicals that your speech and policing laws shrink or enlarge.

### When a revolution wins or loses

A winning revolution takes the state's arsenal, less the warheads lost, as a new regime: it keeps the old doctrine, investments and warhead progress, restarts at 50 credibility, and is not bound by the old government's pledges. If the government wins, the rebels' warheads come back. A winning secession keeps what it seized; a crushed one returns it.

### Deny Them the Bomb

A government losing a revolution while armed (a quarter of its land occupied, or most battles lost) is asked once per civil war, "Deny Them the Bomb". Taking the warheads apart in haste loses some on the way; an accepted foreign custodian can take them apart with none lost; or you keep them, and a winning revolution inherits them. A secession never asks this, because its winner does not take the government's arsenal.

### Foreign powers and a divided arsenal

A week after an armed country's civil war breaks out, every armed major power and armed neighbor gets "A Nuclear Power Divided". It can back either side (+20 relations with it, −20 with the other, +5 war support for that side), offer to secure the arsenal, or stay out. An accepted offer makes the helper the government's custodian for the war.

### The Budapest path

A week after a secession wins while holding warheads, every armed great power gets "A New Nuclear State" and may press it. A month after the first does, the new state gets "The Budapest Offer". Trading the warheads away dismantles them, and the pressers sign treaties with you wherever one can be made: the one with the largest economy a Nuclear Disarmament and a Nuclear Guarantee, every other a guarantee, all binding for ten years. Relations with each rise by 20. Keeping them costs 20 relations with each presser, and for ten years the AI is readier to demand your disarmament.

## 14.9 Loose warheads

Lost warheads stay in the world. The Unaccounted for row in Nuclear Posture counts those from your arsenal.

### Nuclear terror plots

Once warheads are loose, a plot can come up in the yearly terror events of a country with the Terrorism and Antiterrorism technology and at least 10% radicals, a war, or a movement with radicalism of 25 or more. It is real with a 20% chance per loose warhead in the world (at most 80%), using another country's lost warheads. Your security services seize it first with a 10% chance, plus 5% per Ministry of Intelligence and Security level and 15% with the Nuclear Weapons technology (at most 75%). Otherwise it destroys a city in one of your populous states. Once a device surfaces anywhere, no other can for a year.

### Tracing a stolen warhead

A month later, "Where Did It Come From?" reports the result. Your own nuclear expertise, a covert network in the country of origin, a seized device and fewer possible sources make certainty likelier.

| Finding | Your options |
|---|---|
| Confirmed | Hold the origin to account |
| A shortlist of two or three, which may include innocent countries | Name them all in public (−25 relations with each), or ask each privately to open its depots |
| Unknown | Say nothing, or blame a rival anyway |

Blame means negligence, not attack. The origin gets "The Warhead Was Ours" and can pay compensation (up to a tenth of your GDP), let your inspectors in (twenty years of secured custody, Custody Inspections at −5 legitimacy, and a 50% chance of finding one of its missing warheads at once), or deny it (−40 relations, +5 infamy).

### Recovering loose warheads

Recovered warheads are destroyed. Besides the inspection search, three things find them:

- Covert: Secure Loose Material, an operation against a country with warheads missing: a 3% chance a month once established, 6% when fully operational, and more at a higher priority (see Cultural hegemony and covert warfare, which also covers Covert: Nuclear Program Sabotage).
- A Nuclear Security Assistance article: 6% a month for the country helped.
- The Convention on the Physical Protection of Nuclear Material: 3% a month for each party. It reaches the United Nations Assembly only once warheads have gone missing somewhere.

## 14.10 The nuclear taboo

The nuclear taboo is one score for the whole world, from 0 to 100, for how unthinkable nuclear weapons have become. It is born at 20 the week the world's first warhead is built, and from then on every country's Nuclear Weapons entry is active and shows it. It climbs slowly on its own, falls when the bomb is used or brandished, and rises with restraint. The more unthinkable the bomb, the more it costs to use it, threaten with it and, above 40, simply to hold it. It also shapes how the AI builds, threatens and strikes (see How the AI plays nuclear weapons).

## The taboo section and its bands

The Nuclear Taboo section of the entry is open by default.

| Row | What it shows |
|---|---|
| Nuclear taboo | The score, the target it is moving to and its band ("62 ⟶ 68 · Strong"). Hover it for the target's breakdown, part by part |
| First use on a city, First use in battle | The infamy and the relations with every other country that a strategic or tactical first use would cost now |
| A public ultimatum | The infamy a public ultimatum would cost now |
| Burden of our arsenal | Only while you hold warheads: the prestige and leverage your arsenal's burden costs. Hover it for the burden itself |

Below the rows, a line says whether a nuclear weapon has been used in war and, if so, when the last one fell. The Taboo Over Time, open by default, holds two charts, the score and its target, month by month.

The band is named beside the score and marks where the taboo's effects start.

| Band | Score | What it means | What starts here |
|---|---|---|---|
| Normalized | 0–29 | The bomb is treated as one weapon among others | Below 30, AI rulers drop some of their restraint |
| Fragile | 30–49 | The bomb is feared, but its use is still argued for | Above 40, holding warheads is a burden |
| Established | 50–69 | Using the bomb is widely held to be wrong | Above 50, AI countries lean to No First Use and accept arms control and disarmament more readily |
| Strong | 70–89 | The bomb is seen as unusable, and even holding one draws censure | From 70, the AI strikes first only for survival and may dismantle its arsenal |
| Absolute | 90–100 | The bomb is beyond the pale, and holding one marks a state apart | Every cost keeps rising with the score |

## What moves the nuclear taboo

Each month the taboo moves toward its target by a small part of the gap, never more than one point, so it closes about half of a gap in two years. The target is the sum of six parts, which the breakdown lists:

| Part | Range | What sets it |
|---|---|---|
| Base | 20 | Constant |
| Tradition of non-use | 0 to +35 | One point a year, full after 35 years. A first use halves the years counted; an answer to a strike cuts them by a quarter |
| Doctrines | −10 to +10 | The doctrines of the countries that hold warheads, averaged with great powers counting most. No First Use and warheads in storage (Recessed) raise it; Flexible First Use, Compellence, Warfighting and High Alert lower it |
| Restraint | 0 to +15 | Non-use pledges in force (up to +5); countries that gave up an arsenal and have not armed again (up to +10) and countries bound by a Nuclear Arms Limitation treaty, both weighted by rank |
| United Nations | 0 to +18 | The Non-Proliferation Treaty (up to 8), the Physical Protection convention (up to 4) and the Treaty on the Prohibition of Nuclear Weapons (up to 6), in proportion to UN authority |
| Ledger | −30 to +10 | The record of what the world has done lately, which halves every four years |

Base and a full tradition make 55. With no use and nothing else moving it, the target climbs a point a year from 20 to 55 over 35 years, and the score follows a couple of years behind. Anything above 55 has to come from doctrines, restraint, the United Nations and the ledger.

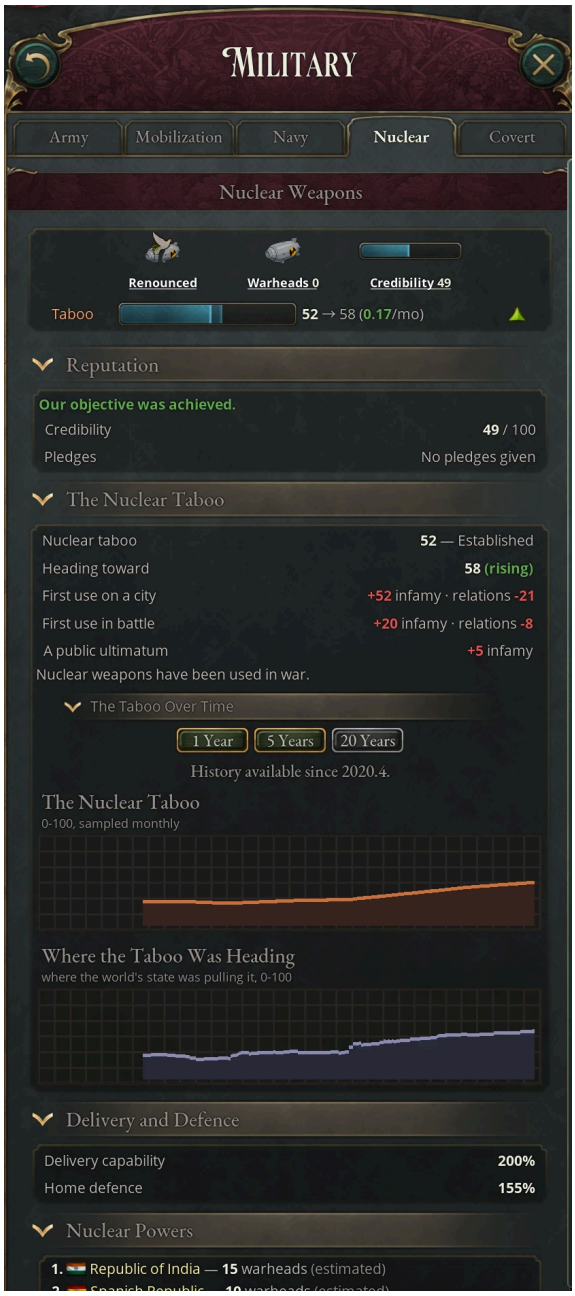


*Figure 22: The Nuclear tab of the Military panel for a country with no warheads and a Renounced posture: Reputation, The Nuclear Taboo with its two charts, Delivery and Defense, and the start of Nuclear Powers. The taboo is Established at 52 and rising toward 58.*

A use that lands hits the score at once as well as the ledger. A first use halves the tradition and takes 15 points off the score and 10 off the ledger for a strategic strike, 6 and 4 for a tactical one. An answer to a strike counts half as much and cuts the tradition by a quarter. Other acts move the ledger only. These lower it:

- Threats: a public ultimatum, and less so a private warning or taking one public.
- Adopting Compellence or Warfighting.
- Repudiating No First Use or breaking a non-use pledge.
- A breakout: a country that gave up an arsenal holding a warhead again.
- Leaving an arms-control treaty (see The Nuclear Arms Limitation treaty).
- Halting a dismantling.
- A condemnation of a nuclear use that fails, or that a veto cuts to a rebuke.

These raise it:

- Adopting No First Use yourself (not when the No-First-Strike Pledge amendment holds you to it).

- A nuclear crisis that ends in a mutual stand-down.
- Giving up an arsenal, by any path, and more for a bigger one: Dismantle the Arsenal, a Nuclear Disarmament article, the Budapest path, or taking the warheads apart in a civil war.
- Each warhead taken apart under a ceiling, yours or a treaty's.
- A condemnation of a nuclear use that carries.
- A loose warhead's terror detonation: no government chose it, and the world closes ranks after it.

Uses count the same whoever makes them. Threats, doctrines, broken pledges, breakouts, walk-outs and halts count by the country's rank: a great power's four times a minor power's.

## What the nuclear taboo costs

Every scaled cost is nothing at a taboo of 0 and grows in step with it.

| Act | Cost | At a taboo of 50 |
|---|---|---|
| Strategic first use | Infamy equal to the taboo, and relations with every country but the victim lowered by four tenths of it | 50 infamy, −20 relations |
| Tactical first use | Four tenths of the strategic figures | 20 infamy, −8 relations |
| A public ultimatum, taking a warning public, adopting Compellence or Warfighting | Infamy of a tenth of the taboo | 5 infamy |

A strike that answers one on you or on a country you cover costs no infamy and no relations at any taboo. At 100 a single first strike on a city costs 100 infamy, enough on its own to reach the Pariah threshold. The taboo section's cost rows show today's figures, and a strike's confirmation repeats them before you launch. The taboo also moves the pressure a threat puts on its target (see Crisis stages, danger and pressure).

Above a taboo of 40, holding warheads is a burden in itself. Your burden, a percentage the Burden of our arsenal row shows on hover, grows with the taboo, from nothing at 40 to full at 100, and with your arsenal, from about a fifth of full for a single warhead to all of it at fifty warheads or more. It does two things:

- The Burden of the Bomb lowers prestige by up to 45% and leverage generation by up to 25%, and the UN Authority Target by up to 1 (see What counts toward Policy). At a taboo of 100, a fifty-warhead arsenal turns Nuclear Power's +30% prestige into −15% and cancels its leverage bonus; five warheads in the same world keep about +17% prestige.
- Interest groups favoring Limited War over Total War object to the arsenal itself: −1 to their view of your posture from a burden of 33%, −2 from 67% (see How interest groups judge your posture). The ceiling's tooltip says how many warheads would ease them a step.

## Reducing or giving up an arsenal

The exits sit under Arsenal Size in the Forces section. An Arsenal row says whether a ceiling holds you or you are dismantling. Below it are the Arsenal ceiling stepper, Lift the arsenal ceiling with its Lift button, Dismantle the arsenal with Begin, and Halt the dismantling with Halt.

The **arsenal ceiling** is the most warheads you will hold. The minus button sets one below your stock, or lowers the one you have, by 1 warhead up to 10, by 5 up to 50 and by 25 above that, and the plus button raises it by the same steps. It goes no lower than 1; below that you dismantle. Warheads above the ceiling are taken apart, a tenth of the excess a month and at least one, and each strengthens the taboo a little. While you hold at least as many warheads as the ceiling, Program Held stops your program: funding goes to zero and can't be raised, and the overview's program icon reads At Ceiling ("Development is held at our arsenal ceiling." on hover). Raising the ceiling above your stock, or lifting it, costs nothing and releases the program at once, unless a treaty's ceiling still holds you.

Dismantle the Arsenal needs warheads, peace, no nuclear crisis and no civil war. It takes 12 months, one more for every 10 warheads above 20, and at most 36. The warheads go at an even pace over that time. Until it ends, your readiness is ordered down to Recessed and can't be raised, and the program is held. Recessed forces can't

launch, so once there you can't answer a strike. A war doesn't stop the dismantling; a civil war of your own pauses it, and it resumes where it left off once every such war is over. Halt the dismantling at any time: what is taken apart stays gone, your credibility falls by 10 and the taboo weakens. If the arsenal goes another way meanwhile (a disarmament treaty, or taking the warheads apart in a civil war), the dismantling ends without its rewards.

When the last warhead goes, "The Last Warhead" fires. Its option gives The Bomb Renounced, a prestige bonus of up to +20% in proportion to the taboo (+10% at 50) that fades over 20 years, and raises relations with every country by a fifth of the taboo (+10 at 50). Restraint-minded interest groups approve and hawks disapprove. You lose Nuclear Power and take Renounced the Bomb: you count as a disarmed country, so no program can run, and the overview's program icon reads Renounced ("We gave up the bomb of our own accord." on hover). Only Dismantle the Arsenal brings these rewards and Renounced the Bomb; every path counts toward the taboo.

Resume the Nuclear Program is a decision you can take while you hold Renounced the Bomb. It ends the renunciation and The Bomb Renounced, and costs infamy of a tenth of the taboo and relations with every country of a fifth of it. It gives back no warheads. A great power, a major power with Intercontinental Ballistic Missiles or a country receiving Nuclear Program Aid can then run a program again; a country without the rank for one is warned that resuming only ends its renunciation. Once any country that gave up an arsenal, by whatever path, holds a warhead again, the taboo books a breakout against it.

## The Nuclear Arms Limitation treaty

Nuclear Arms Limitation is a mutual treaty article, available with the Nuclear Weapons technology, between two countries that each hold warheads or run a program. You set the ceiling in the draft, from 1 up to a quarter above the larger of the two arsenals (at least 5). While the treaty is in force:

- Neither party holds more warheads than the ceiling. Those above it are taken apart, a tenth of the excess a month, and each party's program is held at the ceiling from its next monthly review.
- The Arsenal row reads "Held to N by treaty" when the treaty binds you tighter than your own ceiling, or you have none.
- Each country it binds adds to the taboo's Restraint part, once however many such treaties it has, weighted by rank.

Leaving costs nothing directly, but it weakens the taboo. When a treaty ends, whoever ends it, or is renegotiated to a higher ceiling, each party whose lowest treaty ceiling rises or disappears is booked a walk-out. No walk-out is booked while another such treaty holds the party to the same ceiling or a lower one, or when the partner no longer exists. Your own arsenal ceiling never covers a walk-out. An AI party accepts more readily as the taboo rises above 50 and as its arsenal burdens it, is very reluctant under a militarist government, and resists a ceiling that stops it building the arsenal it wants or cuts it deeper than the other party.

## The United Nations and the taboo

The United Nations can move the taboo, but the taboo works without it. While the Non-Proliferation Treaty is in force it adds up to 8 to the target, the Convention on the Physical Protection of Nuclear Material up to 4 and the Treaty on the Prohibition of Nuclear Weapons up to 6, all in proportion to UN authority (4, 2 and 3 at authority 50).

Once the charter carries Charter Reform II and the Non-Proliferation Treaty is in force, a member without warheads can table the Treaty on the Prohibition of Nuclear Weapons. A party that holds warheads is held to an arsenal ceiling: the warheads it had in its first month as a party, a fifth fewer every five years, none after 25. The schedule stops while the UN is Moribund. Warheads above it are taken apart, each one raising the taboo, and the Nuclear Weapons entry reads "Held to N by the UN ban". A party without warheads may build none. An armed member is bound only by its own ballot: one that cast none, or joins the UN later, is not bound, and leaving the UN while bound counts as walking out of an arms-control treaty. Nuclear powers vote against it and refuse to ratify it.

When the docket takes up a nuclear strike, its appeal can table a condemnation even while the condemnation topic is on its five-year cooldown, provided the striker's case gives grounds. That vote is the Assembly's verdict on the use. A condemnation that carries raises the score 2 points at once and the ledger 3; one that fails, or that a veto cuts to a rebuke, takes 3 off the ledger. Sanctions or a court case over the same strike deliver no verdict. See The UN docket and UN conventions and agencies.

### Nuclear taboo events

When the score crosses a band line, every country gets an event: four as it rises through 30, 50, 70 and 90, and four as it falls back through them. The score has to pass 2 points beyond the line, the band moves at most one step a month, and each of the eight events fires at most once in ten years. The options act on your own country; those that start a real action (a ceiling, dismantling or No First Use) then move the taboo the usual way. Whether you hold warheads decides which options you see.

| Option | Events | Offered when | What it does |
|---|---|---|---|
| Cut our arsenal in half | Rising | You hold more than one warhead, aren't dismantling, and have no ceiling already at half your stock or lower | Sets your arsenal ceiling at half your stock |
| Take our arsenal apart | Rising | You could begin Dismantle the Arsenal | Begins it |
| Pledge never to use it first | Rising | You hold warheads and could adopt No First Use now | Adopts it |
| Our deterrent is not negotiable | Rising | You hold warheads and aren't dismantling | Hawks approve, restraint-minded groups disapprove |
| Champion the norm abroad | Rising | You hold none | Champion of the Taboo (+5% prestige, fading over ten years); −5 relations with each great power that holds warheads |
| Keep out of the powers' quarrel | Rising | You hold none | +5 relations with each great power that holds warheads |
| Modernize the arsenal | Falling | You hold warheads and aren't dismantling | +10 command reliability, +5 survivability; restraint-minded groups disapprove |
| Say publicly that nothing has changed | Falling | You hold warheads | Restraint-minded groups approve, hawks disapprove |
| Seek the shelter of a friendly nuclear power | Falling | You hold none, and an armed country is on amicable terms with you | +15 relations with one such country |
| Put more into our own program | Falling | You hold none and could add a funding step | One more funding step |
| Build shelters and civil defense | Falling | You hold none | Civil Defense, below |

Build shelters and civil defense gives Civil Defense for ten years. It costs a weekly sum set when you choose it, about 0.26% of your GDP a year, and halves the war-support drain of facing a nuclear-armed enemy without the bomb (see War support from mod systems). These events and The Last Warhead are listed in Nuclear taboo event list.

## 14.11 Nuclear treaty articles

Seven treaty articles deal with nuclear weapons. Each costs its payer influence upkeep while the treaty is in force.

| Article | Parties | Effect |
|---|---|---|
| Nuclear Disarmament | The disarmed country concedes; the demander pays 200 | Stockpile and progress go to zero, Nuclear Power is lost, and no program runs while it lasts |
| Nuclear Program Freeze | The frozen country concedes; the demander pays 100 | Funding held at zero; warheads and progress kept |
| Nuclear Program Aid | A nuclear power helps a non-nuclear country and pays 500 | The recipient can run a program at any rank, at double the rate. Refused once the IAEA exists and United Nations authority is 60 or more |
| Nuclear Basing | A nuclear power stations its weapons in a host and pays 100 | Its reach extends from the host's states, and it covers the host as a ready guarantor; not in your own subject. See Range and Nuclear Basing |
| Nuclear Guarantee | An armed guarantor pays 100 | Extended deterrence, as for the umbrella; not for your own subjects. The guarantor needs more warheads than the world believes the protected country has, and an alliance, a defensive pact, a Guarantee Independence or a shared power bloc with it |
| Nuclear Security Assistance | An armed country pays 100 to help one holding or missing warheads | Secured custody and a monthly chance to recover missing warheads |
| Nuclear Arms Limitation | Mutual, between two countries that each hold warheads or run a program; both pay 50 | Neither holds more warheads than the agreed ceiling; see The Nuclear Arms Limitation treaty |

Aid can't be combined with a disarmament or freeze of the same country, and a country whose program is held (at an arsenal ceiling, or while it dismantles) can't agree to a freeze. The United Nations' non-proliferation treaty and IAEA are in The United Nations.

## 14.12 How the AI plays nuclear weapons

The nuclear taboo feeds every one of these decisions: in a world that treats the bomb as ordinary the AI builds more, threatens more and strikes first more readily, and in one that holds it in horror it trims and gives up its arsenal. It retaliates at any taboo.

- It funds its program toward a target stockpile that grows with rank, innovation, war and a rival that seems to hold more, and shrinks as the taboo rises: about a third larger at 0 than at 40, and half the size at 100. From a taboo of 70, an AI at peace that faces no plausible attacker is far slower to fund a program.
- It reviews its posture every six months and when a crisis opens. Most AIs keep Existential Deterrence; cautious rulers and democracies lean to No First Use, and militarist regimes (fascist, or with a jingoist ruler or a powerful Armed Forces in government) to Compellence or Warfighting. Above a taboo of 50, No First Use gains favor; below 30, Compellence and Warfighting tempt any ruler who is not cautious; from 70 they, and Flexible First Use, lose favor.
- It sets its safeguards from one level: one more for a cautious ruler, a restraint-minded government and Delegation or Launch on Warning, one less for a militarist regime or Warfighting and in a budget deficit. It moves to Automatic Retaliation, Launch on Warning (if militarist) or Delegation only when fewer than three of its warheads would survive its main rival and hardening could not change that, and back to Central once that passes.
- It goes to High Alert in a war with an armed enemy, and in an Acute crisis when its armed opponent is at Heightened or higher or fewer than three of its warheads would survive short of High Alert; otherwise to Heightened in any war or Confrontation, and to Recessed only at peace with nothing to deter and nobody to protect, and then only under a cautious ruler, No First Use or a default.
- It strikes first only when its doctrine allows and its survival is at stake, or when it is losing (Flexible), was defied (Compellence) or is at war (Warfighting) against an enemy with no arsenal and no armed protector.

An aggressive ruler losing under Flexible First Use strikes whether or not the enemy can answer; a cautious one strikes first only for survival, unless the taboo is below 30. From a taboo of 70, any AI strikes first only when its enemy means to annex or subjugate it, whatever its doctrine. The lower the taboo, the more readily it takes a strike it is allowed, and that includes carrying out a crisis threat: under Warfighting below a taboo of 25 it does so about twice as readily. It is a fifth as likely to carry out a threat against an insurgency, a revolutionary or secessionist state. It keeps a warhead in reserve unless its survival is at stake, waits six months between first uses, and strikes its own rebels only with 40% of its land occupied under Outlawed Dissent or a Secret Police. None of this holds back retaliation.

- When it strikes, it may fire a salvo instead of one warhead. It aims at the capital of an enemy under Central Authorization when its spare warheads, times the chance each lands there, outnumber the enemy's redundancy as the world sees it; at the forces of an enemy believed armed under Flexible First Use, Compellence or Warfighting; and otherwise at one city. It keeps a third of its warheads, at least one, in reserve. A Forces salvo is sized to destroy the arsenal the world believes the enemy holds; a Leadership salvo to land twice the enemy's redundancy on its capital.
- It warns countries that threaten a protégé or its survival, and countries that struck it or a country it covers outside a war, and a great power warns a rival that is building a bomb. Coercive warnings need a hawkish doctrine or regime and a target that can't answer. It never makes a public bluff. The lower the taboo, the more often it threatens: about twice as often at 0 as at 35, and a third as often at 100, except in defense of a country it covers, its survival or its core territory, or over a nuclear strike. From a taboo of 70 it issues a public ultimatum only in defense of a country it covers, its survival or its core territory, and otherwise warns privately.
- At each posture review it also weighs its arsenal. While its arsenal burdens it and it holds more than half again the stockpile it wants, it sets a ceiling at what it wants; it lifts that ceiling once it wants more, and in a war lifts any ceiling below what it wants. When the burden reaches 30% or the taboo 70, it may begin dismantling, a one-in-five chance each review, provided it is at peace, outside any nuclear crisis or civil war, faces no plausible attacker, has neither a militarist government nor an aggressive ruler, and is covered by a guarantee or umbrella or is not a great power. It halts a dismantling only when an armed enemy at war with it means to annex or subjugate it. A renouncer takes Resume the Nuclear Program only with the standing to run a program, below a taboo of 50, and when an armed country is at war with it or is a rival antagonistic toward it.
- An armed AI proposes Nuclear Arms Limitation when the taboo is 50 or more or its arsenal burdens it, at a ceiling of three quarters of the larger arsenal.
- It usually declines to answer for a protégé that struck first, never withdraws a nuclear umbrella, and demands disarmament or a program freeze from rivals and hostile countries. It makes either demand, in a treaty or as a war goal in a diplomatic play, only of a country with something to stop: a program, the rank of a major power or higher, or, for disarmament, an arsenal the world knows of. It presses hardest against a known arsenal or a program being funded. It accepts a Nuclear Disarmament demand more readily when the taboo is above 50 or its own arsenal burdens it.
- It offers Nuclear Program Aid only to a country that has researched the Nuclear Weapons technology.

CHAPTER 15

# Climate and pollution

Burning coal and oil warms the world in this mod. Every country's emissions add to one global total, the total sets a global temperature anomaly, and the anomaly drives penalties that every country shares. The Global Warming journal entry tracks it and holds the climate policies you can adopt. The Global Warming game rule (on by default) controls all of this; with it off, the world never warms. State pollution, the Environmental Movement and the pollution scandal event work under either setting. The emissions lines in production method tooltips and the phaseout mandate also remain active with the rule off; capture still costs goods.

## 15.1 How emissions become warming

Each country counts the buildings and households in its own states. Buildings contribute their production methods' greenhouse gases, with source capture and synthetic-fuel credits subtracted. The country's own industrial reductions cut its remaining industrial emissions. Households add a heating footprint estimated from population and average standard of living. Direct Air Capture at Carbon Conversion Works subtracts atmospheric removal. Buildings count where they stand, including foreign-owned sites. Markets sum their members; every country's net annual contribution adds once to the world's cumulative total and sets warming.

Exporting synthetic fuels can make the producing market's figure negative: the fuel's production credit counts there, while burning it counts in the importing market. Carbon Captured measures atmospheric removal by Direct Air Capture; synthetic-fuel production does not add to that row.

In the dashboard's units, emissions of one million a year warm the world by 0.1 °C a year. Figures use K for thousands and M for millions. A unit of oil produces 13% less warming than a unit of coal. The mod calls coal Energy and Carbon Minerals; this chapter calls it coal for short.

Production methods that burn coal or oil show Greenhouse Gas Emissions in their tooltips. Fuel methods add to the figure; capture methods subtract from it. Process fuel and automation both contribute. The figure grows with staffing and throughput and feeds the climate calculation. The state owner's industrial policy cut applies afterward.

Coal Mines and Graphite Mines burn some of their own coal. Their Atmospheric Engine Pump, Condensing Engine Pump and Steam Donkey methods take that coal off the mine's output instead of listing it as an input, so the methods show no coal cost but still show the emissions: 2.00 and 3.00 a level for the two pumps and 0.60 for the Steam Donkey, at full staffing and base throughput. An Iron Mine shows the same 2.00 and 3.00 for these pumps; its Steam Donkey, which lists 4 coal as an input, shows 0.80. Oil-fired methods in every mine list their oil as an input and show it as usual.

Household heating demand follows the buy packages at each state's average standard of living, with fossil heating weighted by the coal, oil, wood, fabric and electricity available in the market. Peasants contribute only 5% of the purchased heating estimate. Green Building Codes cuts the estimated household fossil footprint by 60%, Renewable Investment by 25%, and Fossil-Fuel Divestment by 15%. These cuts add together: all three eliminate household fossil-heating emissions. All three policies affect your own households. Industrial emissions still depend on production methods and capture.

Fuel-input discounts reduce goods costs; the custom emissions figure follows the recipe and its staffing and throughput. Public Transit's oil discount has an economic effect, while Fossil-Fuel Divestment also reduces

household emissions. The Environmental Sustainability principle's explicit emissions reduction counts toward warming; its coal and oil input discounts reduce costs. Standalone military-unit and ship fuel is outside this building-based figure.

Electric Arc Process, Aluminum Substitution and Chromium Substitution in Steel Mills use 10 coal per level. Electric Arc Process also uses 50 electricity; the two substitution methods use 170 electricity. Their lower coal use cuts emissions, while electricity from fossil plants adds emissions elsewhere in the market.

Warming does not wear off. The anomaly falls only in a year when the world as a whole captures more carbon than it emits. Cutting your emissions slows the rise; it does not undo what is already there.

Industrial policy reductions come from each country's Carbon Tax, Renewable Investment, Emission Standards, Ministry of the Environment (−5% per level), Environmental Sustainability principle (−5% to −25% by tier) and a few event and United Nations outcomes, such as Climate Summit Leadership. Some event choices, the Environmental Grandfather Clause amendment and the Directed Credit to Heavy Industry banking policy raise industrial emissions instead, by 5% to 10% each. A market member's own ministry reduces its own states' emissions. All of these add together, and the cuts stop at complete elimination. Atmospheric removal retains its full credit and can take a country or market below zero.

Carbon Conversion Works unlock with Carbon Capture and Storage in era 10, after Clean Energy Technologies. Their Direct Air Capture method removes $CO_2$ from the atmosphere without producing goods to sell. At full staffing and base throughput, each level removes 42 a year in the dashboard's units and uses 1,200 electricity, plus engines, steel, chemicals and electronic components. Removal and synthetic-fuel credits follow staffing and throughput. Subsidize the works to keep them staffed; a fossil-powered grid adds emissions from the electricity they use. Genetic Engineering in era 11 unlocks their Synthetic Coal alternative, which produces fuel while capturing carbon.

## 15.2 Capturing carbon at the source

Stationary coal and oil consumers have Carbon Capture controls, including industrial plants, mining equipment, steam-powered automation and eligible company sites. Each eligible fuel-consuming production group has its own control. A Steel Mill can capture its furnace emissions and its automation emissions independently. Each tier comes in a variant for every fuel recipe, named for it, such as Partial Carbon Capture (10 Coal), and only the variants matching the group's current method appear.

| Method | Needs | Captures |
|---|---|---|
| Partial Carbon Capture | Clean Energy Technologies, era 9 | 25% |
| Carbon Capture and Storage | Carbon Capture and Storage, era 10 | 50% |
| Advanced Carbon Capture | Modern Material Science, era 11 | 75% |

Capture uses engines and steel, plus chemicals for most sources. Power plants lose electricity output to run capture equipment; other buildings buy additional electricity. Costs follow the fuel used by that production group. The tooltips show the reduction and goods cost. Capturing a fossil power plant's emissions can make it unprofitable, so subsidies may be needed.

Changing a group's method to one that burns a different amount of coal or oil can reset your capture choice. Under Managed Fossil Phaseout, eligible sources must use at least the middle tier. Without that law, select the tier yourself; the AI generally prefers the cheaper option without capture. A method with no eligible fuel use has Capture Not Applicable and stays valid under the mandate.

Transport exhaust, fishing and whaling fleets, dispersed agricultural and logging machinery, mobile mine excavators and street lighting have no source capture. Oil and coal used as plastics, textile and other chemical feedstock, oil held at refineries and depots, and Synthetic Fuel Works, whose fuel output already counts against their emissions, are also exempt. These consumers still show emissions and contribute to the industrial climate total.

## 15.3 The Global Warming journal entry

With the Global Warming rule on, the entry is listed, grayed out, for every country from the start of the game, with a line saying that it opens once the world has warmed by 0.1 °C and its climate policies at 0.5 °C. It activates for everyone once the anomaly reaches 0.1 °C and then stays active: it never completes and never goes away, even if the world cools again. It also activates early for a country that adopts Fossil Expansion Moratorium or Managed Fossil Phaseout, with the Global Warming rule on or off (The fossil transition). The temperature bar at its top runs to 4 °C, but the penalties keep growing past that.

The same panels appear as a Climate tab in the Market panel, and a change made in one shows in the other. It is grayed until the journal entry is active; hover it for what is still missing. The tab ends with an Open Journal Entry button.

The tab is on the Market panel for every market. On your own market, whether you lead it or joined it, it is the journal entry. On another market, a line names the market's leader, and the parts about the market show that market: in the overview, the leader's role and any Enforce Emissions Reduction treaty that binds it, the market's emissions and carbon captured, and its share and emissions-cut pies; below, the leader's Mitigation Policies, with every Adopt and Repeal grayed, since only that leader can change them. The temperature, Top Emitters, the history charts and the Help page stay your own view. Use it to check what a market you want to bind with Enforce Emissions Reduction emits and which policies it already runs.

The anomaly sets the warming tier, which the entry shows as a thermometer icon with the tier's name beneath it:

| Anomaly | Tier |
|---|---|
| Below 0.1 °C | Negligible |
| 0.1–0.5 °C | Slight |
| 0.5–1.0 °C | Moderate |
| 1.0–2.0 °C | Significant |
| 2.0–3.0 °C | Severe |
| 3.0–4.0 °C | Catastrophic |
| 4.0 °C and above | Apocalyptic |

### The Climate Warming modifier

Each month the entry gives every country the Climate Warming modifier, with every value multiplied by the current anomaly, after any aerosol mask (Aerosol Injection). At 1 °C it is:

- +5% mortality and −1 standard of living in every state;
- −10% farm throughput and −2.5% throughput for every building;
- +5% construction goods used by buildings;
- +1% migration quota, as climate refugees move;
- 25% worse floods, droughts, extreme winds and torrential rains, and 50% worse heatwaves and wildfires, in both impact and duration;
- 15% milder frost, in impact and duration, and 15% weaker hailstorms;
- 25% weaker pollinator surges and 10% weaker moderate rainfall, the two harvest conditions that help.

At 2 °C every line doubles, and at 3 °C it triples. The Warming Penalty box at the top of the entry shows today's standard of living and mortality lines; hover it for every line and the multiplier.

Warming also moves every state's local climate: hot days and winters rise, winters most near the poles, so hot states lose more migration pull to heat and cold states less to cold.

*Figure 23: The Climate tab of the Market panel at 1.90 °C, tier Significant: the overview (tier, Market Leader, Penalty ×1.90, the Temperature bar and its threshold line, the emissions table and the two pies) and the start of Mitigation Policies.*

## 15.4 The climate dashboard

The journal entry opens with an overview that is always shown. The sections below it start open. The "?" at the right of the tab's header, and at the top of the journal entry, opens the Help page, How Global Warming Works. While the entry is active it has no status line: the overview carries the tier and the readings.

The overview opens with boxes, two to a row: Warming, the tier and where the next one begins (hover it for what the tier means); Warming Penalty, today's standard of living and mortality lines of the Climate Warming modifier; and Market, Leader or Member, which says whether you set fossil-fuel tariffs. A Treaty box, Bound with repeals locked, joins them while an Enforce Emissions Reduction treaty binds you.

Below them is the Temperature bar, from 0 to 4 °C, with a headline such as "1.61 → 1.81 °C (+0.02/yr)": the anomaly now, the anomaly in ten years if every year warms as much as the last, and last year's change. The bar shows the same projection. While the world warms, the solid fill ends at today's reading and a red stretch runs on to the projection: red is warming to come. While it cools, the solid fill ends at the projection and a

green stretch runs on to today's reading: green is warming set to be lost. A thin line marks where the next tier begins; hover it for the tier and how many years away it is at last year's rate. Until the first yearly figures arrive in January, the headline shows the anomaly with "(no rate yet)".

Under the bar, a table gives the year's figures: Our Market's Emissions, Carbon Captured, World Emissions and Warming Last Year. While aerosols mask any warming, or a country has just started them, an Aerosol Mask row joins them; hover it for the warming beneath the haze, how many countries fly it and where the mask is heading. Hover a row for what it counts. They update each January; before the first January of a game, a line says when they arrive.

Two pies close the overview. Our Share is your market's share of world emissions. Emissions Cut is how far your country's remaining industrial emissions are cut, by its policies, the Ministry of the Environment and every other source together; hover it for how many of the eleven policies you run. The cut changes the moment a policy is adopted or repealed.

| Section | Starts | Shows |
|---|---|---|
| Mitigation Policies | Open | Eleven policies: market-wide Fossil-Fuel Tariffs first, then ten National policies. Each row has an icon, its name and effect tooltip, an active check, and Adopt or Repeal. Grayed controls explain the warming, authority, technology, treaty, market ownership and customs conditions. Adoption Around the World counts countries using each policy; the tariff row counts market leaders. |
| Top Emitters | Open | The five countries emitting most, largest first, including market members. Each row shows its flag, annual emissions, share of world emissions and national cumulative total. A treaty icon marks a bound country; your row says "Our country". Hover for its industrial reduction and market context, or "+N more countries" for the next 25. Figures update each January; rankings update monthly. |
| Fossil Transition | Open | Your Resource Transition law, what is left of your coal mines, oil rigs and power plants with a retirement control for each, and what could replace them. See The fossil transition. Shown on your own market only. |
| History | Open | Charts of global temperature and of your market's share of world emissions. Both step once a year, when the emissions figures update. |

The Help page explains warming, the warming penalty, emissions, the mitigation policies and geoengineering, then the fossil transition. With the Global Warming rule off it is How the Fossil Transition Works and explains the transition alone.

Top Emitters tells you whom to bind: every country controls its own industrial policies, whether it leads its market or belongs to another country's. A country that captures as much carbon as it emits is not listed. Its cumulative total follows it through market joins, departures and leadership changes; negative years subtract from it. Each member can adopt or repeal its own policies independently.

## 15.5 Climate policies

There are eleven policies. Ten are national, open to every country. Only Fossil-Fuel Tariffs is market-wide and controlled by the market leader. Every policy needs 0.5 °C of warming, except Fossil-Fuel Divestment, which needs 1.0 °C, and Aerosol Injection, which needs 1.0 °C beneath any aerosol haze. A policy that depends on something you must be able to build also needs the technology for it:

| Policy | Technology |
|---|---|
| Renewable Investment | Clean Energy Technologies, era 9 |
| Green Building Codes | Clean Energy Technologies, era 9 |
| Emission Standards | Pollution Control, era 7 |
| Public Transit | Modern Urban Planning, era 7 |
| Carbon Removal Support | Carbon Capture and Storage, era 10 |

| Policy | Technology |
|---|---|
| Aerosol Injection | Geoengineering, era 12 |

Carbon Tax, Climate Adaptation, Reforestation Subsidies, Fossil-Fuel Divestment and Fossil-Fuel Tariffs need no technology.

National policies stay with a country when it changes market, and the leader's adoption or repeal changes only its own country. Most policies carry an Authority Cost for as long as they are in force; adoption requires producing more authority than that cost.

Fossil-Fuel Tariffs raises coal and oil import and export tariffs and costs 100 Authority. It adds 2.5 percentage points at Low, 5 at High and 10 at Maximum. The leader's native tariff levels stay as chosen; a good on No Tariffs or Subventions still collects none. Repeal removes only the rate increase. It can coexist with legislated customs. Losing market control ends the former leader's policy. Tariffs can change trade and fuel costs; they grant no automatic emissions reduction and also apply to traded synthetic coal and oil, which use the same goods as fossil fuels.

| Policy | Scope | Authority | Effects |
|---|---|---|---|
| Fossil-Fuel Tariffs | Market leader | 100 | Coal and oil import/export tariffs +2.5/+5/+10 percentage points at Low/High/Maximum Tariffs; No Tariffs or Subventions stays tariff-free. |
| Carbon Tax | National | none | Industrial emissions −20%; coal and oil output −10%; taxes on oil extraction doubled, manufacturing taxes +10%. |
| Renewable Investment | National | none | Industrial emissions −15%; household emissions −25%; Renewable Energy Plants +20% throughput; conventional power plants −10% throughput; hydro, nuclear, fusion and renewable plants built 25% faster. Costs 0.5% of GDP a year, listed under Climate Policy in the Budget panel's Breakdown tab. |
| Emission Standards | National | 200 | Industrial emissions −10%; generated pollution −25%; −5% throughput for every building. |
| Climate Adaptation | National | 250 | Mortality −2.5% and standard of living +0.5 in every state. |
| Reforestation Subsidies | National | 100 | Farm throughput +5%; droughts and floods 25% weaker and shorter. |
| Carbon Removal Support | National | 100 | Carbon Conversion Works require subsidies and gain +5% throughput; Environmental Movement radicalism −5 percentage points. |
| Public Transit | National | 150 | Transportation output +10%; oil input −5%; infrastructure built 10% faster; pops' automobiles add less infrastructure. |
| Fossil-Fuel Divestment | National | 200 | Taxes on oil extraction +25%; coal and oil input −5%; household emissions −15%. |
| Green Building Codes | National | 100 | Construction goods input +5%; electricity input −2.5%; household emissions −60%. |
| Aerosol Injection | National | none | Masks warming for every country (below); Environmental Movement radicalism +10 percentage points. Costs 1% of GDP a year, listed under Climate Policy in the Budget panel's Breakdown tab. |

Carbon Removal Support funds the works in your own country, whether you lead your market or joined it. The policy itself removes no $CO_2$: staffed works must run Direct Air Capture or Synthetic Coal. Its subsidy

requirement covers both methods. Repealing it ends the requirement and bonus; you can still subsidize the buildings yourself. Direct Air Capture has no sales income, so withdrawing its funding stops it as its workers leave.

Carbon Tax, Renewable Investment and Emission Standards cut your own industrial emissions directly. Green Building Codes, Renewable Investment and Fossil-Fuel Divestment cut your own household footprint. Other national measures protect people and calm the Environmental Movement (The Environmental Movement). Climate Adaptation and Reforestation Subsidies soften warming's harm at home whatever other countries do.

Your industrial policies, household measures and capture work whether you lead your market or are a member. Tariffs are a separate choice for leaders: they can reduce fossil-fuel trade, but higher import tariffs can also protect domestic coal and oil producers. Watch the actual emissions figure when judging them.

### Aerosol Injection

Stratospheric Aerosol Injection hides warming instead of preventing it. Aircraft keep a haze of sulfate aerosols in the stratosphere, and the haze masks part of the warming for every country in the world. The countries running it together mask 1 °C for each fifth of world GDP among them, at most 2 °C and never more than the warming there is. The mask builds up over a year or two, about two thirds of the way in the first year. The temperature the entry shows, the Climate Warming modifier, the tiers and the warming events all read the warming less the mask. Emissions keep accumulating underneath, and the recovery events wait for the warming beneath the haze to fall, not the masked reading.

When no country runs it any more, the mask fades as fast as it built: two thirds of the hidden warming returns within a year, nearly all within three. If at least 0.25 °C is coming back, every country gets Termination Shock, a modifier that fades over five years on top of the returning warming: for each degree returning, −5% farm throughput, heatwaves 25% worse and longer, and droughts 25% worse. Once you start, stopping is expensive unless others keep flying.

The haze has two more costs. A country whose capital lies in South Asia, the Himalayas, Indochina, Indonesia, southern China or West Africa carries Weakened Monsoon while any mask is up: for each degree masked, −5% farm throughput and droughts 25% worse and longer. That takes back half the farm relief the mask brings there and all of its drought relief. And other countries object: those in the monsoon regions, and those that would not run it themselves (their will for it, below, is more than 15 points short of its threshold), each lose 10 relations with a country that starts it, and 5 more every January while it keeps it up.

Geoengineering also brings weather modification, whether or not you fly the haze: in every state, droughts are 10% milder and 10% shorter, and hailstorms 20% weaker.

## 15.6 How the AI adopts climate policy

Each AI country weighs each policy separately, from its own conditions. For every policy it keeps a score, its will, and compares it with that policy's threshold in the table below. Every policy's will starts from a shared core: 20 points per degree of warming beneath any aerosol haze (at most 100), 15 more with the Ministry of the Environment established, and 8 more if an environmentalist leads an interest group in government. Five signals then push it up or down, weighted by who each policy costs or helps:

- laissez-faire economics, against every policy;
- industrialists in government, against the policies that burden industry;
- the Environmental Movement's support, for every policy;
- standard of living against the world average: wealth favors most policies but counts against Climate Adaptation and Reforestation Subsidies;
- how far the market is a net exporter of coal and oil, strongly against Fossil-Fuel Divestment, the Carbon Tax and Fossil-Fuel Tariffs, and mildly against Renewable Investment, Public Transit and Emission Standards.

Aerosol Injection turns most of them around: laissez-faire, industrialists in government and fuel exports count for it, since it asks nothing of industry, and the Environmental Movement counts against it.

The AI adopts a policy once the will reaches the threshold, if it has the authority and the policy's technology; for Carbon Removal Support it also needs a Carbon Conversion Works. It repeals a policy only once the will falls more than 15 points below the threshold, or while its authority is negative, and it also drops Carbon Removal Support when it has no works. It doesn't start Aerosol Injection in default or once the haze is as thick as it can get, and stops it in default. Between the two it leaves the policy alone, so an election doesn't flip policies back and forth.

| Policy | Threshold |
|---|---|
| Climate Adaptation | 25 |
| Reforestation Subsidies | 30 |
| Renewable Investment | 35 |
| Public Transit | 40 |
| Green Building Codes, Emission Standards | 45 |
| Carbon Removal Support | 50 |
| Carbon Tax, Fossil-Fuel Tariffs | 55 |
| Aerosol Injection | 60 |
| Fossil-Fuel Divestment | 70 |

In practice most countries take up Climate Adaptation around 1.25 °C if they have the authority, countries with an environment ministry move early, and Fossil-Fuel Divestment is rare below 3 °C. A laissez-faire oil exporter may never divest, but is the likeliest to fly aerosols, from about 1.6 °C with industrialists in government; most countries wait for 3 °C.

## 15.7 The fossil transition

The Resource Transition laws let you stop building fossil-fuel capacity and then retire what you have. They sit in the Economy laws beside the Ministry of the Environment, and both restrictive laws need that ministry:

| Law | Needs | Effect |
|---|---|---|
| Unrestricted Extraction | | The default. Nothing changes. |
| Fossil Expansion Moratorium | Environmental Movement, Ministry of the Environment | No new Coal Mine, Oil Rig or Power Plant can be built or expanded in your states, by your government, your investors or foreign investors. Those already running carry on. |
| Managed Fossil Phaseout | Carbon Capture and Storage, Ministry of the Environment funded to level 3 | The moratorium, retirement programs, and at least 50% capture for eligible stationary sources. |

Hydro, nuclear, fusion and renewable plants, Synthetic Fuel Works, Carbon Conversion Works and every other mine stay buildable. If you abolish the Ministry of the Environment, the law falls back to Unrestricted Extraction; cutting the ministry's funding later does not. Environmentalist leaders support the transition and Corpocrat leaders oppose it. Otherwise an ideology that favors the Ministry of the Environment approves of the moratorium, and one that opposes the ministry disapproves of the phaseout.

Adopting either restrictive law opens the Global Warming journal entry for you at once, and its Fossil Transition section holds the controls. With the Global Warming rule off, the entry shows only that section; the laws work the same way.

### Retirement programs

Under Managed Fossil Phaseout, the Fossil Transition section has a row for each of Coal Mines, Oil Rigs and Power Plants, with the levels you have left and a Start button. A running program closes the smallest building of its kind in your states every 3 months, starting 3 months after you press Start:

- the whole building goes, every level, whoever owns it: your government, a company, your investors or a foreign investor;
- you pay compensation from the treasury, a quarter of what the building would cost to build at base prices, though no owner receives it;
- its state gets Transition Assistance for 5 years: half the radicals and loyalists that a change in standard of living would make, and +25% qualifications as the laid-off workers retrain.

Hover a row for the next building to close, its state and its compensation, and for the cost of closing all that are left. Stop ends a program; what it closed stays closed. Leaving Managed Fossil Phaseout stops every program.

### Readiness and emissions

Closing mines and rigs cuts supply. Coal and oil get dearer, buildings that burn them earn less and switch to cleaner methods where they can, and anything still short is imported or goes without. Under Replacement Readiness the section shows what could take their place:

| Row | Shows |
|---|---|
| Clean Generation | Hydro, nuclear, fusion and renewable plant levels as a share of all generating levels, power plants included. |
| Synthetic Fuel Plants | Your Synthetic Fuel Works and Carbon Conversion Works levels; hover for whether you have the technologies for oil from grain and electricity (Synthetic Biology) and coal from electricity (Genetic Engineering). Carbon Conversion Works unlock earlier for Direct Air Capture with Carbon Capture and Storage, which removes $CO_2$ without providing replacement fuel. |
| Coal, Oil | What your whole market makes and burns each week; hover for its imports and exports. |
| Our Market's Emissions | With the Global Warming rule on, the overview's figure. |

None of this is a requirement: you can retire capacity you can't yet replace and import the difference, burn less, or accept shortages. The transition earns no emission cut of its own. Your market's emissions come from the coal and oil its buildings and households burn, wherever it was mined. Closing a mine cuts only the fuel that mine burned itself, plus whatever the market then burns less. Power plants are safest to retire once clean generation is most of your fleet, and mines and rigs once your market makes more of the fuel than it burns.

The AI considers the moratorium on the same will as Fossil-Fuel Divestment (How the AI adopts climate policy), and only with a quarter of its power fleet clean; it moves to the phaseout with a stronger will and half its fleet clean. It starts a program only while its market has fuel to spare (or, for power plants, mostly clean generation) and it is not deep in debt, stops one when the fuel or electricity runs short, and returns to Unrestricted Extraction when its market is short of coal, oil or electricity.

## 15.8 Enforce Emissions Reduction

Enforce Emissions Reduction forces its source country to maintain seven national policies: Carbon Tax, Renewable Investment, Emission Standards, Reforestation, Public Transit, Fossil-Fuel Divestment and Green Building Codes. It appears past 0.1 °C and can be signed from 0.5 °C with Intergovernmental Organizations, including against a market member. The source must have researched the technologies those policies need: Clean Energy Technologies, Pollution Control and Modern Urban Planning. Below 1.0 °C it still forces Fossil-Fuel Divestment. It can be a war goal or a requested or offered treaty article. Climate Adaptation, Carbon Removal Support and Fossil-Fuel Tariffs are not forced; the first and last remain freely revocable, while the existing treaty lock also bars repealing removal support if the bound country chooses to adopt it.

Only the source country's policies change. It also suffers −20% power plant throughput, −10% Coal Mine and oil extraction throughput and −2 Industrialists approval. The treaty waives the five forced policies' Authority Cost (−750 in all). The demanding side gains +2% prestige and pays upkeep. Losing a forced policy freezes the treaty. See Diplomacy and The United Nations for treaty mechanics and climate accords.

## 15.9 Climate events

Two kinds of event come with warming. Threshold events fire once for every country as the world crosses 0.5, 1, 2, 3 and 4 °C, and recovery events once if it falls back below a mark it had passed. Each month every country also has a small chance of one recurring event whose conditions it meets. Most need a level of warming, and one, a pollution scandal at large coal mines or oil rigs, fires even with the Global Warming rule off, and less often than the others. Their choices trade money, authority and interest group approval against radicals, mortality and throughput, and four have an extra option for a Ministry of the Environment or of Consumer Protection funded to level 3. The Tide Comes In can leave you with Coastal Flooding or Coastal Population Relocation; while either lasts, a Settlement Authority on Managed Retreat can move people off your coasts (Resettlement programs). Termination Shock fires for every country when the last country flying aerosols stops (Aerosol Injection). Both lists are in Climate event list.

## 15.10 State pollution

Pollution works as in the base game: a state region's buildings generate pollution, and the region's pollution impact moves a little each day toward a level set by that generation and the region's arable land. The mod changes how much of a state's generation counts, adds pollution-control production methods and makes heavy pollution hurt more.

### Generated pollution cuts

The mod adds a Generated Pollution modifier that scales how much pollution a state generates, recalculated monthly. It is lowered by:

| Source | Generated Pollution |
|---|---|
| Pollution Control decree (needs Pollution Control) | −50% |
| Emission Standards policy | −25% |
| Ministry of the Environment | −5% per level |
| Novo Nordisk company, while prosperous | −15% |
| Novo Nordisk Kalundborg Plant, in its own state | Up to −15% |
| Biotechnology company, while prosperous | −5% |
| Electric Vehicles technology | −5% |
| The region's pollution impact (Heavy pollution) | Up to −90% |

The Environmental Grandfather Clause amendment raises it by 5%. The cuts add together, and at −100% nothing the state generates counts. They apply to what is left after the flat cuts below.

Each level of the Ministry of the Environment also cuts emissions by 5% and raises the National Park level cap by one. Each level costs 1% mining and logging throughput, 2% oil extraction throughput and 1% more construction goods.

Other sources take a fixed amount off a state's generated pollution. The pollution-control methods below and National Parks do it per level at full staffing (500 on the National Park and National Wildlife Refuge methods, 250 on National Forest), and a monument dedicated as Botanical Gardens takes 5,000 a step (see Monument dedications). A polluting production method adds between 3 and 100 a level.

### Pollution-control methods

Seven production method groups let polluting buildings clean up, each with a method that does nothing, a basic one and an advanced one:

| Group | Buildings | Basic, per level | Advanced, per level |
|---|---|---|---|
| Tailings and Wastewater Management | Mines | Settling Ponds, −5 | Tailings Treatment, −10 |

| Group | Buildings | Basic, per level | Advanced, per level |
|---|---|---|---|
| Facility Emission Standards | Tooling Workshops, Arms Industries, Munition Plants | Ventilation Systems, −5 | Green Assembly, −10 |
| Industrial Synthesis Scrubbing | Fertilizer Plants | Basic Fume Scrubbers, −10 | Closed-Loop Synthesis, −15 |
| Precision Environmental Management | Rye, Wheat, Rice, Maize and Millet Farms | Improved Crop Rotation, −5 | Precision Agriculture, −10 |
| Effluent Filtration | Paper Mills | Waste Fiber Recovery, −3 | Effluent Filtration, −5 |
| Fume Ventilation | Glassworks | Fume Extraction, −5 | Closed-Loop Recycling, −10 |
| Emissions Control | Power Plants, Automotive Industries, Ports, Steel Mills | Basic Emissions Control, −10 | Advanced Emissions Control, −20 |

The basic methods need Pollution Control. The advanced ones need the Ministry of the Environment and, outside Emissions Control, one of the building's later production methods; the tooltips name them. Once the ministry is established, the methods that do nothing are no longer allowed, so these buildings run at least the basic method. Improved Crop Rotation and Waste Fiber Recovery cost nothing; the others use electricity, tools or fertilizer. The groups cut local pollution only: greenhouse gas emissions have their own controls (Capturing carbon at the source).

## Heavy pollution

Heavy pollution hurts more than in the base game. As a state region's pollution impact rises toward 100%, the mod adds up to +5% mortality, −3 standard of living and −5% workforce ratio on top of the base game's penalty. It also cuts tourism output by up to 200%, so a region at half its maximum pollution has no tourism left.

Pollution impact also damps each state's own generation, adding up to −90% to its Generated Pollution modifier: −45% at 50% impact and −90% at 100%. Pollution health protections don't weaken it. The dirtier a region gets, the less each extra unit of pollution adds, so its impact settles lower than it would in the base game. With no other cuts, generation that would hold a region at 50% impact in the base game holds it near 34%, generation that would just reach 100% holds it near 53%, and it takes ten times that much to reach 100%.

This keeps a dirty region responsive to cleanup. In the base game, a region that generates several times what it takes to reach 100% stays at 100% until its generation falls below that level, so the first cuts show nothing. Here the same region sits below 100%, and every cut lowers it unless the region still generates ten times that level. Cuts take effect gradually: the modifier is recalculated monthly, and impact moves toward its new level a little each day.

## The Environmental Movement

The Environmental Movement is a political movement that forms once you research Pollution Control, if you have not established the Ministry of the Environment. Its support comes from polluted states, from academics and literate middle-class pops, and from warming, rising at 0.5, 1.0, 1.5 and 2 °C; the Environmental Movement technology doubles it, and Genetic Engineering adds a quarter more. Seven climate policies move its radicalism by 10 points: Carbon Tax, Renewable Investment, Climate Adaptation, Emission Standards, Reforestation Subsidies, Public Transit and Fossil-Fuel Divestment. Each one in force lowers it, and each one you could adopt but haven't raises it. Carbon Removal Support only lowers it, by 5 points, Aerosol Injection in force raises it by 10, and Green Building Codes and Fossil-Fuel Tariffs don't count. Each unfunded level of the ministry raises it by 5 points. It disbands once the Ministry of the Environment is established and funded to its maximum. Its support also pushes AI countries toward climate policies.

CHAPTER 16

# The space race

The space race is a sequence of journal entries in which great and major powers compete to reach space milestones, from a first suborbital flight to colonies in the Kuiper Belt. The first country to finish each of the six single-goal milestones, or to found a colony on the Moon, keeps a larger permanent reward than anyone who follows. The entries start to appear once you research Rocketry in era 6, but you need a Space Program building, unlocked by Guided Missiles in era 7, before any of them can make progress. The Space Race game rule controls the system; with it off, the journal entries and their events never appear. Its Full Colonies setting, the default, and its Simple Colonies setting also make colony fleets take months or years to arrive, give colonies settlers and modules, tie Extraplanetary Bases to the room colonies give, let a colony change hands, and let colonies grow restless and break away (see Colony fleets in flight, Colony settlers, Colony modules, Extraplanetary Base room, Ceding a colony and Colony autonomy and free colonies). Full Colonies adds warships in space: fleets you build, move between the planets and fight with, to blockade, seize or destroy colonies, and to hold Earth orbit (see Space fleets).

## 16.1 Joining the space race

Only great powers and major powers can run a space program. If you drop below major power, your running milestones stop and their progress is lost. The one exception is Solar System Colonization once you hold a colony: it keeps running whatever your rank. A player country of either rank always makes progress. An AI country makes progress only while it is in the top three of the global country ranking, so in practice you race against the AI powers that hold those places.

### The Space Program building

The Space Program is a monument in your market capital, and every milestone depends on it. It is a single, government-built building with a very high construction cost, but you don't have to build it: once you research Guided Missiles, the decision Found a Space Program places it in your market capital at once, with no construction cost, running Earth Orbit. Either way it can stand only in the capital of a market you lead, so a country in another country's market can't have one. It is removed if the state stops being your market capital, for example when you lose it in a war or join another market. It employs 40,000 engineers, academics and clerks and consumes Launch Capacity, the good your Aerospace Industry makes. That weekly bill is what the program really costs; the Budget panel's Breakdown tab lists it under Program Costs, in the Space Race row.

Its production method decides which milestones you can run. Each method includes everything the ones before it allow, so you only ever need the highest one your program calls for. Each step up uses far more Launch Capacity and gives more innovation, innovation cap and prestige.

| Production method | Technology | Milestones it allows | Launch Capacity used |
|---|---|---|---|
| Earth Orbit | Guided Missiles (with the building) | Suborbital Flight, Orbital Flight | 500 |
| Moon Mission | Space Exploration | adds Moon Landing and Deep-Space Probe | 1,500 |
| Mars Mission | Knowledge Economy | adds Mars Landing | 5,000 |

| Production method | Technology | Milestones it allows | Launch Capacity used |
|---|---|---|---|
| Solar Colonization | Cislunar Infrastructure | adds Solar System Colonization, the Moon and stages 1 and 2 | 10,000 |
| Deep Space Exploration | Space Colonization | lets a country with no colony beyond the Moon start colonizing at stages 3 to 5 | 50,000 |
| Interstellar Mission | Aneutronic Fusion | adds the Interstellar Probe | 100,000 |

> Lowering the production method, or losing the building, fails every running milestone the new method no longer supports and wipes its progress. The event Program Discontinued (or The Silent Gantry, if the building is gone) lists what was lost. Solar System Colonization only pauses once you hold a colony.

### Moving to the next mission

You can switch the production method in the building panel at any time, but the decisions panel also offers each next mission when you are ready for it. A "Begin" decision appears once you research the method's technology, and you can take it once the milestones before it are complete, while you are a great or major power. It switches the Space Program to that method.

| Decision | Appears while you run | Needs complete |
|---|---|---|
| Begin the Moon Mission | Earth Orbit | Orbital Flight |
| Begin the Mars Mission | Moon Mission | Moon Landing and Deep-Space Probe |
| Begin Solar Colonization | Moon Mission or Mars Mission | Moon Landing |
| Begin Deep Space Exploration | Mars Mission or Solar Colonization | Mars Landing |
| Begin the Interstellar Mission | Mars Mission or any later method | Deep-Space Probe and Mars Landing |

Begin Solar Colonization appears only while there are sites left that Solar Colonization can reach: those of the Moon, and after your Mars Landing those of stages 1 and 2, or of any stage once you also hold a colony beyond the Moon. Every site beyond the Moon needs your Mars Landing. Begin Deep Space Exploration appears only when you need it: every site of stages 1 and 2 is claimed and you hold no colony beyond the Moon. You can begin the Interstellar Mission without finishing colonization, and colonization goes on under it.

Once all 40 sites have been claimed, a site on the Moon that comes free again is settled only by choice, and no program starts on its own. The decision Resume Solar System Colonization (or Begin Solar Colonization) runs your program for the Moon until it is full. A finished program keeps Interplanetary Trade Networks, and it doesn't start again for a site beyond the Moon. While it runs, the program bills its funding and approach, and Commercial Space Industry pauses, as it does whenever one of your programs runs.

## 16.2 Space race milestones

There are eight space race journal entries: six milestones with a single goal, the repeatable Solar System Colonization, and a waiting entry that follows the interstellar probe. Each entry appears, inactive, once you have finished the milestone before it and researched its technology, and it starts when your Space Program runs a method that supports it.

| Milestone | Needs completed | Technology | Progress needed | Setback risk | Cost factor |
|---|---|---|---|---|---|
| Suborbital Flight | nothing | Rocketry (era 6) | 50 | 5% | ×1 |
| Orbital Flight | Suborbital Flight | Guided Missiles (era 7) | 100 | 6% | ×1.5 |

| Milestone | Needs completed | Technology | Progress needed | Setback risk | Cost factor |
|---|---|---|---|---|---|
| Moon Landing | Orbital Flight | Space Exploration (era 7) | 200 | 6% | ×2 |
| Deep-Space Probe | Moon Landing | Space Exploration (era 7) | 150 | 6% | ×2 |
| Mars Landing | Moon Landing | Knowledge Economy (era 9) | 450 | 9% | ×6 |
| Interstellar Probe | Deep-Space Probe and Mars Landing | Aneutronic Fusion (era 13) | 600 | 9% | ×12 |
| Interstellar Probe: Awaiting Data | Interstellar Probe | none | 132 months | none | none |
| Solar System Colonization | Moon Landing (Mars Landing for the sites beyond the Moon) | Cislunar Infrastructure (era 10) | 300 to 800 per colony | 10% | ×8 (×5 before your Mars Landing) |

The setback risk is the base chance each month; the cost factor scales the milestone's innovation cost (see Space race funding and cost). The order is mostly a chain, but it branches after the Moon Landing: the Deep-Space Probe, the Mars Landing and colonizing the Moon can then run at the same time, as long as you have their technologies and methods. The Interstellar Probe needs the Deep-Space Probe and the Mars Landing finished, and colonies beyond the Moon need the Mars Landing. The Deep-Space Probe and Mars Landing show up in the journal as soon as you reach orbit and have their technologies, but neither can start before the Moon Landing.

## The space milestone panel

A running milestone's journal entry shows its panel: an overview at the top and sections below it. An entry that hasn't started yet shows only its status text.

The same panels appear on the Space Race tab of the Timeline Extended window, which the button under the sidebar's Map List opens, and a change made in one shows in the other. The tab is grayed until your first milestone starts; hover it for what is still missing. From then on it stays open, whether or not a milestone is under way. It opens with every milestone you have under way, each under its own name and followed by an Open Journal Entry button; while none is, a note stands in their place. Two sections of its own follow, Space Powers and Orbit (see The Space Race tab), and How the Space Race Works comes once at the end.

The overview is always shown. Its first row is three icons, each with a label under it:

| Icon | Reads | Means |
|---|---|---|
| Approach | Standard, Safe or Ambitious | The approach in force. Under Safe and Ambitious, their weekly innovation cost, as in −30 a week. |
| Setback risk | Risk: 6%/mo | This month's chance of a setback. |
| The first | First: Open or First: Claimed | Whether the larger reward is still there to win. The icon is faint once another power has claimed it. |

Below them, the Progress bar fills toward the goal. Beside it are your progress, the goal and this month's pace, as in "124 / 200 (+3.5/mo)". Hover the bar for the reward for finishing first and for finishing later. Hover the pace for where it comes from: each source that adds progress every month, each that multiplies it, and the pace that results, which is never below 0.5. Every running milestone gains the same pace.

*Figure 24: The Space Race tab of the Timeline Extended window, showing Solar System Colonization: the overview and Program row, Mission Control with the approach buttons and the funding stepper, and Rivals.*

At the foot of the overview, the Program row has an icon for each entry, in order. An icon is lit once you have achieved that milestone, with a flag if you were first. It is half-lit while under way, and faint if you haven't begun it or another power finished it first. The Lunar Colony icon has no half-lit state: it lights when you found your first colony on the Moon. The Interstellar Probe and its wait for data share one icon with four pictures: the probe grayed (not begun), the probe with a wrench (under way), radio dishes with a clock (launched, awaiting data) and the dishes with a check (data received). Hover any icon for its state.

Two sections follow the overview:

| Section | Starts | Shows |
|---|---|---|
| Mission Control | Open | The Approach buttons (Standard, Safe and Ambitious; the one in force is grayed, and each tooltip gives its effect and weekly cost), the Funding stepper with your level and the cap, Setbacks So Far, the Estimate of months left at this month's pace (setbacks not |

| Section | Starts | Shows |
|---|---|---|
| | | counted), the Orbital Debris index on the journal entry (see Orbital debris), and the Mission Profile. |
| Rivals | Open | The other powers running the same milestone, each with an estimate of its progress. |

The "?" at the top of each milestone's journal entry, and in each milestone's header in the Timeline Extended window, opens the Help page, How the Space Race Works. It explains approach and pace, funding, setbacks, the first, rivals, the program, space powers, and satellites and debris, then solar colonization.

A rival's progress is an estimate. Every rival is listed, furthest along first, each with a band reaching 25% of the goal to either side of a marker, which always contains the true figure, with the band's range in percent beside it. Programs are announced but their schedules are not, so where the marker sits in the band is drawn again once a year. An agent network in the rival's country narrows the band to ±10% at strength 50 and ±5% at 75 (see Agent networks). Hover a rival for its figures. The list is drawn afresh at the start of each month.

Solar System Colonization's overview shows its stage (Stage: the Moon, then Stage: 2 of 5) in place of the first, and its bar is Next Colony, the progress toward your next colony. While a full bar waits for a launch window, Staged in Orbit gives the progress built so far in colony fleets ("1.4 fleets"); hover it for each destination's next window. Under the bar, the Sites pie counts the 40 sites: gold for your colonies, red for other powers' and gray for the sites no one holds; hover it for the counts. Its Mission Control adds Next Windows, a table of the destinations your next colony fleet can reach: each one's next window and how long the fleet takes to get there by your drive (see Launch windows). While the program isn't running, because every site is claimed or your Space Program no longer runs the method it needs, the approach reads Idle, the risk and progress rows are hidden, and Mission Control shows only the Mission Profile.

Under Simple Colonies and Full Colonies, a second row of the overview says what needs you, with the detail on hover. A word turns red when something is due:

| Cell | Reads |
|---|---|
| Colonies | Loyal, with the colonies you hold; or, in red, the name of a colony at autonomy 50 or more, which wants self-rule, and how many more do. Hover it for all of them. |
| Defense | Under Full Colonies only. Calm; or, in red, what threatens you (Under assault, Blockaded, Occupied, Interdicted, Earth blockaded or Launch site lost) and the colony it threatens. Hover it for every colony under threat and any hostile fleet on its way. |
| Modules | How many modules are queued, and fabrication's stock. Queue empty turns red once the stock is full and waiting. |
| Next Window | The months to your chemical rockets' next launch window, Open now, Any month once your drive needs no window, or No site left; and the stage. |
| Base Room | Once you have In-Situ Resource Utilization: your Extraplanetary Base levels against your room, red once your built and queued levels pass it, and the grade the next level would work at, queued levels counted (see Extraplanetary Base room). |
| Hostile Fleet | Under Full Colonies, while a fleet of a country at war with you is on its way to Earth: Earth in the months until the first lands, and their strength at and bound for Earth against your fleets there. Red while a landing would tip the balance: you have no fleet at Earth and it isn't blockaded already, or yours holds against what is there now but not against what is coming too. Hover it for how many are coming. |

Below its overview, Solar System Colonization has four tabs, and the one you were last on stays chosen while you play:

| Tab | Holds |
|---|---|
| Mission Control | Mission Control (with Next Windows) and Rivals, as every milestone has. Shown first. |
| Fleets | Colony Fleets, then under Full Colonies Space Command and Space Fleets. |

| Tab | Holds |
|---|---|
| Colonies | Colonies. |
| Solar System | Solar System. The colonies' rules come last on the Space Race's Help page, under How Solar Colonization Works. |

Colonies (open) has a box for every site you hold, with Colonies Held (out of 40) at the top. Each box has a small icon for the kind of body it is on (the Moon, Mars, the asteroids, a moon of Jupiter, Saturn, Uranus or Neptune, Venus, Mercury, or a dwarf planet). Hovering its name shows the colony's permanent effects, and clicking it, or the arrow beside it, opens the colony's page. The page sets out People, Modules, Autonomy and (under Full Colonies) Defenses one under another, each with its control beside its heading (the settler priority, Clear, Abandon), its figures in tiles and every module on a line of its own with its full name, with buttons to queue one more level or take one down. Back returns to the list. The buttons above the rows switch what each box shows:

| View | Shows | Above the rows |
|---|---|---|
| People (the default) | Each colony's people and room, its crews and its settler priority | Emigration, settlers a month and your people off Earth |
| Modules | Each colony's crews' share of its modules, its slots in use (red past them) and its Extraplanetary Base room and grade, then every module's level with a button to queue one more | Fabrication, shipments, module upkeep, Extraplanetary Base room, and the queue in a box of its own with when its next module is built |
| Autonomy | Each colony's autonomy, how much it moves a month (hover it for that colony's terms), what it is building of its own, its charter or garrison, and Abandon | Free colony trade; the free colonies are listed below the rows |
| Defenses | Each colony's Defenses, integrity and marines, and what threatens it | |

The section stays on the panel when the program is Idle, so you can always see where your colony modifiers come from. Under Races Only, Colonies lists your sites and their effects without the views.

Interstellar Probe: Awaiting Data has nothing to control, so it has no Mission Control or Rivals. Its overview is the Transit bar ("40 / 132 months") and the Program row; hover the bar for the months to go and what the probe might find.

A new milestone starts on the Standard approach at funding level 1. It moves at your program's pace and rolls against its base setback risk from its first month.

## The Space Race tab

After your milestones, the Space Race tab of the Timeline Extended window has two sections that don't belong to any milestone, so you can follow the race between milestones too. Both start open.

Space Powers puts your own program first, then every other power that has started the race, furthest along first: the most milestones achieved, then the most colonies and the furthest milestone under way. Each row is the country's name, its program as the same eight icons as the Program row (lit once achieved, with a flag for a first, half-lit while under way, except Lunar Colony, faint otherwise) and its furthest milestone under way with its progress, as in "Moon Landing: 35%–85%". Your own progress is exact. Another power's is an estimate, in the same band as the Rivals rows, which an agent network in its country narrows. Hover an icon for that milestone's state and the name for the milestones achieved and colonies held. The other powers are listed afresh at the start of each month. Unlike Rivals, the list includes AI powers outside the top three of the world ranking, whose programs have stalled.

Orbit gathers what anti-satellite strikes and debris mean for you now:

| Row | Shows |
|---|---|
| Orbital Debris | The index and its tier, as in a journal entry's Mission Control, which on the tab leaves it to this row (see Orbital debris). |
| Our Satellites | Intact, Disrupted or None in Orbit. Hover it for the months a disruption has left, or how long until anyone can strike you again. |
| Anti-Satellite Strikes So Far | The strikes since the game began. Hover it for who struck whom last, and when. |
| Anti-Satellite Moratorium | Not Adopted, Bound or Not Bound (see The anti-satellite moratorium). |
| Our Anti-Satellite Strike | Ready once you have Guided Missiles and a Space Program. Hover it for what is missing, or for the countries at war with you that have something in orbit to strike, each marked if it was struck in the last 12 months. |

Another country's National Info shows its Space Race tab too, less its approach, setback risk, Mission Control, Rivals and strike row. Its progress there is an estimate band, a quarter of the goal either side, that your agent networks don't narrow.

## Standard, Safe and Ambitious approaches

The approach decides how fast a milestone moves, how often it goes wrong and what it costs on top of funding. You can switch a running milestone between the three at any time, back to Standard included.

| Approach | Effect | Innovation cost |
|---|---|---|
| Standard | None. | None. |
| Safe | −50% setback risk. | One extra funding level's worth. |
| Ambitious | +50% progress. | Two extra funding levels' worth. |

The approach label in the overview shows the weekly innovation cost of the approach in force, and each approach button's tooltip gives its own: 15 times the milestone's cost factor for Safe and twice that for Ambitious, so 30 and 60 a week on the Moon Landing.

Approach, funding and mission choices apply to your whole program, not to the milestone they sit on. Their modifiers appear on each milestone's journal entry, but the game adds them up for your country as a whole: every running milestone moves at the same pace, and every milestone's base risk is scaled by the same total. A Safe approach on one milestone cuts the risk on all of them, including those on Ambitious, and an Ambitious one speeds up all of them, including those on Safe. They stack: two Ambitious milestones add +100% progress, and two Safe ones bring every milestone down to the 1% floor. What stays with each milestone is its innovation cost and the kind of setback it suffers, which follows its own approach (see Space race setbacks). Setback risk never drops below 1% a month or rises above 50%. The panel's pace and risk figures are the ones the game uses, so check them after every change.

## Mission profile choices

Thirty days after Orbital Flight, the Moon Landing, the Deep-Space Probe or the Mars Landing starts, an event asks you to choose how to fly it. While that milestone runs, your choice changes the setback risk, and sometimes the pace, of your whole program, in the same way as an approach. When the milestone is finished it also decides a reward that fades over ten years.

| Event | Milestone | Options |
|---|---|---|
| Who Flies First? | Orbital Flight | Civilian volunteer (riskier; prestige, cultural pull); military test pilot (lower risk, faster; military research, influence); scientist (a little riskier; research, innovation cap) |
| Where the Eagle Shall Land | Moon Landing | Shackleton Crater (much riskier, faster; research, innovation cap); Equatorial Plain (lower risk; prestige, research); Sea of Tranquility (a little riskier; prestige, cultural pull); far side of the Moon (riskier, faster; research, innovation cap) |

| Event | Milestone | Options |
|---|---|---|
| Where Shall We Send the Probe? | Deep-Space Probe | Venus (riskier; research, space race progress); Mars (lower risk, faster; the most space race progress, prestige); the asteroid belt (much riskier; space race progress, innovation cap) |
| The Mars Strategy | Mars Landing | Direct landing (much riskier, much faster; prestige, cultural pull); orbital-first (slightly lower risk, faster; research, Space Program throughput); robotic precursors (lower risk; Extraplanetary Base throughput, innovation cap, and Mars Resource Extraction: research speed, space race progress and cultural pull) |

## 16.3 Space race funding and cost

Each milestone has its own funding level, from 0 to six levels past your cap. The cap starts at 3 and rises with technology: thirteen space-related technologies from era 7 to era 11 raise it by 1 each, and Cislunar Infrastructure (era 10), three technologies in era 12 and Microfusion Cells in era 13 raise it by 2 each. With all of them the cap is 13 by the end of era 10, 18 by the end of era 11, 24 by the end of era 12 and 26 in era 13. Tier V of the Advanced Research power bloc principle, its highest, adds another level.

Every funding level up to the cap, on any milestone, adds 0.5 to your monthly pace, and so to every running milestone. It also raises the Space Program's throughput by 25%, which makes the building consume more Launch Capacity. Funding and approach cost weekly innovation: 15 for each funding level, plus nothing for Standard, 15 for Safe or 30 for Ambitious, multiplied by the milestone's cost factor from the table above. A Suborbital Flight at funding 1 on Safe costs 30 innovation a week; an Interstellar Probe at funding 3 on Ambitious costs 900. The Space Program Cost modifier on each journal entry shows the current drain.

You can fund a milestone up to six levels past the cap. Each level past it adds less pace than the one before: the first adds half a level's pace (+0.25 a month), the second a third of one, and so on to a seventh for the sixth, so all six together are worth 1.59 levels. Each still costs the full innovation and still raises the Space Program's throughput by the full 25%, and with it the Launch Capacity bill. Past the cap the Funding stepper reads "9 / 7 (+0.8)": the level, the cap, and the levels' worth of pace the excess adds. If the cap falls below a level you have set, the level stands and the excess counts past the cap.

Your monthly pace is the sum of your progress sources, raised by your progress bonuses, and never less than 0.5.

| Progress source | Monthly progress |
|---|---|
| Space Program: Earth Orbit, or any higher method, fully staffed | +0.5, or +1 |
| Each funding level on each running milestone, up to the cap | +0.5 |
| Space Elevator megaproject, each fully staffed level | +1 |
| SpaceX, Lockheed Martin or Roscosmos company, when prosperous | +0.3 each |
| International Space Partnership, for members that ratify the UN's International Space Cooperation | +0.1 |
| Reusable Rocketry and Space Colonization technologies | +0.2 each |
| Mission choices, probe targets and colony specializations | +0.1 to +0.6 each |

| Progress bonus | Effect |
|---|---|
| Ambitious approach | +50% |
| Stolen Rocket Plans, while your Space Program Espionage operation runs | +10%, and −10% setback risk |
| Outer Space Treaty terms of International Space Cooperation | +10% for lagging powers, −5% for the leader |

| Progress bonus | Effect |
|---|---|
| Advanced Research principle, tier V | +10% |
| Antimatter Engine, each fully staffed level | +5% |
| Nanofabrication Center, each fully staffed level | −1% setback risk |
| Temporary Safety Review, from some failure options | −10%, and −25% setback risk |

The International Space Partnership and Outer Space Treaty figures are those at Established enforcement (×1); they scale with the UN's enforcement, as its other convention effects do.

## 16.4 Space race setbacks

Each month, every running milestone rolls against its setback risk. A hit fires an event about that milestone, and what it costs depends on the approach. Setbacks cost months of progress: a month is your monthly pace when the setback struck, and each option's tooltip gives the months and the progress they come to. Changing funding while the event waits doesn't change either.

On an Ambitious approach you get a mission failure: an explosion on the pad, a lost probe, a habitat breach, a computer that aborts the landing. Most options cost 4 months of that milestone's progress, at the faster Ambitious pace, give Space Mission Failure (−2% prestige and −25 innovation cap, fading over five years) and radicalize some academics. One option usually adds a Temporary Safety Review, which slows progress but lowers the risk of another setback and fades over ten years, and in some events also switches the milestone to Safe. Another usually presses on with 5 or 6 months lost and more radicals. Every milestone keeps rolling for setbacks the month after a failure, as in any other month. Now and then the roll brings An Unexpected Breakthrough instead: no progress is lost, and the Space Program's throughput rises by 15% for 18 or 24 months.

On a Safe or Standard approach you get a minor setback: missing parts, bad weather, a scandal, a quarrel with a foreign supplier. It costs 1 to 4 months of the milestone's progress, usually 2 or 3, with no lasting modifier. A few setbacks, on any approach, also cut the Space Program's throughput for some months. A setback never takes more progress than the milestone has, and in Solar System Colonization never more than one colony's, so fleets staged in orbit are never lost to a setback.

Because the cost is counted in months, setbacks cost the same share of time on every milestone, whatever its goal or your pace; later milestones are harder only because their risk is higher. On Standard, setbacks add roughly a sixth to a milestone's time at 5% risk and two fifths at 10%. Safe roughly halves that. Ambitious loses about as much time as Standard, from a pace half again as fast.

A setback never ends a milestone; it only costs time. Two things do: dropping below major power, or losing the production method or building the milestone needs.

## 16.5 First-to-finish rewards

When a milestone completes, its event plays and every other recognized country gets a notice two weeks later, from which it can push every milestone it has under way by a month of progress or congratulate you for better relations. The finisher keeps a permanent reward. The first country to finish a milestone gets the larger version; anyone who finishes it later gets the smaller one.

| Milestone | Prestige | Innovation cap | Research speed | Influence | Cultural pull |
|---|---|---|---|---|---|
| Suborbital Flight | +2% / +1% | +25 / +10 | none | none | +10% / +3% |
| Orbital Flight | +3% / +1.5% | +50 / +25 | none | none | +15% / +5% |
| Moon Landing | +5% / +3% | +75 / +35 | +5% / +2% | none | +25% / +10% |
| Deep-Space Probe | +4% / +2% | +50 / +25 | +3% / +1% | none | +15% / +5% |
| First lunar colony | +5% / +2.5% | +75 / +35 | +5% / +2% | +5% / +2% | +20% / +8% |

| Milestone | Prestige | Innovation cap | Research speed | Influence | Cultural pull |
|---|---|---|---|---|---|
| Mars Landing | +7% / +4% | +100 / +50 | +5% / +2% | none | +30% / +12% |
| Interstellar Probe | +10% / +5% | +100 / +50 | +10% / +5% | +10% / +5% | +25% / +10% |

The first figure is the first-to-finish reward, the second the later one. The first lunar colony is a milestone of its own: the first colony you found on the Moon with your own fleet plays Foothold on Another World and earns the reward, as the first in the race or a later one. A lunar site gained by treaty, inheritance or seizure doesn't count. Suborbital Flight's completion event also offers a choice: loyalists, or Suborbital Momentum, an extra copy of the later finisher's reward (+1% prestige, +10 innovation cap, +3% cultural pull) that fades over two and a half years. Cultural pull feeds the cultural hegemony competition described in Where cultural pull comes from.

## 16.6 Interstellar probe results

Finishing the Interstellar Probe launches it toward Alpha Centauri and opens Interstellar Probe: Awaiting Data, which counts down 132 months (eleven years). Nothing you do shortens the wait. When it ends, the probe reports one of 30 possible discoveries in four categories, and you keep that category's reward permanently.

| Category | Chance | Discoveries | Reward |
|---|---|---|---|
| Dead worlds and data | 40% | 8 | +3% prestige, +5% research speed, +50 innovation cap |
| Astrophysical wonders | 30% | 7 | +5% prestige, +8% research speed, +75 innovation cap, +10% cultural pull |
| Biological discovery | 20% | 9 | +8% prestige, +12% research speed, +125 innovation cap, +20% cultural pull |
| Intelligence detected | 10% | 6 | +12% prestige, +15% research speed, +200 innovation cap, +30% cultural pull |

Each discovery has its own event, from barren worlds and asteroid maps through ocean worlds and alien vegetation to the ruins of a dead civilization. If another country's probe found the same thing first, the event names it and confirms its discovery; the reward is the same.

## 16.7 Colonizing the solar system

Solar System Colonization is repeatable. Each time its bar fills, a colony fleet leaves for an unclaimed site: the one you chose as its target (see Choosing a colony's site), or otherwise one the program picks among those it can reach that month (see Launch windows). Under Races Only the program picks a random site of the current stage, and the colony is founded the month its fleet leaves. Under Simple Colonies and Full Colonies the fleet travels and founds the colony when it lands (see Colony fleets in flight). Each site can be claimed once, by anyone, so every other country racing for colonies competes for the same 40 sites. Under Races Only a stage opens for everyone once every site in the one before it has been claimed. Each stage needs more progress per colony.

| Stage | Sites | Colonies | Progress per colony |
|---|---|---|---|
| 0 | The Moon (Shackleton Crater, Peary Crater, Tranquility Base, Marius Hills, Daedalus Crater, Mare Imbrium) | 6 | 300 |
| 1 | Mars (Valles Marineris, Olympus Mons, Hellas Planitia, Utopia Planitia, Arcadia Planitia) and the asteroids (Ceres, 4 Vesta, 16 Psyche, 2 Pallas, 10 Hygiea) | 10 | 450 |
| 2 | Jupiter's moons (Io, Europa, Ganymede, Callisto, Himalia, Amalthea) and a Venus cloud habitat | 7 | 525 |
| 3 | Mercury and Saturn's moons (Titan, Enceladus, Rhea, Mimas, Iapetus) | 6 | 600 |

| Stage | Sites | Colonies | Progress per colony |
|---|---|---|---|
| 4 | Uranus's moons (Titania, Oberon, Miranda, Ariel) and Neptune's (Triton, Proteus) | 6 | 700 |
| 5 | The Kuiper Belt and beyond (Pluto-Charon, Eris, Makemake, Haumea, Sedna) | 5 | 800 |

From stage 3, a country that doesn't yet hold a colony beyond the Moon needs the Deep Space Exploration or Interstellar Mission method to colonize there; a country that already holds one carries on through every stage with Solar Colonization. Under Simple Colonies and Full Colonies the same rule applies site by site: a country holding no colony beyond the Moon needs one of those methods to send a fleet to Mercury, Saturn or beyond. A colony on the Moon keeps the program open, but alone it reaches no further than stage 2. Each colony's event offers two specializations, and the one you choose becomes a small permanent modifier on the journal entry. Under Simple Colonies and Full Colonies you can raise it, and add the other, with modules (see Colony modules). The options differ by site: research speed, military research speed, innovation cap, space race progress, Extraplanetary Base, Fusion Plant or Orbital Solar Collector throughput, Advanced Materials or Launch Capacity output, cultural pull, influence, diplomatic reputation or cheaper military goods. Your first colony also brings the event Beyond the Blue.

The entry stays open for as long as you hold a colony, even if you switch off the Solar Colonization method, so your colony modifiers are never lost; the program simply stops (its panel reads Idle), and bills no funding or approach cost, until the method returns. Once all 40 sites are claimed, Solar System Colonization finishes within a month for every country that holds a colony, whoever took the last site and whether or not its program is running. Finishing grants Interplanetary Trade Networks: +10% prestige, +5% research speed, +10% influence and +15% cultural pull.

## The Moon

The Moon is stage 0, settled before Mars. You can colonize it once your Moon Landing is complete, with Cislunar Infrastructure and the Solar Colonization method; the sites beyond it wait for your Mars Landing, and the Target button on one says so until then. Its six sites need no launch window and no transit: under every setting, a colony fleet bound for the Moon founds its colony the month it leaves, and it never takes one of the six places for fleets in flight.

| Site | People room | Slots | Base room | Specializations |
|---|---|---|---|---|
| Shackleton Crater | 25,000 | 2 | 75 | Propellant Depot, Polar Outpost |
| Peary Crater | 15,000 | 1 | 50 | Solar Power, Polar Observatory |
| Tranquility Base | 20,000 | 2 | 100 | Heritage Site, Helium-3 Extraction |
| Marius Hills | 40,000 | 3 | 125 | Lava Tube City, Hardened Installation |
| Daedalus Crater | 10,000 | 1 | 50 | Radio Telescope, Deep-Space Relay |
| Mare Imbrium | 25,000 | 2 | 125 | Mass Driver, KREEP Mining |

The landing site your Moon Landing chose (Where the Eagle Shall Land) takes 25% off the progress of one lunar colony site, for you alone: Shackleton Crater for a Shackleton landing, Tranquility Base for the Sea of Tranquility, Marius Hills for the Equatorial Plain and Daedalus Crater for the far side. A site's hover in the Solar System list says when the discount is yours. Under Races Only it counts when you choose that site as your target, not when the program draws one for you.

Shackleton's Propellant Depot cracks the crater's ice into rocket fuel. While you hold it, with its specialization built and the colony neither blockaded nor occupied, your fleets can leave outside a launch window as they do with Asteroid Mining (see Launch windows).

A lunar colony has no Spaceport: its fleets dock at Earth, the same location, so its slots hold only Habitats and Defenses. A colony on the Moon doesn't count toward the Deep Space rule above. Under Full Colonies the Moon is fought over with Earth orbit (see Battles in space).

## Launch windows

Every site but Mercury and the Moon's six is reached on a transfer orbit that lines up only once in a while. A site's launch window is the month of its best departure and the month on either side. The windows are worked out from the planets' and asteroids' real orbits: Mars opens one every 26 months or so, each asteroid every 15 to 17, Venus every 19, Jupiter every 13, and Saturn, Uranus, Neptune and the Kuiper Belt about once a year. Next Windows in Mission Control lists each destination with a site you can target, its next window, and the months your drive takes to get there. A destination beyond your drive's 15-year reach is left out and counted under the table.

While no destination with a site left is open, the program keeps building. Progress past a colony's goal is fleets staged in orbit, shown on the overview as Staged in Orbit. When a window opens, one fleet departs each month it stays open while staged progress covers a colony, so a three-month window can send three. What is left waits for the next window. It carries on to the next stage if rivals take the stage's last sites, and is lost when the last of the 40 sites is claimed. Cutting funding while you wait gains nothing: whatever you build now leaves in a later window.

Until you hold a colony, the program starts over whenever it stops: when your Space Program leaves the Solar Colonization method, when you fall below major power, or after a revolution. The colony being readied is lost, as an unfinished bar always is, but whole fleets already staged stay in orbit and are back on the bar in the program's first month once it runs again. Once you hold a colony, the program never stops this way and keeps all its progress.

Propulsion technology loosens the windows step by step:

| Technology | Launches |
|---|---|
| Cislunar Infrastructure (opens colonization) | In a window only. |
| Asteroid Mining, or Shackleton's Propellant Depot | In a window as before; outside one, at a premium of extra progress, when your approach judges it worth it. |
| Nuclear Thermal Propulsion | Outside a window at a smaller premium, once Asteroid Mining or Shackleton's Propellant Depot lets you launch there at all. |
| Compact Fusion Reactors | Any month, with no premium. |
| Antimatter Production, with an Antimatter Containment Facility built | Any month, with no premium. |

The premium is the extra propellant an off-window transfer needs, so it climbs fast away from the window: on Mars, about +40% the month after a window closes and +150% the month after that. A nuclear thermal engine burns far less propellant for the same extra speed, so with Nuclear Thermal Propulsion those figures are about +15% and +45%. With Asteroid Mining, Next Windows shows today's figure for each closed destination, at your own engines' premium. A fleet leaves outside a window, for the cheapest destination, when the months it saves (until the stage's next window) are worth the months its premium costs (the goal times the premium, at your monthly pace). Standard launches when it saves at least as many months as it costs, Ambitious when it saves at least half as many, and Safe always waits for a window. A fleet that leaves outside a window spends the goal plus the premium. In practice a fast program launches just after a window closes rather than wait two years for Mars, and almost never just before one, when waiting is cheap.

Fusion Power doesn't change when a fleet can leave, only how long it takes to arrive under Simple Colonies and Full Colonies (see Colony fleets in flight).

## Choosing a colony's site

Colony Fleets, in Solar System Colonization, says where your next colony fleet goes. Click Choose there (Change once you have a target) to open the Colony Target window. It lists the unclaimed sites your program can send its next fleet to this month, under their locations, each with who holds it and its launch window (under Simple and Full Colonies, also its slots, its base room and grade, and the fleets on their way); Show all lists all 40, and a site you can't target says why on its button. Hover a site's name for what it offers: its stage, its two specializations and how long your drive takes to reach it. Click Target to choose that site: Colony Fleets

names it, with Clear beside Change, and its row in the Solar System list reads Our target. The next fleet waits for that site's launch window, even if another site's window opens first. Outside the window it leaves at the premium when your approach judges it worth it, weighing the wait for that site's own window. Under Simple Colonies and Full Colonies a chosen site costs the progress of its own stage. Click Clear to let the program pick again; it also picks again when someone else claims the site first.

Under Races Only you can choose any site of the current stage, or a released site of an earlier stage. The colony is still founded the month its fleet leaves. AI programs don't choose ahead; they pick when a fleet is ready.

## Colony fleets in flight

Under Simple Colonies and Full Colonies, a colony fleet takes months or years to reach its site, and claims it only when it lands. Your drive sets how long:

| Drive | Technology | Leaves | Mars | Jupiter | Uranus | Kuiper Belt |
|---|---|---|---|---|---|---|
| Chemical | (from the start) | In a launch window | 9 months | 33 months | 16 years | 51 years |
| Fusion | Fusion Power | In a launch window | 2 to 3 months | 7 to 9 months | about 3 years | 11 to 25 years |
| Fusion torch | Compact Fusion Reactors | Any month | 1 month | 2 months | 6 months | 1 year |
| Antimatter | Antimatter Production, with an Antimatter Containment Facility built | Any month | 1 month | 1 month | 2 months | 4 months |

Fusion's times are for a departure in the window; a fusion fleet that leaves outside one, at the premium, takes longer, much longer to the outer planets. Mercury takes 4 months by chemical rocket, and the Moon none (see The Moon). Colony Fleets names your drive, lists each fleet in flight with its site and the month it lands, and holds up to six; with six in flight the bar waits.

The stages are not locks: a fleet can leave for any unclaimed site, and distance gates the outer system instead. A site costs its own stage's progress, 300 for the Moon and 450 for Mars and the asteroids, up to 800 for the Kuiper Belt. Without a choice of yours, the program sends each fleet to the best site it can reach within 15 years: nearer, cheaper and inner sites first, and away from sites a fleet already on its way, yours or another country's, will reach first. So a chemical program settles Mars, the asteroids and Jupiter, and waits for better drives before Uranus. AI programs pick the same way. You can still choose a site farther out by hand.

Each site's row in the Solar System list shows the fleets on their way, as the number of fleets and the months until the first lands, in a brighter color when the first is yours. The first fleet to land claims the site. A fleet that lands to find its site taken founds another unclaimed site of the same planet or belt (Mars, Jupiter, Saturn, Uranus, Neptune or the Kuiper Belt) instead. If none is free, or it was bound for an asteroid, Venus or Mercury, it is lost with the progress it cost. Either way an event tells you.

## Colony settlers

Under Simple Colonies and Full Colonies, colonies hold people. A colony starts as an automated outpost with no one in it, so it grows only once settlers arrive: then by about 2.4% a year up to its room (3.7% once you have Self-Directed Evolution, era 13, whose people are adapted to low gravity and radiation), from 10,000 people on the smallest moons and the farthest dwarf planets to 50,000 at Valles Marineris and the Hellas and

Utopia plains. The Colonies section's People view in Solar System Colonization sets how many settlers you send from Earth:

| Emigration | Settlers a month | Launch Capacity bill |
|---|---|---|
| Off | none | none |
| Low | 500 | +5% Space Program throughput |
| Steady | 1,000 | +10% |
| High | 2,000 | +15% |

The settlers leave the state that holds your Space Program, half from the middle strata and half from the lower, and only while your colonies have room; that state's census counts them as emigrants. Each colony's row shows its people and room and a priority button that steps through Normal, High, Closed and Low. A colony's share of each month's settlers is its priority (Closed none, Low 1, Normal 2, High 4) times its free room, and a colony that fills up passes the rest of its share to the others. So set the colony you are building up to High and the ones you don't want settled to Closed.

People are crews: the Crews figure counts how many module levels a colony's people can run, one for every 5,000. A colony's founding specialization runs without a crew. With Artificial General Intelligence colonies need no crews at all, and the figure reads AGI. Habitats, and Terraforming on the worlds that take it, add room for more people (see Colony modules).

Your people off Earth give Off-World Population on Solar System Colonization: +1% prestige and +2% cultural pull for every 100,000 of them, up to a million. A colony's people go with it when it changes hands, stay on it when it becomes a free colony, and are lost if it is released. AI programs set Emigration to Steady while they hold a colony or are in the top three of the space race, with money to spare; to Low otherwise; and to Off in default or when their colonies are full.

## Colony modules

Under Simple Colonies and Full Colonies, you build up each colony with modules, listed in the Colonies section's Modules view in Solar System Colonization:

| Module | What a level does | Room for it |
|---|---|---|
| The two specializations | Adds the specialization's bonus once more: level 3 gives three times the founding bonus | Four levels between the two: one at 4, both at 2, or 1 and 3. The founding choice is level 1 |
| Habitat | Adds half of the colony's starting room for people | The colony's slots, shared with Spaceports, Defenses and facilities: four on Valles Marineris and the Hellas and Utopia plains, one on the smallest moons and dwarf planets |
| Terraforming | Adds 4% of the colony's starting room for people a year while it works, until the world has +300% (see below) | The colony's slots, up to three at once; only on Mars, Venus, Titan, Ganymede and Callisto, with the Terraforming technology |
| Spaceport | Under Full Colonies, a dock for your fleets: they refill supply, repair 10% of their damage a month a level and refit there, and refuel on a site that makes propellant. Under Simple Colonies, nothing | The colony's slots; none on the Moon |
| Defenses | Under Full Colonies, guns that fire on enemy fleets closing in on the colony and make it harder to wear down. Under Simple Colonies, nothing | The colony's slots |
| Facilities | See the table below | The colony's slots |

The facilities each give their bonus for every staffed level, across all your colonies:

| Facility | A staffed level gives |
|---|---|
| Mining Complex | +50 Extraplanetary Base room at the colony |
| Orbital Yard | +4 module fabrication points a month; under Full Colonies, +0.5 Space Command strength a month |
| Research Station | +2% research speed |
| Cultural Center | +3% cultural pull, +1% prestige |
| Off-World Resort | +3% tourism output |
| Black Site | +1 intelligence capacity, +3% covert operation efficiency |
| Training Ground | +4% army experience gain |
| Nuclear Command Center | +4 nuclear command reliability, at most +12 |
| Materials Foundry | +3% advanced materials output |
| Administration | −0.15 a month to the colony's autonomy |

A blockade stops the Mining Complex, Orbital Yard, resort and foundry; an occupied colony's facilities give nothing.

Terraforming (an era 13 technology, after Space Colonization and Geoengineering) works the world itself, on the five Martian sites, the Venus cloud habitat, Titan, Ganymede and Callisto. Each staffed level adds 4% of the colony's starting room for people a year. Up to three work at a colony at once, so a world takes at least 25 years to finish, at +300%, where a Habitat gives +50% at once and holds its slot for good. The room the work has added belongs to the world: it stays when the colony changes hands, is released, or takes a level down. A finished world's Terraforming levels stand down and free their slots. An occupation stops the work, and a colony short of crews works at its staffing share. Once you have the technology, the Modules view shows a Terraforming line under the facilities, with the room added so far and the years to go at this month's pace; the colony's page shows the same in its Terraformed tile.

A colony has 1 to 4 slots by its size, and one more each with Nuclear Thermal Propulsion, Orbital Manufacturing, Advanced Nanofabrication, Space Colonization and Molecular Assemblers, whose tooltips list Colony Module Slots. Past them it still takes up to four more modules, each costing 100 fabrication points more than the last: 200, then 300, 400 and 500. The Modules view's Slots figure turns red past them, and its hover gives the next module's price. Solar System shows each site's slots and Extraplanetary Base room under its name, so you can weigh a site before you settle it.

The founding event has to be answered before you can add a specialization level. Each row's buttons add one more level of a module to the end of your module queue, counting what is already queued and on its way against the colony's room, and its last button takes all of that colony's modules out of the queue. The queue holds up to 20 modules and is listed in order above the rows, each with a button that removes it.

On a colony's page, the minus button beside a module takes one level of it down. That takes 12 months, one level at a time at each colony, and the module works until it comes down; a line under its name counts the months left. A blockade pauses the count, and losing the colony, or its occupation, ends it with the level still standing, so a colony can't be stripped bare once a rival moves on it. The founding specialization keeps its first level. A Habitat can't come down while the colony's people need its room, and while one comes down the colony takes in no more people than the room it will have. Pressing the button again stops the work; the months spent are lost. The freed slot takes a new module once the level is gone.

Modules are built at Earth by your Space Program's third production method group, Module Fabrication, which starts on No Module Fabrication:

| Method | Technology | Launch Capacity and Advanced Materials a week | Points a month |
|---|---|---|---|
| Orbital Assembly | Cislunar Infrastructure | 1,500 and 10 | 5 |

| Method | Technology | Launch Capacity and Advanced Materials a week | Points a month |
|---|---|---|---|
| Orbital Foundries | Space Colonization | 2,000 and 15 | 12 |
| Fusion Shipyards | Compact Fusion Reactors | 2,000 and 10 | 25 |

Each module within a colony's slots costs 100 points, so a fully staffed Space Program builds one every 20, 8 or 4 months, more often with higher funding. Mare Imbrium's Mass Driver adds 3 points a month a level of its own, with or without a fabrication method. Fabrication builds the queue in order, first in first out: the module at the top is next. The arrow on each line from the second on moves that module up a place; Alt+Click moves it to the top. Up to 500 points wait in stock while nothing is queued, enough for the dearest module.

A finished module waits at Earth for its site's launch window, or leaves at once with a fusion torch or antimatter drive, and lands after the transit your drive buys (see Colony fleets in flight). Eight modules can wait or fly at once; while all eight are taken, fabrication waits. A module lands for whoever holds the colony then, and is lost if the site was released or filled to four past its slots in the meantime.

Every level past the first needs a crew of 5,000 people, except Habitats, which house the crews. A colony short of crews runs its modules at the share it can staff: four levels with no crews run at a quarter each, which the Crew column shows. With Artificial General Intelligence every module runs. Each level past each colony's first adds +1% to your Space Program's throughput, raising its Launch Capacity bill (Module Upkeep in the Modules view).

A colony's modules stay with it when it changes hands, and the Cede Colony AI values a colony more for each of them. An AI that holds a colony, or is in the top three, queues one module a year at each colony when it has money to spare: a Habitat once its people fill three quarters of its room, otherwise the next specialization level its crews can run, its founding specialization first, then a facility: an Administration where autonomy has reached 25 and is heading for 45 or more, a Mining Complex when its bases outgrow their room, Terraforming once its people fill half its room, a Nuclear Command Center with nuclear weapons, an Orbital Yard with a Space Command, otherwise a Research Station. It runs the best fabrication method it has while anything is queued, never builds Spaceports or Defenses, and never pays for a module past a colony's slots.

## Extraplanetary Base room

Under Simple Colonies and Full Colonies, Extraplanetary Bases need room. With Asteroid Mining, the near-Earth asteroids give every country room for 10 levels, and each colony you hold adds its own, from 50 levels on the small outer moons to 300 at 16 Psyche, and 50 more for each staffed level of its Mining Complexes. Each colony's room has a base grade (Base Grade, a term you can hover in game), which raises or lowers the output of the levels that work it:

| Sites | Grade |
|---|---|
| 16 Psyche | +30% |
| 4 Vesta, Ceres | +20%, +15% |
| Mars, 2 Pallas | +10% |
| 10 Hygiea, Himalia, Mercury, Tranquility Base, Marius Hills | +5% |
| The near-Earth asteroids, the Moon's other four sites, Ganymede, Callisto, Amalthea | 0 |
| Io, Europa, Venus | −5% to −10% |
| Saturn's moons | −15% to −20% |
| Uranus's and Neptune's moons | −25% to −35% |
| The Kuiper Belt | −35% to −45% |

Your levels fill the best room first, and any past all of your room work at −75%. Before Asteroid Mining, a country with no colony has no room at all. One country modifier, Extraplanetary Base Room, carries the average over all your levels, so only the levels past the room pay the penalty: with room for 10 and 20 levels built, the base works at −37.5%. A new poor colony adds room without lowering what your levels already

earn, and taking a rival's Psyche colony raises your average and lowers theirs. The overview's Base Room cell shows your levels against your room and the grade the next level would work at, counting the levels you have queued; that grade updates monthly and the day after a level finishes, except that it reads −75% at once when your queue passes all of your room. The Colonies section's Modules view shows your levels against your room, any queued, and each colony's Bases figure its own room and grade. Private investors don't build past your room, and AI governments don't either; your government can, at the penalty. Under Races Only, bases have no room limit.

## Who holds a colony

Every site records which country holds it. Solar System Colonization's Solar System section, with Show all, lists all 40 under their places in the solar system: Ours, the holder's flag (hover it for the country's name, click it to open the country), Another power, or Unclaimed.

A colony stays with its holder until the holder is gone, or, under Simple Colonies and Full Colonies, until a treaty moves it (Ceding a colony) or it leaves on its own (Colony autonomy and free colonies). When a holder is annexed, each of its colonies passes, with its specialization and its discoveries, to whoever owns the state region of its last capital. If no country there can take it, the site reads Unclaimed again: the next colony fleet that can reach it claims it, at the current stage's cost, even when its stage is long past or the race is over. Under Simple Colonies and Full Colonies a colony of 1,000 people or more becomes a free colony instead (see Colony autonomy and free colonies). When a revolution wins, the new government holds the old one's colonies at once. Under Full Colonies an enemy can also occupy a colony in a war, which keeps its holder but gives it nothing until the peace (see Blockade, seizure and destruction).

A country that comes to hold a colony this way, or by treaty, gets Solar System Colonization for it even without a space program. Without the Moon Landing it keeps the colony and its specialization but founds no more, and the entry reads Idle and costs nothing.

## Ceding a colony

With the Space Race rule on Simple Colonies or Full Colonies, the Cede Colony treaty article moves one colony from the conceding country to the other. The treaty draft lists the conceding country's colonies by name, such as Valles Marineris Colony. The colony moves when the treaty comes into force and stays moved when the treaty ends, with its specialization and its discoveries. The article costs no influence, and a treaty can carry it once for each colony, in either direction, so two countries can swap colonies, or sell one for a Money Transfer. It can also be demanded as a war goal, for 5 maneuvers and 10 infamy.

AI powers in the race ask for colonies; an AI that holds a colony with no space program of its own offers it. Any AI accepts a cession when the rest of the treaty pays more than the colony is worth to it: 30, plus 10 for each stage past the Moon, 8 for each module level past the first and 1 for each 1,000 people, at most 170. A giver holds out for 30 more besides, so an empty lunar outpost costs 60 points of acceptance and a large, built-up colony 200.

## Colony autonomy and free colonies

Under Simple Colonies and Full Colonies, every colony you hold has an autonomy figure from 0 to 100, listed in the Colonies section's Autonomy view in Solar System Colonization with how much it moves each month; hover that figure for the terms that move that colony. It changes every month:

| What moves it | A month |
|---|---|
| The pull of home | −0.25 |
| Its people | +0.05 for every 10,000 |
| Its distance: the months your drive takes to reach it | +0.01 a month of transit, at most +0.5 |
| No settlers or module has landed for more than a year | +0.25 |
| You are at war | +0.25 |
| A garrison | +0.2 |

| What moves it | A month |
|---|---|
| A charter | −0.4 |
| Its Administration | −0.15 a staffed level |
| Settling | −1% of its autonomy |

Settling makes autonomy level out rather than climb for ever: a colony settles at 100 times the sum of its other terms, and the hover shows where. A colony whose terms add up to +0.3 a month settles near 30; one demands self-government only when they reach +0.5, as for a colony far out that goes a year without contact, or a large one at war. A Mars colony that keeps receiving settlers stays near 0. Better drives bring the outer colonies closer: with a fusion torch the distance all but disappears. Keep settlers or modules arriving, on Low emigration if nothing else, so that no colony goes a year without contact.

At 50, the colony demands self-government, once in five years at most, and you choose:

| Choice | Now | After |
|---|---|---|
| Grant a charter | −30 autonomy | For good: its specializations work at 80% for you, and its autonomy falls 0.4 a month |
| Send a garrison | −25 autonomy | For ten years: +2% Space Program throughput a garrisoned colony (Colonial Garrisons), and the colony's resentment adds 0.2 a month |
| Let it go | It becomes a free colony, on good terms with you | — |

A charter settles a colony for good at a fifth of its output; a garrison buys ten years and costs Launch Capacity. Under Full Colonies a garrison is real troops: it takes effect only once a fleet lands two marine battalions at the colony, within a year of the order, or the order lapses. A colony that reaches 100 declares itself free whatever you chose. A charter stays with the colony when it changes hands; a garrison goes home. AI powers grant a charter, garrison while at war, and let a colony go only when they have no Space Program to supply it.

**A colony's own modules.** Autonomy has its use: each month a colony gains its autonomy ÷ 100 × its people ÷ 10,000 points toward a module of its own, at most 5, shown as Building in the Autonomy view. At 100 it builds one, free and at once, within its slots: a Habitat once its people fill three quarters of its room, a level of its founding specialization (up to 3) or of the other, or a Research Station or Cultural Center. A colony at 50 autonomy with 30,000 people builds one about every five and a half years.

**Abandoning a colony.** Each colony's row has an Abandon button, which asks first. You can bring its people home to your capital, releasing the site and losing its modules, or, with 1,000 people or more, leave them to govern themselves as a free colony. An outpost with almost no one in it can only be closed. The AI never abandons a colony.

**Free colonies.** A free colony keeps its people, modules and discoveries but has no holder, so nobody gets its specialization. The Solar System list shows it as Free, and the Autonomy view lists every one with its people and your ties with it. Click Supply on a free colony to send it supplies:

| Supplying a free colony | |
|---|---|
| Costs | +3% Space Program throughput for each colony you supply (Free Colony Supply) |
| Earns | Free Colony Trade: +1% research speed and +2% influence for every 20,000 of its people, shared among its suppliers, up to five times |
| Ties | +2 a month while you supply it, −1 a month when you don't |

A free colony grows only while someone supplies it. When your ties with it reach 100, it asks to join you, once a year at most: welcome it and it is yours again at autonomy 20, with its people and modules; decline and your ties fall back to 50. A colony you let go or leave free starts at ties 50 with you, so supplying it brings

it back in about two years; one that broke away starts at 0. AI powers in the top three with money to spare supply up to two free colonies and accept a colony that asks.

Covert operations reach colonies too: Incite Colonial Autonomy raises the autonomy of every colony the target holds, Sabotage Colony kills a share of the people of its largest, and under Full Colonies Survey Colony Defenses shows your fleets the target's colony defenses and sharpens their fire on them (see The covert operations).

**Colony life.** Three rare events come with growing colonies. The First Child fires once, when your first colony reaches 2,000 people: prestige, and a choice between pulling the colony closer or letting it be proud of itself. A Strike can come at a colony of 10,000 or more and a Dome Breach, which kills a tenth of a colony's people, at one of 5,000 or more; after either, neither comes again for ten years, so a nation sees one about every fifteen years. Paying for the strike or the repairs costs Space Program throughput for a while and lowers the colony's autonomy; leaving the colonists to it raises it.

### Rare colony discoveries

New colonies can also produce rare discoveries after their founding event. Twenty discoveries cover microbial life in Europa's or Enceladus's oceans, confirmed ancient Martian fossils, buried ice, mineral deposits, ocean layers, atmospheric waves and local geological hazards. Each discovery offers two choices with different small permanent bonuses on the colony journal entry.

Each candidate has a hidden 10% chance of being possible when its colony is founded. A successful founding roll starts a year of surveying. After that, the discovery has a 1% monthly chance for biological finds or 2% for other finds. A country can receive at most one discovery every six months. Each discovery occurs once, and other possible finds wait through that spacing period. Surveys continue while colonization is idle and after all sites are claimed. Colonies founded in an older save receive no new founding rolls.

Each event reports a confirmed discovery by an established colony. The biological finds include living microbes and ancient microbial fossils; other finds establish geological features or chemical processes. Their probabilities are game rules, not estimates of how likely those discoveries are in the real solar system.

## 16.8 Space fleets

Under the Space Race rule's Full Colonies setting, countries build warships in space, and the Solar System Colonization panel gains two sections, Space Command and Space Fleets, and the Colonies section its Defenses view. Nothing here exists under Simple Colonies.

### Space Command

Space Command is a government building, unlocked by Space Militarization. A nation runs one: the first is built in a state with a Space Program, and while you have one the build menu refuses another, with the reason, so after moving your capital you keep expanding the one you have where it stands, or demolish it first. Each level builds fleet strength every month, by its production method:

| Method | Technology | Strength a level a month |
|---|---|---|
| Orbital Shipyards | Space Militarization | 0.5 |
| Fusion Yards | Fusion Power | 1 |
| Torch Yards | Compact Fusion Reactors | 1.5 |
| Antimatter Yards | Antimatter Production | 2 |

The figure is scaled by staffing, and the method's tooltip shows it as Fleet Strength Built a Month. A Space Command in a state you don't control builds nothing. The month's strength is shared among five roles by the build mix, whose buttons step each role's weight from 0 to 3, and waits on the ground in the reserve until it is launched (see Holding Earth orbit):

| Role | Fire on fleets | Fire on colony defenses | Months of supply | Also |
|---|---|---|---|---|
| Warship | 1 | 1 | 18 | Takes losses first |
| Rapid response | 0.5 | 0.5 | 12 | Takes losses first; a fleet of these alone flies one drive faster |
| Siege ship | 0.5 | 2 | 18 | |
| Assault carrier | 0.25 | 0.25 | 12 | Carries 2 marine battalions a point of strength |
| Tender | 0.1 | 0.1 | 36 | Adds propellant and supply to its fleet |

Every Space Command uses a maintenance line, the same whatever its method and however many levels it has: 20 Electronic Components, 20 Wired and 20 Wireless Telecommunication Gear, 10 Software, 20 Industrial Robotics and 40 Chemicals a week at full staff, about 8,400 at base prices. Fleets cost upkeep as +900% of that line for every 10 points of strength at full, the reserve's included, so a fleet of 300 costs about 2.3 million a week. Switching the yards to a cheaper method or cutting levels doesn't lower it. If annexed or formed land brings a second Space Command, the smaller of the two is dismantled, whichever it is (on a tie, the one outside your market capital), and you are told. A large fleet makes Space Command dearer to run; it builds as fast while those goods can be bought. Directed Energy Weapons, Orbital Weapon Platforms and Antimatter Production each raise your fleets' fire by a quarter.

Pause, beside Launch and Auto, stops construction at once: neither Space Command nor your colonies' Orbital Yards build fleet strength, Built a month reads Paused, and the yards buy no Launch Capacity, Advanced Materials or electricity. The maintenance line and the upkeep are still paid, and your fleets, Launch and Auto work as before. Resume waits for the next month's construction, which builds nothing; the button reads Resuming until then, and a click cancels it. From that month the yards buy their inputs again, and the month after builds. A month is built in one go, so it is built only if the yards bought their inputs all month. Pausing costs the inputs bought since the last month's construction, so pause just after Built a month changes.

## Moving fleets

The Space Fleets section lists your fleets, up to nine, with where each is, its strength now and at full (hover it for its ships), its drive, its propellant in km/s, its months of supply and its marines. A fleet can be at one of nine locations: Earth Orbit, Mercury & Venus, Mars, the Asteroid Belt, Jupiter, Saturn, Uranus, Neptune and the Kuiper Belt.

| Order | What it does |
|---|---|
| The destination button | Opens Fleet Destination, a row for each other location: the trip's months and propellant (in red when the fleet carries too little), from Earth its launch window, your colonies there and the enemy's strength there and on its way. Choose picks it; the button's hover then gives that trip |
| Go | Sends the fleet to the destination chosen. It can't leave while it refits, when it is all but destroyed (below the minimum strength), when its propellant doesn't cover this month's trip (leaving Earth outside a launch window takes more), from Earth while Earth is blockaded or your launch site occupied, or without a Space Command (below). You can choose a destination before it can leave, and Fleet Destination says what holds it back |
| Transfer | Opens a window to move ships and marines into a new fleet, or into another of your fleets at the same location. Each role and the marines start at half; – and + change an amount by 1, with Shift, Ctrl and Alt by 10, 100 and 1,000, and a right-click sets none or all. Ships moved into another fleet give it the slower drive and the lower share of propellant and supply; moving all of a fleet merges the two |
| Land, Embark | Puts the fleet's marines at your colony there, or takes them back aboard |

Your fleets take orders only while you have a Space Command level in a state you control. Without one, none can leave, change posture or target, start a refit or be repaired, and a refit under way pauses; Space Fleets says so above its rows. Fleets already on their way land, every fleet fights where it is at the posture it had, at half strength, and a docked fleet still takes on supply, propellant and marines. Demolishing Space Command ends the upkeep and grounds your fleets: one at Earth still fights there at half strength and keeps Earth from being blockaded, but can't leave or be repaired.

A fleet in flight can be neither attacked nor recalled. It docks at Earth and at your colonies with a Spaceport: supply and propellant refill (at a Spaceport, propellant only on a site that makes it, such as Utopia Planitia, Ceres, Europa or Titan), damage is repaired, and a fleet on an older drive refits to your best in three months. Away from a dock supply runs down, and a fleet without it loses 5% of its strength a month. Marines fill up at Earth only. A fleet that arrives where you hold a colony counts as contact for the colony's autonomy. When an enemy fleet sets out for a location where you hold a colony, you are warned with the months until it arrives.

## Battles in space

Each month, fleets of countries at war fight at every location where they meet. What a fleet fights depends on its posture, set by the four posture buttons under its orders. The one in force is lit, and one the fleet can't take is grayed, its hover saying why:

| Posture | Fights | Use it to |
|---|---|---|
| Station | Every enemy fleet there | Guard your colonies. A fleet arriving where you hold a colony stations |
| Stand off | Enemy fleets that stand off or are stationed, out of the colonies' guns | Blockade an enemy's colonies |
| Close in | The target site's Defenses and the enemy fleets stationed there | Seize or destroy a colony |
| Interdict | As Stand off. Offered where a country at peace with you that is not your ally holds a colony; the target button picks its site | Blockade that country's colonies there without a war (see Interdiction) |

The target button beside them opens Fleet Target: the sites where the fleet is, with their holders, and Target on each, grayed with the reason where the fleet's posture can't take that site. Closing in on a site needs it to be an enemy's, a free colony or yours under an enemy's occupation; interdicting, a country's at peace with you that is not your ally. Choosing Close in or Interdict without such a target takes the first site there that fits.

At Earth every fleet is stationed and fights every enemy fleet, an Orbital Battlestation fights for its owner (25 strength a level, regaining 10% a month; none in a state its owner doesn't control), and so do its ground-to-orbit batteries (see Holding Earth orbit). The one exception is a fleet closing in on a colony on the Moon: at Earth the posture buttons show while there is a lunar colony you can close in on (an enemy's, a free one, or yours under an enemy's occupation), and only Station and Close in can be chosen. A fleet closing in there takes fire from every enemy fleet, battlestation and battery at Earth, and no longer guards your own lunar colonies, which can't be seized or destroyed while a fleet of yours or your Orbital Battlestation is stationed at Earth. Losses follow the square law: each side loses in proportion to the other's fire, so concentration pays (100 strength against 50 ends with about 87 left). Warships and rapid response take the losses before the other roles, and half of every loss is damage a dock repairs. A battle at Earth adds to orbital debris. A battle report comes after each month of fighting, and the war support breakdown shows Battles in space.

## Blockade, seizure and destruction

A colony is blockaded when an enemy fleet stands off at its location and none of its holder's fleets holds orbit there. A colony on the Moon is blockaded while its holder's Earth is (see Holding Earth orbit). Settlers and modules stop reaching it, it gives its holder no Extraplanetary Base room, and a specialization that ships goods home (mining, industry, propellant) stops; research and observatory specializations keep running. The Colonies section's Defenses view lists your colonies with their Defenses, integrity, marines and status.

A fleet closing in on a site brings its integrity down, from 100 to 0, faster with siege ships and against fewer Defenses; the site's Defenses fire back. Integrity returns by 5 a month when the attack stops, 10 with a Spaceport. At 0, with no enemy fleet stationed there, the fleet's next step depends on its Seize or Destroy button:

| Choice | Needs | What follows | Price |
|---|---|---|---|
| Seize | More marines than the site's garrison and militia (a battalion for every 20,000 people, at least one) | The site is occupied: it gives its holder nothing, counts as Colonies occupied in space in the war support breakdown, and goes back at the peace unless a Cede Colony article takes it. Its holder, or anyone at war with the occupier, frees it the same way, with more marines than the occupier's garrison | None |
| Destroy | Three months closing in with more fire than its Defenses | The colony and everyone in it are lost, and the site lies ruined for 25 years | 20 infamy, 20 on your UN record of hostile acts in space, +10 autonomy at each of your colonies and their resentment for ten years |

A free colony can be attacked without a war: closing in costs 10 infamy at once, and seizing it makes it yours at autonomy 60. A party to the Convention on the Settlement of Outer Space pays twice the infamy and twice the UN record for destroying a colony or closing in on a free one. Seizing a colony at war costs a party no more than anyone else. Without Survey Colony Defenses against its holder, the target's hover gives a site's Defenses only roughly, and its integrity and marines not at all.

## Interdiction

You can blockade a rival's colonies without going to war, as the United States quarantined Cuba in 1962. Send a fleet to a location other than Earth where a country at peace with you, and not your ally (no alliance or defensive pact), holds a colony, and set its posture to Interdict; the target button picks which country's site. While none of that country's fleets is stationed or standing off there, its colonies at that location are blockaded as an enemy's would be, and its Locations row reads Interdicted. Nobody fires. If its fleet holds orbit there, the two fleets face each other and nothing is blockaded; firing on it needs a war.

| You pay, for each fleet and each country it interdicts | The country you interdict gains |
|---|---|
| 5 infamy, and 10 on your UN record of hostile acts in space | A warning, and a casus belli: for five years, demanding one of your colonies with Cede Colony costs it no infamy |

An interdiction ends, and the fleet stands off, when its target goes to war with you, signs an alliance or defensive pact with you, or no longer holds a colony there. A site that passes to another country at peace with you is a new interdiction, at the same price. In a war against the country you were interdicting, your fleet standing off blockades it anyway, now as an enemy.

## Holding Earth orbit

Earth orbit is where every launch goes up, so it can be fought over. Your Earth is blockaded when a fleet of a country at war with you is at Earth and none of yours is. While it is, nothing of yours leaves Earth: no colony fleet (a full bar waits), no module shipment (modules are still made, and wait at Earth), no settlers and no fleet. Your colonies on the Moon are blockaded, and none of your colonies has contact from home, so their autonomy counts the months. Your Locations row for Earth reads Blockaded, Space Command shows a red note, and you are told which country holds your orbit. Your batteries and your Orbital Battlestation fire on the blockaders but don't lift the blockade: only a fleet of yours at Earth does.

That fleet comes from the reserve. Space Command's construction waits on the ground there, in the five roles, and its Reserve row gives the strength (hover it for the roles). With Auto on, the default, it joins your home fleet at Earth each month while no enemy fleet is there, so in peace the reserve stays empty. While an enemy fleet is at Earth it stays on the ground, and Launch sends all of it up together as one fleet at Earth, with full propellant, supply and marines, blockade or not. A blockade therefore faces everything you have built since it began, not a month's worth at a time. Turn Auto off to keep a reserve in peace and launch it when you choose; it costs the same upkeep on the ground.

You hear of each fleet of a country at war with you that is on its way to Earth, once a leg, within a month of its setting out, or of a war beginning with it already on its way. A notice in the feed tells you of one whose landing would change nothing. A popup, Earth Is Threatened, tells you once of one whose landing would tip the balance: it would find no fleet of yours at Earth (and Earth not blockaded already), or yours holds against what is there now but not against that and what is coming. That can come later in the leg too, if your fleet leaves Earth or theirs grows. One popup a month covers every such fleet that month, and the others you hadn't heard of get the feed notice. The overview's Hostile Fleet cell shows the first landing; hover it for each country's fleets on their way.

Ground-to-orbit batteries are a government building, unlocked by Space Militarization and built in any state. Each level fires on every enemy fleet at Earth every month and takes no losses:

| Method | Technology | Each level a month |
|---|---|---|
| Kinetic Interceptors | Space Militarization | 1 fire on enemy fleets at Earth; each point fired adds 0.02 to low-orbit debris |
| Ground Lasers | Directed Energy Weapons, and electricity | 2 fire, no debris; in peace, 0.01 low-orbit debris swept away |

The method's tooltip and the state's modifiers show both figures, scaled by staffing. A month of fire costs the enemy about a tenth of it in strength: twenty levels take 2 strength a month on Kinetic Interceptors, 4 on Ground Lasers. The technologies that raise your fleets' fire don't raise the batteries'. Ten levels of lasers clear 0.1 low-orbit debris a month: a help against everyday clutter, no cure for a Kessler Cascade, which lives in the high orbits they can't reach.

At war, holding Earth orbit supports your troops: while your fleets at Earth outmatch the enemy's there, or face none, you get Orbital Superiority, up to +10% offense and +5% defense for your units. It grows with the margin, and is full when you have twice the enemy's strength at Earth, or 10 against none; without a Space Command your fleets count half. It stands beside the anti-satellite strike's Orbital Network Disrupted.

Nothing is launched, built or fired from a state its owner doesn't control. While your Space Program's state is occupied, nothing leaves Earth, no module is fabricated and the reserve can't launch, whatever is in orbit; you are told when the launch site is lost and when it is back. Space Command, the batteries and the Orbital Battlestation in an occupied state count for no one: not for you, and not for the occupier, whose gain is that you have lost them. A land war can shut down a space program.

## How the AI fights in space

AI fleet powers keep their fleets at home in peace, send their strongest fleet to defend a location where they hold colonies when an enemy fleet is there or on its way, or when a fleet interdicts them there, bring fleets home short of supply or propellant, and land marines for the garrisons they order. At war, once a month, an AI sends one fleet from a dock to the nearest enemy location within a year's flight, when the fleet is at least half again as strong as the enemy's fleets and its colonies' Defenses there and can fly home again or dock there. It closes in on the enemy colony with the most people and modules, to seize it when its marines outnumber the garrison and militia, and otherwise stands off to blockade. It comes home once the enemy's fleets there outmatch its own. At Earth, while no enemy fleet is there, the AI picks the enemy lunar colony with the most people and modules, and its strongest fleet there whose marines outnumber that colony's garrison and militia closes in on it. Its other fleets at Earth stay stationed. An AI never destroys a colony, never attacks a free one and never interdicts. With an enemy fleet at Earth, it launches its reserve once the reserve and its fleets at

Earth match the enemy's there. It builds batteries only while at war with a country that has fleets, up to five levels, and runs Ground Lasers at war, and in peace once debris is Cluttered. An AI in default, or in deficit while at peace, pauses Space Command's construction, and resumes once out of default with a surplus, or at war, after six months at least.

## 16.9 After the space race

The race is over for you once you have finished all six single-goal milestones, all 40 sites have been claimed by anyone, and no program of yours is still running. A power that gets there after the last site is gone qualifies as soon as its own six milestones are done. From then on, the progress your sources still give each month goes into your Extraplanetary Bases as Commercial Space Industry: each point of monthly progress adds +5% Extraplanetary Base throughput, up to +50% at 10 points. The first time it applies, the event The Commercial Space Age tells you.

Only the flat sources in the table under Space race funding and cost count: the Space Program's method, the Space Elevator, the space companies, the International Space Partnership, the two technologies, and any mission choice, probe target or colony specialization that adds progress. Funding adds nothing, since no milestone is left to fund, and neither do progress bonuses such as Antimatter Engines. The figure is recalculated every month, so a new Space Elevator level or a space company losing its prosperity shows in the next one.

Commercial Space Industry does nothing until you build Extraplanetary Bases (In-Situ Resource Utilization). A Space Elevator of 10 levels reaches the cap on its own. The Space Program's method counts for at most one point, so once the race is over you can switch it to Earth Orbit, which uses 500 Launch Capacity instead of up to 100,000, and give up only half a point. When you choose a colony's specialization, an option that adds progress keeps some value after the race: below the cap, it is worth about a quarter of what the mining options give. AI powers gain Commercial Space Industry in the same way.

## 16.10 Orbital debris

Debris in Earth orbit is one figure for the whole world, the orbital debris index, from 0 to 100. It has two parts. Debris in low orbits is dragged down by the upper atmosphere within a few years: 1.5% of it goes each month. Debris in high orbits stays for a century, losing 0.05% a month. Every running Space Program adds 0.01 to the low orbits each month, so six programs keep about 4 points up, and each anti-satellite strike adds 6 to each part. Mission Control shows the index on every milestone's journal entry, as "Orbital Debris: 12.3 (Clear)"; the Space Race tab shows it once, in its Orbit section. Hover it for the two parts, what the index costs you now and who is cleaning it up.

| Tier | Index | What changes |
|---|---|---|
| Clear | below 25 | Only the costs below. |
| Cluttered | 25 | AI powers start removing debris. |
| Hazardous | 50 | Space Elevators risk ribbon strikes. |
| Kessler Cascade | 75 | Collisions add 0.2 to the high orbits every month until removal brings the index back under 75. |

Every country that uses orbit pays for each point of the index: one with a Space Program, a Space Elevator, an Orbital Solar Collector, an Orbital Battlestation or an Extraplanetary Base.

| Each point | At 50 | At 100 |
|---|---|---|
| +1% setback risk | +50% | +100% (risk still stops at 50% a month) |
| −0.3% space race pace | −15% | −30% (pace never drops below 0.5 a month) |
| −0.2% throughput for Extraplanetary Bases, Space Elevators, Orbital Battlestations and Solar Receivers | −10% | −20% |

Nothing stops: your milestones keep moving, and every cost fades as the debris does. An Orbital Battlestation's lower throughput also lowers the military bonuses it gives, along with its upkeep.

From Hazardous on, a Space Elevator owner has a 0.02% chance per point each month (1.5% at 75) of a ribbon strike. The event Ribbon Strike runs your elevator at −50% throughput for six months, or, if you pay for an emergency repair (the cost grows with your GDP), for two. A ribbon still under repair isn't struck again.

## Clearing orbital debris

Your Space Program has a second production method group, Debris Removal. It starts on No Debris Removal. Each other method removes debris every month, taking it from the high orbits first, and adds a little prestige:

| Method | Technology | Launch Capacity a week | Removed a month |
|---|---|---|---|
| Active Debris Removal | Reusable Rocketry | 300 | 0.05 |
| Laser Broom | Directed Energy Weapons | 300, and 500 electricity | 0.15 |
| Fusion Sweepers | Compact Fusion Reactors | 150 | 0.5 |

Each fully staffed Space Elevator level raises what you remove by 10%, up to +100% at ten levels, and its Launch Capacity makes every method cheaper in your market. Removal also follows the Space Program's throughput, so each funding level raises it along with the bill. Cleaning earns Orbital Cleanup: +1% prestige for each 0.1 you remove a month. The decision Begin Orbital Debris Removal switches your Space Program to the best method you have, and appears again when a better one arrives. Switch removal off in the building's production methods.

Under the Space Race rule's Full Colonies, ground-to-orbit batteries on Ground Lasers also sweep the low orbits in peace (see Holding Earth orbit). Every country in orbit gains from what anyone removes. Fusion Sweepers with a ten-level Space Elevator clear 30 points in about two and a half years. One program on Active Debris Removal can't stop a Kessler Cascade: that takes several cleaners, or better technology. AI powers that use orbit take the decision once debris is Cluttered, more readily at Hazardous, if their weekly surplus covers the method's Launch Capacity.

## The anti-satellite strike

Anti-Satellite Strike is a diplomatic action against a country you are at war with. It needs Guided Missiles (era 7), a Space Program of your own, and a target with a Space Program, a Space Elevator, an Orbital Solar Collector or an Orbital Battlestation. A country can be struck once every 12 months, by anyone. The Orbit section of the Space Race tab shows whether you could strike and whom, the state of your own satellites, and the last strike anywhere.

| Who | What the strike does |
|---|---|
| The target | Orbital Network Disrupted for a year: −10% unit offense and defense, −20% space race pace, and −25% throughput for its Space Elevators, Orbital Battlestations and Solar Receivers. It lasts six months if its Space Program runs the Mars Mission or a later method, and is half as strong if it has an Orbital Battlestation. It ends on its own. |
| You | 5 infamy, 15 case strength at the UN (see Grounds for UN censure), twice both as a party to the Moratorium on Destructive Anti-Satellite Tests, and −30 relations with every other country that uses orbit. |
| Everyone | 6 more debris in low orbits and 6 in high orbits, which your own program flies through too. |

The strike suits a war against a power with more in orbit than you, whose fleet and space race lean on its satellites. Its debris and its diplomatic cost fall on you as well, and a strike in a crowded orbit can tip the world into a Kessler Cascade. AI powers strike rarely, only when the enemy has more in orbit than they do (its Space Program's mission plus its orbital megaprojects' levels), and almost never once debris is Hazardous, unless they are losing the war. Communications Disruption, the covert operation, interferes with a country's networks without the debris (see The covert operations).

## 16.11 Space race events

Besides the choice, setback and completion events above, the space race has yearly events for programs under way. Some are hard science problems, such as radiation or orbital debris; others are discoveries, such as water on Mars. Their options add progress to the milestones they concern, usually 1 to 3 months of it, counted at your pace when the event fires as setbacks are; a few cost a month or more instead. Some also bring loyalists, radicals or better relations. A handful tie the race to other systems: the SpaceX company, a Tourism Industry, the United Nations, the Extraplanetary Base and the Space Elevator. They are listed in Space race event list.

## 16.12 How the AI races

AI great and major powers found their programs and move them up the missions through the same decisions you have. An AI takes them only while it is in the top three of the global ranking, out of default, and only if its weekly budget surplus covers what the step adds to the Launch Capacity bill. AI powers use the same entries, approaches and funding, and start each milestone on Standard as you do; only you can switch a milestone back to Standard. Only an AI in the top three of the global ranking makes progress; a weaker one can open an entry but its bar doesn't move, and the Rivals list leaves it out. Such an AI winds its funding down to 0 and pays nothing for the approach it has chosen until it climbs back into the top three. AI great powers prefer the Ambitious approach, more so when another country is running the same milestone; AI major powers lean toward Safe. The AI raises funding, never past the cap, while it has innovation to spare and cuts it when innovation runs short. Launch windows bind the AI as they bind you, and its approach decides its launches outside a window the same way. Under Simple Colonies and Full Colonies the AI sends its colony fleets where the program picks for you when you choose no site, and sets its emigration and queues its colony modules once a year (see Colony settlers and Colony modules).

A revolution's winner continues the old country's program, with its progress (except progress toward a first colony) and its rewards, and rebels can't start a program of their own during a civil war; see After a revolution.

CHAPTER 17

# Quick reference

This chapter collects facts from the rest of the guide for quick lookup: when each system first appears, and what the mod's own terms mean. Follow the links for the details.

## 17.1 When each system appears

Some systems run from the first day of an 1836 start; most wait for a technology, a law or a rank. The table runs roughly in the order a typical campaign meets them. "Rule" names the game rule that can switch the system off; "none" means no rule can.

| System | When it appears | Rule | Chapter |
|---|---|---|---|
| Construction market | From the start. You set the government's weekly purchase in the construction panel, as a number of points or as a budget. | Free Market Construction | Economy and construction |
| Adaptive standard-of-living expectations, Construction Cost Scaling, transport goods | Always on. | none | Economy and construction |
| The Timeline Extended window's Nation tab | Always on: your standing scores, what the permanent effects the mod's systems give you add up to, with every source on hover, and the effects themselves, which the country panel's Modifiers list leaves out. | none | Introduction |
| Migration crowding, Local Climate, Homeland Dynamics, tourism, world city rankings | Always on, shown in the state panel. Homeland changes also need an enabling law, principle or the Mass Media technology. | none | States and population |
| Grand Monument, the Monuments journal entry and the Timeline Extended window's Grand Monuments tab | From the start, in any state. The journal entry and the tab open once you own one, an interest group asks you for one, or you hold a trophy or have one held abroad. | Grand Monuments | The extended timeline |
| Demographics tab in the Population panel, National Info and the Timeline Extended window, and the Demographics subtab on the state panel | From the start of a new game with the Demographics rule on (Full is the default): every state is counted on the first day and again each year. | Demographics | States and population |
| Plan a New Capital decision | Under the Demographics rule's Full setting, shown from Modern Sewerage; it needs Steel-Frame Buildings and a Dominant or Primate urban pattern. | Demographics | States and population |

| System | When it appears | Rule | Chapter |
|---|---|---|---|
| Releasable city-states | From the start, for whoever holds the city's province: from the Releasable Countries list, or through a Liberate Country war goal. | Releasable Nations (San Francisco only) | Diplomacy |
| Settlement Authority | From the start, in a thinly populated frontier state region. Programs unlock with technologies and laws. | Internal Resettlement | States and population |
| Global Warming journal entry, and the Market panel's Climate tab and the Timeline Extended window's Global Warming tab | Listed from the start; becomes active for every country once warming reaches 0.1 °C. The tab is grayed until the entry is active. On another market it shows that market's emissions and its leader's policies, read-only. | Global Warming | Climate and pollution |
| Strategic Reserve journal entry, and the Market panel's Reserve tab and the Timeline Extended window's Strategic Reserve tab | Shown once you research Logistics; active once you have a Strategic Reserve Hub, built or placed by the decision Establish a Strategic Reserve. The tab is grayed until the entry is active, and shows only on your own market. | none | Economy and construction |
| Banking Cycle journal entry, and the Banking tab in the Budget panel and the Timeline Extended window | Every country, from the start. The financial cycle, the banking tools and the Monetary Policy controls start with Stock Exchange and an Urban Center of level 5 or more. | Banking System | Banking and monetary policy |
| Restructure the Public Debt decision | A financial cycle that has started, with debt past a quarter of your credit limit or in default. | Banking System | Banking and monetary policy |
| Policy rate and monetary policy | The monetary model runs for every country. Your own policy rate needs National Bank Established and Commodity Money, Gold Standard, Fiat Money or Digital Currency; not under Command Economy, dollarized or pegged. | Banking System (Enabled only) | Banking and monetary policy |
| Foreign capital and import-credit interventions | A financial cycle that has started, in a market economy. Requirements vary: your own policy rate, convertible gold, National Bank Established, treasury cash or a financial crisis. | Banking System (Enabled only) | Foreign capital and import credit |
| Collective Governance | The Political Agitation technology and a Distribution of Power law that allows it: any voting franchise, Single-Party State, Technocracy, Oligarchy, Organic Regulation, Anarchy or Algorithmic Governance. | none | Government, laws and characters |
| Military Base | The Trench Works technology; each of its five levels needs a technology of its own. | none | Military and war |
| Tax Code journal entry, and the Tax Code tab in the Budget panel and the Timeline Extended window | From the start, for every country that isn't decentralized. | Legislated Tax Code (off by default) | Taxation (experimental) |

| System | When it appears | Rule | Chapter |
|---|---|---|---|
| Heir Education journal entry | A monarchy with an heir who hasn't been educated. | Heir Education (off by default) | Government, laws and characters |
| Create New Religion journal entry | Players only, while you have no custom religion. | Custom Religion Allowed (off by default) | Government, laws and characters |
| State Collapse journal entry | Appears when your average standard of living falls below 5; the 52-week count runs while it is below 4. Decentralized countries are exempt. | none | Government, laws and characters |
| Cultural Hegemony journal entry, and the Society panel's Hegemony tab and the Timeline Extended window's Cultural Hegemony tab | Any country has researched Mass Media, and you have Romanticism. | Cultural Hegemony | Cultural hegemony and covert warfare |
| Combined arms bonuses | The Combined Arms technology (era 6). | none | Military and war |
| Power bloc principle slots 5 to 8 | On the power bloc panel from the start, locked until the bloc's count reaches them. The count adds to the base game's rank and member slots one for each of Intergovernmental Organizations, Containerization, Globalization and Universal Digital Identity the bloc leader has researched, up to eight. | Power Bloc Principle Slots (Up to Eight only) | Diplomacy |
| Power bloc principle tiers IV and V | A technology researched by the bloc leader, named in the principle's tooltip: Mass Media opens tier IV of ten groups, Combined Arms of the four military ones, Globalization tier V of six. | none | Diplomacy |
| Colonial Empire journal entry, and the Timeline Extended window's Colonial Empire tab | The Decolonization technology (era 6) and at least one overseas colony or colonial subject. The tab opens with the technology alone, showing your stance on colonialism and the colonial powers until the entry runs. | Decolonization | Colonial empires and decolonization |
| Colonial collapse | Once any country has researched Decolonization (era 6); tiny, poor AI countries only, checked yearly. | Decolonization | Colonial empires and decolonization |
| Irredentism and reunification | The Decolonization technology (era 6), a country that shares one of your primary cultures and holds a homeland state of yours, and a government that gives the cause a voice. | none | Diplomacy |
| Ten formable countries | The Decolonization technology (era 6). Each candidate needs major power rank and, except for United Earth, a capital in the region. | none | Diplomacy |
| United Nations journal entry, and the Diplomacy panel's UN tab and the Timeline Extended | Shown at Intergovernmental Organizations (era 6); active for everyone once a great power founds the UN. The tab is grayed until the entry is active. | United Nations | The United Nations |

| System | When it appears | Rule | Chapter |
|---|---|---|---|
| window's United Nations tab | | | |
| Antarctic research stations (the decision Open an Antarctic Research Station) | Space Exploration (era 7), for any country. A country's first station puts the Antarctic Treaty on the UN docket. | Global Commons | The United Nations |
| Nuclear Weapons journal entry, and the Military panel's Nuclear tab and the Timeline Extended window's Nuclear Weapons tab | The Nuclear Weapons technology and Great Power rank (or Major Power with Intercontinental Ballistic Missiles, or program aid by treaty). Once the world's first warhead exists, it is active for every country except decentralized ones and shows the nuclear taboo. The tab is grayed until the entry is active. | Nuclear Weapons | Nuclear weapons |
| The Gathering Storm (World War journal entry) | Great Power rank and Combined Arms, while a great power of an opposed ideological camp exists. | World War (off by default) | Military and war |
| Space race journal entries, and the Timeline Extended window's Space Race tab | Shown at Rocketry for great and major powers; the first milestone needs a Space Program building (Guided Missiles, era 7), built or placed by the decision Found a Space Program. The tab is grayed until your first milestone starts, then stays open. | Space Race | The space race |
| Space fleets, Space Command and Colony Defenses | Under the Space Race rule's Full Colonies setting (the default): Space Command from Space Militarization, in a state with a Space Program, and ground-to-orbit batteries from Space Militarization, in any state; the panel's Space Command, Fleets and Colony Defenses sections in Solar System Colonization. | Space Race | The space race |
| Orbital debris, debris removal and the Anti-Satellite Strike | The index exists once any Space Program runs, and costs every country that uses orbit. Debris removal methods from Reusable Rocketry (era 10); the strike from Guided Missiles (era 7), against an enemy in a war. | Space Race | The space race |
| Covert Warfare journal entry, and the Military panel's Covert tab and the Timeline Extended window's Covert Warfare tab | One covert operation slot more than your rank grants, from the Ministry of Intelligence and Security Established law or the Mainframe Computers technology (era 7). Once active, it stays open. The tab is grayed until the entry is active. | Covert Warfare | Cultural hegemony and covert warfare |
| Civil Rights Movement journal entry | The Civil Rights Movement technology (era 7) and a poorly accepted minority in an incorporated state. | Social Movements | Social movements |
| Digital Rights & Surveillance journal entry | Automated Surveillance or Cybersecurity (era 9). | Social Movements | Social movements |
| Mental Health Crisis and Post-Scarcity Transition journal entries | Mental Health Awareness (era 10) and Post-Work Society (era 12). | Social Movements | Social movements |
| Human Augmentation Debate journal entry | Human Augmentation or Brain-Computer Interfaces (era 11). | Social Movements | Social movements |
| Morphological Freedom Debate journal entry | Morphological Freedom (era 12). | Social Movements | Social movements |

| System | When it appears | Rule | Chapter |
|---|---|---|---|
| Seabed licenses, and the Timeline Extended window's Seabed tab | Abyssal Plain Mining (era 11) and a coastal state. Licenses are shared every January among all coastal countries, those without the technology at three quarters of their weight and their shares waiting for them; a new claimant gets its share as claims the month after, in a sharing that divides every pool again, and the tab opens then. | Global Commons | The extended timeline |
| Megaprojects | Each needs its own era 12 or era 13 technology. | none | The extended timeline |

## 17.2 Glossary

Terms the mod introduces, or uses in its own sense.

| Term | Meaning | Chapter |
|---|---|---|
| Agent network | Your 0–100 presence inside one target country. It gives operations a head start, cover from detection, and intelligence reports, and makes that country's operations against you likelier to be caught. | Cultural hegemony and covert warfare |
| Arsenal ceiling | The most warheads you choose to hold, or a Nuclear Arms Limitation treaty or the UN Prohibition Treaty holds you to. Warheads above it are taken apart, and your program builds nothing while you are at or above it. | Nuclear weapons |
| Bubble pressure | Speculation that builds up during a boom. It sets how likely a banking crash is and how hard it hits. | Banking and monetary policy |
| Case strength | A 0–100 score built from a country's record that decides which punitive UN resolutions can be tabled against it. | The United Nations |
| Charter reform | One of the two amendments to the UN Charter. Each raises the ceiling on UN Authority and opens new business before the General Assembly. | The United Nations |
| Colonial stability | The 0–100 bar of the Colonial Empire journal entry. Its level and the programs you run decide how the empire ends. | Colonial empires and decolonization |
| Colony autonomy | Under Simple and Full Colonies, a colony's 0–100 figure for how far it runs its own affairs. It rises with the colony's people, distance and isolation, your wars and a garrison, and falls with the pull of home, a charter and an Administration; it settles where these balance. A colony builds modules of its own, faster the more autonomy it has. At 50 it demands self-government; at 100 it becomes a free colony. | The space race |
| Colony integrity | Under Full Colonies, a colony's 0–100% hull, domes and power. Enemy fleets closing in wear it down; at 0 the colony can be seized with marines or destroyed. It recovers when the attack stops. | The space race |
| Commencement | The 1st of the month a passed tax bill takes effect, under the Legislated Tax Code rule. Collections change then, not before. | Taxation (experimental) |
| Construction maintenance | The construction good that industry and infrastructure consume each week under the construction market. | Economy and construction |

| Term | Meaning | Chapter |
|---|---|---|
| Construction Services | The good that Construction Sectors sell and Construction Sites turn into construction points. | Economy and construction |
| Contested monument | A Grand Monument that honors something the government is not, after its crown, republic, revolution, leader or faith falls or its state changes hands. It gives only tourism and its local effect until you pull it down, rededicate it or keep it as heritage. | The extended timeline |
| Covert operation slot | Room for one running covert operation at priority 1; an operation fills one slot for each level of its priority. Rank, the Ministry of Intelligence and Security and technologies add slots. | Cultural hegemony and covert warfare |
| Credibility | Whether your nuclear word holds, from 0 to 100. Crises, pledges, guarantees and halting a dismantling move it. | Nuclear weapons |
| Credit standing | The part of your borrowing cost set by what your country is: rank, finance technologies and a stock exchange, institutions and currency credibility. | Banking and monetary policy |
| Cultural share | Your cultural pull as a share of the world's total. It sets your standing in the cultural hegemony race. | Cultural hegemony and covert warfare |
| Cycle value | The 0–100 reading that places your economy in one of the seven banking phases, from Panic to Frenzy. | Banking and monetary policy |
| Detection Risk | The monthly chance that one of your covert operations is caught. Funding, your network in the target and your Efficiency Factor lower it; priority, the target's counterintelligence and any network the target holds inside your country raise it. | Cultural hegemony and covert warfare |
| Docket | The UN's monthly scan of world events, which puts real situations before the countries concerned. | The United Nations |
| Exchange Rate Index | The real value of your currency, where 100 is par. | Banking and monetary policy |
| Flagship building | A company's own unique building, built by the government of the country that owns the state while the company's prosperity bonus is active. | The extended timeline |
| Foreign Cultural Benchmark | The legitimacy penalty on a country whose cultural share trails the hegemon's by more than about three points. It grows with the gap. | Cultural hegemony and covert warfare |
| Free colony | A colony with no holder: it broke away, was let go or abandoned, or outlived its holder. It keeps its people and modules, gives no one its specialization, grows only while supplied, and asks to join a supplier whose ties with it reach 100. | The space race |
| Friend | For covert operations: a country you have cordial relations with plus an alliance, a defensive pact, the same power bloc or an overlord or subject tie, or amicable relations on their own. Having your spying on one exposed costs you at home, and Counterintelligence Support needs one. | Cultural hegemony and covert warfare |
| Funding Stealth | The points covert funding from level 2 up takes off every operation's Detection Risk. | Cultural hegemony |

| Term | Meaning | Chapter |
|---|---|---|
| | | and covert warfare |
| Grand Monument | A building a government raises to what it stands for. Its effects grow with its grandeur, and it can become contested when what it honors falls. | The extended timeline |
| Grandeur | What a Grand Monument's levels were worth when they were built, plus what its years have added: a finished monument grows 2% of its starting grandeur a year. Its effects grow in steps, each taking more grandeur than the one before. | The extended timeline |
| Grid Power | The good Grid Transfer Stations use to move Electricity between the states of one power grid. Its price is fixed at 30. Not traded between markets, but a grid that joins two markets carries it across the border. | Economy and construction |
| Grid Transfer Station | One per state, from Electrical Capacitors. Joins the state to its power grid, which shares Electricity once a month. Always subsidized. | Economy and construction |
| Grid Transmission Efficiency | The share of the Grid Power a receiving station gets that it turns back into Electricity: 75% at first, 100% with Quantum Materials. | Economy and construction |
| Ground reserve | Under Full Colonies, the warships Space Command has built and not yet launched. They join your home fleet each month while no enemy fleet is at Earth (Auto), or go up together when you Launch them, through a blockade too. | The space race |
| Ground-to-Orbit Battery | Under Full Colonies, a government building that fires on every enemy fleet at Earth each month, from Space Militarization. With Ground Lasers it also sweeps low-orbit debris in peace. | The space race |
| Heritage monument | A Grand Monument kept under a government it does not honor. It gives prestige, cultural pull, tourism and its local effect, but no legitimacy and no approval. | The extended timeline |
| HVDC Undersea Interconnector | One per coastal state with a Grid Transfer Station, from Clean Energy Technologies. Lays an undersea cable that joins the station's power grid to others across the sea. Government funded. | Economy and construction |
| Intelligence capacity | The size of your intelligence agency. It sets detection odds in both directions. | Cultural hegemony and covert warfare |
| International standing | Your own record in the UN, from 0 to 100, separate from UN Authority. It starts at 50 when you join, raises or lowers your diplomatic reputation, and sways votes on your resolutions. | The United Nations |
| Interdiction | Under Full Colonies, a fleet standing off in peace at a location where a country that is not your ally holds colonies, blockading them without a shot. It costs infamy and gives that country a casus belli. | The space race |
| Intervention budget | Points that your active banking tools draw on: 1 to 7 from your financial regulation law, and one more each from National Bank Established and a power bloc's banking union. | Banking and monetary policy |
| Kessler Cascade | Orbital debris at 75 or more: collisions add debris on their own every month until removal brings it back under 75. | The space race |
| Launch Capacity | The good made by the Aerospace Industry's rocket production methods, the Space Elevator and Antimatter Engines, used by | The extended timeline |

| Term | Meaning | Chapter |
|---|---|---|
| | space programs, satellites, orbital production methods and megaprojects. | |
| Launch window | The months a site in Solar System Colonization can be reached on a chemical rocket's cheapest transfer: its best departure month and the month on either side. The Moon and Mercury need none. Asteroid Mining, or Shackleton's Propellant Depot, allows launches outside one at a premium, which Nuclear Thermal Propulsion makes smaller; Compact Fusion Reactors end them. | The space race |
| Legislative Override Capacity | A score summed from your laws, institutions and a few technologies and power bloc principles. It decides which Forceful Legislation options you can use; below 0 the event never appears. | Government, laws and characters |
| Local Climate | A state region's climate where most of its people live. A harsh one costs migration pull, up to 80%, and a mild winter adds up to about 7%, until technologies ease them; the state panel's Comfort row gives the cost or bonus. | States and population |
| Megaproject | An era 12 or era 13 structure built one level at a time through construction sites. | The extended timeline |
| Migration Crowding Tolerance | The modifier that raises a state's crowding threshold: +50% lets it hold half as many people again before Population Pressure starts. It multiplies a resettlement frontier's densities the same way. | States and population |
| Momentum | How fast the banking cycle moves: added to the cycle value every month, and losing a tenth of itself each month. | Banking and monetary policy |
| Monument policy | One national setting for all your Grand Monuments: Open to the Public, State Ceremonial, Cultural Promotion or Public Education. Each raises one thing (tourism, legitimacy and authority, cultural pull, education access), and it can change once every five years. | The extended timeline |
| Movement Support | The 0–100 bar of the Civil Rights Movement journal entry. The movement wins at 100 and collapses at 0. | Social movements |
| Nuclear crisis | A threat by one country against another over one dispute, with a deadline. | Nuclear weapons |
| Nuclear doctrine | When your government may order a first strike, from No First Use to Nuclear Warfighting. | Nuclear weapons |
| Nuclear reach | Where your weapons can strike before Intercontinental Ballistic Missiles: countries next to you, to your direct subjects or to a host of your weapons, every country in a strategic region where one of you holds a state, and every coast with Advanced Submarine Technology. | Nuclear weapons |
| Nuclear readiness | How fast your forces can respond: Recessed, Routine, Heightened or High Alert. | Nuclear weapons |
| Nuclear taboo | How unthinkable the world finds nuclear weapons, one score from 0 to 100 born at 20 with the world's first warhead. Use and threats lower it, restraint and the United Nations raise it, and it scales the infamy of nuclear acts and the burden of holding warheads. | Nuclear weapons |
| Nuclear umbrella | The cover you give your direct subjects while you hold Nuclear Power. It spares them the war-support drain of facing a nuclear-armed enemy, weighs your arsenal in crises against them and lets you retaliate for a strike on them under any doctrine. Anyone else | Nuclear weapons |

| Term | Meaning | Chapter |
|---|---|---|
| | needs a Nuclear Guarantee, or to host your weapons under Nuclear Basing. | |
| Open frontier | A state region thin enough in population for a Settlement Authority to be founded there. | States and population |
| Operation phase | How far along a covert operation is: Preparatory for its first six months (no effect), then Establishing (base strength), and Fully Operational from month 12 (double strength). A network in the target shortens the wait. | Cultural hegemony and covert warfare |
| Opportunity | A moment in a country's history, such as a new reign, a victory, becoming a great power or a technology, that brings the chance to build 5 levels of one dedication within 60 months (120 for a major power, 180 for any other). Completing it gives the monument 5 grandeur for free, earns approval and brings a boon for ten years; letting it pass costs nothing. | The extended timeline |
| Orbital debris | One index from 0 to 100 for the whole world: debris in low orbits, which clears in a few years, plus debris in high orbits, which stays for a century. Every country that uses orbit pays for each point in setback risk, space race pace and orbital throughput. | The space race |
| Orbital Superiority | Under Full Colonies, at war, up to +10% offense and +5% defense for your units while your fleets at Earth outmatch the enemy's there, or face none. | The space race |
| Peg Confidence | The market's belief, from 0 to 100, that a gold or treaty peg will hold. | Banking and monetary policy |
| Policy rate | The interest rate your central bank sets under the full Banking System. | Banking and monetary policy |
| Population Pressure | The state modifier that cuts migration pull once a state is crowded for its arable land. | States and population |
| Power grid | Grid Transfer Stations that share Electricity: your stations in one market area, joined across the sea by HVDC Undersea Interconnectors and across borders, into another market too, by Grid Interconnection Agreements. | Economy and construction |
| Reserved half | Under the Regulated seabed regime, the half of each basin's licenses shared only among Law of the Sea parties whose GDP per head is below the UN members' average. A party without Abyssal Plain Mining keeps its share waiting; in a basin where none of them has any claim weight, the half is shared with the open half. | The United Nations |
| Risk premium | The part of your borrowing cost that moves with your circumstances, on top of the world rate and your credit standing. | Banking and monetary policy |
| Seabed claim | Your unused share of a basin's seabed licenses: what you are entitled to and no Deep-Sea Mine level holds yet. Each January divides claims again; Sponsor Contractors keeps at least three quarters of them. | The extended timeline |
| Seabed lease | A Lease Seabed Claims treaty article: some of one country's own claims in a basin leased to another, up to an agreed number, and topped up from the lessor's share at each sharing. Licenses built on it are the lessee's for good. | The extended timeline |

| Term | Meaning | Chapter |
|---|---|---|
| Seabed license | What each Deep-Sea Mine level needs, from one of five ocean basins' fixed pools, shared every January by claim weight among all coastal countries, and again the month after a new claimant appears. A country without Abyssal Plain Mining has a share waiting that no one can build on. A level holds its license, built or queued, for good; it stays with the state the mine stands in. | The extended timeline |
| Seabed regime | How the Law of the Sea governs the seabed's licenses, once a country has Abyssal Plain Mining: Open, Regulated (half of each basin reserved for poorer parties, royalties) or Scramble (everything open, every navy counting double). | The United Nations |
| Secured custody | Protection that halves the warheads lost when an arsenal changes hands. It comes from a foreign custodian in a civil war, inspections after a stolen warhead, a Nuclear Security Assistance article or the Convention on the Physical Protection of Nuclear Material. | Nuclear weapons |
| Setback risk | A running space milestone's monthly chance of a setback that costs progress: a base of 5% to 10% by milestone, scaled across your whole program by your approach and mission choices, never below 1% or above 50%. | The space race |
| Strategic Reserve Hub | The capital building that runs your reserve of strategic goods. | Economy and construction |
| Tax bill | A reform of the tax code under debate. It passes once committed interest groups hold more than half the clout, legitimacy is at least 25 and it has been debated long enough. | Taxation (experimental) |
| Tax code | Under the Legislated Tax Code rule, the tax rates, taxed goods and relief a country's laws hold. It changes only through tax bills. Under the Light Tax Code setting, the rates you set within your taxation law's range, which change at once. | Taxation (experimental) |
| Temperature anomaly | Global warming in °C, driven by the world's cumulative emissions. | Climate and pollution |
| To History | The dedication of the historical monuments that carry their state's heritage tourism, and of some sites once dug up: tourism, prestige and cultural pull, never contested and never pulled down. Every monument policy counts it at half. | The extended timeline |
| Tradecraft | Your intelligence agency's experience, from 0 to 100, which unlocks higher priorities and harder operations. | Cultural hegemony and covert warfare |
| Transportation | The base game's good, now all movement inside a state: pops' travel and industry's freight. Only available in the state that produces it; shipping between markets is Merchant Marine. | Economy and construction |
| Trophy | A Grand Monument, or part of one, carried off from a state occupied in a war, from your own unincorporated state, or by treaty. It stands in its taker's capital and counts for half its grandeur in standing grandeur until it goes home. | The extended timeline |
| UN Authority | How seriously the world takes the UN, from 0 to 100. It sets the tier and the strength of every resolution. | The United Nations |
| UN enforcement | The multiplier the UN's tier puts on every penalty the Assembly imposes and on every convention's effects, from ×0 at Moribund to ×2.5 at Supranational. | The United Nations |

| Term | Meaning | Chapter |
|---|---|---|
| Unique dedication | A dedication of its own that most historical monuments carry on top of the one they stand on, such as The Mandate of Heaven on the Temple of Heaven, and that some opportunities give the monument that completes them. It adds its effect while the monument stands on the dedication it was given on. | The extended timeline |
| Urban Engineering | Points from five early urban technologies. Each adds 1,000 people to every state's crowding threshold, before Migration Crowding Tolerance multiplies it. | States and population |
| Weight in world affairs | A country's share of world prestige against a typical great power's 10% (×1), up to ×5. What its acts add to UN Authority's ledgers is multiplied by it. | The United Nations |
| Wealth Concentration | How much of a country's wealth sits in great family fortunes, from 0 to 100. Each state's score drifts 3% a year toward a target set by the Inheritance law and its amendments, land tenure, who owns its capital, inequality, taxes on wealth, the economic laws, the health, school and intellectual property laws, the slavery law and the return on capital against growth; the country's figure is the states' average, weighted by where property is held. The country's figure above 50 strengthens the Aristocrats and Capitalists and raises the banking cycle's bubble pressure, and below 50 weakens both and lowers it. Each state's own score above 50 raises its owners' investment and the clout their wealth buys, lowers qualifications and education access and, in farming states, sends the young off the land and toward land reform; below 50 it lowers wealth's clout and radicals and raises qualifications and education access. The effects are full at 80 and 20, and grow beyond. | Government, laws and characters |

APPENDIX

# Appendix: social movement details

This appendix holds the numbers and event lists behind Social movements, in the same order as that chapter. You don't need them to play: the journal entries' tooltips show the same figures when they matter.

## 18.1 Civil Rights Movement details

These tables give the exact figures for the Civil Rights Movement journal entry: what moves its bar, what each phase and policy does, and which events fire.

### What moves Movement Support

Each month Movement Support moves by the sum of these lines, which the bar's tooltip lists:

| Source | Change per month |
|---|---|
| Movement decay (always) | −0.4 |
| Social Justice Movements researched (era 9) | +0.4 |
| Protection or Affirmative Action in force | +0.3 |
| Violent Hostility or Ghettoization in force | −0.5 |
| Incorporated states with pops below Provisional Acceptance | Up to +0.5 (at 40 states) |
| Radicals at 10% of your population, and again at 20% | +0.15 each |
| Grassroots Organizing | +0.2 |
| Federal Protection | +0.3 |
| Gradualist Accommodation | +0.1 |
| Cooptation / Token Reform (first 12 months) | +0.15 |
| Suppression / Crackdown | −0.6 |
| Segregationist Consolidation | −0.9 |
| Support below 5 | +0.4 |

The last line cancels the decay near the bottom, so an ignored movement hovers around 5.

### Civil rights phases

The bar's level sets a phase, applied as a modifier on the journal entry:

| Phase | Support | Effects |
|---|---|---|
| Marginal | Below 20 | −10% radicals from political movements |
| Growing | 20–40 | +5% turmoil penalties |
| Active | 40–65 | +5% turmoil penalties, +10% radicals from political movements |
| Pressuring | 65–90 | +10% turmoil penalties, +20% radicals from political movements |
| Imminent Reform | 90 and above | +15% turmoil penalties, +20% radicals from political movements, +10% enactment success chance |

### Civil rights policy effects

| Policy | Support per month | While active |
|---|---|---|
| Grassroots Organizing | +0.2 | +5% enactment success chance, +10% radicals from political movements |
| Federal Protection | +0.3 | +5 acceptance, −5% Bureaucracy, −5% Authority |
| Gradualist Accommodation | +0.1 | −5% turmoil penalties, +2 acceptance |
| Cooptation / Token Reform | +0.15 for 12 months | −10% turmoil penalties, +3 acceptance, +5% legitimacy penalty from ideological incoherence |
| Suppression / Crackdown | −0.6 | +5% Authority, +10% turmoil penalties |
| Segregationist Consolidation | −0.9 | +10% Authority, +30% radicals from political movements, −10 acceptance |

Acceptance figures apply to pops outside their homeland.

### Civil rights event list

Four milestones each bring one event, once, when the bar first crosses their mark:

| Support reaches | Event | Instead, when |
|---|---|---|
| 25 | First Mass Rally Draws Notice | The Underground Railroad, under Violent Hostility or Ghettoization |
| 50 | Coalition Forms with Trade Unions | Blood of a Martyr, under a discriminatory minority law |
| 75 | Mass Disobedience under a discriminatory minority law, otherwise Equal on Paper | Federal Commission Recommends Action, if Federal Protection has run over 24 months |
| 90 | The March on the Capital | Always this one |

Equal on Paper comes to a country whose laws no longer discriminate, where the movement now demands that they be enforced. Sending in inspectors and prosecutors gives +5 acceptance outside homelands, fading over five years, and loyalists among discriminated pops; it pleases the Intelligentsia and angers the Devout, Rural Folk and Armed Forces. Leaving it to the courts brings fewer loyalists and a milder, shorter version of the same reaction. Refusing pleases the Devout, Rural Folk and Armed Forces, angers the Intelligentsia and radicalizes the discriminated pops.

The random events, about one month in seven, are Civil Rights March, Segregation Incident (under a discriminatory minority law), Cultural Renaissance (at 40% literacy), The Boycott and the international condemnation described in the chapter.

## 18.2 Details of the movements carried by events

The options of the events in the chapter's table of movements carried by events, movement by movement.

### Second-wave feminism events

Two events need Second Wave Feminism (era 7), and each fires once. The Second Shift, under Propertied Women or Women in the Workplace, trades research speed against legitimacy: welcoming women into the workforce angers aristocrats, keeping traditional roles slows research. Half the Salary, under Women in the Workplace, asks for equal pay: passing it pleases the Intelligentsia and angers the Devout, Rural Folk and Armed Forces, and shelving it does the reverse.

### LGBTQ+ rights events

Pride and Protest's options, from protecting the march to breaking it up, depend on your LGBTQ+ Rights law. The Marriage Equality Question, under Basic Protections or Anti-Discrimination Laws, offers civil partnerships, marriage equality, a traditional definition or waiting for the courts.

### Secularization and faith events

Both events are in the yearly draw of social-tension events, which has a 65% chance each year of picking one event from a large pool.

Religious Revival Sweeps the Nation needs a marginalized Devout group and Decline of Organized Religion (era 10), Sexual Revolution or Social Media: you embrace it (Devout political strength up, Authority down), stay secular (a small research bonus) or channel it into charity (a little standard of living). Faith Against Modernity needs Devout clout of at least 5% and either Total Separation or one of Second Wave Feminism, Sexual Revolution and LGBTQ+ Rights Movement: you side with tradition, modernize, or seek a theological compromise.

## 18.3 Religious revival event list

Each revival event is tied to a technology. State Atheism blocks all but The Preferential Option and The Faithful Hand, which then tell of underground faith.

| Event | Technology (era) | Embracing it grants, for 20 years |
|---|---|---|
| The Moral Majority | Television Broadcasting (7) | The Religious Right: +35% Devout political strength, +30% Devout attraction, −50 Authority |
| The Electronic Pulpit | Pop Culture (7) | Televangelism Movement: +20% strength, +50% attraction |
| The Preferential Option | Civil Rights Movement (7) | Liberation Theology: +30% strength, +25% attraction, +0.5 standard of living |
| One Nation Under God | Globalization (9) | Religious Nationalism: +50% strength, +40% attraction, +10% assimilation, −5% prestige (fading) |
| The Digital Pulpit | Social Media (9) | Digital Faith Movement: +25% strength, +45% attraction |
| The Culture War | Sexual Revolution (8) | Culture War: +40% strength, +35% attraction, slower research (fading) |
| The Faithful Hand | Social Justice Movements (9) | Faith-Based Social Services: +25% strength, +30% attraction, +0.5 standard of living (fading) |

The Devout clout ceiling is 15% instead of 10% for One Nation Under God, The Culture War and The Faithful Hand.

Islamic and Dharmic countries get their own text for all seven events, and some titles change: The Electronic Pulpit becomes The Satellite Minbar for Islamic countries, and One Nation Under God becomes One Ummah, One Law (Islamic), One Civilization, One Dharma (Dharmic) or The Promised Land (Jewish).

APPENDIX

# Appendix: events

For social movement events, see Appendix: social movement details.

This appendix lists the events of the other systems, in the order of the chapters. Each chapter's event section says what kinds of events a system brings, when they come and what is at stake, and links to its list here. You don't need the lists to play: an event's options show their effects in the tooltip when it fires.

## 19.1 Resettlement event list

These are the events that a Settlement Authority can trigger, behind Resettlement events in States and population. Each month that one of your Authorities receives settlers, you have a 10% chance of one of these events, set in one such Authority's state. After one fires, none comes for eighteen months, however many Authorities you run. Money costs are a share of your yearly gross income.

| Event | Fires for | First choice | Second choice |
|---|---|---|---|
| Land Rush | Voluntary programs | "Open the land at once": +2,000 settlers a month there for a year, and more pressure on the people already living there | "Survey it first": +10% construction there for two years |
| The Speculators | Voluntary programs | "Send in the inspectors": −50 authority for two years; Rural Folk approve | "The market will sort it out": 500 fewer settlers a month there for two years; Landowners and Industrialists approve |
| A Hard Winter on the Frontier | Any program | "Send relief": 1% of yearly income | "They will have to endure": 1,000 lower-strata deaths and radicals there, and −10 migration pull there for a year |
| Dust Storms | Homesteads, once per state after ten years | "Fund a conservation program": 2% of yearly income, and −5% farm and ranch throughput there for ten years | "Keep plowing": −20% farm and ranch throughput there, fading over five years |
| The Reform Campaign | Coercive programs | "End the coercive programs": every coercive program switches to your best voluntary one; +5% prestige for five years; Intelligentsia approve | "These accounts are exaggerated": −5% prestige for five years; Intelligentsia disapprove |
| Famine in the Settlements | Special Settlements | "Send grain and tools": 2% of yearly income | "Let the settlements fend for themselves": 5,000 farmers die and many turn radical; Rural Folk strongly disapprove |

| Event | Fires for | First choice | Second choice |
|---|---|---|---|
| A Petition to Return | Coercive programs, once per state after five years | "Let them go home, and end the program here": one in twenty of the lower strata there who are at least second-class citizens return to the state that last sent settlers, which gains loyalists; that Authority switches to your best voluntary program | "The program continues": many radicals among the lower strata there |
| Land Disputes | States where settlers arrive among people below second-class citizens | "Negotiate reserves for them": 1% of yearly income, −25 authority for two years and 300 fewer settlers a month there for ten years; the cultures there below second-class citizens gain loyalists | "Back the settlers' claims": those cultures turn radical |

In the Petition to Return, only pops of at least 2,000 people send anyone home, so small communities stay. The Reform Campaign's first choice ends coercive programs in every Authority, while the Petition ends the program only in the Authority where it fires.

## 19.2 Demographic event list

These are the events behind Demographic events in States and population, under the Demographics rule's Full setting. Each fires when the census crosses its threshold, at most once in 25 years (the Last Generation? once a game). Money costs are a share of your yearly gross income; modifiers last ten years unless the table says otherwise.

| Event | When it fires | Choices |
|---|---|---|
| The Demographic Dividend | 62% of your people are aged 15 to 64, a tenth more than in 1836 | Schools: more education access, a costlier Ministry of Education. Industry: construction and manufacturing throughput. Savings: cheaper loans and more tax capacity |
| The Baby Boom | A war lasting two years or more ends, after its dead passed a hundredth of your men of fighting age | Births +15% fading over ten years, with standard of living +1 for the lower strata for five years at 5%. Or births +20% and fewer women at work. Or births +10% and more women at work |
| The Lost Generation | Your war dead pass a twentieth of your men of fighting age | Of three harms (legitimacy and radicals; lower throughput; poorer widows and fewer births), memorials at 2% spare the first, women in the factories spare the second at the Devout's and Landowners' cost, and widows' pensions at 5% spare the third |
| The Graying of the Nation | Pensioners reach a fifth of your people; with no pension law, 14% of your people are past the age where old age begins | With Old Age Pension: raise the pension age to 70 (grayed while your pension setting's five-year term runs), cut benefits (income of a fifth of the pension bill, poorer lower strata, Trade Unions angry), raise contributions (income of a seventh, Industrialists and Trade Unions displeased), open the borders (more immigrants, a nativist movement grows), or keep the pension. Without it: open the borders, almshouses (fewer deaths, a costlier Ministry of Health), or leave the old to their families (fewer women at work) |
| The Youth Bulge | More than a fifth of your people are aged 20 | For five years: public works at 5% (fewer radicals, more construction), national service (more conscripts, fewer |

| Event | When it fires | Choices |
|---|---|---|
| | to 29 and a tenth of your workforce has no work | radicals), emigration (more people leave, less prestige), or nothing (more radicals and turmoil) |
| Missing Girls | 110 or more boys are born per 100 girls (Birth Limit after Microprocessor) | Ban sex selection (5 fewer boys per 100 girls for twenty years, 25 authority for five), pay for daughters (3 fewer for twenty years, 5%), relax Birth Limit (one child back to two, or end it once its term has run), or call it a private matter (the feminist movement grows) |
| Bare Branches | 110 or more men per 100 women aged 20 to 59 | Pay women's passage (more immigrants, 2%), call the bachelors up (more conscripts, fewer radicals), more police (less turmoil, 25 authority for five years), or nothing (more radicals) |
| The Empty Villages | A state where a quarter of the people farm, the median age is 40 or more and the population fell by half a percent or more in a year | Of three harms there (idle fields, closed schools and clinics, the old alone), farm subsidies at 2% spare the first, keeping the services open for 25 authority over five years spares the second, and moving the old into the market town spares the third at the Rural Folk's cost |
| The Overspill | One of your states reads Metropolis (60% or more of its urban people in its largest city) and is at Migration Crowding's ceiling | New towns at 5% (fewer newcomers to the city, which builds faster; your Settlement Authorities switch to Planned Towns where you have Modern Urban Planning), a green belt for twenty years (standard of living +1, crowding sooner, slower construction), or sprawl for twenty years (crowding later, worse infrastructure, more deaths) |
| The Last Generation? | You research Biological Immortality | For twenty years: birth licenses (families want 30% fewer children, 50 authority), leave children a right, or pay for the next generation at 5% (families want 10% more) |

## 19.3 Diplomacy event lists

These lists are behind the event sections of Diplomacy: the international relations events, the chains they start between rivals, and the irredentist events.

### International relations event list

These are the Cold War events behind International relations events.

| Event | When it can happen | What you decide |
|---|---|---|
| Plans on the Table | Radar; a rival that is a major power or greater | Build up against the rival, or shelve the plans. |
| The Shadow War | Covert Warfare off; Intergovernmental Organizations; a rival that is a major power or greater and has subjects | Fund an insurgency in the rival's sphere, or stay out. |
| International Summit | Intergovernmental Organizations; great power | Host (+10 relations with every other great power), attend (+5) or boycott (−5). |
| The War of Words | Covert Warfare off; Mass Media; a rival that is a major power or greater | Open a propaganda offensive against the rival, or not. |
| The Iron Purse | Keynesian Economics; a rival that is a major power or greater | Impose an embargo (−30 relations), or keep trading. |
| An Agent in Place | Covert Warfare off; Cryptography; a rival that is a major power or greater | Plant an agent in the rival's defense ministry, or not. |

| Event | When it can happen | What you decide |
|---|---|---|
| The Heavens Beckon | Space Race off; Space Exploration; great power | Celebrate a space milestone, or spend on the people. |
| Nuclear Brinkmanship | You and an opponent in a play both hold nuclear weapons, outside a nuclear crisis | Stand firm, open back channels, or mobilize. |

### Rival-choice chain list

These are the chains behind Rival-choice event chains. Plans on the Table, The War of Words and An Agent in Place go to the rival first, and you hear only if it acts: "The Gathering Storm" (an event, not the World War journal entry) if it arms, "Hearts and Minds" if it broadcasts, and "The Spy Who Was Caught" about half the time it plants an agent. Your answer goes back to it: matching a buildup or a campaign sends it a follow-up, and a caught spy can be tried, traded quietly or answered by expelling its diplomats. Your own choices in The Shadow War, The Iron Purse and Nuclear Brinkmanship reach the other country the same way.

### Irredentist event list

These are the events behind Irredentist events. "The Diaspora Calls" fires on a country with a candidate. You can stoke the movement (a claim on another homeland state, 1 infamy, −10 relations), fund cultural ties (+15 relations and a five-year bond that makes a later union likelier), demand a state back, or wait. A demand gives the target a choice: cede the state, or refuse and face your diplomatic play to take it. "An Opportunity for Union" fires when relations with a candidate are warm and lets you propose a union.

## 19.4 Cultural hegemony event list

These are the events behind Cultural hegemony events in Cultural hegemony and covert warfare.

Rising powers get opportunities as their share climbs, from Soft Power Dividend at 8% to Media Dominance and Lingua Franca at 20%, mostly trading authority or prestige for pull, research or migration. Two milestones fire once: The World is Watching when you first pass 10%, and The Cultural Hegemon when you first pass 25% while ranked first.

Trailing countries under the benchmark get events such as Foreign Cultural Infiltration, Brain Drain and Foreign Ideological Influence. The harshest is Hegemonic Convergence: when the hegemon holds at least 25% and your share is at most a tenth of its share, you may be pressed to adopt one of its laws. Yielding enacts the law at once, with radicals, a decaying −10 legitimacy and +10 relations with the hegemon. Refusing costs 5 infamy, −10 relations and a decaying modifier with −5 legitimacy and worse turmoil effects.

Four events follow the political models:

| Event | Who gets it | Choices |
|---|---|---|
| Our Model Abroad | The hegemon, when its model holds 40% of world culture across three or more countries | Champion it (prestige and a stronger push abroad, paid in influence and relations) or lead by example (legitimacy). |
| Between Two Suns | A trailing country when the top two cultural powers run different models | Lean to either power (relations, research, a push toward its model) or stand apart (legitimacy). |
| The Standard Falls | Countries still running a model the leading power has abandoned | Hold course (legitimacy falls, governing interest groups approve) or adapt (relations with the new leader, a push toward its model). |
| Across the Border | Neighbors of a country that has just adopted the hegemon's model | Close the border to its ideas (costs authority and relations) or let them travel (relations, a push toward the model). |

## 19.5 Decolonization event lists

These lists are behind Decolonization events and Former colonies in Colonial empires and decolonization.

## Colonial power event list

These events fire for the colonial power while the Colonial Empire journal entry runs. With the Neocolonialism law, four of them offer independence on your terms. A colonial subject freed this way leaves with Neocolonial Dependency, and in The Price of Empire a 25-year treaty also gives you investment rights and a trade privilege in the new country, whose upkeep you pay.

| Event | When it can fire |
|---|---|
| Winds of Change | A colonial subject reaches 50 liberty desire. |
| The World is Watching (not the cultural hegemony milestone of the same name) | An anti-colonial great power presses you over a colonial subject (needs Intergovernmental Organizations). |
| The Price of Empire | A colonial subject at 75 liberty desire rises in armed revolt (needs Civil Rights Movement). |
| Trouble in (the colony's name) | A colony has a severely unaccepted population. |
| Blood in the Colonies | Armed resistance in a colony, once two or more colonies are badly unaccepted. Below 40 stability its suppression is a war of independence. |
| Declaration in (the territory) | Below 40 stability, a colony with severely unaccepted people declares independence; also each declaration of a collapse. See Wars of independence. |
| The Captains' Revolt, The Republic in Danger | A war of independence has run four years with your war support in it at 25 or below for a year. |
| Judged by New Standards | An anti-colonial great power condemns you in international forums. |
| Powerful Friends | A pro-colonial great power offers its backing. |
| These Dark Satanic Mills of Empire, The Intelligentsia Petition | An Anti-Colonialist leads the Intelligentsia. |
| The Treasury Says No | Your debt is high while you hold colonies. |
| Conscription Crisis | You are fighting a war of independence (against a colony's rebels, a rebelling colonial subject or a colony's secessionists), or are at war while Colonial Crackdown runs. |
| The Universities Speak | The entry has run for ten years and a powerful Intelligentsia is out of government. |
| The Veterans' Protest | Your garrison has run for more than three years, or you lost a war of independence. |

## Former colony event list

A former colony can propose closer ties only to an old ruler with no colonial subjects; a dominion or other subject doesn't stand in the way. If the old ruler accepts in Old Ties, New Terms, relations improve by 30 and the old ruler gains Post-Colonial Partnership. When a former colony demands reparations of a tenth of the old ruler's yearly revenue, capped at a tenth of its own GDP, in Debts of Empire the old ruler either pays, taking Colonial Reparations Paid (a ten-year loss of prestige, influence and bureaucracy) for 40 better relations, or refuses and loses 20.

The young-state events of a former colony's first 20 years need only that some country in the world has researched Decolonization; the former colony doesn't need the technology itself.

| Event | Who gets it | Choices |
|---|---|---|
| Year One | A former colony of minor rank or below | Schools and roads, foreign capital (better relations with the old ruler) or self-reliance (worse relations) |
| The Partition Question | A former colony bordering a newly freed country of shared heritage, or | Press territorial claims, propose a fraternal federation (the neighbor may |

| Event | Who gets it | Choices |
|---|---|---|
| | a colonial power that rules people of its heritage | accept a five-year truce in A Fraternal Proposal), or accept the borders |
| Whose Country Is This? | A former colony bordering another newly freed country | Demand new borders, ask for mediation, or build within them (the neighbor may accept a five-year truce in The Line on the Map) |
| The Strongman's Promise | A former colony with a revolutionary movement, primary cultures below half its population, or radical soldiers | Military rule, democracy or a one-party state |
| The Non-Aligned Path | A former colony while one great power condemns colonialism and another backs it | Non-alignment, or side with either power |
| Whose Language Do We Dream In? | A former colony with no homeland of its own cultures, whose capital is the homeland of a culture living there | Make the capital's native cultures your primary cultures, or keep the colonial-era identity |

The Nationalization Question reaches every country formed by a release or a collapse, a week after it forms, and any subject that becomes independent once the era has begun, whenever a foreign country of minor rank or above owns more than 5% of its GDP. It can seize everything, nationalize strategic sectors or protect foreign property. If it seizes everything, a great-power old ruler still running the Colonial Empire entry may face The Crisis: send the fleet, impose sanctions or accept. Sending the fleet brings Gunboats in the Harbor to every other great power, which can condemn the intervention, back it or stay out.

## 19.6 World War event list

These are the leadup, wartime and post-war events of The World War journal entry in Military and war. The Brink of War, The Hour of Decision and the Peace Conference are described in that chapter.

Leadup events come at random. In Ideological Demands you decide whether to press an opposed great-power rival to renounce its system; if you do, it answers in Ideological Confrontation. Border Incident, with an opposed rival on your border, lets you demand an apology, downplay it, or mobilize (Rearmament for five years). Diplomatic Crisis fires when an opposed rival opens a play against a smaller country in your power bloc or under your protection: guarantee its independence for five years, abandon it, or mobilize in its defense.

Wartime events include Home Front Rally, Strategic Bombing (once an enemy has Bombing Aircraft; air defense costs a spell of Home Front Strain) and Resistance Movement (after a year of war, when the enemy holds 5% of your land). From two years, War Weariness asks whether to push on or seek terms, and Stalemate on the Front, when your war goal has stalled after 24 months and ten large battles, offers +5 war support to hold the line or −10 to open armistice talks.

In the 36 months after peace, three events can reach great powers. War Crimes Tribunal lets you prosecute (Liberation Hero), move on, or bury the evidence, which brings War Crimes Exposed (−30% prestige for ten years). The Post-War Order lets a victor take a further spell of Victor's Peace, and any power that didn't lose champion self-determination (Liberation Hero) or turn to recovery. The New Rivalry comes when a great power you fought beside now holds an opposed ideology; either answer closes the entry at once.

## 19.7 Nuclear taboo event list

These are the events behind Nuclear taboo events in Nuclear weapons. The band events reach every country as the nuclear taboo crosses a band line, once the score is 2 points past it, and each fires at most once in ten years:

| Event | When |
|---|---|
| Something Different About This Weapon | The taboo rises through 30 |
| A Line the World Will Not Cross | It rises through 50 |
| The Unusable Weapon | It rises through 70 |
| Beyond the Pale | It rises through 90 |
| The First Crack | It falls back through 90 |
| Talk of the Unthinkable | It falls back through 70 |
| The Taboo Frays | It falls back through 50 |
| Just Another Weapon | It falls back through 30 |
| The Last Warhead | You finish dismantling your arsenal |

## 19.8 Climate event list

These are the events behind Climate events in Climate and pollution, in two lists.

### Warming threshold event list

The threshold events fire once for every country as the world crosses each mark, and the recovery events fire once if it falls back below a mark it had passed:

| Event | When |
|---|---|
| The Mercury Rises | Warming reaches 0.5 °C |
| The Sweltering Season | 1.0 °C |
| When the Levees Break | 2.0 °C |
| The Reckoning | 3.0 °C |
| Four Degrees | 4.0 °C |
| Out of the Furnace, Off the Brink, The Heat Recedes, A Cooler Decade, Below the Line | Warming falls back below 4.0, 3.0, 2.0, 1.0 and 0.5 °C |
| Near Baseline | Warming falls back below 0.1 °C, after having reached 0.5 °C |
| Termination Shock | The last country running Aerosol Injection stops, with at least 0.25 °C of warming masked |

The recovery events read the warming beneath any aerosol haze, so a mask alone never fires them.

Four Degrees offers no good choice. Each option spares you one of three harms (mortality and standard of living, farm throughput, or building throughput) and adds a cost of its own, all fading over ten years.

### Recurring climate event list

Each month every country has a small chance of one recurring event whose conditions it meets. Three have an extra option for a country that funds its Ministry of the Environment to level 3, and The Fever's March has one for the Ministry of Consumer Protection at level 3.

| Event | Conditions |
|---|---|
| These Dark Satanic Chimneys (a pollution scandal) | Coal Mines or Oil Rigs of level 3 or more; fires even with the Global Warming rule off |
| A Greener Shade of Politics | 0.5 °C and Pollution Control |
| The Uprooted (climate refugees) | 1.0 °C |
| The Tide Comes In (coastal flooding) | 1.0 °C and a coastal state with a Port |
| The Congress of Smoke (a climate summit) | 1.0 °C, a major or great power with Pollution Control, and no United Nations |

| Event | Conditions |
|---|---|
| The Barren Harvest | 2.0 °C, any grain farm or livestock ranch, and a Ministry of the Environment below level 5 |
| The Wells Run Dry | 2.0 °C |
| The Fever's March | 2.0 °C and a Ministry of Consumer Protection below level 5 |
| The Great Thaw | 2.5 °C and a coast |
| Green and Gold | 3.0 °C, Clean Energy Technologies and industrialists in government |
| Quiet Power | The Environmental Movement technology and a Renewable Energy Plant |
| Friends in High Places | A Carbon Tax, Emission Standards or Fossil-Fuel Divestment, and Coal Mines or Oil Rigs of level 5 or more |

## 19.9 Space race event list

These are the yearly events behind Space race events in The space race. Some are hard science problems, such as radiation, the rocket equation, communication delay, closed-loop life support, solar storms and orbital debris; others are discoveries, such as water on Mars or lunar helium-3. The table lists the handful that connect the space race to other systems:

| Event | When it can fire | What it offers |
|---|---|---|
| The Private Sector Steps Up | You have the SpaceX company and a milestone under way | 1 to 3 months of progress on every milestone under way, more the more you hand to the company |
| Tickets to the Stars | You have reached orbit and own a Tourism Industry | Space Tourism Revenue (+5% tourism output, fading over ten years), a research bonus instead, or both for two and a half years |
| A Station for All Nations | You are a UN member between Orbital Flight and the Moon Landing, and another member has reached orbit | A month of progress on every milestone under way, and a ten-year, fading bonus from either a joint station (research, prestige, influence, better relations with spacefaring members) or a national one (research, prestige) |
| Off-World Industry Meets the Space Program | You are colonizing and own an Extraplanetary Base | Full integration: 2 months of colonization progress, plus research and prestige, fading over ten years. Staying commercial: 1 month, plus Extraplanetary Base throughput and advanced materials output, fading over ten years |
| The Elevator Advantage | You own a Space Elevator and are landing on Mars or colonizing | 1 or 2 months of progress on those missions |
| The Kessler Threshold | Orbital Flight is under way, or any milestone is once orbital debris is Cluttered or worse | 1 or 3 months of progress, the larger with radical academics, or better relations with spacefaring powers |

The Space Elevator and the Extraplanetary Base are covered in The extended timeline, tourism in States and population, and International Space Cooperation and its agency in The United Nations.

APPENDIX

# Appendix: reference lists

This appendix holds the longer lists behind the chapters, in the order of the chapters that summarize them. You don't need them to play: the technology screen, the building panels, the military screens and the ship designer show the same details in the game.

## 20.1 Wonder list

These are the 29 wonders tied to one state, the landmark's real location, so only that state's owner can build them. What wonders do, and the eight that any country can race to build, are in Wonders.

| Wonder | State | Unlocked by |
|---|---|---|
| Golden Gate Bridge | California | Marketing Research |
| Empire State Building | New York | Modern Skyscrapers |
| CN Tower | Ontario | Modern Skyscrapers |
| Sydney Opera House | New South Wales | Art Deco Architecture |
| Hoover Dam | Nevada | Public Works Programs |
| The Pentagon | Virginia | Combined Arms |
| Azadi Tower | Mazandaran | Decolonization |
| Kenyatta International Convention Centre | Kenya | Decolonization |
| Kennedy Space Center | Florida | Space Exploration |
| Aswan High Dam | Upper Egypt | Nuclear Energy |
| Tokyo Tower | Kanto | Television Broadcasting |
| Moscow State University | Moscow | Mainframe Computers |
| Berlin TV Tower | Brandenburg | Transistor Technology |
| Lotus Temple | Delhi | Pop Culture |
| Basilica of Our Lady of Peace | Ivory Coast | Pop Culture |
| Burj Khalifa | Trucial Coast | Computer-Aided Design |
| World Trade Center | New York | Containerization |
| Itaipu Dam | Paraná | Industrial Robotics |
| Petronas Towers | Malaya | Cellular Networks |
| Three Gorges Dam | Western Hubei | Clean Energy Technologies |
| Gardens by the Bay | Malaya | Clean Energy Technologies |
| Channel Tunnel | Home Counties (sorry, France) | Globalization |
| Abraj Al-Bait Clock Tower | Hedjaz | Globalization |
| Taipei 101 | Formosa | Digital Telecommunications |
| Lotte World Tower | Gyeonggi | World Wide Web |

| Wonder | State | Unlocked by |
|---|---|---|
| FAST Telescope | Guizhou | Machine Learning |
| Shanghai Tower | Jiangsu | Advanced Structural Engineering |
| Statue of Unity | Gujarat | Advanced Structural Engineering |
| Golden Bridge | Tonkin | Advanced Structural Engineering |

## 20.2 Banking tool list

Every market economy banking tool, with its category in the banking panels, the intervention points it holds and what it needs. What each does and when to use it is in Market economy banking tools and the sections after it.

| Category | Tool | Points | Needs |
|---|---|---|---|
| Monetary Policy | Open-Market Operations | 4 | Keynesian Economics; under the full Banking System, Fiat Money or Digital Currency with the rate at its floor |
| Prudential Regulation | Moral Suasion | 1 | – |
| Prudential Regulation | Raise Reserve Requirements | 2 | – |
| Prudential Regulation | Enable Counter-cyclical Buffer | 2 | International Exchange Standards |
| Prudential Regulation | Raise Margin Requirements | 2 | – |
| Prudential Regulation | Expand Deposit Guarantee | 4 | Consumer Credit |
| Directed Credit | Directed Credit to Infrastructure | 3 | – |
| Directed Credit | Directed Credit to Heavy Industry | 4 | – |
| Directed Credit | Directed Credit to Agriculture | 2 | – |
| Directed Credit | Directed Credit to Armaments | 3 | – |
| Directed Credit | Directed Credit to Electrification | 1 | Rural Electrification |
| External & Currency | Export Credit Facility | 2 | Corporate Governance |
| External & Currency | Capital Controls (Outflows) | 2 | a law that allows it, or war |
| External & Currency | Restrict Speculative Inflows | 2 | full Banking System; your own policy rate; a law that allows capital controls, or war |
| External & Currency | Sterilize Capital Inflows | 2 | full Banking System; convertible Gold Standard with your own policy rate; treasury cash |
| External & Currency | Limit Foreign-Currency Borrowing | 2 | full Banking System; National Bank Established |
| External & Currency | Foreign Exchange Surrender Requirement | 2 | full Banking System; convertible Gold Standard with your own policy rate; exports, cash and room in the vault; a law that allows capital controls, or war |
| External & Currency | Emergency Import Financing | 3 | full Banking System; National Bank Established; a financial crisis and treasury cash |
| Crisis Response | Declare a Bank Holiday | 2 | Downturn or Panic; none in the last five years |
| Crisis Response | Emergency Liquidity Program | 4 | Investment Banks; Downturn or Panic |
| Crisis Response | Asset Relief Program | 5 | Keynesian Economics |
| Crisis Response | Bail-in Regime | 3 | Globalization |

## 20.3 Amendment petition list

These are the interest-group requests described in Amendment petitions. Each also needs a law the amendment attaches to, and none comes while you are enacting another law in that law's group. A group asks only if it isn't marginal; where two are named, the second asks when the first is marginal or absent. The chance is per year while you qualify.

| Amendment (petition) | Comes when | Chance | Asked by |
|---|---|---|---|
| Minority Script Preservation (*The Old Letters*) | A separatist movement is growing, or a cultural minority movement has 5% support | 5% | Intelligentsia |
| Language Requirement for Naturalization (*Spoken Like a Citizen*) | A cultural majority movement has 5% support, or under 70% of your population belongs to your primary cultures | 2.5% | Petite Bourgeoisie, or Rural Folk |
| Official Bilingualism (*A Second Official Tongue*) | Under 80% of your population belongs to your primary cultures, and the minority stirs as for The Old Letters | 5% | Intelligentsia, or Trade Unions |
| National Champion Exemption (*Too Large to Break*) | You have a company, and a banking-cycle panic or downturn, a fall in power rank or a default | 10% | Industrialists |
| Worker Cooperative Preference (*The Workers' Own Firms*) | A labor movement has 5% support, and the Trade Unions are powerful or in government | 5% | Trade Unions |
| Mandatory Minimum Sentences (*Order in the Courts*) | Country-wide turmoil above 20%, or more than 15% of your population radical | 10% | Petite Bourgeoisie, or Armed Forces |
| Community Service Alternative (*The Cost of Cells*) | Debt at a quarter of your debt ceiling or more, or a default | 5% | Intelligentsia, or Trade Unions |
| Corporate Data Access Exemption (*The Data Lobby*) | Hard times as for Too Large to Break, and the Industrialists powerful or in government | 10% | Industrialists |
| Whistleblower Protection Act (*Those Who Speak Up*) | Legitimacy below 40 | 5% | Intelligentsia |
| No-First-Strike Pledge (*Never the First*) | Nuclear Weapons researched and at peace, with a nuclear-armed rival or an anti-war movement at 5% support, and either no arsenal yet or a doctrine already at No First Use | 5% | Intelligentsia, or Trade Unions |
| Military Necessity Clause (*Necessity Knows No Law*) | At war, with the Armed Forces powerful or in government | 25% | Armed Forces |
| Small Donor Matching (*Pennies Against Millions*) | An election campaign, with the Industrialists or the Landowners powerful | 25% | Trade Unions, or Intelligentsia |

| Amendment (petition) | Comes when | Chance | Asked by |
|---|---|---|---|
| Religious Exemption Clause (*The Pulpit's Exception*) | The Devout are powerful or in government, and their approval is below zero | 5% | Devout |
| Spoils System (*Rotation in Office*) | One group holds at least half of your government's political strength | 2.5% | That group |
| Corporate Security Powers (*Company Men*) | A labor movement has 5% support, or unrest as for Order in the Courts; from Corporate Charters on | 5% | Industrialists |
| Civilian Oversight (*Complaints Against the Force*) | A civil rights movement has 5% support, or unrest as for Order in the Courts; from the Civil Rights Movement technology on | 5% | Intelligentsia |
| Estate Duties (*What the Dead Owe*) | At war, or a reformist or social democratic group in government; from Political Agitation on, on a law with private estates and no Birthright Endowment | 2.5%, and 25% more at war | Intelligentsia, or Trade Unions |
| Perpetual Dynastic Trusts (*Dynasty Trusts*) | Freedom of Testation, with the Industrialists powerful or in government; from Globalization on | 2.5% | Industrialists |
| Birthright Endowment (*The Duty and the Stake*) | A law carrying Estate Duties, which the Endowment replaces, with a reformist or social democratic group in government; from Universal Basic Income on | 2.5% | Trade Unions, or Intelligentsia |
| Undivided Farm Succession (*The Farm Is Not to Be Cut*) | Forced Heirship or Customary Inheritance, with the Rural Folk powerful or in government | 2.5% | Rural Folk |
| Suspension of Habeas Corpus (*The Writ Suspended*) | Statutory Courts or Constitutional Review, at war, with the Armed Forces powerful or in government | 25% | Armed Forces |
| Judicial Elections (*Let the People Choose Their Judges*) | Statutory Courts or Constitutional Review under an elected government, with a group led by a Jacksonian Democrat or populist powerful or in government | 5% | That group |
| Mixed Commercial Courts (*Courts for the Merchants*) | Religious Courts or Executive Tribunals, not under Isolationism, with the Industrialists or Petite Bourgeoisie powerful or in government | 2.5% | The stronger of the two |
| Jury Trials (*Twelve Good Citizens*) | Statutory Courts or Constitutional Review, with the Intelligentsia powerful or in government | 2.5% | Intelligentsia |
| Military Tribunals for Civilians (*Courts-Martial for Rebels*) | Executive Tribunals or Statutory Courts, unrest as for Order in the Courts, and the Armed Forces powerful or in government | 10% | Armed Forces |
| People's Tribunals (*Justice of the Party*) | Executive Tribunals under a Single-Party State, with a governing group led by a vanguardist or fascist | 5% | That group |
| Human-in-the-Loop Appeals (*A Human Signature*) | Algorithmic Jurisprudence, with legitimacy below 50 or the Intelligentsia or Trade Unions powerful or in government | 5% | Intelligentsia, or Trade Unions |

| Amendment (petition) | Comes when | Chance | Asked by |
|---|---|---|---|
| First Past the Post (*Two Parties Are Enough*) | Two parties whose groups hold at least 60% of your clout, on a voting law without it | 2.5% | The strongest group in either party |

Refusing Order in the Courts or Those Who Speak Up also costs up to 2 legitimacy for a few years, and refusing The Pulpit's Exception costs the Devout 5 approval instead of 3. The No-First-Strike Pledge holds your nuclear doctrine at No First Use, so its petition comes only when that changes nothing today.

## 20.4 Government type list

These government types follow from the governance laws in Collective Governance and New laws in base-game law groups.

| Government type | When | Ruler title |
|---|---|---|
| Direct Democracy | Collective Governance with a voting franchise | Speaker |
| Collective Leadership | Collective Governance with Single-Party State | Chairman or Chairwoman |
| Collegial Administration | Collective Governance with Technocracy | Coordinator |
| Free Federation | Collective Governance with Anarchy | Representative |
| Patrician Council | Collective Governance with Oligarchy or Organic Regulation | Syndic |
| Noble Commonwealth | As Patrician Council, with Traditional Vassalage | Marshal |
| Algorithmic Commons | Collective Governance with Algorithmic Governance | Steward |
| Collective Governance | Collective Governance under any other combination, until the laws settle | Speaker |
| Corporate Democracy | Neocameralism with a voting franchise | Director |
| Corporate Syndicalism | Neocameralism with Single-Party State | Director |
| Corporate Autocracy | Neocameralism without a voting franchise | Director |
| Algorithmic Directorate | Algorithmic Governance, except under Collective Governance or Neocameralism | Director |

## 20.5 City-state list

The 22 city-states from Releasable city-states, with the state each is carved from and its owner at the 1836 start. Dakar and Calabar start inside decentralized nations, which can't release them, so they become releasable once another country holds the city.

| City-state | State | Held at the start by |
|---|---|---|
| New York | New York | United States of America |
| San Francisco | California | Mexico |
| Singapore | Malaya | Great Britain |
| Hong Kong | Eastern Guangdong | Great Qing |
| Shanghai | Suzhou | Great Qing |
| Constantinople | Eastern Thrace | Ottoman Empire |
| Macau | Western Guangdong | Portugal |
| Kuwait | Basra | Ottoman Empire |

| City-state | State | Held at the start by |
|---|---|---|
| Fiume | Croatia | Croatia-Slavonia |
| Bombay | Bombay | East India Company |
| Goa | Bombay | Portugal |
| Pondicherry | Carnatic | France |
| Aden | Yemen | Lahej |
| Dubai | Trucial Coast | Trucial States |
| Tangier | al Rif | Morocco |
| Lagos | Yoruba States | Oyo |
| Dakar | Senegal | Cayor |
| Djibouti | Eritrea | Aussa |
| Calabar | Niger Delta | Igbo |
| Guayaquil | Ecuador | Ecuador |
| Cartagena | Antioquia | New Granada |
| Buenos Aires | Buenos Aires | Argentine Confederation |

## 20.6 Combat unit list

The mod's battalion types by unit group, with the technology that unlocks each. How the new units fight and what they cost is in New combat units.

| Group | Unit | Technology | Era | Battalion size |
|---|---|---|---|---|
| Infantry | Armored Infantry | Combined Arms | 6 | 1000 |
| Infantry | Robotic Soldiers | Autonomous Weapons | 11 | 600 |
| Infantry | Swarm Bots | Swarm Technology | 11 | 400 |
| Infantry | Utility Fog Phalanx | Molecular Assemblers | 13 | 250 |
| Marines | Combined Arms Marines | Combined Arms | 6 | 1000 |
| Marines | Stealth Marines | Stealth Technology | 8 | 800 |
| Marines | Networked Marines | Joint All-Domain Command and Control (JADC2) | 10 | 600 |
| Marines | Bioenhanced Marines | Bioenhanced Soldiers | 12 | 500 |
| Artillery | Motorized Artillery | Motorized Artillery | 6 | 1000 |
| Artillery | Guided Artillery Projectiles | Guided Missiles | 7 | 1000 |
| Artillery | Networked Guided Artillery Projectiles | Network Centric Warfare | 9 | 800 |
| Artillery | Orbital Bombardment | Space Militarization | 11 | 100 |
| Artillery | Orbital Precision Antimatter Strikes | Antimatter Production | 13 | 100 |
| Cavalry | Modern Scout Tank | Inertial Navigation Systems | 7 | 1000 |
| Cavalry | Stealth Reconnaissance Drone Carrier | Rapid Deployment Forces | 9 | 1000 |
| Cavalry | Holographic Ambush Combat Vehicle | Augmented Reality Warfare | 11 | 1000 |
| Heavy Tanks | Main Battle Tanks | Guided Missiles | 7 | 1000 |
| Heavy Tanks | Reactive Armor Tank | Advanced Materials Armor | 8 | 1000 |
| Heavy Tanks | Railgun Tank | Hypersonic Weapons | 10 | 1000 |

| Group | Unit | Technology | Era | Battalion size |
|---|---|---|---|---|
| Heavy Tanks | DEW Tank | Compact Fusion Reactors | 12 | 1000 |
| Aircraft | Propeller Aircraft | Bombing Aircraft | 6 | 200 |
| Aircraft | Jet-Powered Aircraft | Jet Engine Technology | 7 | 200 |
| Aircraft | Stealth Aircraft | Stealth Technology | 8 | 200 |
| Aircraft | Orbital Tactical Vehicles | Space Militarization | 11 | 100 |
| Aircraft | Orbital Weapons Platforms | Orbital Weapon Platforms | 12 | 100 |

## 20.7 Ship list

The mod's ship types in the order they arrive, with the technology that makes each one obsolete. See Ships and ship modifications.

| Ship | Class | Role | Technology | Era | Obsolete with |
|---|---|---|---|---|---|
| Fleet Carrier | Cruisers | Carrier | Bombing Aircraft | 6 | Nuclear Energy |
| Amphibious Assault Ship | Cruisers | Transport | Combined Arms | 6 | Precision Guided Munitions |
| ASW Destroyer | Torpedo Craft | Escort | Sonar | 6 | Network Centric Warfare |
| Nuclear Submarine | Torpedo Craft | Submarine | Advanced Submarine Technology | 7 | Rapid Deployment Forces |
| Guided Missile Ship | Torpedo Craft | Escort | Guided Missiles | 7 | Directed Energy Weapons |
| Nuclear Supercarrier | Cruisers | Carrier | Nuclear Energy | 7 | Hypersonic Weapons |
| Marine Assault Ship | Cruisers | Transport | Precision Guided Munitions | 8 | Rapid Deployment Forces |
| Arsenal Ship | Capital Ships | Battle line | Network Centric Warfare | 9 | Hypersonic Weapons |
| Stealth Battlecruiser | Capital Ships | Battle line | Network Centric Warfare | 9 | Hypersonic Weapons |
| Networked Air Defense Cruiser | Cruisers | Escort | Network Centric Warfare | 9 | Directed Energy Weapons |
| Expeditionary Fast Transport | Cruisers | Transport | Network Centric Warfare | 9 | Orbital Weapon Platforms |
| Stealth Attack Submarine | Torpedo Craft | Submarine | Rapid Deployment Forces | 9 | Directed Energy Weapons |
| Expeditionary Sea Base | Cruisers | Transport | Rapid Deployment Forces | 9 | Orbital Weapon Platforms |
| Hypersonic Missile Platform | Capital Ships | Battle line | Hypersonic Weapons | 10 | Antimatter Production |
| Pulsed Laser Escort | Cruisers | Escort | Directed Energy Weapons | 11 | |
| Swarm Coordination Vessel | Cruisers | Escort | Swarm Technology | 11 | |
| Autonomous Swarm Submarine | Torpedo Craft | Submarine | Swarm Technology | 11 | |
| Antimatter Battleship | Capital Ships | Battle line | Antimatter Production | 13 | |

| Ship | Class | Role | Technology | Era | Obsolete with |
|---|---|---|---|---|---|
| Orbital Support Mothership | Cruisers | Transport | Orbital Weapon Platforms | 12 | |

## 20.8 Ship utility modification list

The eight utility modifications the mod adds to the ship designer, and the new ships that can mount them.

| Utility modification | Technology | Effect | Mounted on |
|---|---|---|---|
| Sonar Suite | Sonar | +50% accuracy against every submarine, +100 detection | ASW Destroyer, the three new submarines |
| Ballistic Missile Bay | Precision Guided Munitions | +50% blockade strength, +15% hull damage | Guided Missile Ship, Arsenal Ship, Hypersonic Missile Platform, Antimatter Battleship |
| Underway Replenishment | Containerization | +100 supply capacity, +25% supply efficiency, +30% distance from port | Fleet Carrier, Nuclear Supercarrier, Arsenal Ship, Stealth Battlecruiser |
| Aegis Air Defense | Network Centric Warfare | +200 screening, +30% accuracy against Guided Missile Ships, +20% against Arsenal Ships | Guided Missile Ship, Networked Air Defense Cruiser |
| Helicopter Pad | Rapid Deployment Forces | +0.1 marine capacity, +0.2 carrying capacity, +15% naval invasion efficiency | The five transports |
| DEW Point Defense | Directed Energy Weapons | −25% vulnerability, +200 screening | Nuclear Supercarrier, Arsenal Ship, Stealth Battlecruiser, Hypersonic Missile Platform, Pulsed Laser Escort, Swarm Coordination Vessel, Antimatter Battleship, Orbital Support Mothership |
| Drone Complement | Swarm Technology | +30% accuracy, +25% screening, +200 detection | Nuclear Supercarrier, Expeditionary Sea Base, Stealth Battlecruiser, Hypersonic Missile Platform, Pulsed Laser Escort, Swarm Coordination Vessel, Antimatter Battleship, Orbital Support Mothership, the two later submarines |
| Composite Armor Plating | Modern Material Science | +200 armor, −15% hull damage and −20% critical hits taken | Pulsed Laser Escort, Swarm Coordination Vessel, Antimatter Battleship |

## 20.9 Mobilization option list

The mod's mobilization options with their unlocks and effects. The transport and medical options, and which of them exclude each other, are in Mobilization options. The principles come from your power bloc (New principle groups); the augmentation laws are in Rights and society laws.

| Option | Unlocked by | Effect |
|---|---|---|
| Logistical Support, then Extensive, Modern and Advanced Logistical Support | Military Statistics, Central Planning, Reconnaissance Satellites, Rapid Deployment Forces; each needs the one before | −40% attrition risk, then a further −30%, −15% and −10% |
| Home Communications | Cellular Networks | +25% morale recovery |
| Robotic Assistance | Network Centric Warfare | +25% morale recovery, −20% morale loss, +5% offense and defense |
| Coffee, Appliances | The good on your market | +10% morale recovery |
| Air Transport | Commercial Aviation | +150% movement, +100% mobilization speed |
| Space Transport | Rapid Deployment Forces | +300% movement and mobilization speed |
| Entrenchment | Engineering and Logistics principle, tier III+ | +25% defense, half the provinces lost and captured, −50% movement |
| Space Recon | Reconnaissance Satellites | +100% occupation, more surprise maneuvers |
| Molecular Scanners | Molecular Assemblers | +100% occupation |
| Radar Equipment | Radar | +10% defense, +5% offense, −10% morale loss |
| Night Vision Gear | Infrared Night Vision | +5% offense and defense, −10% morale loss, +10% morale recovery |
| Missile Defense System | Missile Defense Systems | +20% defense, +5% offense, −10% morale loss |
| Directed Energy Defenses | Directed Energy Defenses | +20% defense, +5% offense |
| Cyberwarfare Team | Cyber Warfare | +20% morale damage, −20% morale loss, +5% offense and defense |
| Electronic Warfare Team | Electronic Warfare | +20% morale damage, +10% offense, +5% defense |
| Exoskeleton Suits | Advanced Body Armor | +15% offense and defense |
| Medevac Helicopters | Advanced Military Aircraft | +80% recovery rate |
| Voluntary Bioenhancement | Regulated Augmentation Market or Unrestricted Augmentation law | +25% offense and defense |
| Mandatory Bioenhancement | Mandatory Augmentation law | +50% offense and defense |
| Jungle Combat Training | Military Training principle, tier III+ | +25% offense and defense in forest and hazardous terrain |
| Mountain Combat Training | Military Training principle, tier III+ | +25% offense and defense in hills, mountains and hazardous terrain |
| Amphibious Warfare Training | Engineering and Logistics principle, tier III+ | +25% offense and defense in water terrain |
| Flight Simulators | Military Training principle, tier V | +10% offense and defense for Jet-Powered Aircraft and later |

## 20.10 Military base production method list

The Military Base's production methods, with the technology that unlocks each and its effect for a fully staffed level. See Military bases.

| Group | Production method | Unlocked by | Effect per level |
|---|---|---|---|
| Fortification Type | Earthwork Fortifications | Trench Works | +1 fortification level |
| Fortification Type | Concrete Bunkers | Concrete Fortifications | +2 fortification level, uses steel |

| Group | Production method | Unlocked by | Effect per level |
|---|---|---|---|
| Fortification Type | Hardened Positions | Guided Missiles | +3 fortification level, uses steel and explosives |
| Base Purpose | Garrison Duty | Trench Works | −15% turmoil effects, monthly war support |
| Base Purpose | Territorial Defense | Trench Works | +1 fortification level, less monthly war support |
| Base Purpose | Logistics Hub | Trench Works | +10 infrastructure, −3% supply consumption for your whole army |
| Missile Defense | Missile Defense Battery | Missile Defense Systems | +2% Nuclear Strike Defense Chance in this state |
| Missile Defense | Directed Energy Point Defense | Directed Energy Defenses | +3% Nuclear Strike Defense Chance in this state |